POWER SYSTEM OPERATION & CONTROL

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Module 1 Introduction

Lecture 1 Modern Power Systems

Objectives

In this lecture you will learn the following

- Evolution of power systems: significant milestones.
- Structure of a bulk power system network.
- What is a synchronous grid? Why is grid frequency standardised?

Evolution

Electrical Technology was founded on the remarkable discovery by Faraday that a changing magnetic flux creates an electric field. Out of that discovery, grew the largest and most complex engineering achievement of man: the electric power system. Indeed, life without electricity is now unimaginable. Electric power systems form the basic infrastructure of a country. Even as we read this, electrical energy is being produced at rates in excess of hundreds of giga-watts (1 GW = 1,000,000,000 W). Giant rotors spinning at speeds up to 3000 rotations per minute bring us the energy stored in the potential energy of water, or in fossil fuels. Yet we notice electricity only when the lights go out!

While the basic features of the electrical power system have remained practically unchanged in the past century, there are some significant milestones in the evolution of electrical power systems:

- First complete DC power system built by Edison (1882): Incandescent lamps supplied by steam driven DC generators (electrical cable system at 110V). 59 customers spread over an approximate area with 1.5 km radius.
- Development of transformers led to supersession of DC systems by AC systems **(why?)**.
- Nikola Tesla - polyphase induction motors: led to development of AC 3 phase systems.
- Interconnection of systems led to standardization of frequency **(why?)**, 60 Hz in North America and 50 Hz in most other countries. Use of higher and higher voltage levels (up to 1000 kV line-line rms AC). Standardization of voltage levels.
- Development of Mercury Arc Valves, and subsequently thyristors led to high voltage dc transmission (HVDC): DC transmission suited for very long distance bulk transmission and underwater cable links. First commercial DC link in 1954.
- Several new developments : Gas turbines, static excitation systems, fast acting circuit breakers, microprocessor based relaying, use of communication technologies etc.
- Need for better utilisation and operation of AC transmission systems by use of high power electronic converters suggested. Several such converters are now in operation.

Structure

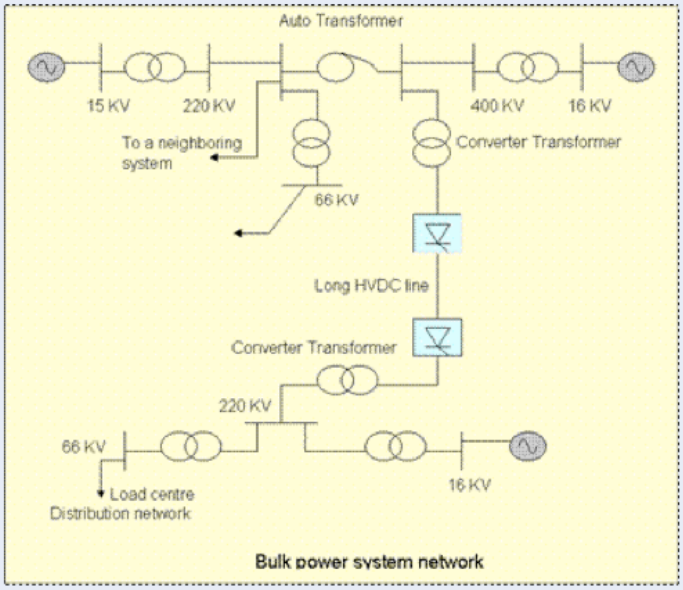
Present day power systems are characterised by:

- AC generation (3 phase synchronous machines using hydro, fossil and nuclear as primary energy sources)
- AC transmission (a limited number of DC transmission links are present in some systems)
- 3 phase AC utilization (some loads are single phase, but they are usually distributed equally in all phases to obtain a balanced system)

The transmission network consists of an extra high voltage (220 kV and above) lines interconnecting large generators to major load centers. These lines are capable of transmitting large amounts of power over large distances (1000 km is not unusual!).

For example, several 400 kV lines may be required to transmit, say, 2000 MW from a large thermal power plant to a distant load center.

Note : high voltage dc lines are suitable if distances are very large (we shall see why later in the course). Generators, typically having terminal voltages of around 15 kV and are connected to these EHV lines via transformers. Interconnecting transformers connect EHV lines at different voltage levels (they are usually auto-transformers - **why?**).



Bulk power system network

A sub-transmission network, consisting of transmission lines at 66 kV and 132 kV are used to deliver power received from EHV network to various distribution substations. Finally, a distribution network consisting of lines at less than 66 kV are used to transfer power to each consumer.

A large power system may also be connected to neighbouring power systems via ac transmission lines or dc links. The power exchange between such neighbours is usually *regulated* (you will learn how this is done later).

Power Grids

"*Except for a few islands and some small isolated systems, the entire electric grid in North America is really one big circuit. The humble wall outlet is actually a gateway to one of the largest and most complex objects ever built. The grid encompasses billions of individual components, tens of millions of miles of wire and thousands of individual generators* " - **Thomas Overbye, Reengineering the Electric Grid, American Scientist, 2000, Vol. 88, Iss. 3.**

The Indian Power Grid is not quite as big as North American power systems, but still has a large generating capacity of about 100 GW (100,000 MW). It is actually made up of *three* synchronised grids:

- Northern Regional Grid
- Southern Regional Grid
- Western + Eastern + North-eastern Regional (WR-ER-NER) Grid

Update: A very recent development has been the synchronisation of the WR-ER-NER grid with the NR grid (August 2006)

A synchronised grid implies that the generators in that grid are connected to one another by at least one AC transmission path. This also means that the synchronous generators in that grid are all operating at the same electrical frequency (what happens if they do not ?).

For a well designed and operated system, inter-connected synchronous generators are seen to "stick together" in synchronism (by virtue of the physical equations which govern their motion). However they may lose synchronism if subjected to large disturbances. A generator when connected to a power system has to be connected by a special procedure known as 'synchronisation'. You will learn some of these things in forthcoming lectures.

A load can be connected in a synchronous grid by simply connecting it in shunt (after suitable voltage transformation using transformation and ensuring that the system is capable of servicing that extra load).

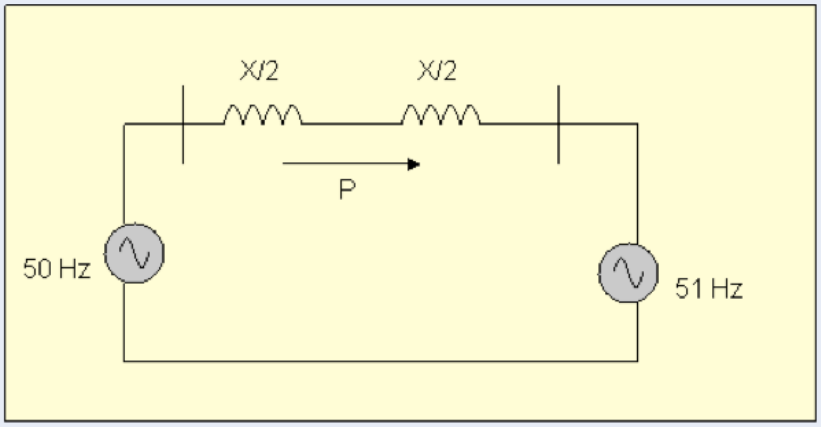
Did you know? In the WR-ER-NER grid, a synchronous machine in Panandhro Thermal Power Station (Kutch, Gujarat) operates "in synchronism" with Ranganadi, a hydro-generator in Arunachal Pradesh, a distance of more than 2000 km!

There may be a very small number of induction generators in a synchronous grid. Induction generators do not run at synchronous speed and are sometimes used in conjunction with wind turbines. However, since induction generators always absorb reactive power, they are limited to small applications wherein reactive power is supplied externally by capacitors and synchronous generators.

The three synchronous grids in India have the same nominal frequency (50Hz) but may not operate exactly at the same frequency. They are operated independently. However, the synchronous grids are connected by ***DC links*** which allow for a limited and controlled power exchange. It is planned to make a national grid consisting of large capacity links between all grids in the future (**why is there such a high degree of interconnection?**).

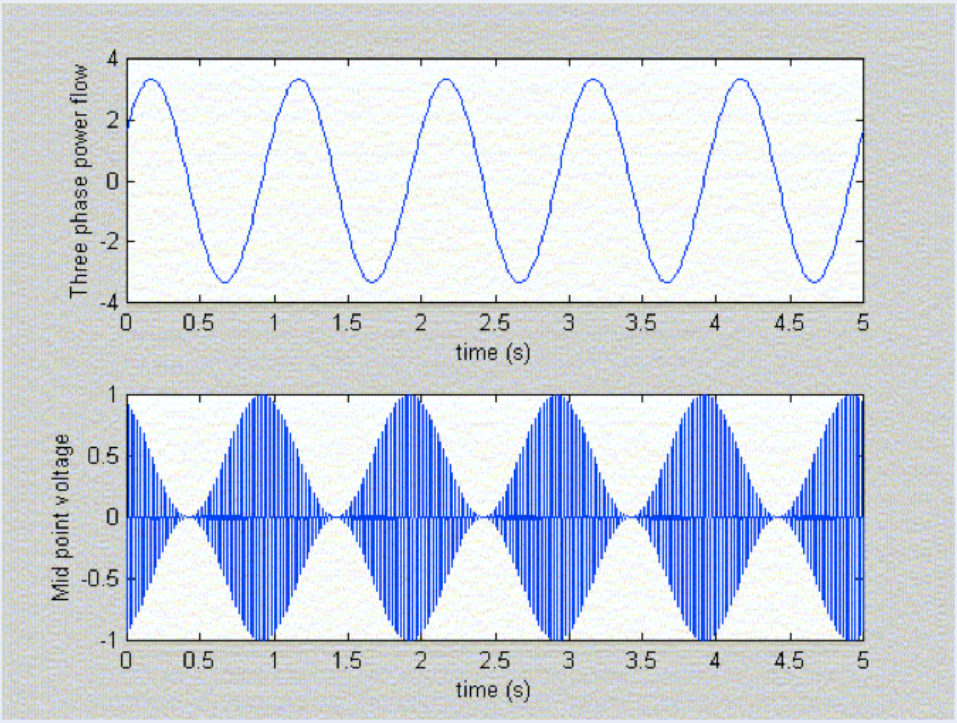
Example

Consider 2 generators connected by a transmission line. If the frequency of both sources is not equal but their voltage magnitude are the same, what will be the nature of power flow P ? Is this acceptable?



Answer : Three phase ac power flow through a transmission line is proportional to the sine of the phase angular difference between the 2 sources. If the sources are of different frequencies, this phase angular difference can be thought of as continuously changing. Therefore, power will pulsate at the difference frequency. Such power pulsations are unacceptable as they can damage the shafts of the generators. If such a situation arises in a power system, the generators are immediately disconnected by "out-of step" relays. You can also show that voltage magnitude at the mid point of the line will also pulsate.

This is shown below. It is assumed that the voltage magnitude at both ends is 1.0 pu and X=0.3 pu.



Note : In the above example, power oscillates at 1 Hz which would otherwise have remained constant

Thus an important requirement of ac system operation is that all inter-connected generators operate at the same frequency. Therefore, when electrical power systems were first interconnected (we shall see why in the next lecture), it was important to *standardize* the frequency to one value. Electrical equipments are designed to operate at that frequency.

If different equipment are designed for, or operated at the different frequencies, it would be impossible to connect them unless frequency converters are used.

In India, the standard operating frequency is 50 Hz.

Recap

In this lecture you have learnt the following

- Power systems have evolved from being DC systems to highly interconnected three phase AC systems
- Bulk power systems consist of a backbone of a extra high voltage transmission grid with generators pooling their generation into it. The pooled power is withdrawn by loads at lower voltage levels.
- AC interconnected power systems have an important property: frequency all over the grid is the same in steady state.

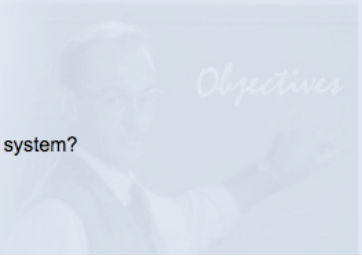
Congratulations, you have finished Lecture 1. To view the next lecture select it from the left hand side menu of the page

Lecture 2 Why Make Interconnections?

Objectives

In this lecture you will learn the following

- Why are power systems interconnected?
- What is the concept of reliability? Why is it more economical to interconnect a power system?
- Identifying the major elements of a practical interconnected power system.



Reliability and Economy are the main reasons for interconnecting power systems

Reliability implies that with a large interconnected grid, the loss of a system component like a major transmission line or generator will have minor impact on system. When one device fails, another one makes up for the loss. In a practical grid, there exists more than one path connecting a load to each generator. Thus the loss of one transmission line or tripping of one generator does not usually interrupt power to a load.

Economy implies that electricity can be obtained from where it is cheap. It is more economical to operate large generators 24 hours a day at full capacity (base load stations), catering not to a particular load, but pooling power into the grid and to be used by many connected loads. A few generators which can be started almost instantaneously can then act as reserves to cater to sudden increases in load (peak load stations). Generally steam stations are run as base load stations while reservoir based hydro-stations and gas stations act as peak load stations. The possibility of sharing reserves in interconnected systems also results in smaller reserve requirements. Different regions in a grid may face peak demand at different times of the day. Therefore the total system peak demand is smaller than the sum of the individual peaks of various regions.

However, extensive interconnections also mean that a disturbance in one part of the system may quickly spread to the entire system, leading to tripping of loads/generators, and may even make interconnected system operation unviable. For example, some large disturbances may make it impossible for generators to run in synchronism. In case interconnected system operation becomes unviable, the system must "gracefully" split into smaller systems (islands). However, if the control and protection systems are inadequate to face this eventuality, a complete blackout may also occur, leading to loss of service to millions of consumers.

However, grid failures are rare ("the lights are always on") and one may justifiably ask a question : how does the whole system work so well? After all, for well designed power systems, power is available on demand and can be obtained by simply "paralleling" the load on the grid. Similarly, a synchronous machine driven by a prime-mover can be synchronised with a grid and may supply power to it. Of course, all this is subject to certain constraints which we shall study in the next module.

A great deal of prior planning and control during operation is required to make an inter-connected network capable of catering to a certain level of power flow and prevent blackouts.

Example of a large power system

Western Region of the WR-ER-NER Synchronous Grid: A description

This region has the largest inter-connected network in the country, comprising the states of Chattisgarh, Goa, Gujarat, Maharashtra, Madhya Pradesh, besides Union territories of Daman and Diu, and Dadra and Nagar Haveli. In 2002, the installed capacity was 31.5 GW with 13.8% hydro, 66% thermal- coal, 15.9% gas/liquid fuel, 2.4% nuclear and wind 1.9%.

The generation is owned by State Electricity Boards, NTPC(National Thermal Power Corporation) and NPC(Nuclear power corporation), besides some Independent Power Producers (IPPs).

Generation

The western region has major generating stations of Korba TPS (Thermal Power Station)-2100 MW, Vindyachal TPS–2260 MW, Korba (West) –840 MW, Korba (East) –400 MW, Sanjay Gandhi TPS –820 MW, Chandrapur TPS –2340 MW, Amarkantak TPS –300 MW etc., located in the eastern part of the region.

The major hydro stations in the region are at Koyna –1960 MW, Bansagar –315 MW, Ukai –300 MW, Tata –444 MW. The western part of the grid has thermal generating stations of Trombay –1330 MW, Wanakbori –1470 MW, Ukai –850 MW, Nasik –910 MW, and nuclear generating units at Tarapur –320 MW and Kakrapur –440 MW.

The major load centers in the region are located at the western part of the region and bulk of the power flow is from eastern to western part of the region.

Bulk (EHV) Transmission

The 400 kV system consists of three main corridors.

The upper corridor links Vindyachal TPS-Itarsi via Satna-Bina-Bhopal. The second evacuation corridor is from Korba TPS-Bhilai/Raipur and bifurcates thereafter into middle and lower (third) corridors.

The middle corridor links Bhilai-Koradi-Bhusawal-Bableshwar-Padghe-Kalwa.

The lower (third) corridor consists of Raipur-Chandrapur-Parli-Karad-Lonikhand and Kalwa/Padghe linked together.

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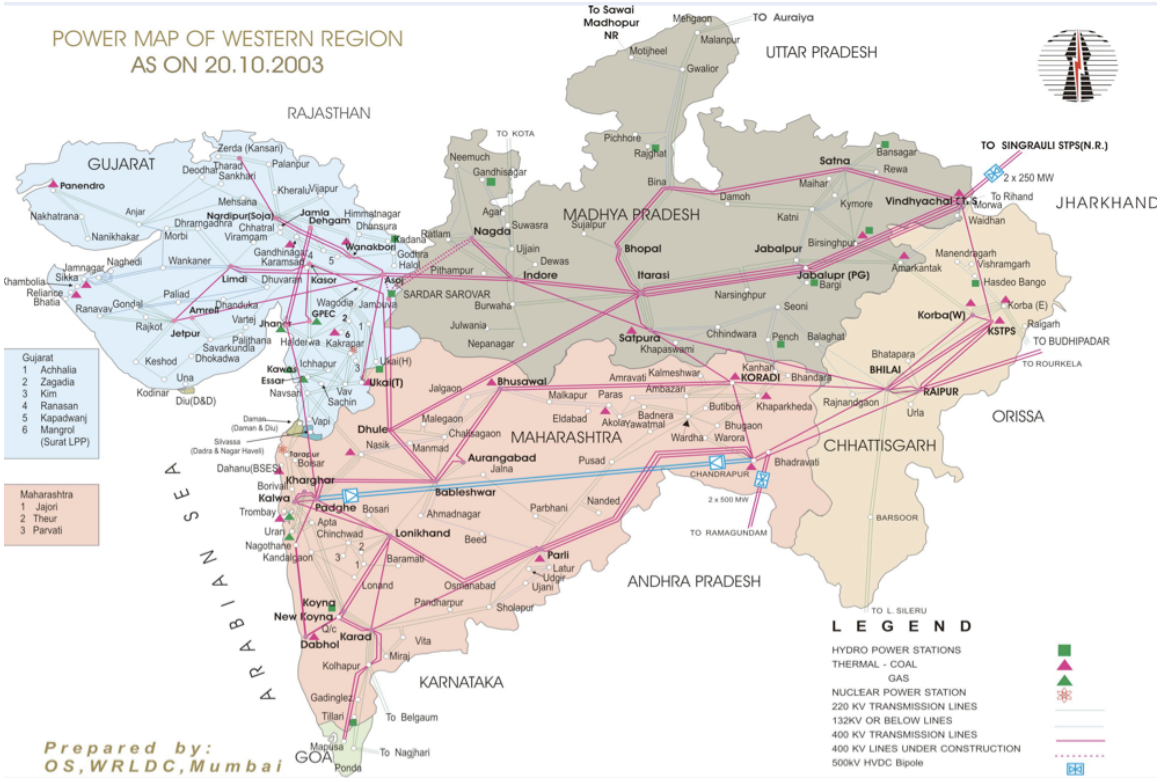
The upper and middle corridor are interlinked by 400 kV Itarsi-Satpura-Bhilai, Itarsi-Dhule links and also Vindyachal TPS-Korba TPS links. Further North MP loads are fed on 400 kV Vindyachal-Satna-Bina links and 400 kV Itarsi-Bhopal-Bina links. The generation of Koyna stage IV is connected to Karad and Lonikhand by 400 kV lines.

Another important bulk transmission corridor in the region is ± 500 kV, 1500 MW HVDC from Chandrapur-Padghe. The Western region is connected to southern region and northern region through DC systems *(asynchronous links)* at Bhadrawati (1000 MW) and Vindyachal (500 MW) respectively. Note that the operating frequency of the northern and southern grids need not be the same as the western region.

The western regional grid operates *in synchronism* with the eastern grid and is interconnected to it by 400 kV lines between Raipur and Rourkela.

Can you identify the lines and generators mentioned in the map of western region of the WR-ER-NER grid as shown on the right- only major lines are shown ? (courtesy: WRLDC Mumbai, http://www.wrldc.com/WR%20map-20Oct03.jpg)

(click to enlarge figure)



Recap

In this lecture you have learnt the following

- Power Systems are interconnected to ensure reliability and economy
- Reliability ensures that a loss of an element has a minor impact on system operation
- Interconnected power systems can utilise the economy associated with larger power plants (as opposed to many small ones). Also diversity in time of peak demand within a grid ensures that total peak demand is less than sum of peak demand of individual loads.
- A practical power system in our country is studied as an example.

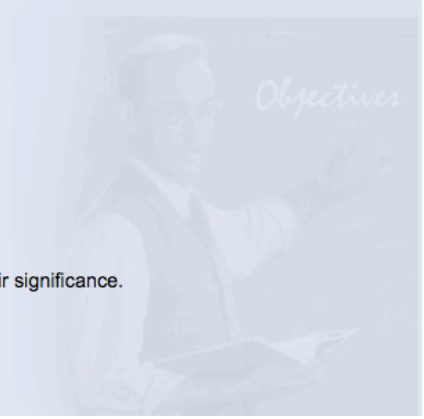
Congratulations, you have finished Lecture 2. To view the next lecture select it from the left hand side menu of the page.

Lecture 3 Power System Controls

Objectives

In this lecture you will learn the following

- Operational Objectives of a Power System.
- Hierarchical Control in Power Systems.
- Issues of ownership and co-ordination.
- Definition of various operating states, nature of various control actions and their significance.
- An overview of what you will learn in this course.



Operational Objectives

"*Most people give little thought to the source of power that comes out of the electrical outlet. And why should they? The American electric power system was designed as the ultimate plug and play convenience, seemingly as dependable as the sun rising in the morning.* " - **Thomas Overbye, Re-engineering the Electric Grid, American Scientist, 2000, Vol. 88, Iss. 3.**

Before we discuss issues in control of a power system, it is important to understand the objectives.

The operational objectives of a power system are:

- Continual matching of load demand (plug and play!): Electrical energy cannot be conveniently stored in sufficient quantities. Therefore a readily available reserve of generation should be available and controlled. However in many developing countries, there are shortages of generation resulting in load curtailment.
- The power system should provide for a certain level of reliability and quality (frequency and magnitude of voltages should lie in a narrow, pre-defined range).
- The electrical energy should be at low cost and have a low environmental impact.

Hierarchy of controls in a power system

Does a power system require continuous monitoring and control?

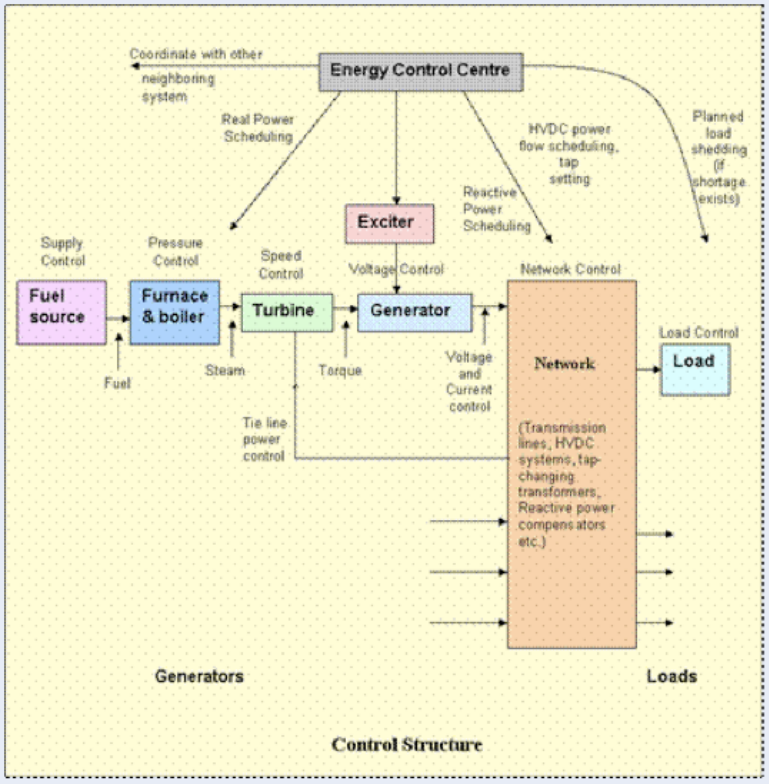
Yes. Like many other engineering systems, a power system operator has to continuously monitor the health of a power system and perform control actions *when necessary*. A power system has a large number of *automatic* controllers too. Manual actions may be necessary to supplement these controllers. In fact, there is a well defined hierarchy of controls which are deployed.

Equipments like generators have a wide array of automatic feedback control systems which not only serve to implement real and reactive power scheduling commands, but also regulate key parameters like voltage & frequency, prevent thermal overloads, over-speed etc. Most of the controllers are decentralised, i.e., they use locally available feedback and control a locally accessible quantity. An example is generator terminal voltage regulation by controlling its field voltage. Parameters of some decentralised controllers require system wide co-ordination (e.g. governors which control generator speed).

The reference values of some regulators can be set by other slow acting closed loop controllers. Some references are also set or can be changed by system or power plant operators for optimal or secure system operation.

A large power system may consist of autonomous areas ("control areas") which are controlled independently and exchange pre-decided amounts of power with other areas through interconnecting tie lines or DC links. However during transients and abnormal conditions, they are expected to act in co-ordination with each other.

In the figure given on the right, we illustrate the various controls in a power system (this includes some important controllers but not *all* controllers).



Ownership and Co-ordination

We have seen in the previous slide that the reference of many controllers ('scheduling') are set by an energy control center.

The scheduling of references may have economic and technical consequences. For example, increasing the scheduled power of a costly generator while reducing that of a cheap one, will increase the cost of power. Moreover, large power plows beyond transmission system capability may endanger inter-connected system operation (we shall see why later in the course).

The question which then arises is : Who directs the whole show? In the previous slide, we assumed *all* directives emanate from an "energy control center" or an omnipotent "system operator". However, considering that ownership of generation, transmission and distribution systems maybe with several different entities, there may not be one energy control center, but several of them obeying a certain hierarchy with a strictly defined authority. Thus, ownership of power system components is an important issue in grid co-ordination.

A typical power system consists of several "control areas". In India, a control area can be a state. Most of the transmission and generation in an area is usually owned by state electricity boards. They have an obligation to serve customers in their respective areas and have complete authority over all activities in generation, transmission and distribution in their domain of operation ("vertically integrated utilities").

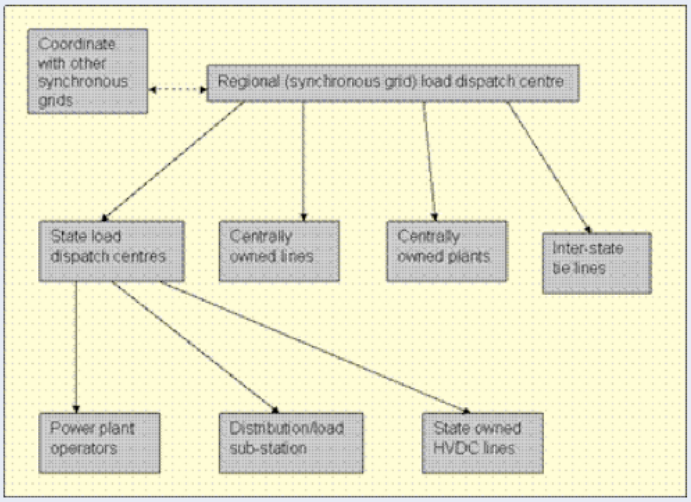
Ownership and Co-ordination (con td...)

Inter-state and some other major transmission links may be owned by another independent entity (e.g. Power Grid Corporation of India). At present, the vertically integrated utilities (state electricity boards) can import or export a pre-decided amount of power from neighboring states or generators owned by other entities like National Thermal Power Corporation or independent power producers.

However, this ownership structure is gradually changing. The functions and ownership of the hitherto vertical integrated utilities (like state electricity boards) are being divided by forming separate companies (GENeration COmpanies (GENCOs), a TRANSmission COmpany(TRANSCO) and DIStribution COmpanies(DISCOs)). While transmission will remain a natural monopoly, generation and distribution can be open to competition with private players. These new entities can have open access to the transmission system (subject to technical constraints). We will discuss these developments in detail later in the course.

The overall operational co-ordination in an area is done by a load dispatch center. Inter-area exchanges are coordinated by a central load dispatch center.

A synchronous grid made up of many control areas and many entities, i.e., state electricity boards, state owned or independent power producers and transmission companies, requires a high degree of co-ordination. In spite of diverse ownership, no part of an interconnected grid is totally independent of another. All entities are required to co-operate to run a grid smoothly. Planning, design and co-ordination issues are important and are looked after by a Coordinating Council (e.g. Western Regional Power Committee).



Operating states and nature of control actions

The state (or condition) of a power system can be judged from the answers to the following questions:

- Is the demanded load being met? (i.e., is there a real and reactive power balance?)
- Are all equipment within their current and voltage limits?
- Can the system withstand stresses due a possible contingency (leading to a loss of equipment)?

Definition of states and control actions

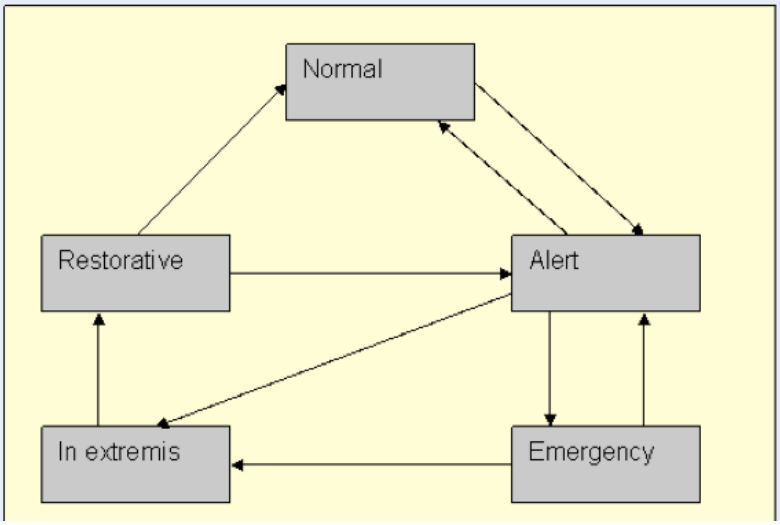
A *normal* (secure) state is the ideal operating condition, wherein all the equipment are operating within their design limits and the demanded load is being met. Also, the power system can withstand a contingency without violation of any of the constraints. This means that the power system has reserve capacity (generation and transmission) available in this state. If this reserve capacity is reduced, or the possibility of major multiple contingencies increases due to adverse weather, then the system is said to be in the *alert* (insecure) state. Although equipment are within their limits and load demand is met, the system is "weaker" and may not be able to withstand a contingency. **Preventive Control** actions like shifting generation (re-scheduling) are required to get the system back to the normal state.

Definition of states and control actions (con td...)

If preventive control actions do not succeed, a power system remains insecure (in the alert state). If a *contingency* occurs, the system may go into the *emergency* state where overloading of equipment (above the short term ratings of the equipment) occurs. The system can still be intact and can be brought back to the alert state by **Emergency Control** actions like fault tripping, generator tripping, load tripping, HVDC power control etc. If these measures do not work, integrated system operation becomes unviable and a major part of the system may be shutdown due to equipment outages (In *extremis* state). Load shedding and "graceful" or controlled system separation (islanding) is necessary to prevent spreading of disturbances and a total grid failure.

If there is a widespread or partial blackout, the surviving or restarted generators are connected to local loads;the restarted / surviving small power systems (islands) are reconnected to restore the power system to alert or normal state **(Restorative Control).**

Note that preventive and restorative actions are "directed" by a system operator and usually involve manual control actions. However, emergency control actions are usually done by fast acting control or protective relays



What will you learn in this course?

You must have already completed a first course in power engineering wherein you learnt about basic power system equipment, its modelling and power system analysis. In this course, we shift our attention to the following issues:

- What are the operational constraints due to equipment and stability?
- How is frequency and voltage control achieved in a coordinated but decentralised fashion?
- Can power flow in transmission paths be controlled and how?
- How is real and reactive power scheduling done ?
- How do Black-outs occur and what are preventive, emergency and restorative control procedures?

Recap

In this lecture you have learnt the following

- A power system has a wide array of automatic controllers which are also supplemented by manual scheduling actions.
- Hierarchy of controls and their co-ordination is strongly tied to the issue of ownership
- Various operating states in a power system, viz, normal, alert, emergency and in-extremis
- Nature of various control actions, viz, preventive, emergency and restorative

Congratulations, you have finished Module I.

Module 2

Equipment and Stability Constraints in System Operation

Lecture 4 Introduction

Objectives

In this lecture you will learn the following

- What are the nature of constraints faced in power system operation ?
- What are the various types of equipment constraints ?

Power System Operating Constraints

Can you use the 50 Hz power supply at your home to drive a 100 kW load? Obviously, the answer is no!

The reason is that the power supply in your house is not *designed* to handle such power. If you tried to connect a motor of that rating, you would find that the fuse (or miniature circuit breaker) will trip and disconnect the power supply. If it did not do so, the wiring would be damaged.

In a similar fashion, a power system is designed to handle several load demand scenarios. This design is done much in advance (planning stages) based on the expected demand, while keeping some reserve "margins" for situations in which one or more equipment is out of service. A planner will not *over-design* (i.e., have unreasonably large margins) since it will cost a lot of money.

However during operation, it is not possible to requisition or install equipment at short notice. Therefore, an operator is forced to ensure that the system is operated within the existing design constraints.

Nature of constraints

There are two types of constraints which limit the capability of a power system:

a) **Equipment Constraints:** An equipment must be operated within the specified ratings otherwise it may result in damage. Examples of such ratings are the maximum current handling capability of a conductor, the maximum voltage across an insulator before it breaks down etc. Equipment like generators may have a relatively large number of constraints. An equipment which is designed to have a larger capability is also costlier (e.g. a higher current ability will require one to use thicker conductors). Therefore, system and equipment designers do not over-design an equipment. Under abnormal or unforeseen situations, an equipment may get overloaded. If the overloading exceeds limits, the equipment is tripped out by protection systems.

b) **Stability Constraints:** A power system may not be able to cater to power flows beyond a certain point due to stability constraints. An unstable system is a one which cannot withstand disturbances, i.e., it may not settle to an equilibrium although a post-disturbance equilibrium condition may exist. This is due to the basic physical characteristics which define the behavior under transient conditions. Improvement of stability may require system reinforcement (like adding new transmission lines) and/or improving/augmenting existing automatic controllers. Inability to come to an equilibrium may *eventually* lead to equipment constraints being violated too. This will cause operation of protection systems.

Loss of equipment due to stability or equipment constraints may take the system into an emergency or in-extremis state wherein interconnected operation may become unviable.

Therefore, it is important to characterize the capability of the system to handle load power demands and power flows in a transmission network without violation of the above constraints.

Equipment Constraints:

There are two major equipment constraints:

1) Thermal: Excessive heat produced by current carrying conductors results in unacceptable sags in transmission lines and degradation of insulation in other equipment. Depending on the thermal time constants (note that the temperature does not jump instantaneously), an equipment may have larger short time thermal ratings.

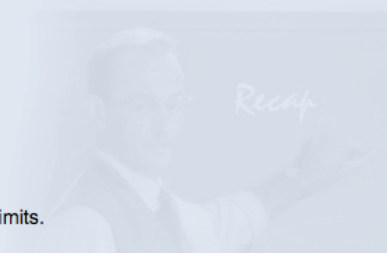
2) Dielectric: Over-voltages result in large electric fields causing dielectric breakdown. Dielectric breakdown may also occur due to aging or degradation of insulation due to thermal limit violations. Typically +/- 10% variation in the rated voltage is often permissible.

For mechanical equipment, parameters like steam pressure and temperature (which may restrict for instance, the rate of ramping the mechanical power in steam plants) have to be monitored to prevent overheating. Vibration of turbine blades also needs to be monitored especially during off-nominal frequency operation.
We shall now consider equipment constraints of 2 major power system components:
(1) Generators
(2) Transmission lines

While it is understood that every equipment has certain limits, we shall restrict our discussion to these two components in this course.

Recap

In this lecture you have learnt the following



- A power system has to operate within its capability
- There are two types of constraints : equipment and stability
- Electrical equipment constraints are mainly those related to dielectric and thermal limits.

Lecture 5 Generator Constraints

Objectives

In this lecture you will learn the following

- Nature of generator constraints.
- Generator Capability curves.
- Typical rating of a large synchronous generator.

The Generator Capability Curves

Generator Limits: Generator Capability is constrained mainly by the following limits:

- **Voltage limits:** The terminal voltage of a generator is limited due to 2 reasons : 1) Dielectric 2) Heating in core due to excess magnetic flux. However, the maximum continuous limit due to excess flux is lower than that due to dielectric breakdown considerations. Therefore the limit due to excess flux is the "determining" limit. The flux in the core is also affected by the frequency (core flux is proportional to voltage/frequency).
- **Armature Winding (heating) Limit:** Armature winding heating results due to the resistive loss in armature windings.
- **Field Winding (heating) Limit:** Ohmic loss and consequent heating in the field winding, imposes a restriction on the maximum field current. Since field winding current is proportional to the field voltage (after electrical transients have died down), this limit is equivalent to a field voltage limit. Field current is higher when the generator supplies reactive power and is over-excited.
- **Core-end heating limit:** Core-end heating results when field current is low (under-excitation). During under-excitation conditions, the axial flux in the end region is enhanced. This results in heating which may limit the capability of a generator.

The heating limits are dependent on the efficacy of cooling. A higher pressure of the cooling medium (hydrogen) results in higher heating limits. Armature winding current limit is essentially an MVA limit since terminal voltage magnitude is maintained near the rated value. Therefore armature winding limit locus is a circle on the P-Q plane with origin as the center**(why ?)**

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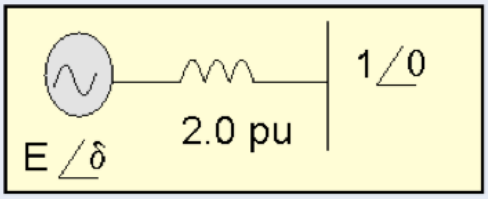
In practice, the field current of a generator is measured or estimated, and the excitation system of a generator is controlled so as to avoid exceeding field current limits. Excitation system controls are covered later in the course.

We shall now visualize a typical capability curve of a generator by an **example**.

Consider a generator connected to a fixed voltage bus (1.0 pu) as shown in the figure below. The synchronous reactance of the generator is 2.0 pu.



Draw the capability curve of the generator.

Solution:

The **armature current limit** implies that : $I < I_{max}$, therefore,
$VI < VI_{max}$, which implies that $S < VI_{max}$

Since terminal voltage (V) is usually near its rated value (1.0 pu) ,
VI_{max} is taken to be equal to the MVA rating of the generator.

If real and reactive power of a generator (P and Q) are expressed in per-unit on the MVA rating of the generator, then

$$S = \sqrt{(P^2 + Q^2)} < 1$$

Thus the armature current limit locus is a circle on the P-Q plane.

The **field winding limit** implies that : $I_f < I_{fmax}$

Since the field current is proportional to the field voltage
(and therefore the 'internal voltage' E) in steady state it follows that

$E < E_{max}$.

Since $P = \frac{EV\sin\delta}{X}$

and $Q = \frac{EV\cos\delta}{X} - \frac{V^2}{X}$, it is clear that

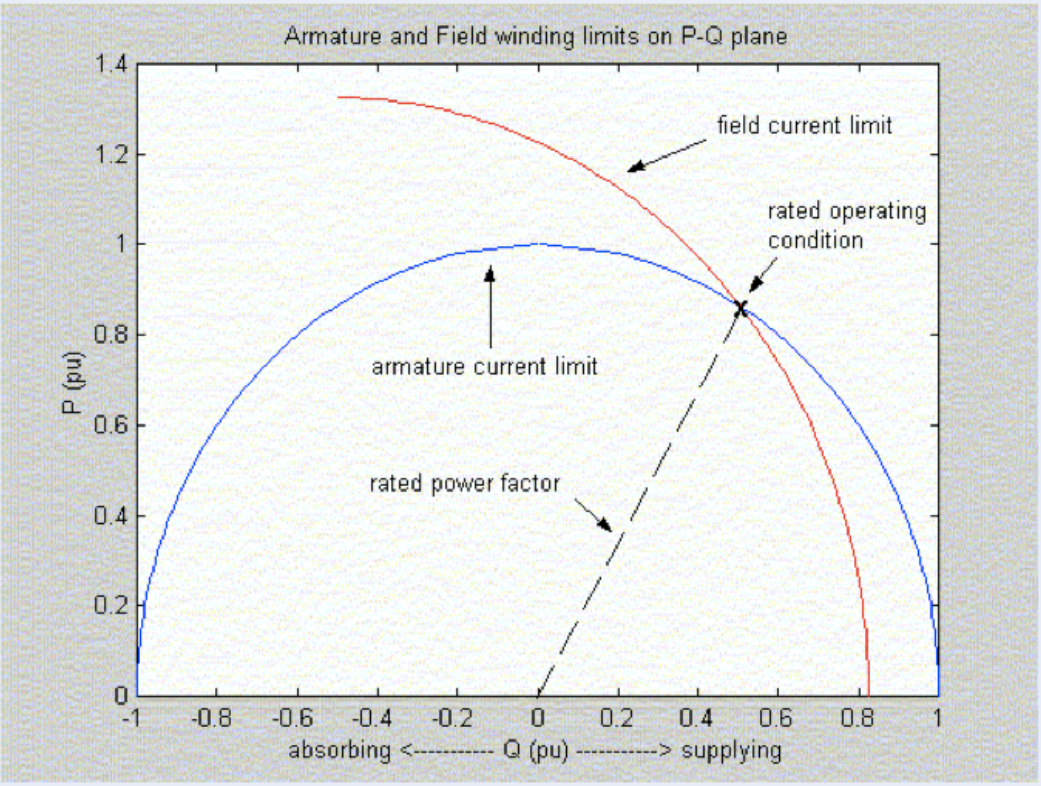
$$P^2 + (Q + \frac{V^2}{X})^2 = (\frac{EV}{X})^2,$$

and since voltage V is near about 1.0 pu, this implies that at the limit :

$$P^2 + (Q + \frac{1}{X})^2 = (\frac{E_{max}}{X})^2$$

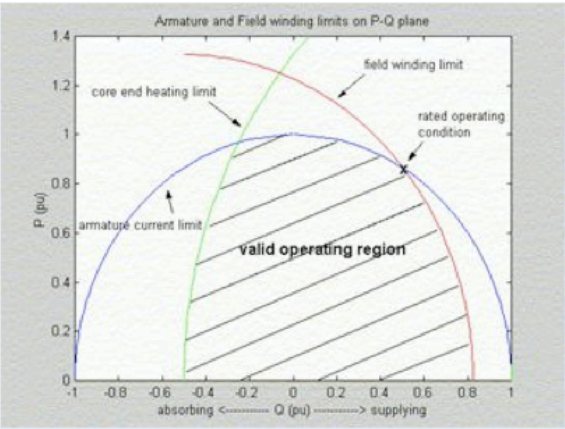
Typically, the field winding limit locus and armature winding limit locus intersect at a point at which power factor equals the rated power factor (say 0.85).

With the value of X = 2.0 pu, it is possible to plot these loci, as shown below :



The core end heating limit is caused by axial flux which tends to be large when Q is absorbed by a generator. It is difficult to obtain a direct expression for this limit in terms of P and Q, since it involves obtaining the flux distribution in the generator for various operating points. We state without proof that it is approximated by a circular locus in the P-Q plane as shown in the figure on the right.

The capability in the Q absorption region is more severely limited than in the Q supply region. Typically, the core end heating limit locus and armature winding limit locus intersect at a point at which power factor equals about 0.95 leading. The hashed region in the figure below defines the valid operating region of a synchronous generator.



(click to enlarge)

Remarks:

1. For simplicity, the limits shown above are for a cylindrical rotor machine (no saliency). The locus of constant excitation voltage and constant MVA are not circles in the P-Q plane if saliency is present.

2. Only limits determined by the generator *alone* are depicted above. When a generator is connected to other generators and loads, stability may also constrain the maximum power as well as the minimum excitation voltage. We shall see this later in this module. Moreover, the prime-mover which drives the generator itself will have a maximum power limit.

Voltage Limits

As mentioned in the previous slide, the voltage limit of a generator is mainly due to excess flux considerations. While the maximum continuous limit is about 1.05 pu at nominal frequency, a larger voltage can be tolerated for a short while (why?).

The typical allowable voltage (at nominal frequency) for different intervals of time are given below

V/Hz (pu)		1.25	1.2	1.15	1.10	1.05
Damage Time (in min.)	GEN	0.2	1.0	6.0	20.0	Infinity

Typical Name Plate Data of a Generator

Typical Data for a 247 MVA generator is given below:

MVA maximum continuous rating: 247 MVA

MW maximum continuous rating: 210 MW

Rated voltage: 15.75 kV

Rated stator current : 9050 A

Rated power factor : 0.85 lagging

Field current (at maximum continuous rating) : 2600 A

Field Voltage (at maximum continuous rating) : 315 V

Efficiency (at maximum continuous rating) : 98.55 %

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Rated power factor : 0.85 lagging

Field current (at maximum continuous rating) : 2600 A

Field Voltage (at maximum continuous rating) : 315 V

Efficiency (at maximum continuous rating) : 98.55 %

Speed : 3000 rpm

Cooling System : Hydrogen at 3.5 kg/cm^2

Recap

In this lecture you have learnt the following

- Generators are constrained by heating limits on the armature windings, field winding and core end region
- Field winding heating limit decides the capability in the over-excited region (lagging power factor) while core end heating determines the under-excitation capability (leading power factor)

Lecture 6 Transmission Line Constraints

Objectives

In this lecture you will learn the following

- Thermal and dielectric limitations of a transmission line.
- Effect of loading on voltages, for long distance ac transmission.
- AC line loadability (power transmission capability) versus distance.

Transmission line thermal limits

A large current flow increases the losses in the form of heat. This results in increased conductor temperatures. Excessive temperature may result in expansion and resultant sag of conductors causing decreased clearance to ground. Temperature extremes have an "annealing effect" causing reduced mechanical strength of aluminum.

Since thermal time constants are large, it is useful to distinguish between steady state and transient thermal ratings.

Thermal Capability is a function of 1) ambient temperature 2) Wind conditions 3) Condition of conductor 4) Conductor type and 5) ground clearance.

Typical *steady state* thermal specifications of conductors used for 400 kV overhead transmission:

ACSR (aluminum conductor steel reinforced) *Moose* Conductor (520 sq mm): For an ambient temperature of 40° C, and a maximum conductor temperature of 75° C, ampacity is 700 A (approx.). Typically, two moose conductors form a bundle of 400 kV line so that the total ampacity is 1400A for such a line.

If **AAAC (all aluminum alloy conductor) of 520 sq mm** is used, higher conductor temperatures as compared to ACSR are possible. For an ambient temperature of 40° C, and a maximum conductor temperature of 85° C, ampacity is 850 A (approx.).

The thermal ratings are a strong function of **ambient conditions** (wind flows, solar radiation) and absorption and emmisivity of the conductor material ; it makes sense to evaluate the actual ampacity during real time operation (season to season, day to day or hour to hour).

Dielectric Limits

Exceeding dielectric limits (maximum electric field strength) results in failure of insulation, causing faults. See a photograph of dielectric breakdown in an insulator **here**. Electric fields may be excessive (due to overvoltage) under low loading conditions on long ac transmission lines (Ferranti Effect) or during abnormal conditions like lightning strokes.

Deviation of voltages beyond certain limits can also be considered to be an unacceptable compromise on the quality of power being supplied to consumers. Low or high voltages can also damage electrical equipments.

Voltages and reactive power demand of transmission lines are affected by:

- Line parameters
- Length of line
- Power transfer

Given a nominal voltage rating, a steady state overvoltage of about 10% is often permissible. Shunt reactors (inductors) are often connected in shunt on transmission lines to prevent overvoltages under low loading. If permanently connected to a line, however, they may excessively reduce the voltages during heavy load conditions.

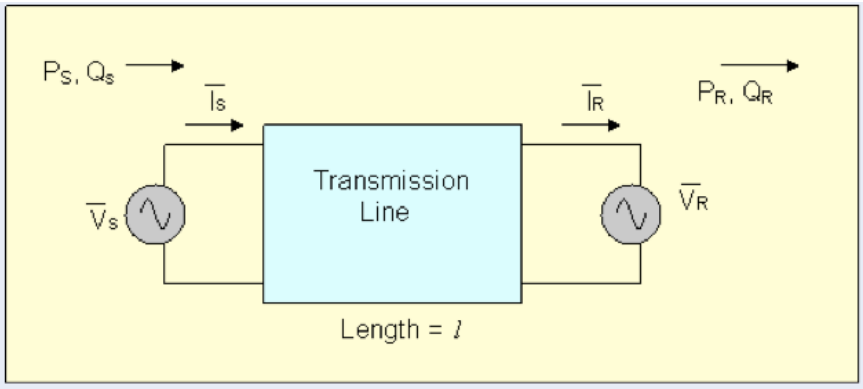
Line Parameters: Line parameters are dependent on the conductor dimensions and relative placement. The surge impedance of most overhead lines is around 250-350 ohms whereas it is 30-50 ohms for cables.

Typical *positive sequence* inductance and capacitance parameters for a 400 kV overhead line: L = 1.044mH/km, C = 12 nF/km. For EHV lines , X/R ratio is large. The resistance per unit length of this line = 0.0296 ohm / km

For a 400 kV paper-insulated lead-covered(PILC) cable, typical *positive sequence* parameters are: L = 0.78 mH/km, C = 0.95 uF/km.
Note the high capacitance as compared to an overhead line.

What is the Surge Impedance Loading (SIL) of the overhead line and cable?

To analyse the effect of various parameters, we now have a quick look at the basic **equations of a lossless transmission line**.



For a transmission line of length l (or a transmission segment of length l), the equations relating the two ends are given by:

If $\beta = \omega\sqrt{LC}$ and $Z_C = \sqrt{\frac{L}{C}}$

where L and C are inductance and capacitance per unit length , then:

$\bar{V}_S = \bar{V}_R cos\beta l + jZ_C\bar{I}_R sin\beta l$ and $\bar{I}_S = \bar{I}_R cos\beta l + j\frac{\bar{V}_R}{Z_C} sin\beta l$

if $\bar{V}_S = V_S\angle\delta_S$ and $\bar{V}_R = V_R\angle\delta_R$

then expression for sending and receiving end real/reactive power are:

$$P_S = \frac{V_S V_R sin(\delta_S - \delta_R)}{Z_C sin\beta l} \qquad P_R = -\frac{V_S V_R sin(\delta_R - \delta_S)}{Z_C sin\beta l}$$

$$Q_S = \frac{-V_S V_R cos\delta + V_S^2 cos\beta l}{Z_C sin\beta l} \qquad Q_R = \frac{V_S V_R cos\delta - V_R^2 cos\beta l}{Z_C sin\beta l}$$

where $\delta = \delta_S - \delta_R$

where b is the propagation constant in rad/m and Zc is the characteristic impedance in Ω

If sending end and receiving end voltage magnitudes are equal to the nominal voltage ($V_S = V_R = V$) then

$P_S = P_R = \frac{P_{SIL} sin(\delta)}{sin\beta l}$ where, $P_{SIL} = \frac{V^2}{Z_C}$ is the Surge Impedance Loading of the line.

Note: if the line at one end is open circuited i.e. $\bar{I}_R = 0$ then

$\frac{\bar{V}_S}{cos\beta l} = \bar{V}_R$ and $\bar{I}_S = j\frac{\bar{V}_S}{Z_C} tan\beta l$

obviously the receiving end voltage exceeds the sending end voltage (**ferranti effect).**

The expression for the magnitude of the voltage at the midpoint of the line is:

$$V_{mid} = \frac{V cos(\delta/2)}{cos(\beta l/2)}$$

How does voltage vary in an AC transmission Line?

Important results for lossless lines (EHV lines have low resistance, so lossless line approximation is valid).

- Voltage profile along the line is flat only if R= Zc is connected at receiving end **(SIL)**. Under this situation, there is no demand of reactive power from either sending or receiving ends. Loads are decided by consumers and not by a system operator. As such, this exact condition is practically never met although it is ideal.
- If voltage at both ends is maintained at 1.0 pu (say, by controlling excitation of generators connected at both ends), the voltage tends to sag as we move towards the midpoint if Ps > SIL. The line absorbs reactive power.
- If Ps < SIL, voltage swells and the line generates reactive power.
- For a line which has one end maintained at 1.0 pu but kept open-circuited at the other end, overvoltage will be evident at the open end.
- All these effects are very prominent for long lines.
- *Cables have very high SIL* **(why?)**. The current at SIL usually exceeds the thermal rating of a cable. Therefore a cable is invariably loaded below its SIL. Cables, therefore, generate reactive power and usually voltages can be very high at low loading levels, especially for long lengths. Besides this, cables are much costlier than overhead lines and are more "unforgiving" towards overloads. Therefore, cables of lengths exceeding 30-40km are rarely used for ac transmission.

Due to voltage problems (and stability problems, which we shall study in the following lecture), underwater cable transmission and bulk transfer over very long distances is carried out using DC transmission technology. We shall study the essentials of this technology later in the course.

We now will illustrate voltage variation in a long transmission line by an **example**.

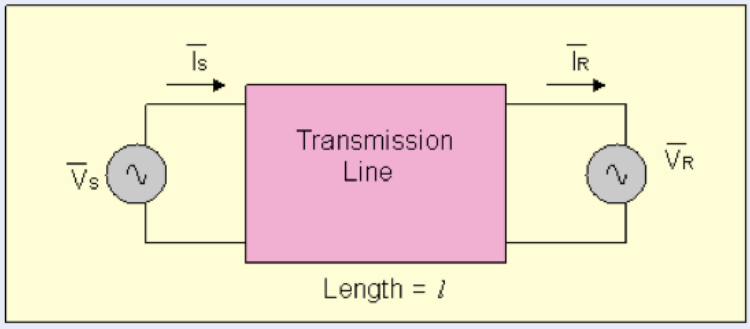
Consider a 50 Hz, 400 km, 400 kV loss less transmission line, with the following parameters:

L=1.04 mH/km C=12 nF/km

Compute

a. Characteristic Impedance
b. Surge Impedance Loading
c. If the line is open circuited at one end while the other end is maintained at 400 kV, compute the voltage at the open circuited end and the reactive power generated by the line at the 400 kV end.
d. Plot the reactive power absorbed by the line at sending (Qs) and supplied at receiving end (Qr) of the transmission line if power flow in the transmission line is varied from 0.2 P_{SIL} to 1.5 P_{SIL} . The voltages at both ends are maintained at 400 kV.

Also plot the voltage at the midpoint of the transmission line.



Solution:

Refer to the lossless line formulae for this example.

a) The characteristic impedance of a line is given by the formula:

$$Z_c = \sqrt{\frac{L}{C}} = 294.4\ \Omega$$

b) Surge Impedance Loading is given by

$$P_{SIL} = \frac{V^2}{Z_c} = 543.5\ \text{MW}$$

c) Receiving end voltage on open circuit is given by:

$$V_R = \frac{V_S}{\cos \beta l} = \frac{V_S}{\cos(\omega \sqrt{LC})l} = 443\text{kV}$$

d) Plots of real and reactive power at both ends of the line and the midpoint voltage is shown below.

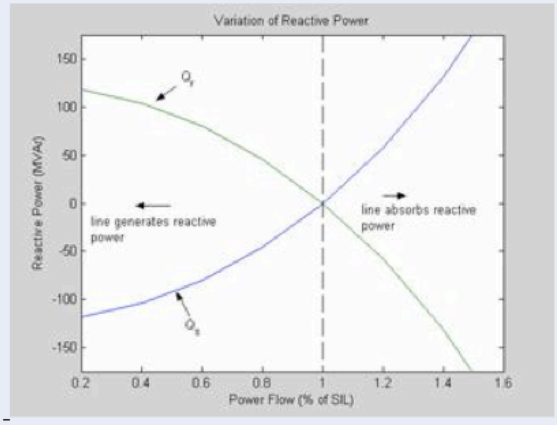
Note

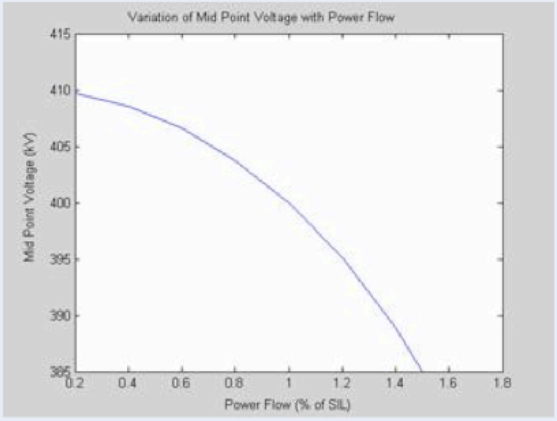
Reactive power is generated by the line for loadings less than SIL

Reactive power is absorbed by the line for loadings greater than SIL

Mid-point voltage is 1 pu at SIL.

It swells at lower loads and sags at higher loads





Line Loadability in AC lines

In order to prevent overvoltages at light loads, it is necessary to have devices for absorbing reactive power (like shunt reactors) not only at either end of a long line line but even at intermediate points. Generators connected at the ends of the line have limited reactive power absorption capability as defined by their capability curves. If transmission redundancy exists (i.e., parallel transmission paths exist), then a very lightly loaded long line may be tripped to avoid overvoltage. However this may be detrimental to system security if some additional line trippings take place due to faults. If shunt reactors are permanently connected, they result in large sags in the voltage under heavy loading conditions. Moreover, reactive power demanded by long transmission lines under these situations may be excessive and may lead to system-wide low voltage conditions.

Compensation of a line involves changing the effective line parameters by connecting (lumped) capacitors in series and shunt. These compensating elements effectively reduce the line reactance and increase the shunt susceptance, thereby decreasing the surge impedance. Thus the effective SIL of a capacitor compensated line is higher than an uncompensated line. This increases the loadability of a long line.

Since total conductor cross-sectional area for EHV lines is mainly decided by electric field considerations (corona), these lines have large thermal capabilities, much in excess of the SIL. For long EHV lines, one cannot deviate much from SIL due to voltage constraints. Therefore, the thermal limit of a long EHV line is *not* the key limiting factor. However, thermal limit is the main limiting factor for short lines (< 100 km) wherein voltage constraints are not violated even for large deviations from SIL.

The line loadability as a function of distance can be obtained by considering thermal and voltage limits as given in this **example**.

Additional limits are imposed by stability and are discussed in the next lecture.

Consider a 50 Hz, 400 kV loss less transmission line, with the following parameters:
L=1.04 mH/km C=12 nF/km

Assume that both ends of the line are maintained at 400 kV.
Compute line loadability vs distance if the following constraints are imposed:

1. Current in the line should be within the thermal limits (say 1400 A).
2. Reactive power demand should not exceed 100 MVAr at each end (this ensures that excessive reactive power is not required to maintain voltage magnitudes).

Note: Voltage magnitudes at the ends can be maintained by controllable sources (like generators). Usually, these sources may not be right at the transmission line terminals but connected via other lines or transformers. Thus, in actual practice, a more rigorous methodology is required to specify the voltage/reactive power constraints when computing maximum loadability.

Vary the distance from 0 to 600 km.

For distances beyond this, it is necessary to consider **angle stability constraints** which will be discussed separately in the next lecture.

Solution:

Refer to the **lossless line formulae**

1) **Thermal Limit :**

If current is limited to 1.4 kA, the MVA limit is given by

$$S = \sqrt{3}VI = \sqrt{3} * 400 * 1.4$$

Current magnitude is not the same throughout the line (show this !).

However, for simplicity we use the expression for current and MVA at the sending end of the line.

Thus $P_s^2 + Q_s^2 \leq S^2$, and it follows that

$$\left(\frac{V^2 \sin\delta}{Z_c \sin\beta l}\right)^2 + \left(\frac{V^2 \cos\beta l - V^2 \cos\delta}{Z_c \sin\beta l}\right)^2 \leq S^2$$

Thus $$\delta \leq \arccos\left(\frac{1 - \left(\frac{S \sin\beta l}{P_{SIL}}\right)^2 + (\cos\beta l)^2}{2\cos\beta l}\right)$$

Therefore we can obtain the maximum allowable P using the formula $P = \frac{V^2 \sin\delta}{Z_c \sin\beta l}$

We can plot maximum allowable P as a function of distance by varying the length.

2) **Reactive Demand Limit**

Maximum magnitude of reactive power equals 100 MVar

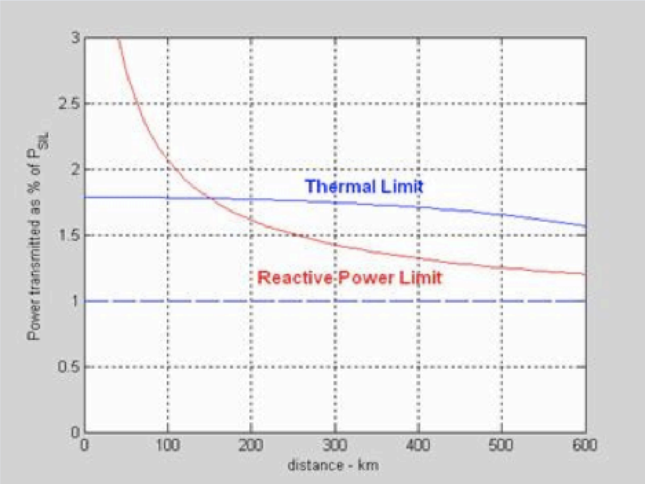
Therefore,

$$Q_s = -Q_r = \frac{V^2 cos\beta l - V^2 cos\delta}{Z_c sin\beta l}$$

again, we obtain

$$\delta \leq arccos\left(cos\beta l - \frac{Q_s sin\beta l}{P_{SIL}}\right)$$

Therefore we can obtain the maximum allowable P using the formula $P = \frac{V^2 sin\delta}{Z_c sin\beta l}$



Recap

In this lecture you have learnt the following

- Power transfer capability of a ac line is restricted by thermal and voltage constraints
- Short lines are limited by thermal constraints
- Long lines are limited by voltage constraints. Stability constraints are also important for long ac lines and are studied in the next lecture

Lecture 7 Stability Problems in Power Systems

Objectives

In this lecture you will learn the following

- What does one mean by stability?
- What is angular stability?
- Dynamical equations for a single machine -infinite bus system. Equilibrium points.

What is Stability?

Stability is essentially the ability of a system to recover from disturbances, both large and small, and settle to an acceptable equilibrium. Randomly occurring load changes, faults resulting in line or generator tripping and changes in reference values of regulating controllers, are examples of disturbances.

One should note that equipment constraints are distinct from stability constraints in the sense that even though an equilibrium condition may exist (which is within equipment constraints), a system may not be able to "settle down" to it if it is perturbed or initially away from it.

If a system is not stable for even small disturbances, *it cannot be operated at all* since there are always small and random perturbations in the system due to load variations

If a system is stable for small disturbances but unstable if the disturbances are "large", then the system can be operated. However, the system may not be *secure*, i.e., it may be unstable if a large enough disturbance does actually occur .

The major stability problems which are inherent in AC interconnected grids are discussed next.

Angle Stability

An interesting physical characteristic of interconnected synchronous generators is their ability to generate restoring torques when disturbed from an equilibrium. These torques ensure that all machines stay in synchronism --- generator electrical speeds become equal in steady state. Equivalently, the phase angular differences between ac voltages at various points become constant if machines stay in equilibrium.

However, the restoring torques can become zero or negative for very large disturbances. This can result in machines falling out of step (i.e., they lose synchronism - the machines do not settle to the same electrical speed; this makes operation unviable - see **Module 1**). Sometimes, due to presence of automatic controllers, damping of the rotor oscillations is inadequate or negative, causing growing or sustained oscillations ('hunting') and may also lead to loss of synchronism.

The problem of loss of synchronism between synchronous machines is also known as the "angular stability problem". Note that if machines lose synchronism, then the phase angular difference between ac buses in the system will not settle to constant values.

To study the angular stability problem, consider the governing equations of the system.

The equations of motion of a synchronous machine rotor are given by:

$$\frac{d\delta}{dt} = (\omega - \omega_o)$$

$$\frac{d(\omega - \omega_o)}{dt} = \frac{\omega_B}{2H}(T_m - T_e)$$

where δ, ω are the rotor angle (with respect to a frame which rotates at ω_o) and rotor speed respectively

T_m, T_e are the mechanical and electrical torques respectively expressed in per-unit.

ω_B is the base angular frequency, and H is the inertia constant in MJ/MVA

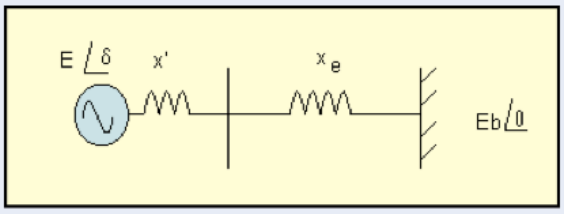
T_e is a non-linear function of δ. It is also a function of several other variables.

However, for simplicity we concentrate on the dependence on δ alone by considering a simple system.

SMIB System

A Single - Machine Infinite Bus system (SMIB), is an oversimplified model of a power system but it helps us to understand the essence of the large disturbance angular stability problem. The SMIB system represents a small generator connected to a large power system. The large power system is represented by an infinite bus (fixed voltage source with a constant frequency). The generator itself is represented as a constant magnitude voltage source behind its transient reactance. The infinite bus voltage is assumed to have a frequency wo. The phase angle of the internal voltage E with respect to the infinite bus is given by d. Note that this angle will change if the relative frequency between the generator and infinite bus changes.

Thus, if d does not reach a steady state value after a disturbance, then it implies that the generator has lost synchronism, or equivalently, is angular unstable.



For this simplified model, the electrical torque in per-unit is approximately given by,

$$T_e = \frac{EE_b \sin\delta}{X}, \quad X = x' + x_e$$

Thus,

$$\frac{d\delta}{dt} = (\omega - \omega_o)$$

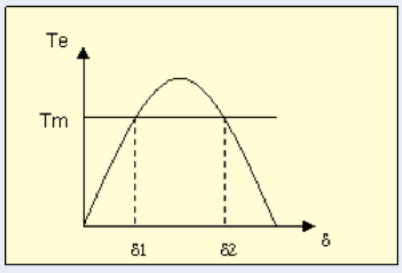
$$\frac{d(\omega - \omega_o)}{dt} = \frac{\omega_B}{2H}\left(T_m - \frac{EE_b \sin\delta}{X}\right)$$

The set of equations are non-linear because of the $\sin\delta$ term.

x_e is the transmission line reactance in Ω

The values of δ, ω at which $\frac{d\delta}{dt}$ and $\frac{d(\omega - \omega_o)}{dt} = 0$, represent the equlibrium values.

There are 2 equilibria $\delta = \delta_1, \omega = \omega_o$ and $\delta = \delta_2, \omega = \omega_o$, between 0-180 degrees as seen in the figure below.



However, $\delta = \delta_1, \omega = \omega_o$ is a stable equilibrium.

This is because, i.e., $(T_m - \frac{EE_b \sin\delta}{X})$ is negative if the rotor angle is slightly perturbed from δ_1 in the positive direction, and is positive if the perturbation is negative. Thus the torque is restorative in nature.

On the other hand, $(T_m - \frac{EE_b \sin\delta}{X})$ is positive if the rotor angle is slightly perturbed from δ_2 in the positive direction, and is negative if the perturbation is negative. Thus the torque enhances the disturbances!

Thus δ_2 is an unstable equilibrium.

Another interesting aspect is that $(T_m - \frac{EE_b \sin\delta}{X})$ is a nonlinear function of δ, and may change sign with increasing value of δ.

Even if the system is stable for small perturbations around d1, it does not imply that it is stable for large perturbations. This is because, for large perturbations from d1, i.e., when d is large, the torque may not be restorative in nature. We will try to undertand this using simple analogies and numerical techniques in the following lectures.

Analyzing a system described by nonlinear differential equations is not easy. Therefore we take a minor diversion and learn about techniques to analyze dynamical systems. Those who are aware of numerical techniques for analyzing differential equations may skip this lecture.

Recap

In this lecture you have learnt the following

- A system is said to be stable if it can withstand disturbances and come to an equilibrium
- Angular stability refers to the study of whether machines remain in synchronism after a disturbance
- The equations of a single machine infinite bus system reveal that torque is a nonlinear function of rotor angle

Lecture 7a Numerical Solutions of Differential Equations

Objectives

In this lecture you will learn the following

- How does one simulate the behaviour of a dynamical system numerically ?
- Runge-Kutta Fourth order method

Why numerical methods ?

The reason we have taken a minor diversion here is that we wish to know the techniques to "understand" how systems described by differential equations behave. For very simple systems like the one below:

$$\frac{dx}{dt} = ax$$

It is clear that x=0 is an "equilibrium" solution of the system, since the LHS of the above equation equals zero at this value of x. In general, the behaviour of x, when its value at time t = 0 is x(0), is given by :

$$x(t) = e^{at}x(0)$$

Verify that the above "solution" satisfies the differential equation. Note that the solution is in terms of a well known exponential function. If a > 0 , then the magnitude of x(t) keeps increasing with time if x(0) is not zero. On the other hand, if a<0, then x(t) tends to go to zero as time progresses. Thus we are able to gain an insight into the behaviour from the solution given above.

However it turns out that for many systems, it is not possible to write down the solution in terms of well understood simple functions. This occurs when the RHS of the differential equations have terms which are non-linear or time variant functions of the variables. For example, the behaviour of rotor angle and speed deviation for a synchronous machine is described by the non-linear differential equations:

$$\frac{d\delta}{dt} = (\omega - \omega_o)$$

$$\frac{d(\omega - \omega_o)}{dt} = \frac{\omega_B}{2H}\left(T_m - \frac{EE_b \sin\delta}{X}\right)$$

Numerical Solution of Differential Equations

Since it is difficult to obtain the solution for a nonlinear set of differential equations, we try to utilize a computer program which numerically computes the solution at *discrete* points in time. An alternative would have been to implement a setup using scaled physical elements which mimic the differential equations, i.e., an analog computer. However, given their flexibility, numerical evaluation using computers is convenient and economical.

So how do we solve the differential equations numerically ? Let us consider a simple example:

$$\frac{dx}{dt} = ax$$

If we wish to obtain the solution denoted by x0, x1, x2, ...xk at the closely spaced time instants t = 0, h, 2h, ..., kh, then we can approximate the above equation as:

$$\frac{(x_{k+1} - x_k)}{h} = ax_k$$

Therefore if x_k is known, x_{k+1} can be obtained by

$$x_{k+1} = (1 + ah)x_k$$

Therefore, if the initial condition x0 is known, we can recursively find the solution of x at discrete instants of time. The approximation is likely to work well only if h is "small enough".

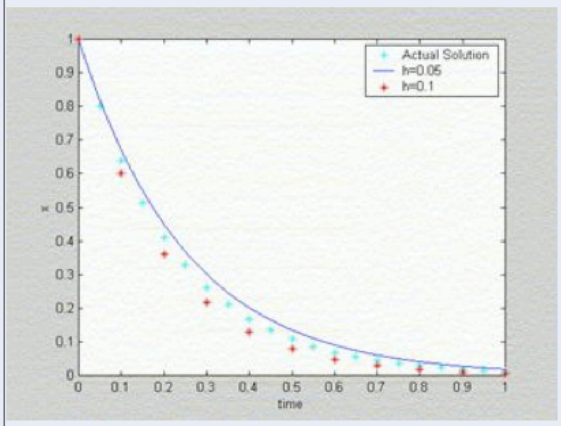
To illustrate this, consider the numerical solution at discrete points and compare it with the correct solution:

$$x(t) = e^{at}x(0)$$

Let us take the constant a = -4.

We see from the figure above that the approximation works better for smaller h.

The method that we have presentated is in fact the simplest possible one and is called "Euler's Method" of numerical integration.



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t is not difficult to see that another possible approximation of the differential equation could be:

$$\frac{(x_{k+1} - x_k)}{h} = ax_{k+1}$$

This is called the "Backward Euler" method. The numerical solution obtained using different approximations and different values of h could vary. Inappropriate selection of the method and the time step h can lead to errors which can cause erroneous conclusions.

Comparison of the two methods

We have considered two methods: Euler Method and Backward Euler method. Euler's method is simple to implement on a computer program, but is known to have poorer numerical properties. In particular, it may wrongly show a system which is actually stable to be unstable. As an example consider the following equations:

$$\frac{dx_1}{dt} = -5x_1 + 0.01x_2$$
$$\frac{dx_2}{dt} = -100x_2$$

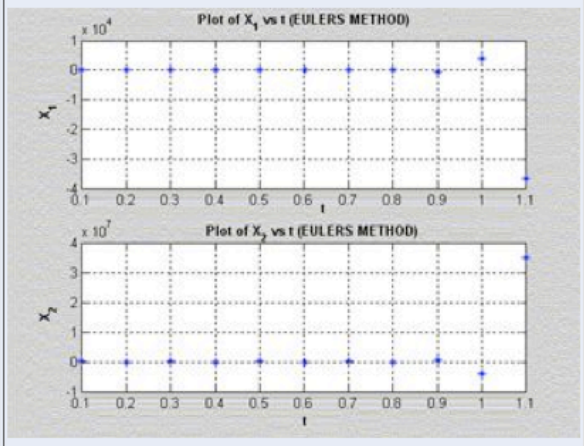
If we use Euler Method, we descretise the equations as follows.

$$\frac{(x_{1_k+1} - x_{1_k})}{h} = -5x_{1_k} + 0.01x_{2_k}$$
$$\frac{(x_{2_k+1} - x_{2_k})}{h} = -100x_{2_k}$$

For a time step, h=0.1, with initial conditions x1(0)=1 and x2(0)=0.01, **the numerically evalauated response is unstable**, although this system is actually stable and has a time response given by:

$$x_1(t) = e^{-5t}x_1(0) + \frac{e^{-5t}}{9500}x_2(0) - \frac{e^{-100t}}{9500}x_2(0)$$
$$x_2(t) = e^{-100t}x_2(0)$$

One has to use a very small time step (say, h = 0.005s) if one has to avoid getting a grossly incorrect numerical solution. However, this would mean more computation time, even one may be interested only in capturing the slow response corresponding to the exp(-5*t) term in the response.



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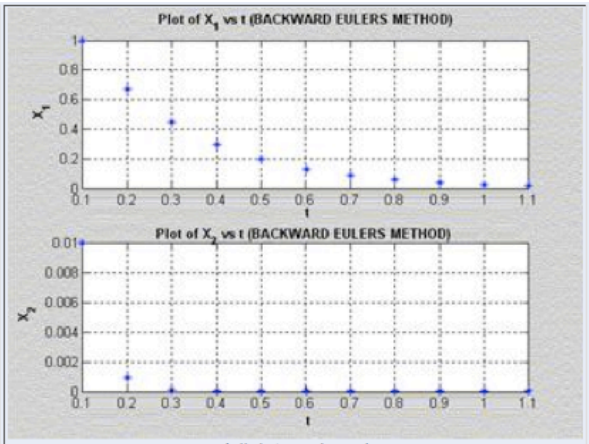
Let us repeat this exercise with Backward Euler with the same h=0.1. The discretised equations are:

$$\frac{(x_{1_k+1} - x_{1_k})}{h} = -5x_{1_k+1} + 0.01x_{2_k+1}$$
$$\frac{(x_{2_k+1} - x_{2_k})}{h} = -100x_{2_k+1}$$

Clearly, the solution obtained from Backward Euler Method does not "blow up". It also captures the slow transient corresponding to exp(-5*t) quite accurately with this value of time step.



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The problem with Backward Euler Method is evident when we apply it to the solution of swing equations:

$$\frac{\delta_{k+1} - \delta_k}{h} = (\omega_{k+1} - \omega_o)$$
$$\frac{\omega_{k+1} - \omega_k}{h} = \frac{\omega_B}{2H}(T_m - \frac{EE_b \sin\delta_{k+1}}{X})$$

Clearly, getting the values of $\mathbf{d}$ and $\mathbf{w}$ at the k+1 th instant from the values at the k th instant is tough as we have to solve nonlinear *algebraic* equations (how does one solve them?).

Clearly there is a trade-off between numerical accuracy and complexity of implementation! We now consider a method which is not too difficult to implement and also gives reasonably good accuracy.

Runge-Kutta Fourth Order Method

Consider the differential equation:

$$\frac{dx}{dt} = f(x,t)$$

The Runge-Kutta 4th order algorithm uses intermediate points in an interval of duration h to calculate the state x_{k+1} from x_k as follows:

$$k_1 = f(x_k, t_k)$$
$$k_2 = f(x_k + \frac{h}{2}k_1, t_k + \frac{h}{2})$$
$$k_3 = f(x_k + \frac{h}{2}k_2, t_k + \frac{h}{2})$$
$$k_4 = f(x_k + hk_3, t_{k+1})$$
$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Note that the number of calculations per time step is larger than that for Euler Method. However, this method is more accurate than Euler method (for the same time step) for most systems.

Can you program an algorithm which implements this method for the swing equations ?

Recap

In this lecture you have learnt the following

- Numerical Integration Methods
- Euler, Backward Euler and Runge Kutta methods

Lecture 8 Large Disturbance Angle Stability

Objectives

In this lecture you will learn the following

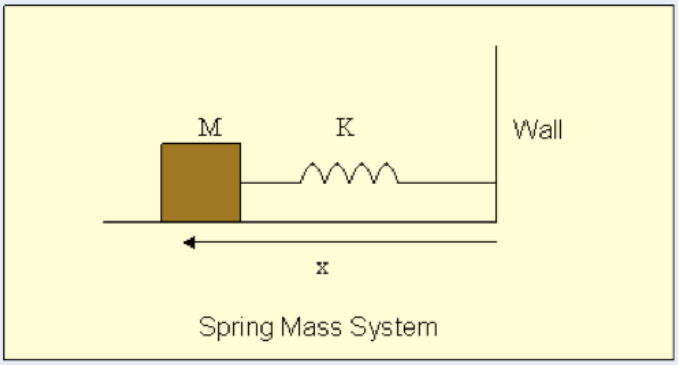
- Spring Mass analogy of a single machine infinite bus system.
- Large disturbance angular instability is demonstrated using an example.
- Angular stability in multimachine systems.
- Impact of angular stability problems on system operation.

A Simple Analogy of a single machine infinite bus system: A spring mass system

In this lecture we first understand the behaviour of the dynamical system qualitatively using a simple analogy and then use a numerical technique to simulate the behaviour of the system in an example.

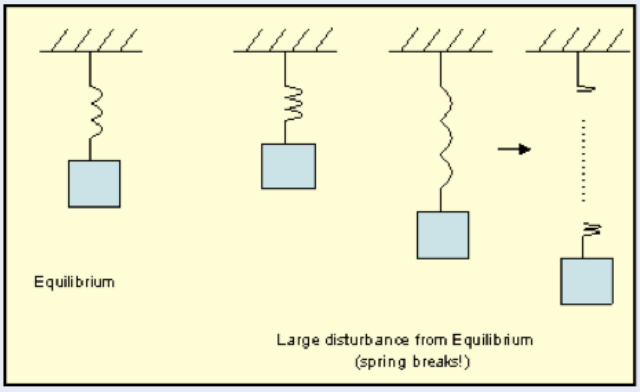
The differential equations of a single machine infinite bus(SMIB) system are similar to that of a "mass connected to a wall by a nonlinear spring" system.

For a *linear* spring, the restoring force is directly proportional to the stretch of the spring. If the mass is displaced from its equilibrium, then oscillatory motion results irrespective of the extent of the disturbance.



If the restoring force of the spring is a *nonlinear* function of the stretch, then the behavior may be different and depends on the extent of the disturbance.

For example, if the restoring force of the spring is almost a linear function upto a certain stretch but reduces to zero beyond that value, then for a large disturbance it is possible that the spring will break (as shown in the figure below).



To avoid instability (breaking of the spring), the following can be done:

1) Make the spring more stiff.
2) Reduce the disturbance.
3) Keep the equilibrium stretch of the spring small (so that it has more margin to stretch in case of disturbances).

The behavior of SMIB system is similar. Under normal operating conditions, there are restoring torques which tend to keep the synchronous machine in synchronism with the infinite bus. However when disturbances are large, the torque may not be able to "pull back" the machine to synchronism. We now illustrate this by an **example**.

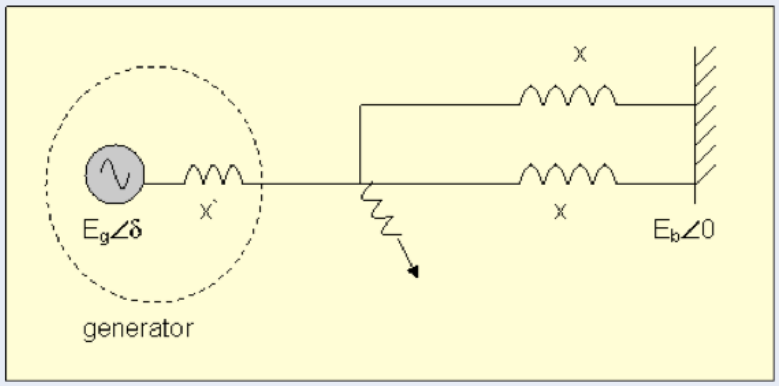
Consider a single machine infinite bus system shown below. The generator is connected via 2 parallel lines to an infinite bus. The generator is initially at equilibrium supplying a certain amount of power (Pe) to the infinite bus. There is a sudden fault on one of the parallel lines, very close to the generator. The fault (a three phase short circuit) is cleared in Tclear seconds. The fault is cleared by tripping the faulted line. Therefore the post fault system consists of the generator connected to the infinite bus by a single line.

Simulate this disturbance by a numerical integration method (e.g. Runge- Kutta method 4th order) with a time step of 0.01 s

Data:

$H = 3$ MJ/MVA, $x = 0.5$ pu, $E_g = E_b = 1.0$ pu, $x' = 0.1$ pu,

Initial power generated $P_{eo} = 1.0$ pu ($= T_{eo}$ in pu), $\omega_o = 314$ rad/s



Solution:

Before the fault occurs equilibrium exists. This implies

$T_e = T_{eo} = T_{mo} = \frac{E_g E_b \sin\delta_o}{X}$, $X = x' + x_e = x' + \frac{x}{2}$, and,

$\omega = \omega_o$

Thus, equilibrium value of $\delta = \delta_o = arcsin(\frac{T_{eo}X}{E_g E_b}) = 0.35$ rad

During the fault, the voltage at the generator bus is zero.

Therefore, $T_e = 0.0$ during the fault which lasts for T_{clear} seconds.

After the fault, line clearing takes place. The torque is given by :

. $T_e = \frac{E_g E_b \sin\delta}{X_{pf}}$, $X_{pf} = x' + x_{epf} = x' + x$

The new equlibrium is given by:

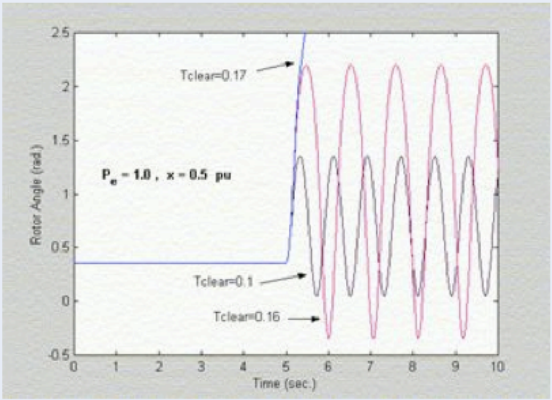
$T_{epf} = T_{mpf} = \frac{E_g E_b \sin\delta_{pf}}{X_{pf}}$, $\omega = \omega_o$

Will the system settle to this new equilibrium ?

To answer this, we will simulate the system with a numerical integration method. The results of the simulation are shown in the next slide.

Will the system settle to this new equilibrium ?

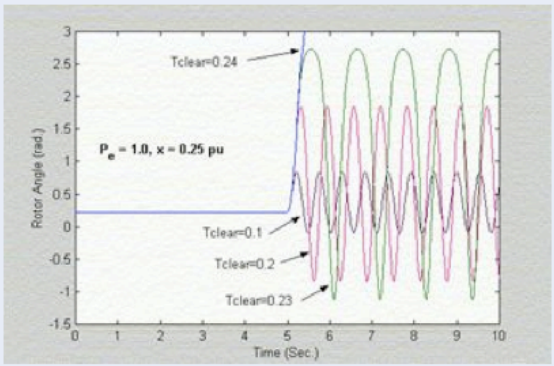
To answer this question, let us simulate the system using 4th order Runge-Kutta numerical integration method with a time step of 0.01 s. **A MATLAB/SIMULINK simulation model file to do the same can be downloaded from here (smib_sim.mdl)**. The rotor angle with respect to the infinite bus varies as shown in the figure on the right.



If the disturbance magnitude is increased by increasing Tclear, then a point is reached when the rotor angle increases monotonically (i.e., loses synchronism). Under this situation the rotor does not come back to equilibrium and is unstable.

Let us now try a variation:

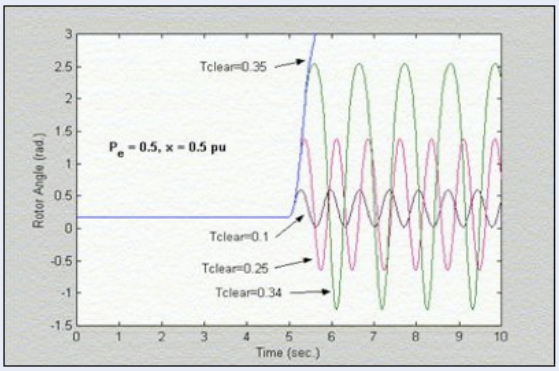
reduce the line reactance by half. Under this situation, the system can sustain a larger clearing time without losing synchronism (i.e., stability is improved) as is seen from the figure on the right.



Let us try another variation.

Let us **decrease the initial power flow** to 0.5 pu. Again we see from the simulations shown on the right, that stability is improved.

Note: You can also find the critical clearing time for this simple system without doing numerical integration using the well known *Equal Area Criterion.*



(click on figure's to enlarge)

We can infer the following from this example:

a) For large disturbances, the system is *inherently* unstable.

b) The stability of the system can be improved by making it more 'stiff' or strong (i.e., reduce effective line reactance - by having more parallel paths , shorter lines or by connecting series capacitors in transmission lines).

c) The stability of the system for a given disturbance can also be ensured by reducing the level of power transfer (i.e., by operating conservatively). Thus stability is an important constraint in power system operation.

In this example, we did not completely answer the question of whether the system will settle to the new equilibrium, i.e., even if the rotor angle does not increase monotonically, does the oscillation around the equilibrium die down ? With the given model, there is no inherent 'damping' , therefore oscillations do not die out.

In this example, we did not completely answer the question of whether the system will settle to the new equilibrium, i.e., even if the rotor angle does not increase monotonically, does the oscillation around the equilibrium die down ? With the given model, there is no inherent 'damping' , therefore oscillations do not die out.

When detailed models considering effect of rotor field and damper windings are considered, then damping is usually present. Presence of automatic controllers like voltage regulators in generator excitation systems can often cause 'negative damping' - resulting in growing oscillations. Damping can be improved by augmenting or re-tuning existing controllers. However, this issue is not considered here.

Machine Synchronisation

We have seen how the system behaves when subject to large disturbances in the previous **example**. In the example, we had assumed that the machine was already connected and was initially in equilibrium (i.e., was in synchronism with the infinite bus).

It is natural of you to ask : How was the machine "synchronised" when it was first connected to the grid?

When a synchronous machine is connected to a grid, it is important to follow a particular procedure so that the transients which occur are minimised and the incoming generator is "pulled into synchronism".

This includes ensuring that the generator voltages and the system voltages are in the same phase sequence and the voltage magnitude of the generator and the grid at the point of interconnection are nearly equal.

Additionally:

a) The electrical speed of the incoming generator should be almost equal to the grid frequency.

b) The phase angular difference between the generator voltage and the grid voltage at the point of interconnection at the instant of interconnection should be small.

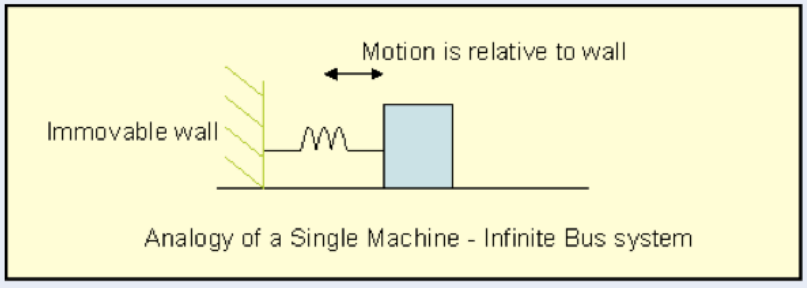
The situation is analogous to connecting two *moving* masses by a spring. To avoid excessive stretch in the spring when the masses are inter-connected (which may cause the spring to break), one would need to have the two masses to be close to each other and to be moving at nearly the same speed at the time of interconnection. If the interconnection is done smoothly, without breakage of the spring, then the masses move together at the same speed after transients die out. Thereafter, the masses are held together by the spring unless a large disturbance causes the spring to break.

In practice, a plant operator starts rotating the generator by introducing a prime mover torque. He ramps up the speed to a value which is very close to the grid frequency by adjusting the prime mover torque. He then switches on the field excitation and brings voltage close to the grid voltage at the point of interconnection by adjusting the field voltage. Using an instrument called a **synchroscope**, he monitors the phase angular difference and switches on the interconnecting circuit breaker at the instant when the phase angular difference is very small.

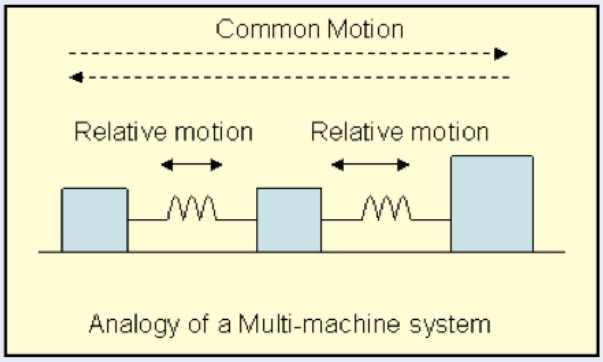
You can simulate the process of synchronisaton by downloading **all** of the following MATLAB/SIMULINK simulation files: 1 , 2, 3, 4,5 ,6. Run only the sync.mdl (1) file for simulation purpose.

Multi-generator systems

In a realistic system, there are many interconnected generators. Moreover, in practice there is no "infinite bus". The rotor angles swing relative to one another during transients although they settle to the same electrical speed if stable. Aggregate motion of the entire system (all masses move together) is also possible in addition to the relative motion. This is illustrated below by the spring mass analogy.



Analogy of a Single Machine - Infinite Bus system



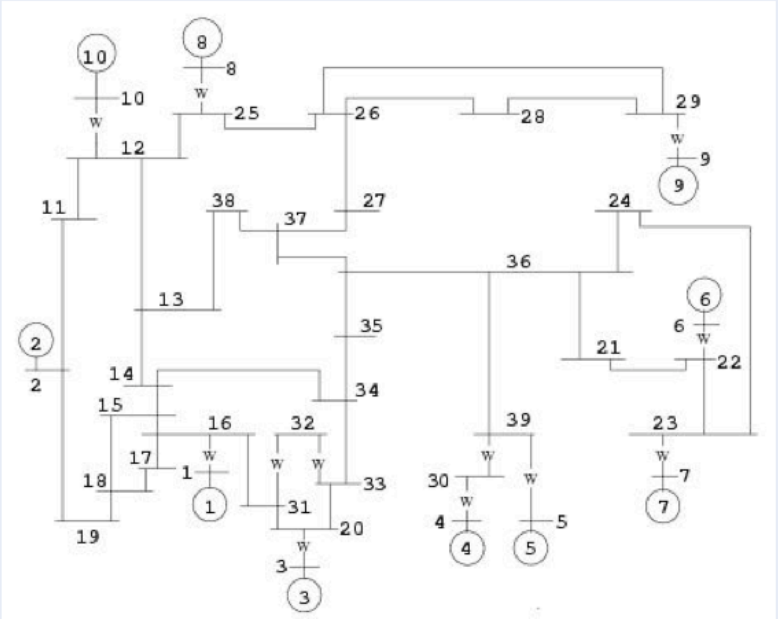
When a practical multimachine system is subjected to a large disturbance, loss of synchronism may take place if one or more groups of machines "separate out" from each other. It is not meaningful to see rotor angles relative to an infinite bus, since there isn't one! Instead they are plotted relative to the rotor angle of one machine, or to the *center of inertia* of the system which is defined as follows:

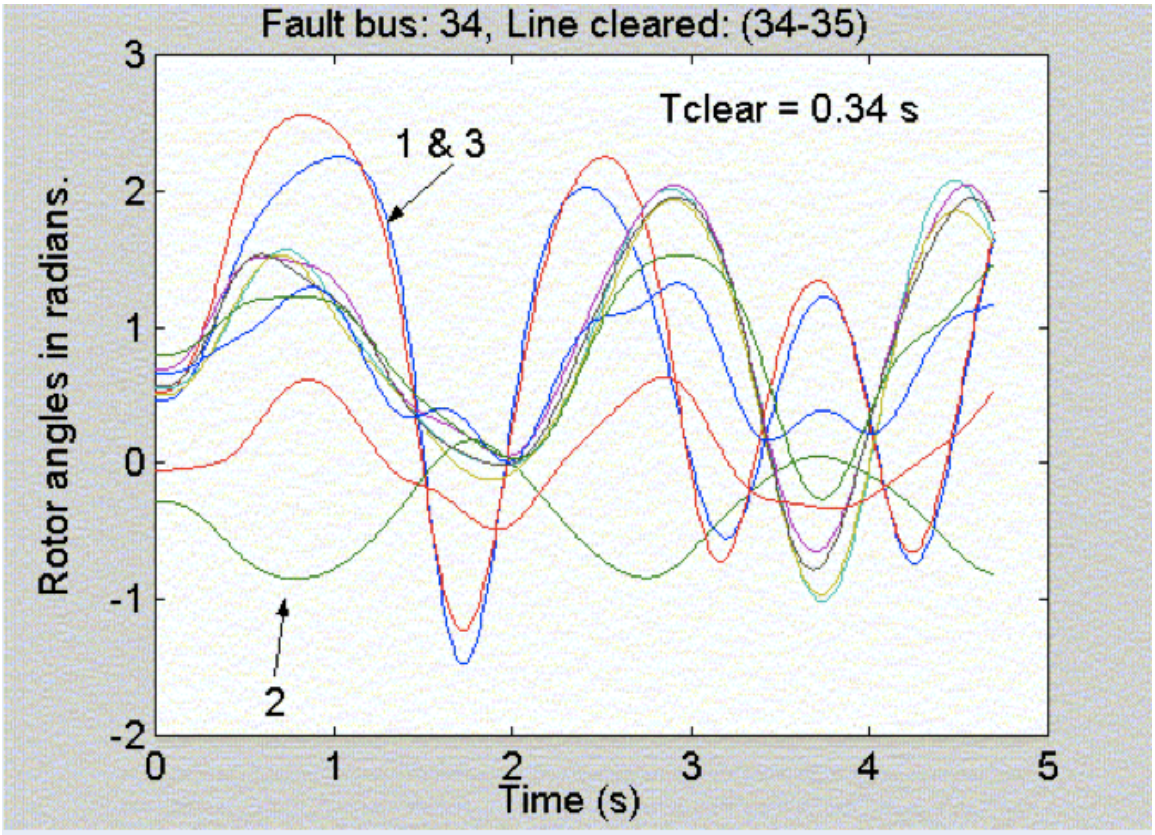
$$\delta_{COI} = \frac{H_1\delta_1 + H_2\delta_2 + \ldots..}{H_1 + H_2 + \ldots.}$$

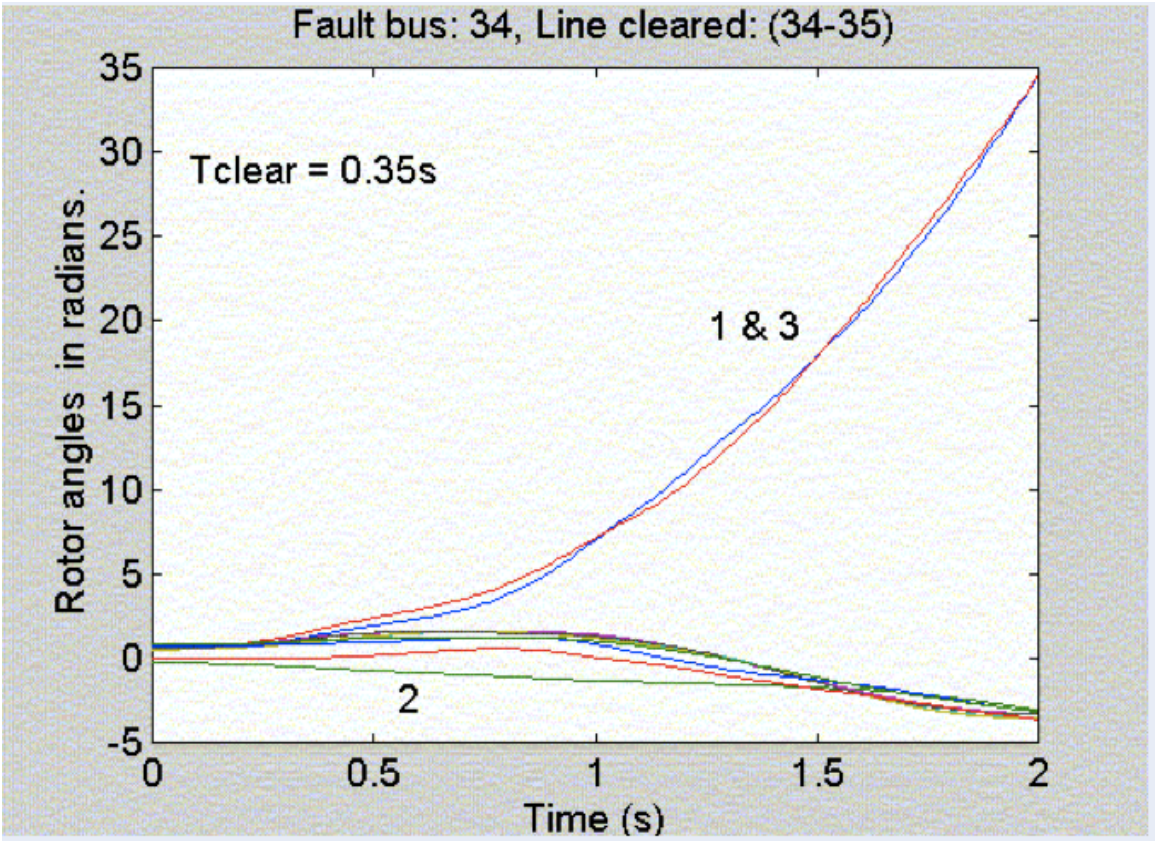
Note: Movement of all generator rotors together (or equivalently, movement of the center of inertia) *does not* indicate loss of synchronism. Instead it tells us that overall load and generation in the system are not balanced. This issue is discussed in the next module. Loss of synchronism means that the ***relative motion*** between generator rotors is unstable.

This is illustrated by the plot of rotor angles for a **10 machine system** for a large disturbance (detailed machine model and constant impedance loads are considered). Machines 1 and 3 separate out from the other machines if the clearing time is greater than 0.34 s.

10 machine system




Fault bus: 34, Line cleared: (34-35)
Tclear = 0.34 s
1 & 3
2
Rotor angles in radians.
Time (s)
3
2
1
0
-1
-2
0
1
2
3
4
5


Fault bus: 34, Line cleared: (34-35)
Tclear = 0.35s
1 & 3
2
Rotor angles in radians.
Time (s)
35
30
25
20
15
10
5
0
-5
0
0.5
1
1.5
2

Angular Stability and System Operation

Operation under loss of synchronism conditions is not feasible as voltage, torques etc. undergo large fluctuations and may damage equipment.

How does one prevent loss of synchronism due to disturbances?

Some of the methods used to reduce the possibility of loss of synchronism are:

a) **Reducing the magnitude of disturbance:** Large disturbances are typically due to faults which are cleared by line tripping. Reducing the duration of fault is one method of preventing loss of synchronism. These days, it is possible to clear faults within 3-5 cycles (taking into account relay and circuit breaker opening times).

b) **Planning:** (for a "stronger" transmission network). This would mean augmenting the transmission network with new transmission lines. Increasing the power transfer capability of transmission lines by compensating transmission reactance with series capacitors and shunt capacitors.

c) **Control:** The short term or excess capacity of many equipment can be used to enhance stability by appropriate controls. For example, excitation systems can allow for short term boosting of field voltage. Also, capacitors can be switched into the network in series and shunt. The switching may be done by means of circuit breakers or power electronic controls (power electronic controllers also allow for continuous control; this will be discussed in later modules).

d) **Operation:** During operation, it is ensured that the system is operated such that there exists sufficient "margin" for the system to withstand a credible disturbance (like a fault). This imposes a constraint on the level of power transfers which are permissible -- for larger levels of power flow, the synchronous generators are more likely to lose synchronism after a large disturbance. A system operator continuously monitors a power system and ensures that this is unlikely to happen. If the system is unlikely to withstand a credible disturbance, then *preventive control actions* like reducing the power transfer levels are taken. This may be achieved by *re-scheduling* power generated at various generators.

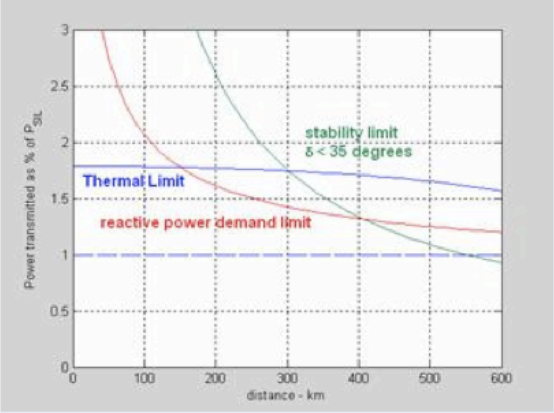
Line Loadability and Stability Constraints:

We mentioned in the previous slide that during operation, it is ensured that the system is operated such that there exists sufficient margin for the system to withstand a disturbance. It is ensured that phase angular difference across transmission paths is not too large. Thus angular stability puts a limit on the levels of power which can be transferred *securely.* One of the constraints for long distance AC transmission (other than thermal / voltage constraints) is the large phase angular difference which is required to transmit a given amount of power.

The modified loadability of a line (see a previous **example** on line loadability) by taking into account stability limits is shown in the figure on the right.

It is not easy to specify stability limits as they depend on the parameters of power system components and the operating conditions. However, for convenience it is specified by the maximum phase angular difference which is allowed across the line.

It is clear that stability is *the* major constraint for lines more than 300 km long.



A thought which may come to your mind at this point is: if ac interconnections have these constraints (voltage/ angular stability), why not go in for DC transmission especially when transmitting power over long distances? This is an important planning option.

To answer this question dispassionately, let us consider the following points:

1. Long distance transmission requires us to use high voltages and lower currents (otherwise resistive losses will be too large). So step-up will still be required (it is not feasible (as yet) to generate power by ac or dc machines at high voltage). So transformers will be used to step-up generated ac voltage to higher ac voltage, and then rectify it to DC using power electronic converters. DC is again inverted to ac at the receiving end of the line. Thus a dc interconnection will require 2 large ac to dc converters at either end.

2. Voltage profile along a line is not an issue since series inductive and shunt capacitive effects are not manifested in steady state for DC transmission. Resistive drop is usually small.

3. Power Flow **can be controlled** using firing angle of thyristors used in the rectifier and inverter. Power flow is *independent* of the phase angular difference between sending(rectifier) and receiving(inverter) ends! So angular stability constraint is not there. In fact if two areas are connected by only dc inter-connections, then even frequency of the two areas need not be the same.

Thermal limit is the only limit for DC transmission lines and loadability is practically independent of distance.

The main disadvantage of dc links is the *high cost* of the power electronic converters. However, in many cases DC may be most suitable option inspite of the cost. Therefore, you do see several applications of dc links in our country (as an exercise, can you do a survey of this ?).

Recap

In this lecture you have learnt the following

- AC systems face an inherent angular stability problem: Synchronous machines lose synchronism with each other when subjected to large disturbances
- System operators avoid the angular stability problem by operating the system conservatively
- Operating conservatively implies that power flow in long ac transmission is severely curtailed due to angular stability constraints. HVDC transmission is an option for transmitting bulk power over large distances

Lecture 8a A Brief Review of Control Feedback Systems

Objectives

In this lecture you will learn the following

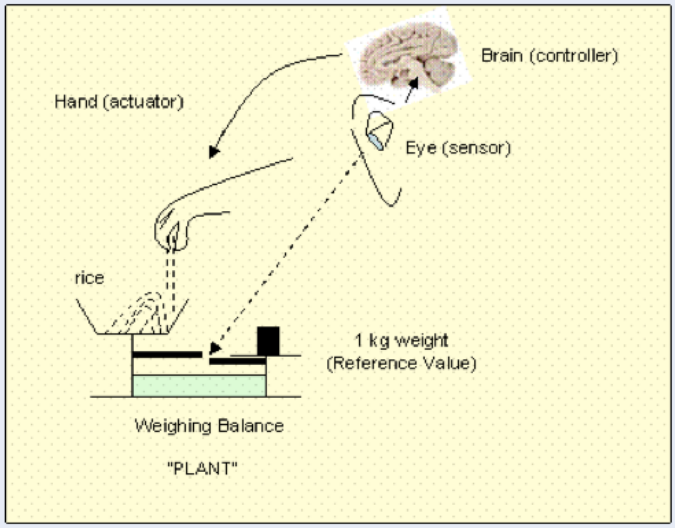
- What are feedback control systems ?
- Transfer Function Representation of linear systems
- Proportional and Integral Controllers

What is a feedback control system ?

Before we move on to study frequency control of a power system, we will study about another phenomena known as **voltage instability**. To get a better understanding of frequency control and voltage instability, we will first briefly review the topic of **feedback control systems**.

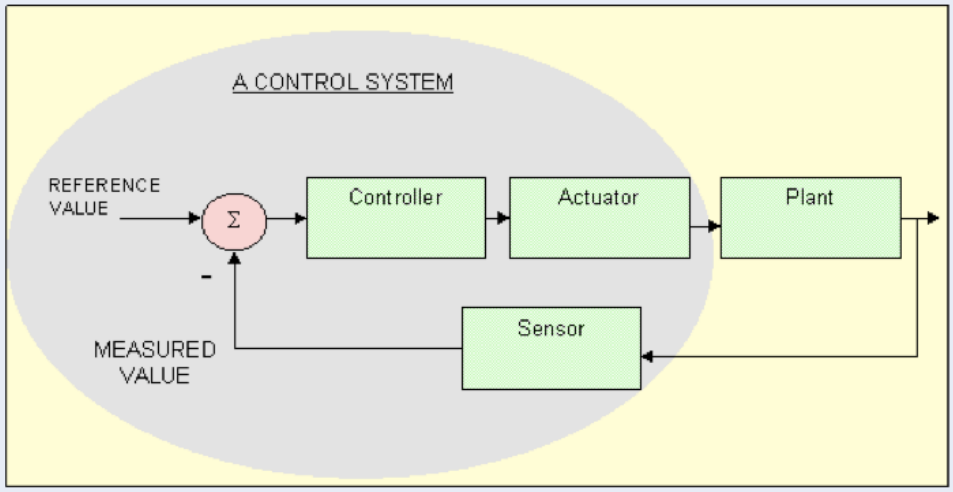
What does one mean by a feedback control system and why/when is it necessary? To understand this let us take a simple example.

Let us suppose that you wish to buy 1 kg of rice from your grocer. If all rice grains were of the same weight, it would be possible (although by a very tedious process!) to calculate how many grains are required. However, not only will this be time consuming, but it is also true that there are uncertainties in the grain weight. Therefore, rather than follow this procedure the grocer will put some rice into his weighing pan (counter balanced by 1 kg standard weight) and depending on the extent of imbalance, he will incrementally add or reduce grain in the pan. A seasoned grocer may even hold some grain in his fist and gradually pour it by monitoring the position and rate at which the pan moves. By doing this, he can speed up the process of weighing the rice. This is a manual feedback control system. The major components are shown below:



Components of a feedback control system

We saw in the previous slide that a feedback system can be used to quickly perform the task of weighing 1 kg of rice.It was a manual feedback which involved a human being as the controller and actuator. While a controller decides on what control actions are to be taken based on the feedback obtained by a sensor, an actuator is a device that converts the command of a controller into appropriate actions.

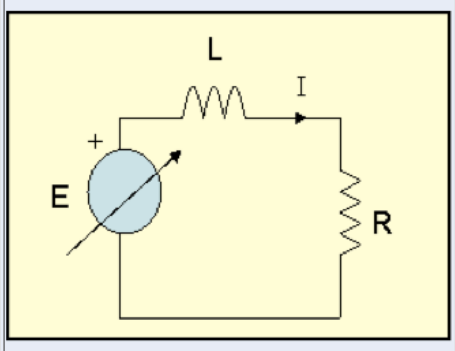


In an *automatic* feedback controller, the controller actions are decided by some mathematical function which is implemented by a computer or some electronic devices. The nature of mathematical function is decided based on the static and dynamic characteristics of the plant itself.

The main aim of the control system above is to ensure that the steady state value of the variable is equal to the reference value. This kind of a control system is referred to as a **regulator**. Note: some control systems do not necessarily have a steady state regulation objective, but use feedback to enhance the stability of a plant.

Note that an automatic controller (which is a mathematical function), if designed improperly may worsen the stability of the system. For example, in the rice weighing control system, if the grocer over-reacts to a small imbalance shown by the weighing machine, and pours in (removes) a lot of rice into (from) the weighing pan, then he may be caught in an endless cycle of removal and pouring in rice. This is an unstable situation. Therefore the controller actions should be designed with care.

Example : Suppose we wish to regulate the current in an R-L circuit by controlling the input voltage:



The equation describing the "plant" is : $\frac{dI}{dt} = -\frac{R}{L}I + \frac{E}{L}$.

If we wish to get a current $I = I_{REF}$ in steady state, then we can set the input voltage to be $E = RI_{REF}$. If we do this the transient response if such a voltage is applied to the circuit, assuming initial current is zero is:

$$I = (1 - e^{-\frac{R}{L}t})\frac{E}{R} = (1 - e^{-\frac{R}{L}t})I_{REF}$$

This kind of control is **"open loop"** and does not require continuous feedback of the current. However, the main problem is that one should know R accurately and also that the transient response is determined by R and L values -- it cannot be modified.

Proportional Control

An alternative to open loop control is feedback control wherein the input voltage E is adjusted continually based on the measured current (feedback). For example let us assume that the controller behaves as per the following law:

$$E = K(I_{REF} - I)$$

We assume that the current measurement is available without any delay. The resulting differential equation can be re-written as follows:

$$\frac{dI}{dt} = -\frac{R}{L}I + \frac{K}{L}(I_{REF} - I) = -\frac{(R+K)}{L}I + \frac{K}{L}I_{REF}$$

The response of the current if the controller comes into action at time t=0, and when initial current is zero is:

$$I = (1 - e^{-\frac{(R+K)}{L}t})\frac{KI_{REF}}{R+K}$$

In steady state, $I = \frac{KI_{REF}}{R+K}$. However, if K is very large compared to R, then in steady state (approximately), $I = I_{REF}$. This will also ensure that response will be faster and practically independent of R.

In practice K cannot be made too large, because it is possible that some component of transient response due to the *unmodelled* measurement (sensor) block may get destabilized.

The controller described above is called a proportional controller. An alternative to having large K to obtain $I = I_{REF}$, is to modify the nature of the controller. We can also use an *integral controller* for performing regulation function.

Integral Control

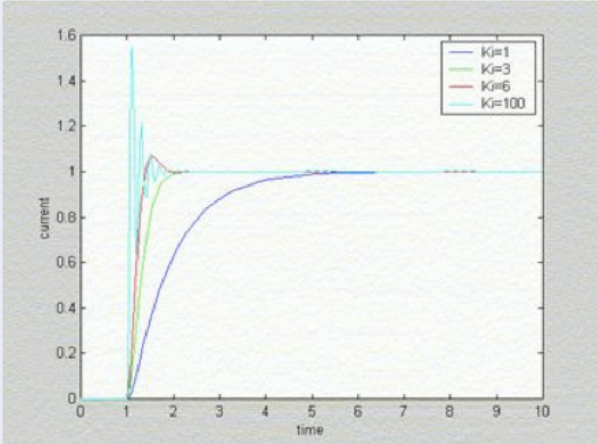
An alternative to using a proportional control law is to vary the input E using the following rule:

$$\frac{dE}{dt} = K_i(I_{REF} - I)$$

For an integral controller, in steady state $I = I_{REF}$. If this were not true, E would go on increasing, as a consequence of the above equation (therefore contradicting the statement that the system is in steady state).

Therefore Integral control ensures perfect regulation in steady state. .

However transient response can be poor with an integral controller. A step change in the reference value of current can lead to oscillatory behaviour. This is depicted for a integral controller for which the reference current is increased from 0 to 1 A at t=1s. The initial current in the circuit is zero and R=1 W, L=0.1 H. It is clear that a very large value of integral controller gain gives an overshoot and oscillatory behaviour



(click to enlarge)

To obtain good regulation as well as good transient response, one can use a combination of a proportional and an integral controller. The following control law achieves this:

$$\frac{dE_I}{dt} = K_i(I_{REF} - I)$$
$$E_P = K_p(I_{REF} - I)$$
$$E = E_P + E_I$$

Alternatively, one could also have an additional component of E which varies proportional to the rate of change of current. Such a controller is called a proportional-integral-derivative controller. Note that in practice one may only be able to obtain an approximate derivative (an exact derivative requires future information which is not possible to implement due to causality conditions).

While we have discussed the nature of control laws, we have not really discussed the *design* of the individual parameters of controllers. The choice of actual values of gains (e.g. K, Ki, Kp), are chosen based on the satisfaction of certain specifications like maximum steady state error, stability, overshoot etc. For a detailed account of design issues, you may refer to any book on basic control systems (see, for example, Nagrath I J & Gopal M: Control System Engineering – New age International Publishers).

We now discuss the transfer function representation which is commonly used to represent such systems.

Transfer Function Representation of Systems

Many control systems are depicted as transfer function block diagrams. Transfer function representations are obtained by taking the Laplace transformation of all the variables and obtaining relationships between them. For example, the circuit equation of the R-L circuit given by:

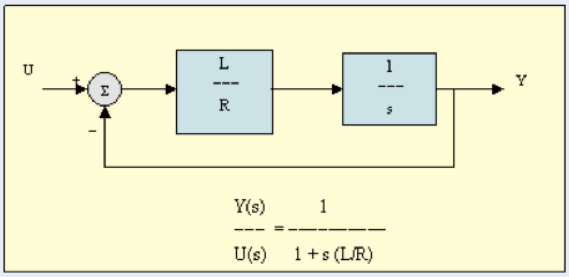
$$\frac{dI}{dt} = -\frac{R}{L}I + \frac{E}{L}$$

can be written down as: $sI(s) = -\frac{R}{L}I(s) + \frac{E(s)}{L}$ by taking the Laplace Transforms of the variables. The "transfer function" between I(s) and E(s) is given by:

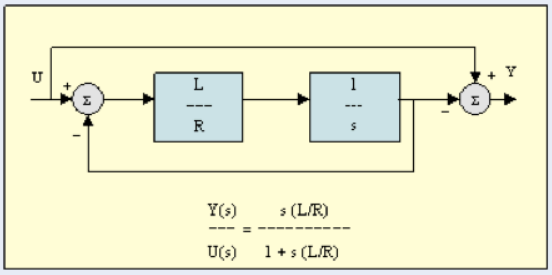
$$\frac{I(s)}{E(s)} = \frac{1}{R + sL}$$

Note here that if $X(s)$ is the laplace transform of $x(t)$, then $\frac{X(s)}{s}$ is the laplace transform of $\int x(t)dt$.

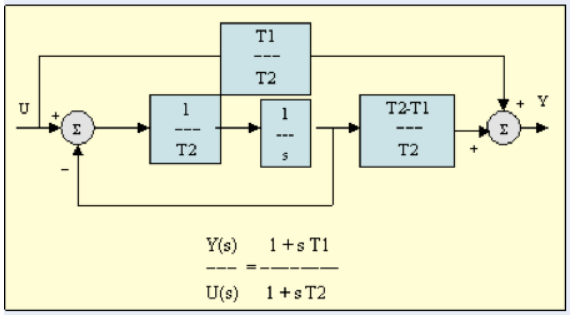
The block diagram equivalent (this is not unique) using the transformed variables, of various transfer functions are shown in the figures below:



1. A first order transfer function. Steady state gain =1

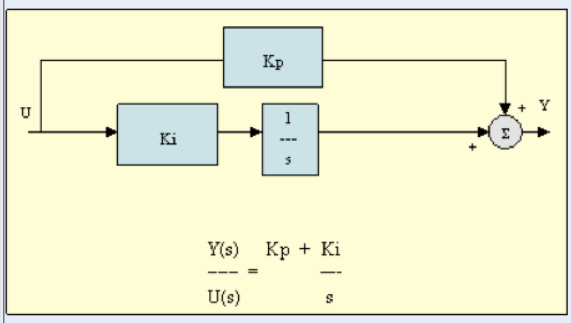


2. A "washout" transfer function. Steady state gain = 0



3. A lead (if T1>T2) or lag (if (T2<T1) function.

Steady state gain = 1



4. A Proportional-Integral Controller. Steady state gain = infinity

Recap

In this lecture you have learnt the following

- What are Feedback Control Systems
- Proportional and Integral Controllers

Lecture 9 Voltage Instability

Objectives

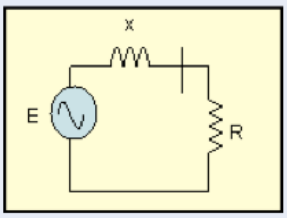
In this lecture you will learn the following

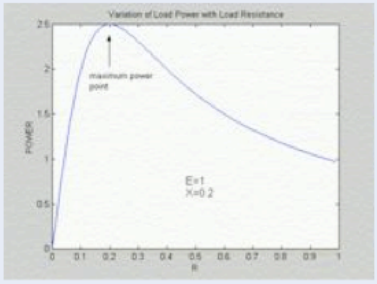
- What is voltage instability ?

Voltage Instability

Voltage Instability is another stability problem that may manifest under heavy loading conditions. This problem causes extremely low voltages. It stems due to the actions of loads to demand power which is beyond the capability of the generators and the transmission network.

To understand this, consider a system shown below. As the load resistance decreases, the voltage at the load bus falls while power is expected to increase. However, a point comes beyond which the load power *decreases* as resistance falls. This occurs at R=X (the maximum power point) --- show this!





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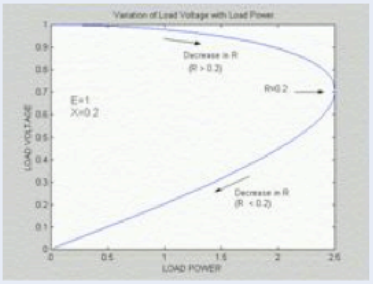
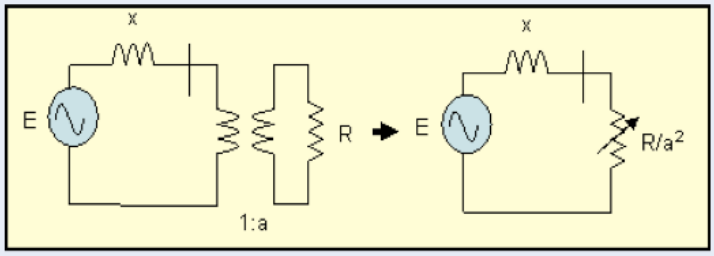


(click to enlarge)

Now, suppose the load is not a constant resistance, but is a *controlled* resistance which tries to keep load power constant by *decreasing* resistance if power is less than the desired value. This is all fine if the desired power is less than the maximum power point. If the desired power is greater than the maximum power point, the system will never reach an equilibrium because beyond the maximum power point, the controller action reduces the resistance, which *reduces* power and further pulls down the voltage. Usually any controller will have limits because of which controller action will cease beyond a point. However, the "damage has been done", in the sense that load voltage will be very low.

Voltage Instability (Con td..)

A *controlled* resistance which maintains constant power can be implemented using a tap changing transformer feeding a constant resistance. The tap value is changed if voltage at load is not equal to the desired value. Therefore the reflected resistance on the system side is controlled.



Normally, a power system has connected loads which are lesser than the maximum power transfer capability of the generation and transmission network. However, loss of lines may significantly increase transmission reactance. Generators may also hit their reactive capability limits resulting in inability to maintain voltage at key points in the network. This results in decrease of effective "E" and increase of effective "X" seen by loads.

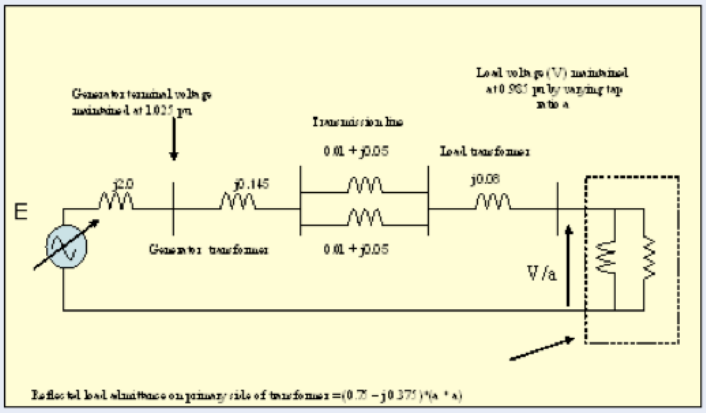
We illustrate this with an **example.**

A stronger transmission network and adequate reactive power reserves, to maintain voltages at key points in the network, are needed to avoid voltage instability.

We will now see how voltage instability can occur in a stressed system with load restoration dynamics.

Consider the following system, i.e., a single generator supplying a load through a transmission system and transformers. The load transformer has provision for taps which allows it to maintain a voltage V at the load. Due to tap changing transformer action, the load attempts to draw a constant power irrespective of the voltage on the transmission line side of the transformer.

In the diagram given on the right, the load is reflected on to the transmission line side of the transformer.



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The main parameters for the system are as follows:

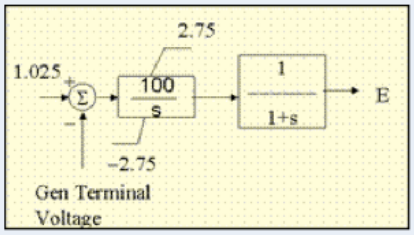
Generator Transformer reactance : 0.145 pu , Load Transformer reactance : 0.08 pu

Transmission line impedance (each line) : 0.01+ j 0.05 pu,

Load Admittance : 0.75 - j 0.375

Generator is represented as a voltage behind its synchronous reactance (2.0 pu). The magnitude of the generator terminal voltage is maintained at 1.025 pu by changing the "internal voltage", E. This internal voltage is proportional to the field voltage. Due to field current limits, the maximum value of E is 2.75 pu, beyond which the generator is unable to regulate the terminal voltage at 1.025 pu. The generator terminal voltage regulator, along with exciter and field dynamics is assumed to be represented by the block diagram on the right.

In actual practice, excitation system and field flux dynamics are more complicated but we use simple models here.



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Tap Changing Transformer: A tap changer tries to keep voltage at the load a constant. This also implies that tap changer tries to maintain the load power irrespective of the voltage on the transmission line side of the transformer

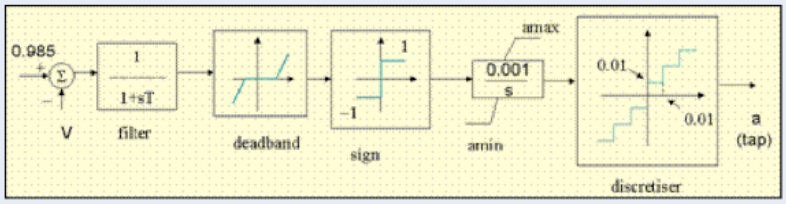
Initial ratio of (1:a) load transformer is 1:1.1 so as to obtain an approximate initial voltage of 0.985 pu at the load.

Maximum : a=1.2, Minimum : a=0.9

The tap changer control system is assumed to be as follows:

Taps ratio is changed in discrete steps.

The tap ratio of the transformer is incremented every 10 seconds by 0.01 if voltage at load is less than the load reference voltage of 0.985 pu. It is decremented by 0.01 every 10 seconds if it exceeds the reference voltage. Since tap values are discrete, this may result in "chattering" between two values. A filter (time constant T=1s) and a deadband (+/- 0.01) is provided to avoid frequent operation. The block diagram is shown on the right.



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Simulate the following cases by Runge Kutta method with time step of 1 second.

a) A loss of one transmission line (at 1000s)

b) A step change in load admittance to 0.85 - j 0.375 at 3000 s

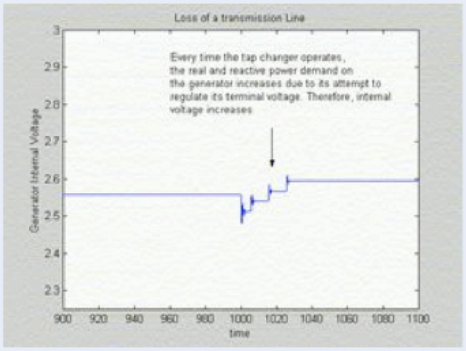
Simulate the behavior of tap ratio, voltage at load & generator terminals and the internal voltage E.

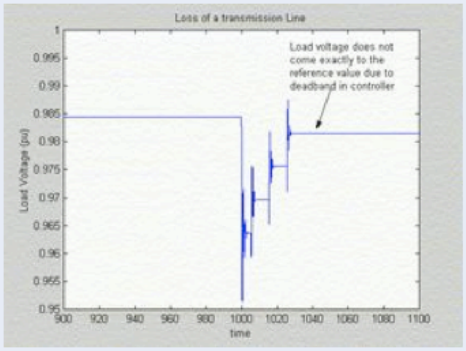
For solution refer to the next page.

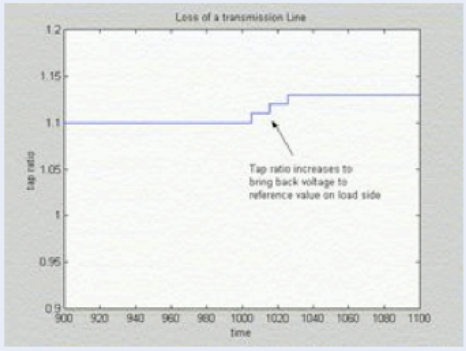
Solution:

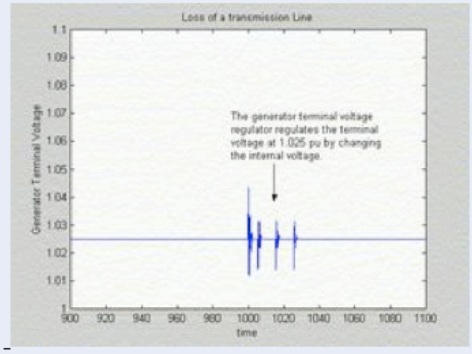
The system given above has been simulated with MATLAB/SIMULINK. **The model file tap.mdl can be downloaded from here.**

The simulated responses for the loss of one transmission line at 1000 s are given below :









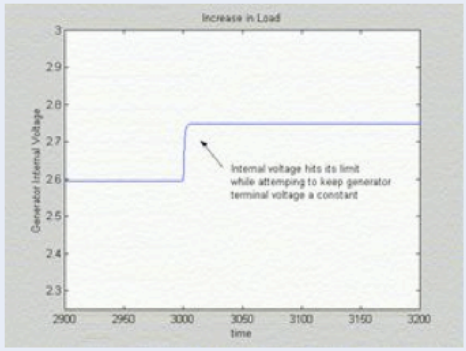
It is clearly seen that the voltage at the load recovers to its reference value due to tap changer action. Also note that the field voltage of the generator increases to keep the terminal voltage of the generator a constant. Thus the network and generator are capable of meeting the load demand even though the transmission is depleted due to loss of a line.

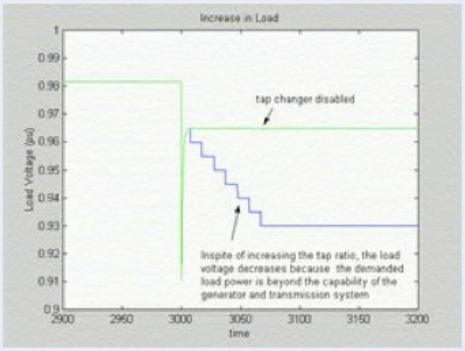
But can the same be said if in addition, there is a load increase ?

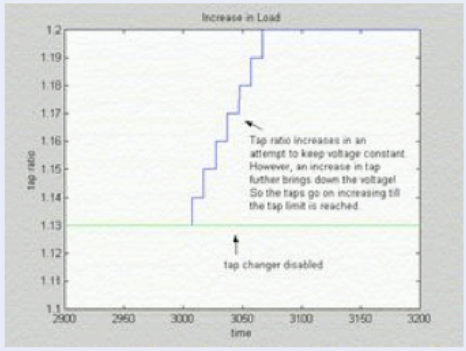
The simulated response for the increase in load at 3000s is shown in the next slide.

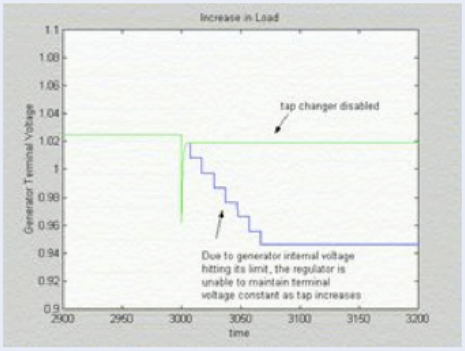
Solution (con td...):

We now simulate an increase in the load at 3000 s.









It is clearly seen that due to the increased load the field voltage tries to increase in order to keep the terminal voltage constant. However, due to the field voltage limit, the generator cannot do so. As the tap ratio of the transformer is increased in order to maintain load voltage at the reference value, the situation is worsened. The voltage at the load keeps falling (instead of increasing) with increasing tap ratio, since the generation and transmission system cannot supply the demanded power. Interestingly, if the tap changing of the transformer is disabled, then the load voltage settles to a better value (although not at the original one)!

If the generator field voltage limit is increased, then it is possible to cater to the increased load (try this out in the simulation program).

This example shows that the attempt to draw constant power (by use of tap changing transformer) beyond the capacity of the generator and transmission system results in instability and low voltages. In this case the major limitation is the field current capability of the generator. If one wishes to cater to this load, one would require to augment the system with devices which can provide additional support to the voltage of the system at various locations, like shunt capacitors.

To see this, you may reduce the inductive susceptance(B_L=0.375) of the load by half. You will see that for the same disturbances, the system is voltage stable.

Recap

In this lecture you have learnt the following

- Voltage stability is triggered by having loads which attempt to draw power which is beyond the maximum capability of generation and transmission.
- Limited power transfer capability occurs due to a weak network aggravated by generators hitting their reactive power capability limits

Concluding Remarks

1) Power Transfer Capability is affected by equipment and stability limits.

2) Equipment limits of Generators and Transmission Lines are essentially thermal and dielectric limits.

3) Generator has thermal limits due to armature heating, field winding heating and core -end heating.

4) Power transfer of short lines is limited due to thermal constraints, while voltage and stability considerations prevail for long EHV lines.

5) Interconnected Generators may lose synchronism if subjected to large disturbances.

6) A stressed power system with inadequate reactive power capability and loads with restoration dynamics can experience voltage instability.

Congratulations, you have finished Module II.

Module 3: Frequency Control in a Power System

Lecture 10 Introduction

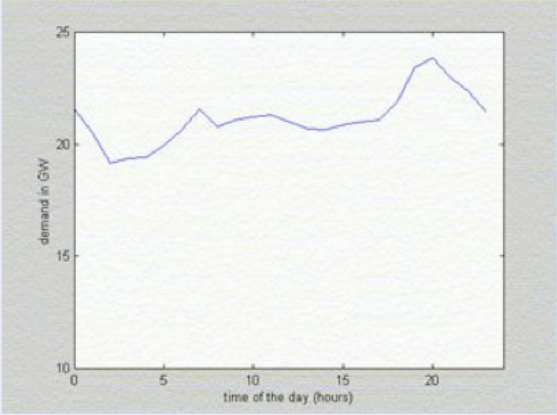
Objectives

In this lecture you will learn the following

- Why is maintenance of frequency important?
- An example of typical load and frequency variation in an Indian grid.

Introduction

Frequency all over a synchronous power grid is the same in steady state. Maintaining a near-constant frequency (one may allow frequency to vary over a very narrow band) is considered an important requirement of power system operation.
Frequency in a power system is intimately related to the electrical speed of synchronous generators. The difference between mechanical and electrical torques govern acceleration of a rotor of a generator. Therefore to maintain a constant speed, mechanical input and electrical output power need to be continually matched. Electrical load can vary randomly, but the total load versus time *roughly follows a trend*. For example, total load in a grid can vary over twenty four hours as shown in the figure on the right.



(click to enlarge)

There is a distinctive peak at about 8 pm. This is the typical variation for a particular season of the year. For other seasons, the load profile is different. Also, power systems are not immune to sudden large load or generation throw-offs due to contingencies.

Frequency variation is dependent on several factors which includes the load characteristics and generator prime mover controls. We shall study these issues in detail in the following lectures.

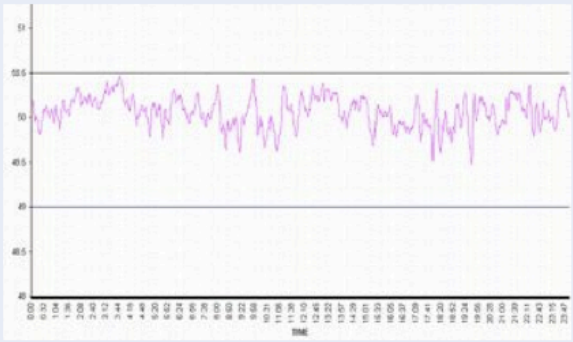
Why is maintenance of frequency in a power system important?

Frequency needs to be maintained near 50 Hz for the following reasons:

- Steam turbine blades are designed to operate in a narrow band of frequencies. Deviation of frequency beyond this band may cause gradual or immediate turbine damage. Consequently, protective and control equipment take corrective action in case of under/over frequency. A 50 Hz steam turbine may not be able to withstand frequency deviation of +2 Hz to -2.5 Hz for more than an hour in its entire life!

- Loads and other electrical equipment are usually designed to operate at a particular frequency. Off-nominal frequency operation causes electrical loads to deviate from the desired output. The output of power plant auxiliaries like pumps or fans may reduce, causing reduction in power plant output.

At the present time, grid frequency deviation in some Indian grids is generally between 50.5 and 49.0 Hz under normal operating conditions. The figure below shows the typical frequency variation in the Western Grid of India (taken from the Western Regional Load Dispatch Center website - www.wrldc.com).



Recap

In this lecture you have learnt the following

- Why is maintenance of frequency important ?
- Typical load and frequency variation in an Indian grid

Lecture 11 Definition of Frequency

Objectives

In this lecture you will learn the following

- What determines the frequency at which all generators in an inter-connected system settle to?
- What is the relationship between generator rotor speeds and "system frequency" during transients?

What determines grid frequency?

While it has been stressed that frequency is the same in steady state throughout a synchronous grid, two very interesting questions need to be answered:

- What is the precise value of frequency to which a system will settle?
- How does the "system frequency" vary during transients? How does one obtain a useful definition of frequency when individual generators speeds can be different during transients?

To understand this we need to look at the basic governing equations of synchronous machines in a system.

We know that for the **ith** generator,

$$\frac{d(\omega_i - \omega_o)}{dt} = \frac{d\omega_i}{dt} = \frac{\omega_B}{2H_i}(T_{mi} - T_{ei})$$

where ω_i is the rotor speed and ω_o is the nominal (or rated) speed.

T_{mi}, T_{ei} are the mechanical and electrical torques respectively expressed in per-unit.

ω_B is the base frequency and H is the inertia constant in MJ/MVA.

By summing up these equations we obtain,

$$\sum \frac{2H_i}{\omega_B}\frac{d\omega_i}{dt} = \sum(T_{mi} - T_{ei})$$

If frequency deviations are assumed to be small, then in per-unit $(T_{mi} - T_{ei}) \approx (P_{mi} - P_{ei})$

Therefore,

$$\sum \frac{2H_i}{\omega_B}\frac{d\omega_i}{dt} = \sum(P_{mi} - P_{ei})$$

If natural transients in the transmission network are neglected, $\sum P_{ei} = \sum P_{Lk} + losses$

where, $\sum P_{Lk}$ is the total load on the system.

$$\sum \frac{2H_i}{\omega_B}\frac{d\omega_i}{dt} = \sum P_{mi} - \sum P_{Lk} - losses$$

Note : P_{mi} and P_{ei} denote the per unit mechanical and electrical powers respectively.

Since in general, P_{Lk}, P_{mi} and even losses are functions of the frequency:

$$\sum \frac{2H_i}{\omega_B}\frac{d\omega_i}{dt} = \sum P_{mi}(\omega_i) - \sum P_{Lk}(\omega_{bk}) - losses(\omega_1, \omega_2 \ldots)$$

ω_{bk} is the bus frequency at the k^{th} load.

Note that frequency all over the system is equal in steady state.

Therefore, in steady state

$$\sum P_{mi}(\omega_e) - \sum P_{Lk}(\omega_e) - losses(\omega_e) = 0$$

The solution of the above equation gives the value of steady state frequency ω_e.

It was seen that steady state frequency is determined by the load-generation-loss balance equation. For analysis under transient conditions, consider the analytical development and illustrative analogy given below.

There are two types of motion to be considered when generator rotors are subject to transients :

a) **Relative motion between generators:** Rotor speeds may not be equal during transients. Exchange of power between generators during transients causes relative rotor movement ("swings"). This phenomenon was considered in the previous module. Relative motion dies down if the system is *angular stable*. Relative motion needs to be stable so that eventually all rotors move together at a common speed.

b) **Motion of the center of inertia** or the "**common motion**": If we sum up the swing equations of individual generators, we obtain:

$$\sum \frac{2H_i}{\omega_B}\frac{d\omega_i}{dt} = \frac{2(\sum H_i)}{\omega_B}\frac{d\left(\frac{\sum H_i\omega_i}{\sum H_i}\right)}{dt} = \frac{2H_{coi}}{\omega_B}\frac{d\omega_{coi}}{dt} = \sum P_{mi} - \sum P_{Lk} - losses$$

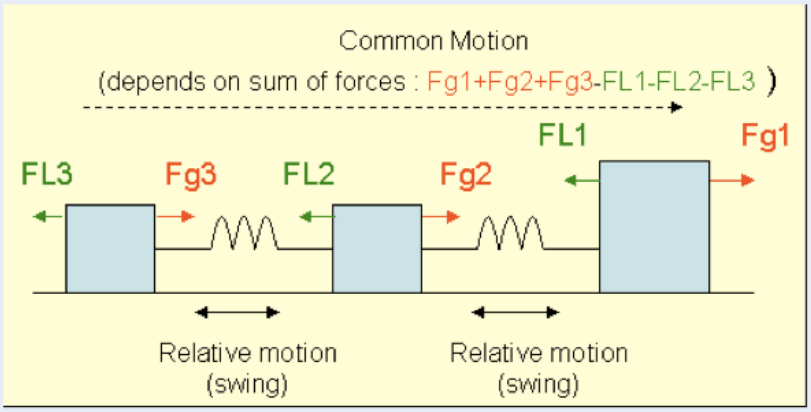
Note $\omega_{coi} = \left(\frac{\sum H_i\omega_i}{\sum H_i}\right)$ represents the centre of inertia speed of the system.

Interestingly, **center of Inertia** movement is determined **ONLY** by the load-generation-loss balance. Motion of center of inertia may exist even if there is no *relative* movement between rotors. Conversely it is possible to have relative motion between generator rotors without center of inertia movement.

Center of inertia frequency can be considered as a definition of "system frequency" under transient conditions. You can verify that under equilibrium conditions, i.e., when all machines settle to a common speed we, center of inertia frequency also equals we.

In this module, we will concentrate on center of inertia movement and not on relative motion between rotors. We shall assume that relative motion is stable (machines stay in synchronism).

We illustrate the concept of "system frequency" as a quantity affected by the aggregate of forces acting on the system by the familiar mass - spring analogy.



Recap

In this lecture you have learnt the following

- A system settles to a frequency at which generation is equal to the load plus losses.
- It is important to distinguish between relative motion between generator rotors and center of inertia (common) movement.
- Center of inertia movement is dependent **only** on the load-generation balance and is a useful definition of frequency under transient conditions.

Lecture 12 Load Characteristics

Objectives

In this lecture you will learn the following

- Load - Frequency characteristic of typical loads.
- An example of the characteristics of an induction motor driven fan type load.

Load Frequency Characteristics

Normally, loads are a function of frequency. For example, a fan driven by an induction machine slows down and consumes lesser power if frequency is reduced. In the previous lecture we saw that a system will settle to a frequency at which load(+losses) and generation is balanced. Therefore it is instructive to examine the load behavior as a function of frequency.

For simplicity, let's assume voltage and frequency changes are small. In such a case, load can be represented (in steady state) as follows:

$$P_L = P_{L0}\left(1 + k_{pv}\frac{V - V_0}{V_0}\right)\left(1 + k_{pf}\frac{f - f_0}{f_0}\right)$$

$$Q_L = Q_{L0}\left(1 + k_{qv}\frac{V - V_0}{V_0}\right)\left(1 + k_{qf}\frac{f - f_0}{f_0}\right)$$

where P_{L0}, Q_{L0} are the real and reactive power at nominal voltage V_0 and frequency f_0.

Load Frequency Characteristics (Con td..)

Typical values for various loads are given below:

Component	Power factor	k_{pv}	k_{qv}	k_{pf}	k_{qf}
Refrigerator	0.80	0.77	2.5	0.53	-1.5
Incandescent lights	1.00	1.55	0.0	0.00	0.0
Fluorescent lights	0.90	0.96	7.4	1.00	-2.8
Industrial motors	0.88	0.07	0.5	2.50	1.2
Fan motors	0.87	0.08	1.6	2.90	1.7
Agricultural pumps	0.85	1.40	1.4	5.00	4.0
Arc furnace	0.70	2.30	1.6	-1.00	-1.0
Transformer (unloaded)	0.64	3.40	11.5	0.00	-11.8

We now consider an example to show how the parameters of an induction motor driven fan load can be obtained.

Induction machine Load

Consider an induction motor driving a fan type load.

The load torque for a fan is a function of speed ω_r.
Let us assume that the per-unit torque of the fan (on machine VA base) is given by :

$$T_L = 0.9\left(\frac{\omega_r}{\omega_{so}}\right)^2$$

where ω_{so} is the synchronous speed at nominal frequency (50 Hz).
The motor torque in per-unit as a function of slip is

$$T_m = \frac{V_e^2 \frac{R_r}{s}}{\left(R_e + \frac{R_r}{s}\right)^2 + (X_e + X_r)^2}$$

where

$$s = \frac{\omega_s - \omega_r}{\omega_s}$$

where, ω_s is the synchronous speed at the supply frequency.
R_r is the rotor resistance in p.u.

Induction machine Load

$$V_e = \frac{jX_m}{R_s + jX_s + jX_m} V$$

$$R_e + jX_e = \frac{jX_m(R_s + jX_s)}{R_s + jX_s + jX_m}$$

X_s, X_r and X_m are the stator leakage, rotor leakage and magnetising reactances in p.u. respectively at the supply frequency. Thus,

$$X_s = X_{so}\frac{\omega_s}{\omega_{so}},$$

$$X_r = X_{ro}\frac{\omega_s}{\omega_{so}},$$

$$X_m = X_{mo}\frac{\omega_s}{\omega_{so}}$$

X_{so}, X_{ro} and X_{mo} are the stator leakage, rotor leakage and magnetising reactances in p.u. respectively at the nominal frequency.
V is the supply voltage magnitude.

Induction machine Load

The operating speed is obtained from the equation

$$T_m(\omega_r) = T_L\ (\omega_r)$$

If the parameters of the induction motor (in p.u. on machine VA base) are :

$$X_{so} = 0.065 \qquad R_s = 0.078$$
$$X_{ro} = 0.049 \qquad R_r = 0.044$$
$$X_{mo} = 2.67$$

Let $V = 1.0$ p.u.

Compute the change in power drawn by the induction machine for 1 Hz decrease in frequency

from the nominal (50 Hz).

Solution :

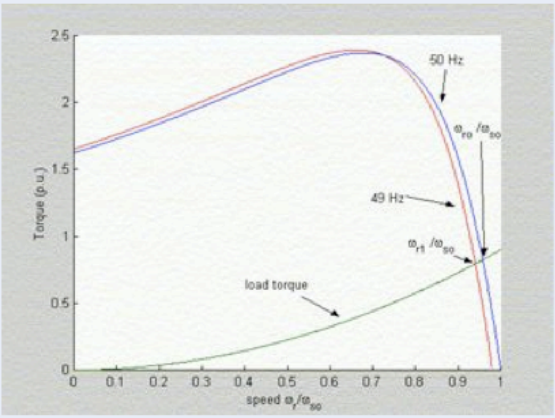
Using the formulae given above we can plot T_m and T_L vs $\frac{\omega_r}{\omega_{so}}$ as shown below

Clearly there is a decrease in the operating speed ($\frac{\omega_{ro}}{\omega_{so}} = 0.9579$ and $\frac{\omega_{r1}}{\omega_{so}} = 0.9405$).

and torque as the frequency decreases.

The percentage change in power (drawn from source) for percent change in frequency is given by :

$$k_{pf} = \frac{\Delta P}{P_o}\frac{\omega_{so}}{\Delta\omega_s} = \frac{0.8558 - 0.8232}{0.8558} * \frac{50}{1} = 1.90$$



Recap

In this lecture you have learnt the following

- Motor loads like fans, agricultural pumps usually have a significant frequency dependence.

Lecture 12a Solution of Non-Linear Algebraic Equations

Objectives

In this lecture you will learn the following

- A revision of how to solve nonlinear algebraic equations.

Non-linear algebraic equations and their solution

In the next lecture, we will compute the steady state frequency of a power system, given the load characteristics. We shall see that in general, this will require us to carry out a "loadflow". A loadlfow involves the solution of a set of non-linear algebraic equations. Therefore, in this lecture we revise the basic methods to solve non-linear algebraic equations.

We are aware that a transmission network in sinusoidal state state can be modelled by linear algebraic equations in the node voltage phasors(**V**) and the nodal current phasor injections (**I**):

$$I = Y_{bus} V$$

where Ybus is the bus admittance matrix.

However, in power system studies, nodal injections are not specified as current phasors but as real and reactive power injections (nonlinear functions of **V** and **I**) , and/or voltage magnitudes of some nodes. We have also seen that real and reactive power can be a function of frequency. In such a situation, obtaining the steady state solution (i.e. node voltage phasors and frequency) will require us to solve a set of *non-linear* equations.

Therefore we take a silight diversion from the main theme and review why and how we use numerical techniques for solving non-linear algebraic equations.

Let us consider the "why" question first. If we wish to solve an equation of the form:

$$e^{-x} - x = 0$$

Perhaps, "simplifying" it will help us solve it ?

$$e^{-x} = x$$

Perhaps, if we take the natrual logarithm of both sides we may be able to do something ?

$$-x = \log x$$

Fixed Point Iteration Method

Since we have some idea of how the exponential function behaves we can try to guess the solution. We know that:

$$e^{-x} = \frac{1}{e^x}$$

and

$$e^1 \approx 2.7$$

and

$$e^{0.5} \approx \sqrt{2.7} \approx 1.6 \approx \frac{1}{0.6}$$

We can guess that the solution for $e^{-x} = x$ should lie between 0.5 and 1.

However, this is a rough estimate. Surprisingly if we take an initial guess value :

$$x_0 = x_{init} = 0.5$$

and iterate as follows starting with k=0:

$$x_{k+1} = e^{-x_k}$$

then x_1=0.606, x_2=0.545 , x_3= 0.579, x_4=0.5600, x_5=0.571, x_6=0.565, x_7=0.568, x_8=0.566, x_9=0.567 ....

We seem to be "converging" to a solution which satisfies the equation $e^{-x} = x$!

Why does the Fixed Point Iterative Method Work ?

We can try to understand why we converge to the right solution by examining the behaviour of the iterative method near the solution. Suppose the correct solution to the equation $e^{-x} = x$ is x = x_s , i.e.,

$$x_s = e^{-x_s}$$

Suppose the value of x at the kth iteration is near the solution x_s and differs from it by a small amount Dx_k, i.e.,

$$x_k = x_s + \Delta x_k$$

then:

$$x_s + \Delta x_{k+1} = e^{-(x_s + \Delta x_k)} = e^{-x_s} e^{-\Delta x_k} \approx e^{-x_s}(1 - \Delta x_k)$$

which yields :

$$\Delta x_{k+1} \approx -e^{-x_s} \Delta x_k$$

therefore if at k=0, x = x_{init} then,

$$\Delta x_k = (-e^{-x_s})^k \Delta x_{init}$$

Since:

$$|-e^{-x_s}| < 1$$

Therefore as k tends to infinity, Dxk tends to zero. This means that if we start close enough to the solution, we will converge to the correct solution after many iterations, i.e., $x_k = x_s$ if k is large. We say that the solution has converged if :

$$|e^{-x_k} - x_k| < \epsilon$$

where ε is the desired tolerance.

The convergence is affected by the properties of the function at the solution point. You can check that if we wish to find out the solution of :

$$\frac{1}{x^2} - x = 0$$

by the iterative algorithm:

$$x_{k+1} = \frac{1}{x_k^2}$$

the solution will *diverge* for any value of the initial guess which is not the true solution (x = 1).

What happens if we formulate the iterative algorithm for the same equation as follows ?

$$x_{k+1} = \frac{1}{\sqrt{x_k}}$$

Fixed Point Iteration Method

We now formally write the algorithm down. If we wish to find the solution for a set of n algebraic equations in n variables:

$$\begin{aligned} f_1(x_1, x_2, \ldots x_n) &= 0 \\ f_2(x_1, x_2, \ldots x_n) &= 0 \\ &\cdot \\ &\cdot \\ f_n(x_1, x_2, \ldots x_n) &= 0 \end{aligned}$$

We first write the equations in the form (there is no unique way to to this!):

$$\begin{aligned} x_1 &= g_1(x_1, x_2, \ldots x_n) \\ x_2 &= g_2(x_1, x_2, \ldots x_n) \\ &\cdot \\ &\cdot \\ x_n &= g_n(x_1, x_2, \ldots x_n) \end{aligned}$$

We then guess the initial values of all n variables and iterate as follows:

$$x_{1_{k+1}} = g_1(x_{1_k}, x_{2_k}, \ldots x_{n_k})$$
$$x_{2_{k+1}} = g_2(x_{1_k}, x_{2_k}, \ldots x_{n_k})$$
$$\cdot$$
$$\cdot$$
$$x_{n_{k+1}} = g_n(x_{1_k}, x_{2_k}, \ldots x_{n_k})$$

Unfortunately, this method does not converge always to the true solution for all nonlinear equations (as seen in the previous slide) and even if it does it sometimes converges very slowly. Some variations are proposed to improve convergence. For example, for the solution of $x = g(x)$, instead of using $x_{k+1} = g(x_k)$, one may first compute:

$$\bar{x}_{k+1} = g(x_k)$$

$$x_{k+1} = \alpha \bar{x}_{k+1} + (1 - \alpha) x_k$$

where a is a value which is between 0 and 1. It is found that this may speed up convergence for certain nonlinear equations.

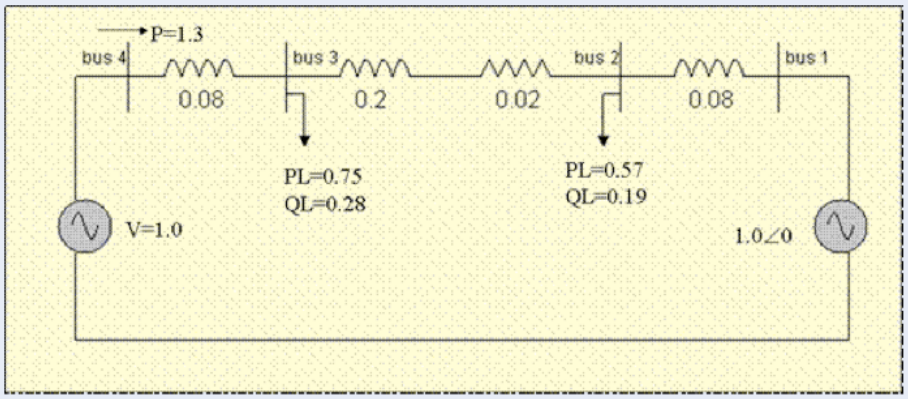
Alternative methods like **Newton Raphson(N-R)** can also be used for solving nonlinear equations. For an equation $f(x) = 0$, N-R algorithm is :

$$x_{k+1} = x_k + \left[\frac{\partial f}{\partial x} \right]^{-1} f(x_k)$$

where the partial derivative is **evaluated** at x=xk.

Exercise:

Can you write down the nonlinear algebraic equations corresponding to the real and reactive balance equations at each node for the system shown below ? The numerical values of the "known quantities" are indicated on the figure.



Can you formulate the iterative algorithm to find out the unknown quantities, i.e., the bus voltage phasors at buses 2 and 3, and the bus voltage phasor angle at bus 4, using the **fixed point iterative method**? Note that the bus voltage phasor at bus 1, voltage magnitude and real power injection at bus 4 is indicated on the figure.

Recap

In this lecture you have learnt the following

- A recap on how to solve non-linear algebraic equations.

Lecture 13 Calculation of System Frequency

Objectives

In this lecture you will learn the following

- How is equilibrium system frequency calculated if load parameters are frequency dependent ?

Calculation of System Frequency

The equilibrium system frequency $\left(f = \frac{\omega_e}{2\pi}\right)$ is a solution of the following equation :

$$\sum(P_{mi}(\omega_e) - \sum P_{Lk}(\omega_e) - losses(\omega_e)) = 0$$

If it is assumed that:

a) Loads are not voltage dependent but are frequency dependent. Load real power variation is characterised by:

$$P_L = P_{L0}\left(1 + k_{pf}\frac{f - f_0}{f_0}\right)$$

b) P_{mi} is not a function of frequency (generator mechanical power input is constant).

c) Losses are a small fixed proportion of the total generation and are frequency independent. Strictly speaking, losses *are* frequency dependent because transmission line current flows are dependent on the loads, which are themselves frequency dependent. However this effect is indirect and small.

d) Network parameters (reactance/ susceptances) are not frequency dependent.

Calculation of System Frequency (Con td..)

then equilibrium frequency is given by:

$$\sum P_{mi} - \sum P_{L0k} - losses = \sum P_{mi}(1-l) - \sum P_{L0k} - \sum P_{L0k} k_{pfk} \left(\frac{f - f_0}{f_0} \right) = 0$$

where *losses* are assumed to be l times the total generation.

therefore :

$$\frac{f - f_0}{f_0} = \frac{\sum P_{mi}(1-l) - \sum P_{L0k}}{\sum P_{L0k} k_{pfk}}$$

Let $\sum (P_{mi}(1-l) - P_{L0k}) = 0$, i.e., $\sum P_{L0k} = \sum P_{mi}(1-l)$.

then it is clear that $f = f_0$.

Now suppose 10% of the load is switched out, i.e. $\sum \hat{P}_{L0k} = 0.9 \sum P_{L0k}$, and

it is assumed that load frequency characteristics are the same for each load, i.e., $k_{pfk} = k_{pf}$,

then

$$\frac{\hat{f} - f_0}{f_0} = \frac{\sum P_{mi}(1-l) - \sum \hat{P}_{L0k}}{\sum \hat{P}_{L0k} k_{pf}} = \frac{\sum P_{L0k} - \sum \hat{P}_{L0k}}{\sum \hat{P}_{L0k} k_{pf}} = \frac{0.1}{0.9 k_{pf}}$$

if $k_{pf} = 1.5$, it is clear that $\left(\frac{\hat{f} - f_0}{f_0} \right) = 0.074$ implying that $\hat{f} = 53.7 Hz$.

Consideration of Real & Reactive Power Losses

If we consider voltage dependence of real and reactive loads and the marginal effect of change in losses, then the calculation of equilibrium frequency is a bit more complicated. The solution for equilibrium frequency has to consider real and reactive power balance at all nodes to obtain estimates of voltages and losses. Therefore a complete load-flow solution is required.

To illustrate this, we shall now consider an example in which we tackle three cases:

a) A "usual" load flow for a 4 bus system.

b) A load flow for a 4 bus system considering voltage dependence of loads, load at nominal voltage is specified and load is assumed to vary as follows:

$$P_L = P_{L0} \left(\frac{V}{V_0} \right)^2 \approx P_{L0} \left(1 + 2 \frac{V - V_0}{V_0} \right)$$

$$Q_L = Q_{L0} \left(\frac{V}{V_0} \right)^2 \approx Q_{L0} \left(1 + 2 \frac{V - V_0}{V_0} \right)$$

It is assumed that voltage deviation from the nominal is small.

Consideration of Real & Reactive Power Losses (Con td..)

c) A load flow in which **all** generator (including slack bus) powers are specified and frequency dependence of loads is considered.

$$P_L = P_{L0}\left(\frac{V}{V_0}\right)^2\left(1+2\frac{f-f_0}{f_0}\right) \approx P_{L0}\left(1+2\frac{V-V_0}{V_0}\right)\left(1+2\frac{f-f_0}{f_0}\right)$$

$$Q_L = Q_{L0}\left(\frac{V}{V_0}\right)^2\left(1-2\frac{f-f_0}{f_0}\right) \approx Q_{L0}\left(1+2\frac{V-V_0}{V_0}\right)\left(1-2\frac{f-f_0}{f_0}\right)$$

Cases (b) and (c) require minor modification of a "conventional" load flow program.

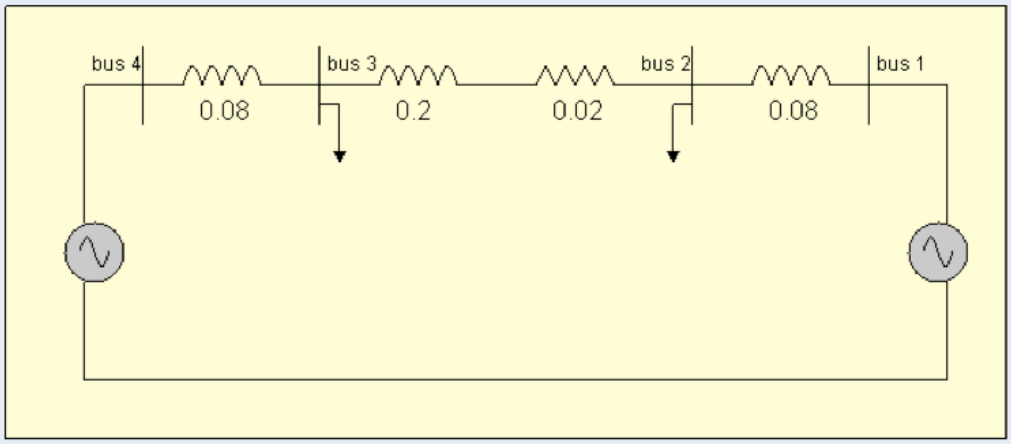
For (b), a simple modification is to *vary* the P-Q specification at load buses in accordance with the voltage magnitude at that iteration step.

For (c), at every step the calculated slack generator power is compared with the specified slack generator power, and frequency is *incremented* by some proportion of the difference. The P-Q specification at load buses is varied in accordance with the voltage magnitude and frequency at that iteration step.

With these two hints, can you write a load flow program which includes voltage and frequency dependence of loads?
You will need to use a method to solve non-linear algebriac set of equations, as discussed in the previous lecture.

We now give the **solved load flow cases (along with the specifications) for a 4 bus system**.

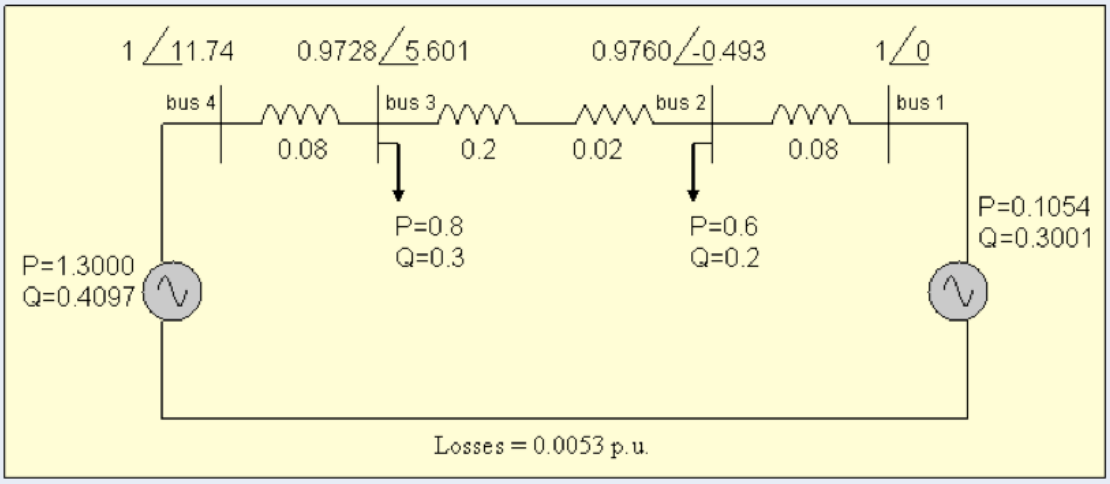
Consider a 4 bus system shown below:



Case (a): Carry out a "conventional" load flow. You may use the following downloadable MATLAB file lf.m. Once the file is downloaded, it can be invoked by the command lf in the matlab workspace.
Specifications:

Bus No.	Voltage	Phase Angle	PG	QG	PL	QL
1.	1.0	0	-	-	-	-
2.	-	-	-	-	0.6	0.2
3.	-	-	-	-	0.8	0.3
4.	1.0	-	1.3	-	-	-

The load flow solution is indicated on the figure below. Angles indicated are in degree.

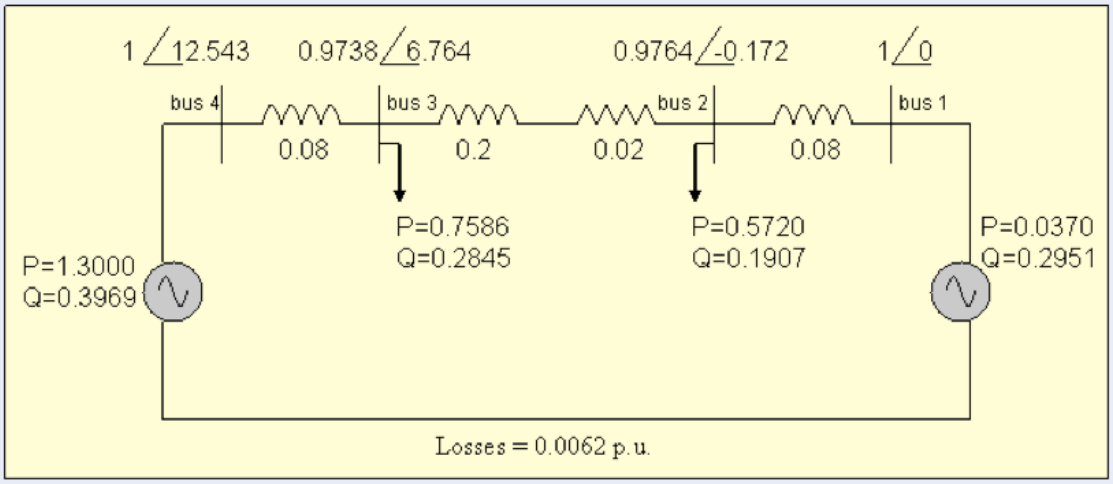


Case (b): Carry out a load flow when load is voltage dependent – load at nominal voltage is given. The factors k_pv and k_qv are 2 and 2 respectively. You may use the following downloadable MATLAB file lf_VD.m. Once the file is downloaded, it can be invoked by the command lf_VD in the matlab workspace.

Specifications:

Bus No.	Voltage	Phase Angle	PG	QG	PL0	QL0
1.	1	0	-	-	-	-
2.	-	-	-	-	0.6	0.2
3.	-	-	-	-	0.8	0.3
4.	1	-	1.3	-	-	-

The load flow Solution is indicated on the figure below - note that actual load is smaller than the nominal load because of lower voltage magnitude at the load buses. Angles indicated are in degrees.

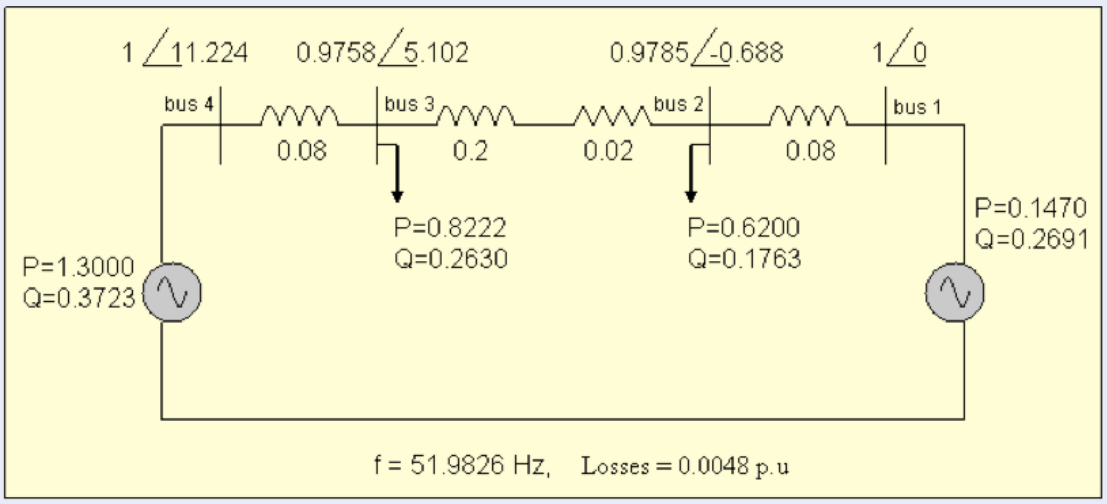


Case (c): Carry out a load flow when load is voltage *and* frequency dependent – load at nominal voltage and frequency is given. Also, slack bus power is specified. The factors k_pf and k_qf are 2 and -2 respectively. You may use the following downloadable MATLAB file: lf_VDFD.m. Once the file is downloaded, it can be invoked by the command lf_VDFD in the matlab workspace.

Specifications:

Bus No.	Voltage	Phase Angle	PG	QG	PL0	QL0
1.	1	0	0.1470	-	-	-
2.	-	-	-	-	0.6	0.2
3.	-	-	-	-	0.8	0.3
4.	1	-	1.3	-	-	-

The solution for system frequency is such that load-generation-loss balance is achieved at that frequency. With the specifications given above, the total generation is greater than the load and losses at the nominal frequency. Therefore, frequency will be higher than the nominal frequency.



Recap

In this lecture you have learnt the following

- We have calculated system frequency for a simple case of loads being frequency dependent but not voltage dependent.
- If both voltage and frequency dependence is to be considered, then we need to use a modified load flow program. This is demonstrated with an example.

Lecture 14 Frequency Control

Objectives

In this lecture you will learn the following

- How can frequency be controlled? What is a speed governor?
- How do parameters of a governor affect generated power?

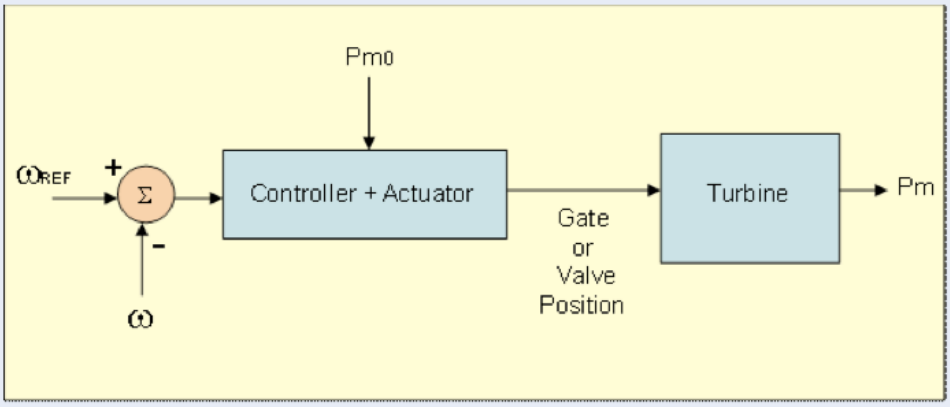
Speed Governor

We saw in a previous example that frequency deviation can be quite substantial -- more than 3 Hz -- for 10% change in load. In fact this deviation is considered unacceptably large (at this frequency, the generators will trip due to over-speeding).

Therefore, depending on the frequency dependence of load (which is quite weak) to bring about load generation balance is not a good operating practice.

Therefore it is important to introduce frequency dependence of generated power. This is done by a speed control mechanism known as **speed governing**.

Speed dependence can be introduced by having a feedback control system as shown in the figure below.



Speed Governor (Contd..)

The above schematic is **oversimplified**, especially in the case of thermal/nuclear reactors, which have a more complicated control structure involving a boiler and reactor. These additional controllers regulate other parameters like steam pressure by control of fuel, air and feed water. Therefore, strictly speaking, output power for steam turbines is not a function of control valve position alone. However, we shall use this simplified model to illustrate the concepts of speed control.

The exact nature of generator power - frequency steady state relationship can be "set" by the controller parameters and Pm0 - the initial setting.

Typically, **in steady state**, the generated output power (in per -unit) is related to change of speed by the following relationship:

$$P_m = P_{m0} - k_G\left(\frac{f - f_{REF}}{f_0}\right) = P_{m0} - k_G\left(\frac{\omega - \omega_{REF}}{\omega_0}\right)$$

f_0 is the rated frequency in Hz.

ω represents the angular (electrical) speed in rad/s.

$\left(\frac{f - f_{REF}}{f_0}\right)$ and $\left(\frac{\omega - \omega_{REF}}{\omega_0}\right)$ are equivalent and will be used interchangeably in what follows.

Note that W_0 and W_{REF} need not be equal.

Speed Governor (Con td..)

k_G is the per unit change in power for per unit change in frequency.

If the base value of power is taken to be the generator rated active power, then the reciprocal of k_G is also known as the "droop".

P_{m0} is the output when $f - f_{REF} = 0$.

k_G, P_{m0} and f_{REF} (equivalently ω_{REF}) can be set by system / plant operators.

Normally k_G is "set" after proper co-ordination among various generators.
It is not usually "tinkered with" during operation. However, P_{m0} and f_{REF} are adjusted by system operators or by slow acting automatic controllers to make changes in generated power.

The following questions naturally come to one's mind:

What is the effect of the droop on generator load sharing?

We try to understand this using an **example**.

Consider a three generator *lossless* power system operating as given below :

Frequency : 50 Hz

Generation at 50 Hz in MW:

Generator 1 : $P_{m01} = 200$ MW

rating : $P_{R1} = 250$ MW,

droop : $\frac{1}{k_{G1}} = 0.05$ (in per-unit calculated using generator rating as the base)

Reference frequency: $f_{REF1} = 50Hz$

Generator 2 : $P_{m02} = 200$ MW

rating : $P_{R2} = 250$ MW,

droop : $\frac{1}{k_{G2}} = 0.05$ (in per-unit calculated using generator rating as the base)

Reference frequency: $f_{REF2} = 50Hz$

Generator 3 : $P_{m03} = 500$ MW

rating : $P_{R3} = 600$ MW,

droop: $\frac{1}{k_{G3}} = 0.05$ (in per-unit calculated using generator rating as the base)

Reference frequency: $f_{REF3} = 50Hz$

Total Load at 50 Hz in MW: $P_{L0} = 900$

Since the losses in the network are assumed to be zero, load-generation-loss balance exists at 50 Hz. If the load increases by 10% i.e., by 90 MW, compute the new generation and frequency.

Solution:

Previous Slide Next Slide

We have from the load-generation balance in MW:

$$900 = P_{m01} + P_{m02} + P_{m03}$$

after the load increase we have

$$990 = P_{m1} + P_{m2} + P_{m3}$$

Now generator power in MW is

$$P_{m1} = P_{m01} - k_{G1} * P_{R1} * \left(\frac{f - f_{REF1}}{f_0}\right)$$

$$P_{m2} = P_{m02} - k_{G2} * P_{R2} * \left(\frac{f - f_{REF2}}{f_0}\right)$$

$$P_{m3} = P_{m03} - k_{G3} * P_{R3} * \left(\frac{f - f_{REF3}}{f_0}\right)$$

Since $f_{REF1} = f_{REF2} = f_{REF3} = 50\,\text{Hz}$

$$f = \frac{P_{m01} + P_{m02} + P_{m03} - 990}{k_{G1} * P_{R1} + k_{G2} * P_{R2} + k_{G3} * P_{R3}} f_0 + 50$$

$$= \frac{-90}{20 * 250 + 20 * 250 + 20 * 600} * 50 + 50$$

$$= 49.795\,\text{Hz}$$

The change in generation for each machine is as follows:

$$P_{m1} - P_{m01} = k_{G1} * P_{R1} * \left(\frac{f - 50}{50}\right) = 20.454\ \text{MW}$$

$$P_{m2} - P_{m02} = k_{G2} * P_{R2} * \left(\frac{f - 50}{50}\right) = 20.454\ \text{MW}$$

$$P_{m3} - P_{m03} = k_{G3} * P_{R3} * \left(\frac{f - 50}{50}\right) = 49.09\ \text{MW}$$

Notice that since the droops in pu on machine power rating are equal, the generators share the increased load in proportion to their MW rating. See the effect of droop.

Effect of change in droop:

To verify that if the pu droop of one machine is decreased then its share increases, let us reduce the droop of generator 1 to 0.04 and increase that of generator 2 to 0.066. For this change, it can be easily seen that frequency after the load change is approximately the same as before : 49.795 Hz. However, the sharing of among generators 1 and 2 changes.

$$P_{m1} - P_{m01} = k'_{G1} * P_{R1} * \left(\frac{f - 50}{50}\right) = 25.625\ \text{MW}$$

$$P_{m2} - P_{m02} = k'_{G2} * P_{R2} * \left(\frac{f - 50}{50}\right) = 15.53\ \text{MW}$$

Recap

In this lecture you have learnt the following

- Speed governor is a mechanism to introduce frequency dependence of generated power. This is required to prevent unacceptable deviation in frequency when load changes.
- The droop setting of speed governors allows us to impose a certain sharing of excess load amongst generators.

Lecture 15 Speed Governer

Objectives

In this lecture you will learn the following

- What are the transient processes before system frequency settles down to steady state ?

We have seen the effect of speed droop settings on sharing of excess load between generators in steady state.

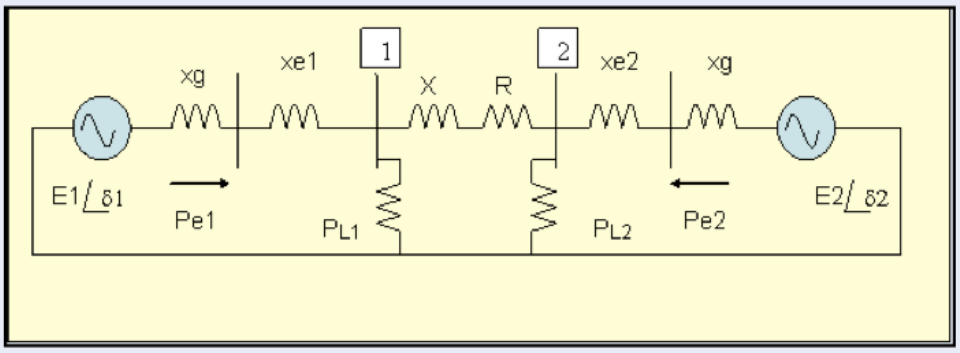
Although we have considered only the steady state conditions, it is to be noted that governors, turbines and generators (and practically all constituents of a power system) are governed by dynamic equations. This means that steady state is not reached instantaneously. Indeed, if the transient processes are not stable, the system will not settle down to the steady state. Thus it is important to consider the dynamical equations which govern the physical structures, especially the prime mover system. While dynamical modelling of components is beyond the scope of this course, we will briefly discuss the working of a steam governor.

A speed governor is a device which senses speed deviation from its reference value and appropriately changes control valve position in a steam turbine or gate position in a hydraulic turbine. This is achieved in older units using mechanical and hydraulic arrangements (mechanical hydraulic governors), while in modern units, the sensing and computing functions are done electrically (electrohydraulic governors). A speed governor should allow us to change the governor gain (or equivalently the droop) and also change the speed and/or load reference. A governor may also have additional parameters which tailor its dynamic response so that instability does not take place.

A student is referred to the book "Power System Stability and Control" by P. Kundur, McGraw Hill, New York, 1994, for further reading on this subject.

We study the frequency *transients* for a simplified two generator system with speed governing, by an **example**.

Consider 2 identical generators supplying 2 loads as shown in the figure below.



Consider the 2 generator, 2 load system shown in the figure below.

The generator field voltage is controlled to maintain its terminal voltage. A simplified model of a generator is used (a voltage source behind an equivalent reactance -xg).
The loads are voltage dependent but not frequency dependent, and are of unity power factor.

$$\text{Load 1}: P_{L1} = 0.5 * (\frac{V_1}{V_o})^2$$

$$\text{Load 2}: P_{L2} = 0.5 * (\frac{V_2}{V_o})^2$$

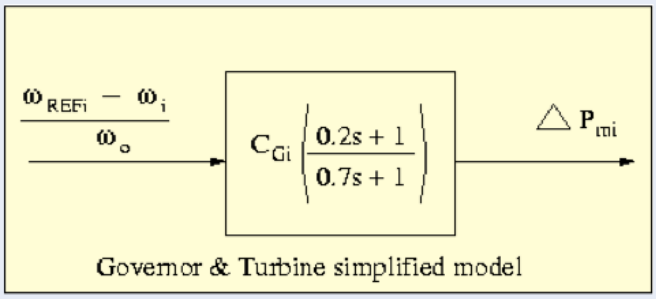
$$V_o = 1.0$$

A simplified composite transfer function of governor and turbine for both generators is assumed to be as follows:

$$\Delta P_{mi} = C_{Gi}\left(\frac{0.2s+1}{0.7s+1}\right)\left(\frac{\omega_{REFi}-\omega_i}{\omega_o}\right)$$

$$P_{mi} = P_{m0i} + \Delta P_{mi}$$

All values are on a common base.



Governor & Turbine simplified model

Assume that for both generators, $P_{m0i} = 0.5$ and $C_{Gi} = 20$; $\omega_{REFi} = \omega_o = 2 * \pi * 50$
Here, C_{Gi} represents the gain of a ***proportional speed controller***. Note that in steady state :

$$\Delta P_{mi} = C_{Gi}\left(\frac{\omega_{REFi}-\omega_i}{\omega_o}\right)$$

Thus C_{Gi} is related to the droop of the generator characteristic.
Plot the variation in generator speeds if Load 1 is increased by 10 %, i.e.,

$$P'_{L1} = 0.55 * (\frac{V_1}{V_o})^2$$

Assume inertia constant of each generator $H = 3$ MJ/MVA
$x_g = 0.5, x_e = 0.2, R = 0.02, X = 0.2$

Initially, the voltage at the load buses 1 and 2 are 1.0 pu.

The generator mechanical ("swing") equations have already been discussed before. The natural transients associated with the electrical network/loads are assumed to be fast compared to the electro-mechanical transients; therefore the network and loads are represented by their sinusoidal steady state equations.

The solution of the problem is given in the following slides.

Solution:

The variation in generator speeds can be obtained by numerically integrating the swing equations and the dynamical equations corresponding to the speed governors and turbines. They are :

$$\frac{d(\omega_i - \omega_o)}{dt} = \frac{d\omega_i}{dt} = \frac{\omega_B}{2H_i}(P_{mi} - P_{ei}) = \frac{\omega_B}{2H_i}(P_{m0i} + \Delta P_{mi} - P_{ei})$$

$$\frac{d\delta_i}{dt} = (\omega_i - \omega_o)$$

The dynamical equations of the turbine -governor for each generator can be written as follows:

$$\frac{d\Delta P_{si}}{dt} = \frac{1}{0.7}(-\Delta P_{si} + C_{Gi}\left(\frac{\omega_{REFi} - \omega_i}{\omega_o}\right))$$

$$\Delta P_{mi} = C_{Gi}\left(\frac{\omega_{REFi} - \omega_i}{\omega_o}\right) * \frac{0.2}{0.7} + \Delta P_{si} * (1 - \frac{0.2}{0.7})$$

where ΔP_{si} is a state of the system.

Verify that these yield the transfer function

$$\Delta P_{mi} = C_{Gi}\left(\frac{0.2s + 1}{0.7s + 1}\right)\left(\frac{\omega_{REFi} - \omega_i}{\omega_o}\right)$$

Solution (Con td..):

Also,

$$P_{ei} = \mathrm{Re}(\bar{V}_{Gi}\bar{I}^*_{Gi})$$

where $\bar{V}_{Gi}$ and $\bar{I}_{Gi}$ are the generator terminal voltages and current respectively. These may be obtained by solving the network (in terms of δ_1 and δ_2).

The initial conditions of the various state are as follows:

Initial Conditions:

$$\Delta P_{si} = \Delta P_{mi} = \frac{\omega_{REFi} - \omega_i}{\omega_o} = 0,$$

$$P_{m01} = P_{m02} = 0.5$$

There is no power flow initially on the line connecting bus1 and 2. Thus $\theta_1 = \theta_2$ initially. If the initial phase angle at load bus 1 is taken to be zero, then $\theta_1 = \theta_2 = 0.0$.

$$\delta_1 = 19.29; \quad \delta_2 = 19.29 \text{ deg}$$

Verify that $E_1 = E_2 = 1.0595$, so that the load bus voltages are 1.0 pu.

Solution(Con td...):

The results of the numerical integration (Runge-Kutta 4th order method with time step of 0.01) are shown below. **A MATLAB/SIMULINK simulation model files to do the same can be downloaded from here (init.m, governor.mdl)**. First run the file init.m by typing init at the MATLAB prompt. After that, run the SIMULINK file.

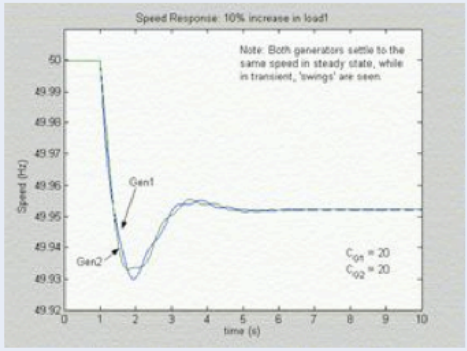
For the step increase in load 1 by 10% (i.e., the load resistance at bus1 reduces by 10%), note the following:

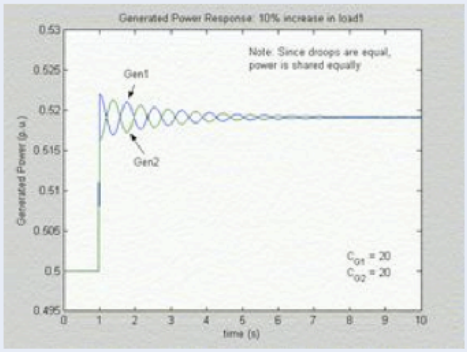
a) Changes in generator speeds. Notice the aggregate movement as well as the relative motion (swings).

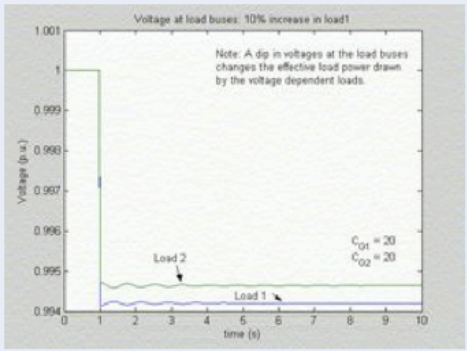
b) Power output of both machines

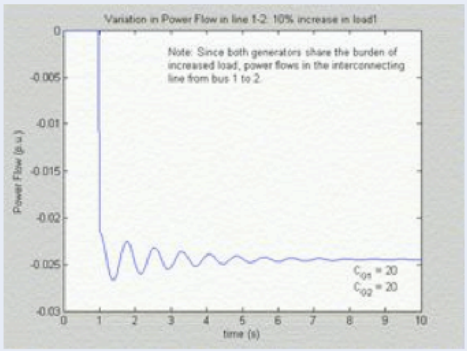
c) Voltages at both load buses

d) Power Flow through the tie line between buses 1 and 2 (it changes : why?)



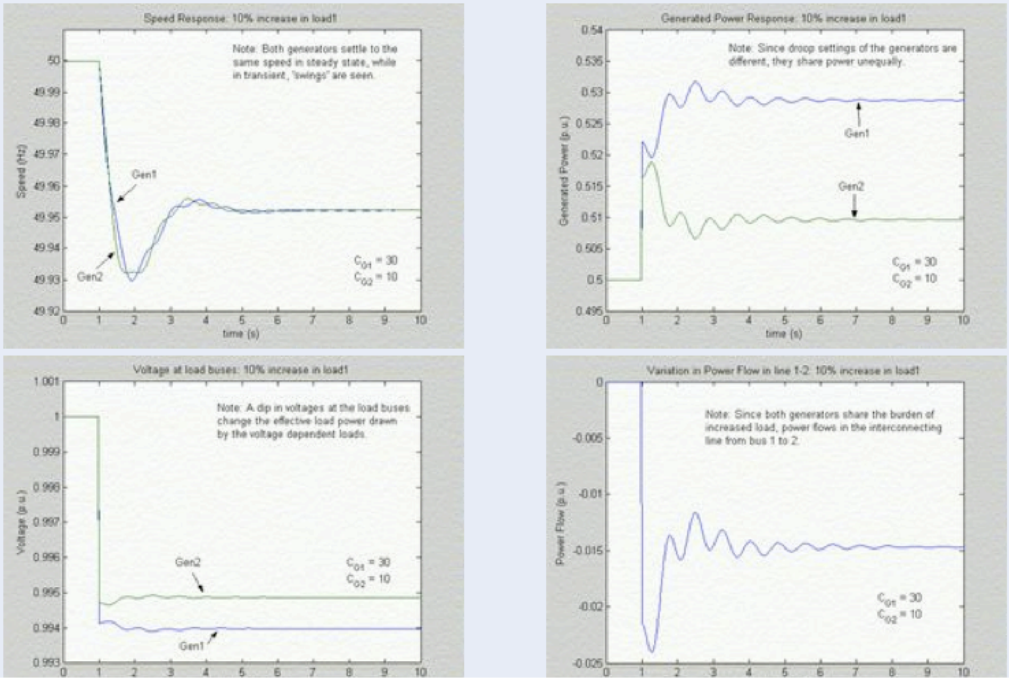






(click on images to enlarge)

Let $C_{G1} = 30$ and $C_{G1} = 10$ and let us repeat the exercise.
The change in the generation share is obvious. The generator with a larger C_G takes on more load. The values of C_{Gi} not only determine the load sharing, but also the speed of response.



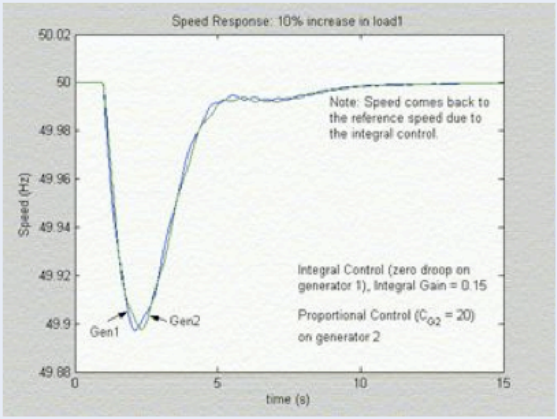
We now see the effect of having integral controller (i.e., zero droop characteristic) on one governor.

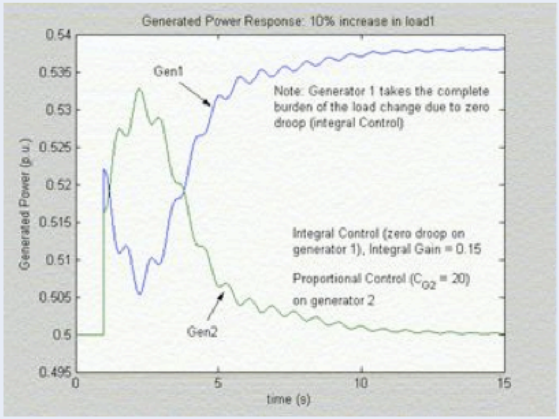
Now, if we use an ***integral speed controller***, the overall governor turbine transfer function is

$$\Delta P_{mi} = C_{Gi} \frac{1}{S} \left(\frac{0.2s+1}{0.7s+1} \right) \left(\frac{\omega_{REFi} - \omega_i}{\omega_o} \right)$$

Since $C_{Gi}\frac{1}{s}$ is infinite in steady state, $\omega_{REFi} - \omega_i$ has to be equal to zero for ΔP_{mi} to be finite.

In other words, the droop of this speed governor characteristic is zero.
We have discussed in detail the consequences of having one or more than one generator having droop zero.
If generator 1 has the integral speed governor, then it takes on ***all*** the load change as shown in the figures below.





However we emphasize the point that it is not feasible to have integral type speed governors (or equivalently zero droop characteristics) on *more than one* generator in the system; non-zero droops on all machines is preferable for load sharing among generators.

Recap

In this lecture you have learnt the following

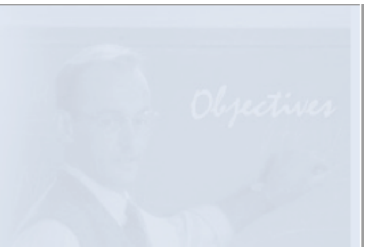
- The transient processes before a system settles to equilibrium have been discussed.
- This is illustrated using an example wherein numerical integration of system equations is done.
- It is clear that droop decides the load sharing; Zero droop on *one* generator causes that generator to take on the complete burden of excess load.

Lecture 16 Automatic Generation Control (AGC)

Objectives

In this lecture you will learn the following

- What is automatic generation control ?
- In what way is it different from governor action ?



Automatic Generation Control (AGC)

We saw in the previous **example** how load sharing in a multi-generator power system can be achieved using droop characteristics of governors. The sharing according to droop is *irrespective of load location*.

However if non-zero governor droops are used (which is necessary for appropriate sharing), a steady state frequency error will remain which needs to be corrected. Moreover, since all the governors respond to the load change irrespective of load location, there may be undesirable exchange of power between different areas of the grid. This is manifested as a change in the flows of lines interconnecting these areas.

To ensure that frequency steady state error is corrected and generators in a particular area take on the burden of their own load, the load reference (P_{m0}) of governors is adjusted ***slowly***. This control is also called "secondary control". This correction may be done over several minutes as opposed to 5-10 seconds for initial or "primary" control action of governors.

Thus, while primary control (governor action) ensures that a large and sudden frequency fall or rise is prevented, secondary control or Automatic Generation Control ensures that frequency is brought back to the nominal value and inter-area power flow is regulated.

Automatic Generation Control (AGC) (Con td..)

Any change of reference value will lead to a change in sharing among the generators. Thus by slowly changing the reference of speed governors we can over-ride the sharing which is imposed by the droop characteristics.

The question then arises: which generators should change their governor references and what the exact value of these changes ?

In a 2 - area system, if governors are present on some machines in both areas

and

a) power flow in certain ac line, which connects 2 areas is to be regulated.

and

b) Frequency has to be brought back to its nominal value after any load generation change

then

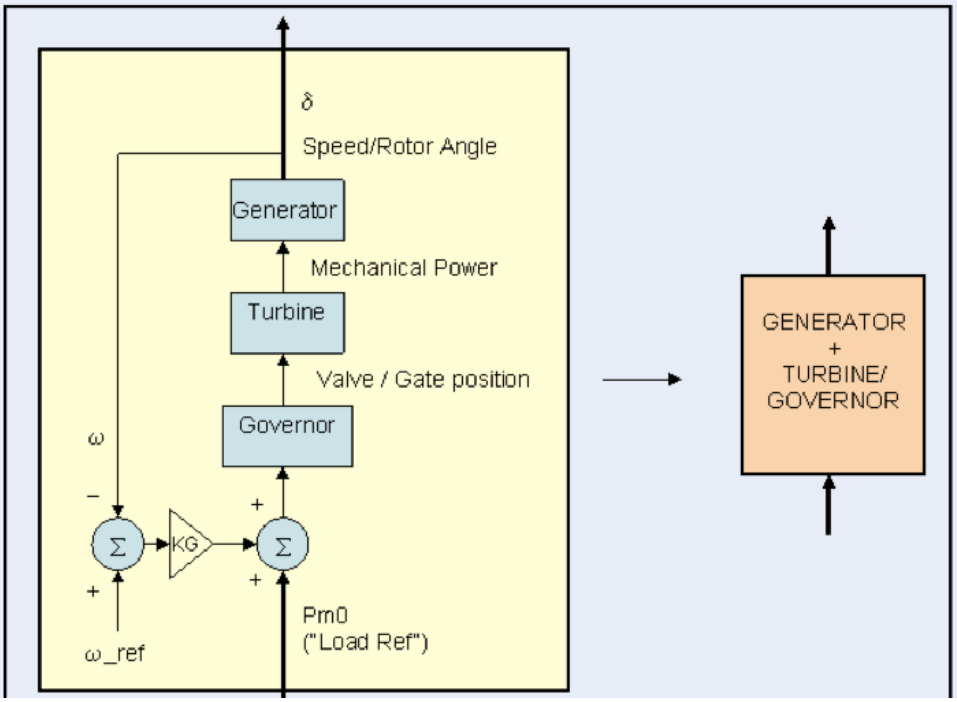
the references of atleast 2 governors (one in each area) in the system should be changed to achieve these objectives.

Note: It is not feasible to *independently* change more than one governor reference in one area, otherwise there is no unique value of reference change for different governors. Thus if more than one governors in an area are "on AGC", then their actions have to be in a pre-decided proportion and not independent of one another.

The AGC concept is illustrated by the following schematic in the next slide.

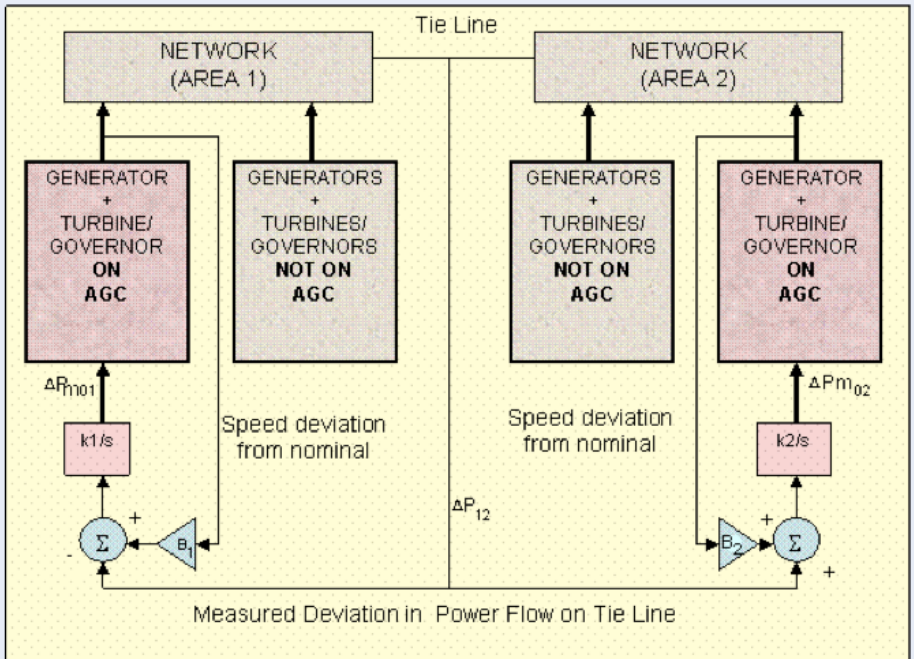
Automatic Generation Control (AGC) (Con td...)

Each generator which has a governor can be represented as follows. Note that the value of the load reference P_{m0} is adjusted by AGC.



Automatic Generation Control (AGC) (Con td...)

The following figure shows how a generator in each area in a 2 area system receives feedback from a tie line (P_{12}). The feedback is combined with the speed deviation signal in a certain proportion (decided by the constant B). It is then fed to an integral controller, the output of which changes P_{m0}.



Automatic Generation Control (AGC) (Con td..)

An integral controller acts on this error to change the load reference of both governors. Since an integral controller drives its input to zero in steady state **(why?)**, it follows that in steady state:

$$B_1(\frac{\omega_{REF} - \omega_1}{\omega_0}) - \Delta P_{12} = 0$$

$$B_2(\frac{\omega_{REF} - \omega_2}{\omega_0}) - \Delta P_{21} = 0$$

$\Delta P_{21} \simeq -\Delta P_{12}$ (losses are assumed to be small), $\omega_1 = \omega_2$ in steady state and typically $\omega_{REF} = \omega_0$

Then it follows that if B_1 and B_2 are > zero,

$$\omega_1 = \omega_2 = \omega_0$$

and

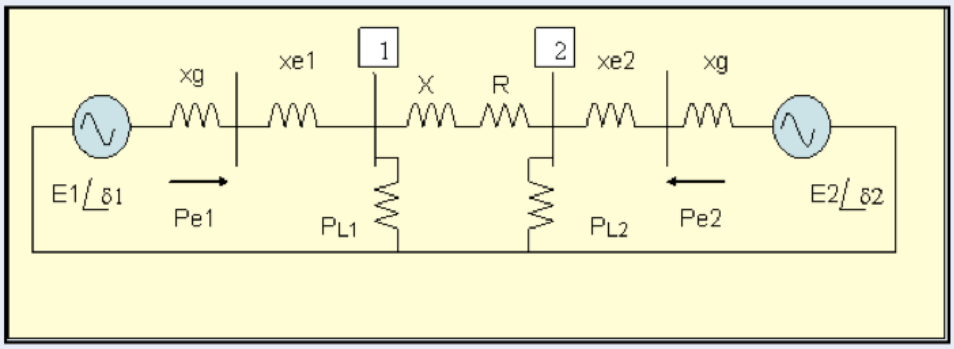
$$\Delta P_{12} = 0$$

which means that in steady state, both frequency deviation and tie line power flow deviation are made to go to zero by the AGC.

While any value of the weighting factors which are non-zero gives the same result in steady state, they are chosen such that the transient response is good.

We **illustrate this concept** on the same system as in the previous example.

We consider the same 2 generator, 2 load system as in the previous example. Let Generator 1 and load at bus 1 be in Area 1 and Generator 2 and load at bus 2 be in area 2. The line between bus 1 and 2 is the inter-area tie line.



Suppose that the aim of the AGC is to keep interchange between the 2 areas via the tie line zero as well as bring the deviation in frequency to zero. **MATLAB/SIMULINK simulation model files to do the same can be downloaded from here (init.m , agc.mdl).**

The action of AGC and Governor is represented as follows:

Generated Power is given by

$$P_{mi} = P_{m0i} + \Delta P_{mi}$$

Governor action on both generators causes a change in power ΔP_{mi} as follows:

$$\Delta P_{mi} = \left(C_{Gi}\left(\frac{\omega_{REF} - \omega_i}{\omega_o}\right) + \Delta P_{m0i}\right) * \left(\frac{0.2s + 1}{0.7s + 1}\right)$$

while, ΔP_{m0i} represents the change in the load reference due to AGC, which is given by the following transfer function:

$$\Delta P_{m01} = \frac{k_1}{s}\left[B_1\left(\frac{\omega_{REF} - \omega_1}{\omega_o}\right) - \Delta P_{12}\right]$$

$$\Delta P_{m02} = \frac{k_2}{s}\left[B_2\left(\frac{\omega_{REF} - \omega_2}{\omega_o}\right) - \Delta P_{21}\right] \approx \frac{k_2}{s}\left[B_2\left(\frac{\omega_{REF} - \omega_2}{\omega_o}\right) + \Delta P_{12}\right]$$

The values k_1, k_2, B_1 and B_2 are decided based on obtaining good time response (no overshoots and undershoots). Note that

$\Delta P_{21} \approx -\Delta P_{12}$ (losses are assumed to be small)

Please read the notes given on the figure to understand the behavior of the AGC.

We now see what happens for a 10% load change at bus 1 by numerical integration of the differential equations.

Assume that : $B_1 = B_2 = 20$ and $k_1 = k_2 = 0.01$

$$B_1\left(\frac{\omega_{REF} - \omega_1}{\omega_o}\right) - \Delta P_{12} = 0$$

$$B_2\left(\frac{\omega_{REF} - \omega_2}{\omega_o}\right) - \Delta P_{21} = 0$$

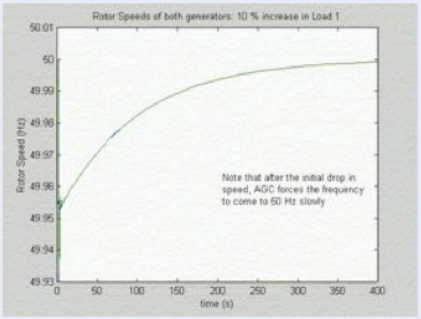
$\Delta P_{21} \approx \Delta P_{12}$ (losses are assumed to be small), $\omega_1 = \omega_2$ in steady state and typically $\omega_{REF} = \omega_o$

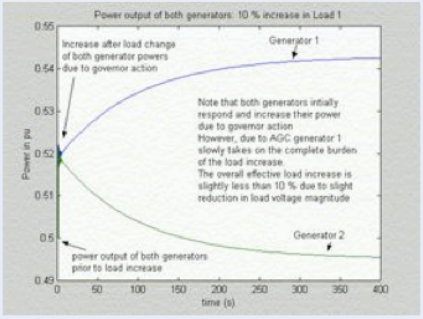
Therefore, it follows that if B_1 and B_2 are nonzero,

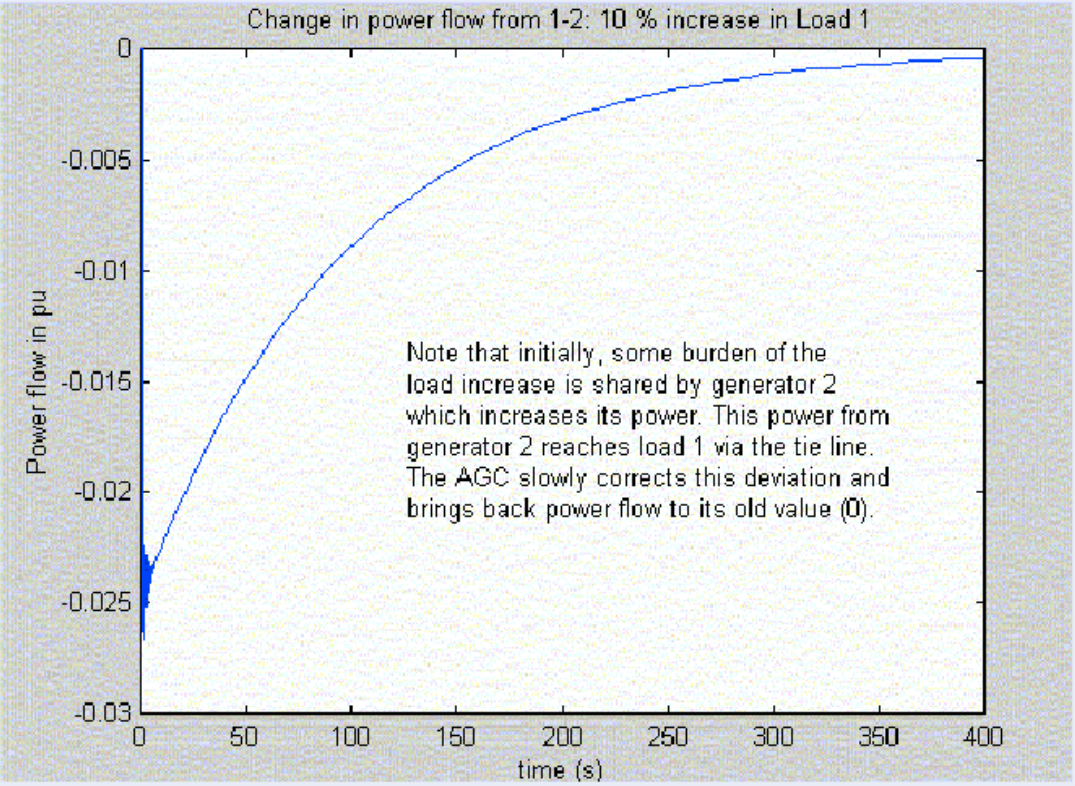
$$\omega_1 = \omega_2 = \omega_o$$

and

$$\Delta P_{12} = 0$$







Economic Dispatch:

The change in generation in steady state after a load change (due to governor and AGC action), may not be the most optimal one from an economic standpoint. Note that different generators may have different cost of generating power. A power plant or system operator may wish to re-adjust the generated power in various units for maximum economic benefit. This adjustment of generation is generally done manually and is often called "tertiary control".

Of course, how this re-adjustment is done is dependent on generation cost, power -pricing mechanism and the ownership of generation. Interestingly, if power pricing itself is made a function of frequency, then tertiary control also may contribute to (slow) frequency control ! This idea (called **Availability Based Tariff**) is implemented in some regions of our country.

Limitations of prime mover systems:

Generation reserves available for control of frequency are typically only a fraction of the existing generation. Moreover, there are practical constraints like limits on the rate of rise of prime mover power (in a steam turbine) to avoid rapid heating. While an initial sudden change of about 10% can be tolerated, subsequently, rate of rise is limited to 2% (of plant MW rating) per minute. The boiler in a steam prime mover is relatively slow in maintaining the steam pressure by increasing fuel input. Thus as control valves open, restoration of steam pressure is slow. For some types of hydro-turbines, there are forbidden operating zones due to cavitation effects in turbines.

In countries like ours, where substantial generation shortage exists, load shedding may be done to keep frequency within bounds.

Recap

In this lecture you have learnt the following

- AGC is control scheme which ensures that frequency deviation from the nominal value is brought to zero, and the power flows between different areas in an interconnected system are regulated.
- While governors act relatively fast to arrest frequency decline, AGC is slow acting. While governors can be present on almost all generators in a system, AGC is present only in selected units.
- Tertiary control involves adjusting generator powers due to economic reasons. This is the slowest generator power control action.

For a stable synchronous power system, frequency all over the system is equal in steady state.

a) Load Generation Imbalance causes decline or rise in the speeds of synchronous machines connected in the system.

b) If prime movers deliver a constant power, the equilibrium value of frequency is determined by the frequency dependence of loads. Since this dependence is weak, the frequency decline or rise due to load-generation imbalance can be quite high.

c) Prime mover control is done by making prime mover power a function of the speed. This mechanism is called a speed governor. Speed Governors prevent sudden large rise or fall in speed.

d) The sharing of extra load by generators is determined by the droop of speed governors. The droop is related to the steady state proportional gain given to the speed error. Due to proportional gain feature, a steady state error in frequency will remain when a governor causes a steady state change in prime mover power. Therefore frequency regulation by a governor is not perfect, but permits a droop so that sharing can be achieved conveniently.

e) Frequency can be brought back to the nominal value and inter-area power flow can be regulated by another slow acting control loop which controls prime mover output in a few generators in each area. This is called Automatic Generation Control. It is important to note that an AGC is much slower acting than a speed governor.

f) The steady state scheduling of generators based on economic factors is done by changing the load reference of the prime mover manually.

Therefore there is a hierarchy of prime mover controls, operating on three different time scales which decide the final generator output.

In emergency situations wherein large load-generation imbalance exists generator or load tripping may be required to prevent large frequency changes.

Congratulations, you have finished Module III.

Module 4

Voltage and Power Flow Control

Lecture 17 Reactive Voltage and Power Control

Objectives

In this lecture you will learn the following

- Means of voltage control in power systems.
- Generator Excitation Systems and reactive power characteristics of power system components.

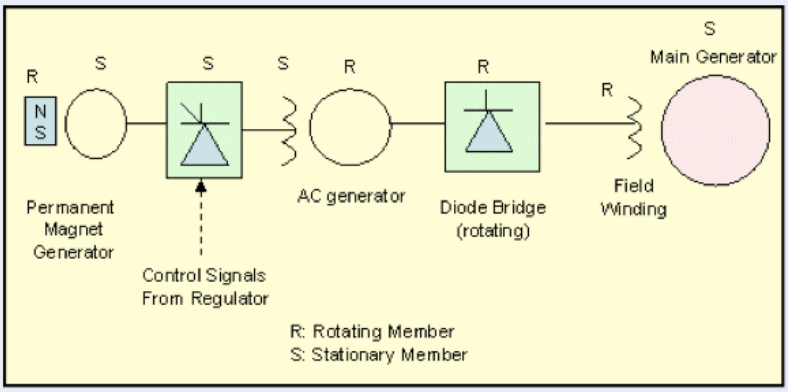
Production and Absorption of Reactive Power

Since voltage magnitudes in a system are intimately related to the ability of various components to absorb or supply reactive power, we shall begin by reviewing these characteristics.

Synchronous Generators: Can generate or absorb reactive power by control of field voltage. The amount of reactive power which can be exchanged by a generator is given by the reactive power capability curve (discussed in **Module 2**).

In rotating-type excitation systems, field voltage is obtained using separate generators connected to the same shaft as the main generator. The excitation for these excitation generators themselves is obtained by self-excitation or permanent magnet generators. Control is achieved by controlled rectification using thyristor bridges.

A schematic of a brushless rotating excitation system is shown on the right.

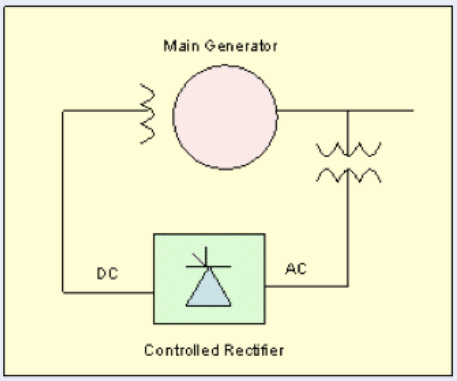


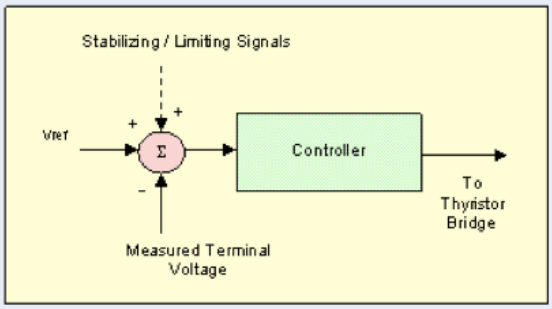
Production and Absorption of Reactive Power (contd.)

In static excitation systems, the field voltage is obtained directly by rectifying (using controlled rectifiers) the output voltage of the main synchronous generator. Therefore in these systems, the generator is practically self excited with some initial voltage being obtained temporarily using station batteries (called field flashing).

In static excitation systems, it is necessary to use slip rings to convey the field voltage (output of the controlled rectifier) to the rotating field winding of the main generator.

An exciter is usually controlled so as to *regulate* generator terminal voltage. The general structure of a voltage regulator system is shown on the right. While the regulator maintains voltage close to the reference value, an operator may also slowly change the reference value itself if a change in reactive power is desired. A typical excitation system also utilizes limiting signals which change the reference voltage so that the generator works within its reactive power capability.





Production and Absorption of Reactive Power: Other Power System Components

Overhead Lines: Absorb or supply reactive power based on loading conditions.
Overhead transmission lines may draw or supply reactive power depending on the loading conditions. We saw in Module 2 that a line draws reactive power if loading is higher than the Surge Impedance Loading and generates reactive power otherwise.

Cables: Surge Impedance of cables can be quite high (higher than the thermal limits) and they are typically loaded much below their Surge Impedance Loading. This leads to significant generation of reactive power and overvoltage problems if cable length is greater than 30-40 km.

Transformers: Transformers generally absorb reactive power due to shunt magnetising and series leakage reactances.

Loads: Normally absorb reactive power (see power factor of some loads in the previous module).

In the next lecture, we will consider special reactive power absorption/supply characteristics of special devices which are used to control voltages in a power system.

Recap

In this lecture you have learnt the following

- The major means of voltage control in a power system is by generator excitation control.
- Power system components like loads and transformers generally absorb reactive power. Overhead transmission lines supply or absorb reactive power depending on their loading. Cables generally supply reactive power.

Lecture 18 : Reactive Power Characteristics of Special Devices

Objectives

In this lecture you will learn the following

- Reactive power characteristics of HVDC converters (line commutated thyristor rectifiers/inverters)
- Transmission network based voltage control devices.

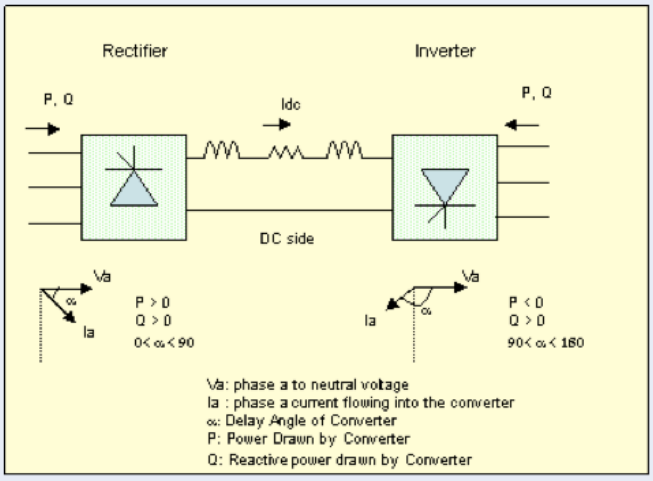
HVDC Terminals

You must have learnt about various kinds of AC/DC converters in a Power Electronics Course. Line Commutated Converters (Thyristor Rectifier / Inverter Bridges) always absorb reactive power.The reactive power absorbed is substantial and can be as much as 30-40% of the power transmitted by the HVDC link. Usually shunt capacitors are connected to provide reactive power on the ac side of the converters.

If the thryistors are fired *as soon as they get forward biased*, then we say that the "delay angle" is zero. In such a case a thyristor bridge is identical to a three phase diode bridge. However, by delaying the firing, we can control the average voltage which appears on the DC side.

The delay angle associated with these controlled rectifiers results in lag of the fundamental of AC side current by approximately the same angle as the delay angle.

This is also true for inverter stations where the firing angle is greater than 90 degrees but less than 180 degrees. This is illustrated by the figure given on the right.



You may simulate HVDC link behavior using the MATLAB/SIMULINK file (HVDC.mdl).

HVDC systems of lower ratings may also be implemented using **Voltage Source Converters.** These converters use devices like Gate Turn off Thyristors which can be *turned off* by a gate signal. Unlike in a line commutated converter, the voltage on the DC side is unidirectional while current can flow in either direction on the DC side. **These converters can absorb or supply reactive power independent of the real power flow in the line.** As an exercise, read about AC/DC converters in a book on Power Electronics (e.g. Power Electronics by M.H.Rashid, Prentice Hall, 2002).

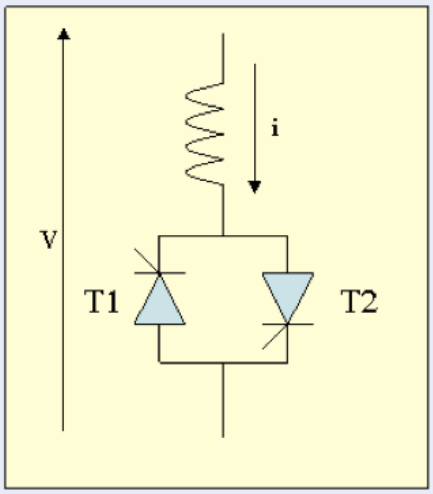
Static Var Compensators

While the major means of control of reactive power and voltage is via the excitation systems of synchronous generators, devices may be deployed in a transmission network to maintain a good voltage profile in the system.

Shunt connected devices like shunt capacitors or inductors or synchronous condensers can be used for reactive power control. Shunt Capacitors/Inductors may be fixed or switched (using circuit breakers). *Vernier* or smooth control of reactive power is also possible by varying effective susceptance characteristics, by use of power electronic devices. As an example let us consider a "Static Var Compensator (SVC) " which employs thyristors. An SVC is a parallel combination of switched or switched capacitors and a **Thyristor Controlled Reactor (TCR).**

The basic principle of operation of a TCR (see figure on the right) is as follows:

A reactor is connected in series with a back-to-back connected pair of thyristors. The thyristor does not conduct if gate pulses are not provided. In such a case, the effective susceptance is zero.



Static Var Compensators (contd..)

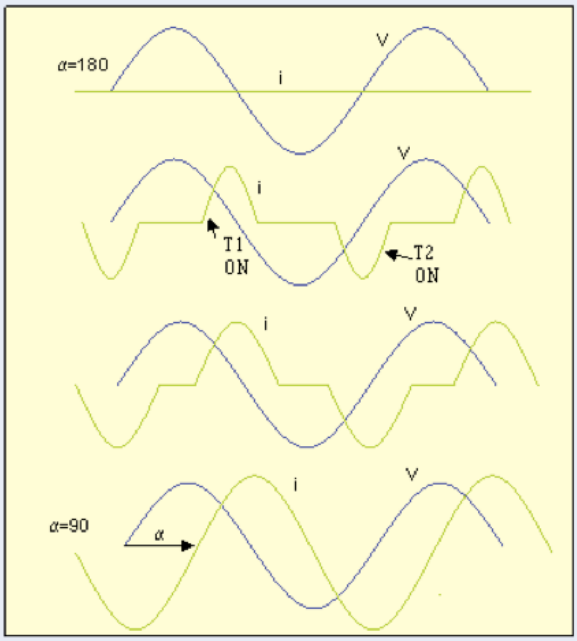
On the other hand if thyristors are always on, the effective susceptance is equal to that of the inductor. Now if the each thyristor is fired at a delay angle (as measured from the point where it gets forward biased) between 90 and 180 degrees, the resulting current waveform is as shown on the right. The figure shows the current waveform for various values of delay angle. Note that this waveform is obtained by integrating the voltage from the point at which the thyristor switches on using the rule:

$$L * i = \int v \, dt$$

The thyristor switches off when the current through it falls to zero. The reverse connected thyristor behaves similarly for the other half cycle.The instantaneous current *i* over half a cycle is give by

$$i = \frac{\sqrt{2}V}{X_L}(\cos\alpha - \cos\omega t), \quad \alpha < \omega t < \alpha + \sigma$$
$$= 0, \quad \alpha + \sigma < \omega t < \alpha + \pi$$

where *V* is the rms voltage applied, X_L is the fundamental frequency reactance of the reactor. w is the radian frequency of the voltage and s is the angle for which a thyristor conducts. Show that $s = 2(p-a)$.



The current is non-sinusoidal (except when a =90) and contains odd harmonics which are functions of the conduction angle s. The fundamental component of the TCR (Thyristor Controlled Reactor) current, I_1 is obtained by Fourier analysis and given by (Show this!) :

$$I_1 = B_{TCR}(\sigma)V$$

where

$$B_{TCR} = \frac{\sigma - \sin\sigma}{\pi X_L}$$

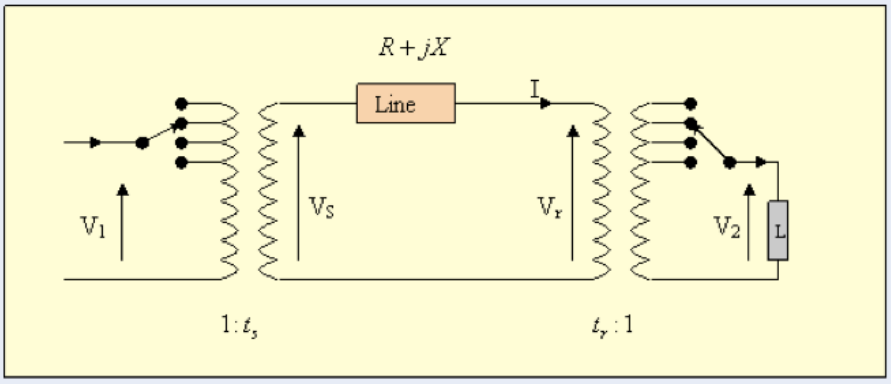
You may simulate a SVC using the MATLAB/SIMULINK file (SVC3.mdl).

Other Devices

Series Connected Devices like fixed, switched or controllable series reactance compensators can also be used for reactive power control. While the basic aim of connecting series capacitors is for power flow control and enhancement (to be discussed later), they also affect the voltage profile in a system.

Tap Changing Transformers provide a means of controlling voltage and reactive power flows in a system. Typically, all generator step-up transformers and auto-transformers (which inter-connect EHV lines which are at different voltage levels) are provided with a facility for on-load tap changing. The change of tap is effected by motors and follows a manual command or the output of an automatic voltage regulator (as in a generator). Tap changing is a slow process and there are several inherent and deliberate delays and dead-bands in the control system to avoid frequent changes in tap. Tap changing can also be effected by power electronic switches.

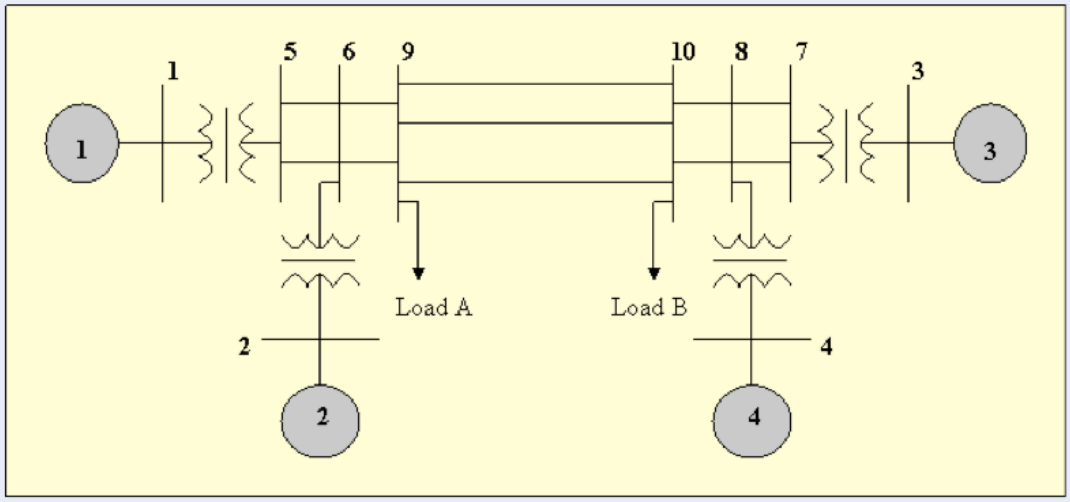
A schematic of a tap changing transformer scheme is shown in the figure below.



- Voltages at all buses should be within acceptable bounds.
- Reactive Power flow should be reduced (generate or absorb reactive power locally) otherwise voltage profile is poor and losses increase. Unlike frequency control wherein the cumulative load and generation determines the system frequency, the voltages are usually amenable to control by local injection/absorption of reactive power.
- Reactive Power control done at 2 levels:
  1. Transmission System Compensation or Voltage Control
  2. Load Compensation or Power Factor Correction / Distribution System Voltage Control

We now consider an **example**, of voltage control in an power system.

Consider the 4 machine 2 area system shown in the figure below:



The network data (pu) is given below:

From Bus	To Bus	R	X	B(total)
9	10	0.022	0.22	0.330
9	10	0.022	0.22	0.330
9	10	0.022	0.22	0.330
9	6	0.002	0.02	0.030
9	6	0.002	0.02	0.030
10	8	0.002	0.02	0.030
10	8	0.002	0.02	0.030
5	6	0.005	0.05	0.075
5	6	0.005	0.05	0.075
7	8	0.005	0.05	0.075
7	8	0.005	0.05	0.075

Transformer Data (pu):

From Bus	To Bus	R	X	Tap
1	5	0.001	0.012	1.00
2	6	0.001	0.012	1.00
3	7	0.001	0.012	1.00
4	8	0.001	0.012	1.00

Consider the following (solved) load flow scenario:

Bus	Voltage Magnitude	Voltage Angle(deg.)	P_gen	Q_gen	P_load	Q_load
1	1.0100	8.6096	7.0000	1.9039	;0.0000	0.0000
2	1.0100	-1.9178	7.0000	4.9406	0.0000	0.0000
3	1.0100	0.0000	7.4035	2.2852	0.0000	0.0000
4	1.0100	-11.5692	7.0000	5.8729	0.0000	0.0000
5	0.9838	3.8704	0.0000	0.0000	0.0000	0.0000
6	0.9476	-6.6561	0.0000	0.0000	0.0000	0.0000
7	0.9793	-5.02056	0.0000	0.0000	0.0000	0.0000
8	0.9365	-16.3072	0.0000	0.0000	0.0000	0.0000
9	0.8995	-15.6492	0.0000	0.0000	11.5900	2.1200
10	0.8784	-25.8224	0.0000	0.0000	15.7500	2.8800

Losses : 1.06350 pu

It is clear that voltage at buses 9 and 10 are quite low. How does one correct this situation?
Let us consider 2 alternatives:

a) Increase the reactive output of all generators by increasing their terminal voltage (which is achieved by increasing the generators field voltages).

b) Provide shunt compensation at the load itself.

The solved load flow under these circumstances are given in the following slides.

Alternative 1

Increasing the generator terminal voltage by increasing the field voltage using an exciter.

Bus	Voltage Magnitude	Voltage Angle(deg.)	P_gen	Q_gen	P_load	Q_load
1	**1.0500**	8.2685	7.0000	1.7032	0.00	0.00
2	**1.0500**	-1.3729	7.0000	4.5421	0.00	0.00
3	**1.0500**	0.0000	7.2876	2.0231	0.00	0.00
4	**1.0500**	-10.3291	7.0000	5.3792	0.00	0.00
5	1.0269	3.8910	0.0000	0.0000	0.00	0.00
6	0.9943	-5.7377	0.0000	0.0000	0.00	0.00
7	1.0232	-4.5608	0.0000	0.0000	0.00	0.00
8	0.9847	-14.6901	0.0000	0.0000	0.00	0.00
9	0.9506	-13.8756	0.0000	0.0000	11.59	2.12
10	0.9318	-23.1844	0.0000	0.0000	15.75	2.88

Losses: 0.94756 pu

Alternative 2

4 pu capacitive shunt susceptance is provided at load buses 9 and 10.

Bus	Voltage Magnitude	Voltage Angle(deg.)	P_gen	Q_gen	P_load	Q_load
1	1.0100	7.8176	7.0000	0.7190	0.00	0.00
2	1.0100	-2.2769	7.0000	1.3373	0.00	0.00
3	1.0100	0.0000	7.2226	1.0189	0.00	0.00
4	1.0100	-10.6720	7.0000	2.1331	0.00	0.00
5	0.9979	3.0780	0.0000	0.0000	0.00	0.00
6	0.9906	-7.0163	0.0000	0.0000	0.00	0.00
7	0.9944	-4.8924	0.0000	0.0000	0.00	0.00
8	0.9811	-15.4111	0.0000	0.0000	0.00	0.00
9	**0.9881**	-15.1190	0.0000	0.0000	11.59	2.12
10	**0.9694**	-23.8214	0.0000	0.0000	15.75	2.88

Losses: 0.88263 pu

A few observations can be made:
Compensating the reactive power at the load bus itself reduces the losses and improves the voltage profile, hence it is preferable.

Recap

In this lecture you have learnt the following

- Special devices like Static VAR Compensators and tap changing transformers can be deployed in the network to improve voltage profile of a system
- Line -commutated rectifiers or inverters, employed in HVDC links, always absorb reactive power.
- Voltage control of a bus can be done by local reactive power injection.

Lecture 18 a : HVDC converters

Objectives

In this lecture you will learn the following

- AC-DC Converters used for HVDC applications.
- Introduction to Voltage Source Converters.

Thyristors

In the previous lecture, we have introduced the use of controlled rectifiers (AC-DC converters) to provide controllable field voltage to a synchronous generator. In the following lecture we shall consider their use again as a part of of HVDC systems.

Therefore in this lecture, we revise the operation of a typical AC-DC converter used in these applications. The treatment given in this lecture is not exhaustive, but is given to bring out some important functional characteristics of these converters which are important from a power systems perspective. A reader is encouraged to read a power electronics text for a rigorous analysis of AC-DC converters.

Power Electronic controllers are now very widespread and are present in computers and satellite power supplies, drives used in traction etc. Their use in extremely high power applications is not so well known. In fact, in many HVDC links, the total power handled ("converted") by a power electronic converter system exceeds 1 GW !

Power Electronic converters use semiconductor devices which are operated in " ON " and "OFF" states. In ON state, the voltage across the device is negligible, while in an OFF state, current flow through it is negligible. The state of a device is decided by the external circuit conditions and a (a low power) signal which is provided at the GATE terminal of the device. Diodes do not have a GATE terminal and their state is wholly determined by the external circuit (forward or reverse bias).

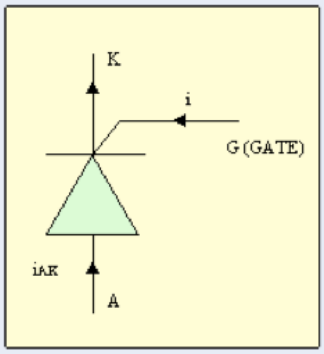
Currently, most power electronic converters used in power system applications, like HVDC, use a device known as a thyristor. Thyristors are available at ratings in excess of 10 kV, 8000 A. Many thyristors can be connected in series to yield larger equivalent voltage ratings which are required for power system applications.

An IDEAL thyristor has the following characteristics:

1. A thyristor turns ON if $V_{AK} > 0$ *(forward bias)* and $i_g > 0$ *(firing)*.
2. Once a thyristor is ON, it remains ON even if i_g becomes 0.
3. A thyristor which is ON, switches OFF if i_{AK} falls to 0.

A practical device has the following limitations:

1. A thyristor requires some time to turn on after the application of the gating signal, known as turn on time, ton. This is usually quite small.
2. Once i_{AK} falls to 0, positive V_{AK} should not be applied for a duration, toff (called turn off time), otherwise the thyristor may turn on again even without application of a gate signal !
3. A thyristor should be protected against high di_{AK}/dt and dv_{AK}/dt otherwise it may damage the device.

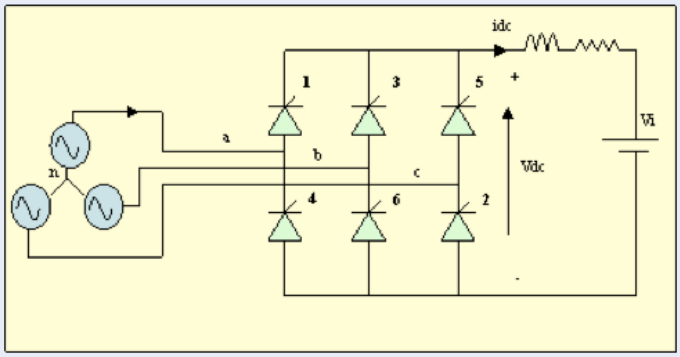


AC-DC converters

Most HVDC AC-DC converters use line commutated thyristor bridges. These converters use the AC voltages to provide the required forward voltages which are essential for thyristor turn on. A "six pulse" thyristor bridge can implement an AC-DC converter, and is shown in the figure below. It is assumed here that the AC side voltages are balanced and the phase sequence is a-b-c.

The basic idea is to construct the DC side voltage waveform by parts of the three phase AC voltages. For example, when thyristors 1 and 2 are ON, the voltage across the DC side 7 is Vac.

Of course, one cannot switch on the thyristors randomly: they need to be forward biased in order that they may be switched ON. Moreover if thyristors are fired in a particular sequence, indicated by the numbers 1-6 on the side of the devices (see the figure on the right), **and fired at the instant when they get forward biased**, then the resulting steady state waveforms are as shown on the right. These waveforms are similar to those of a diode bridge. Note that a switching takes place every 60° in a cycle.

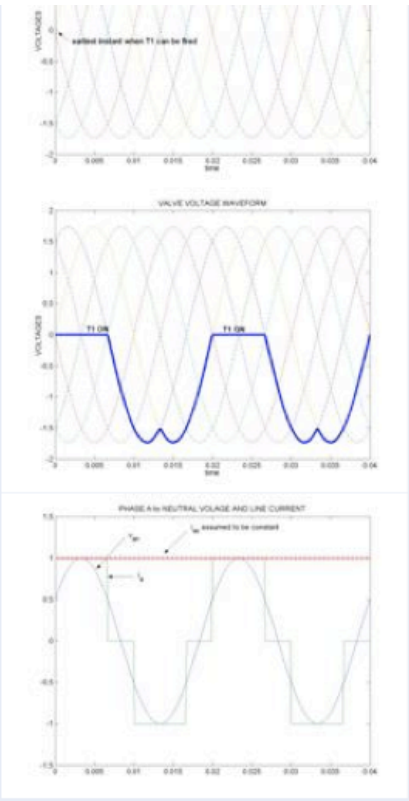


It is assumed here that the dc side inductance value is large, dc resistance is small , and the DC source voltage Vi is smaller than the average of Vdc(t) over a cycle. This can ensure that the current on the dc side does not touch zero and is practically a constant. In fact, for most applications, the inductance value is large enough to ensure a smooth dc side current.

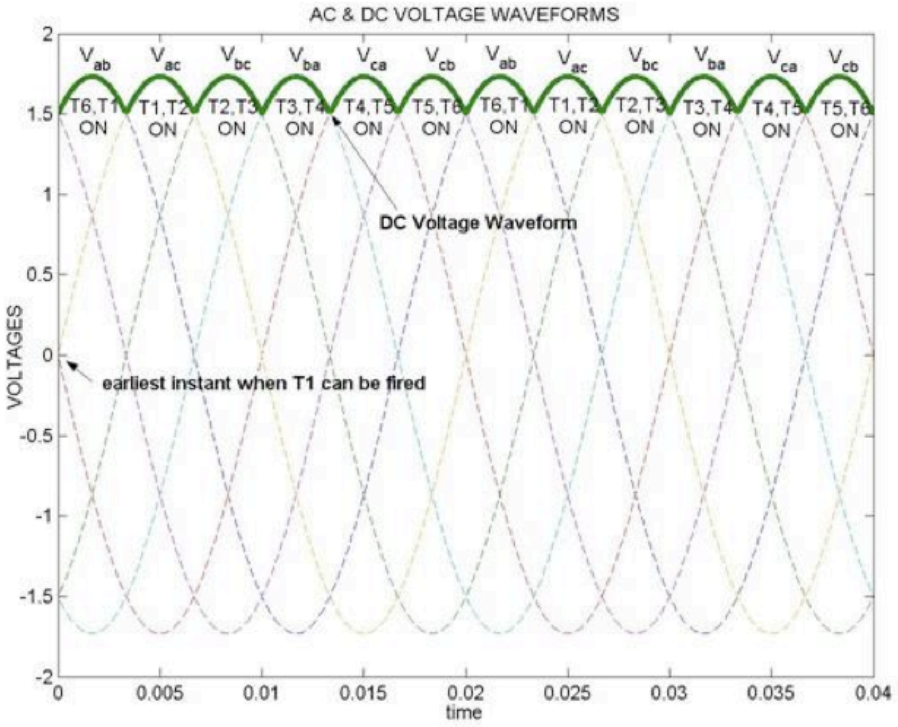
It is clear that if the firing sequence is maintained as given above, then each device conducts for 120° in a cycle. The fundamental component of the current on the AC side is in phase with the corresponding phase to neutral voltage.

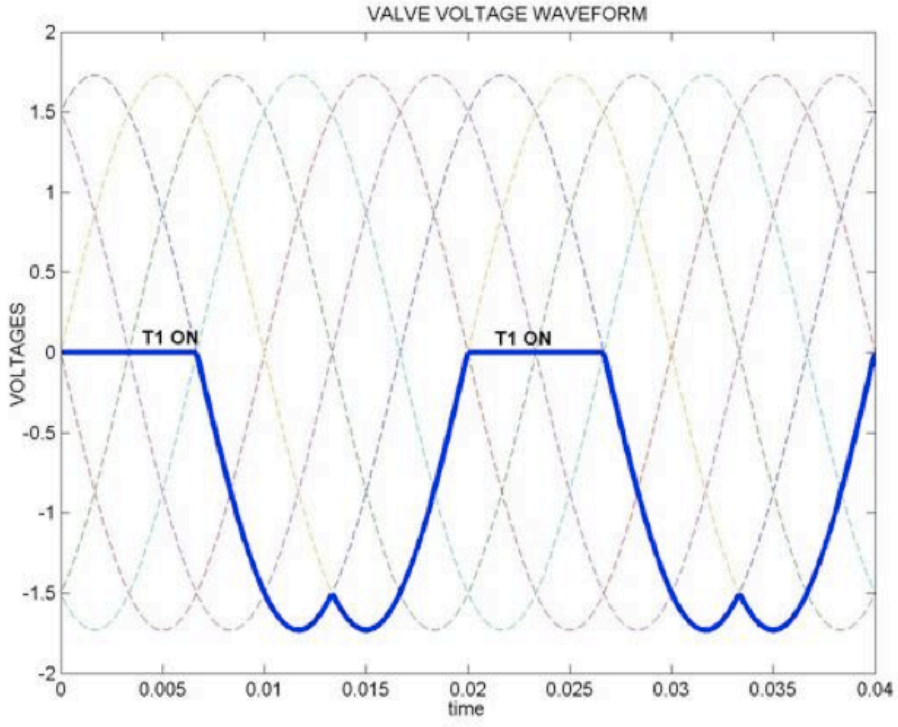
Note: The waveforms shown are representative of the steady state and not the transient behaviour. When a bridge is started, current will gradually build up. For current to start flowing on the DC side, it is important that *two* thyristors be on (one each in the top and bottom half of the bridge). Therefore, while the thyristors are fired in sequence T1 to T6 in a cycle, the gate signals are maintained for more than 60° so that a couple of thyristors will be in a position to turn on when a bridge is started. Also during starting, Vi should aid buildup of current.

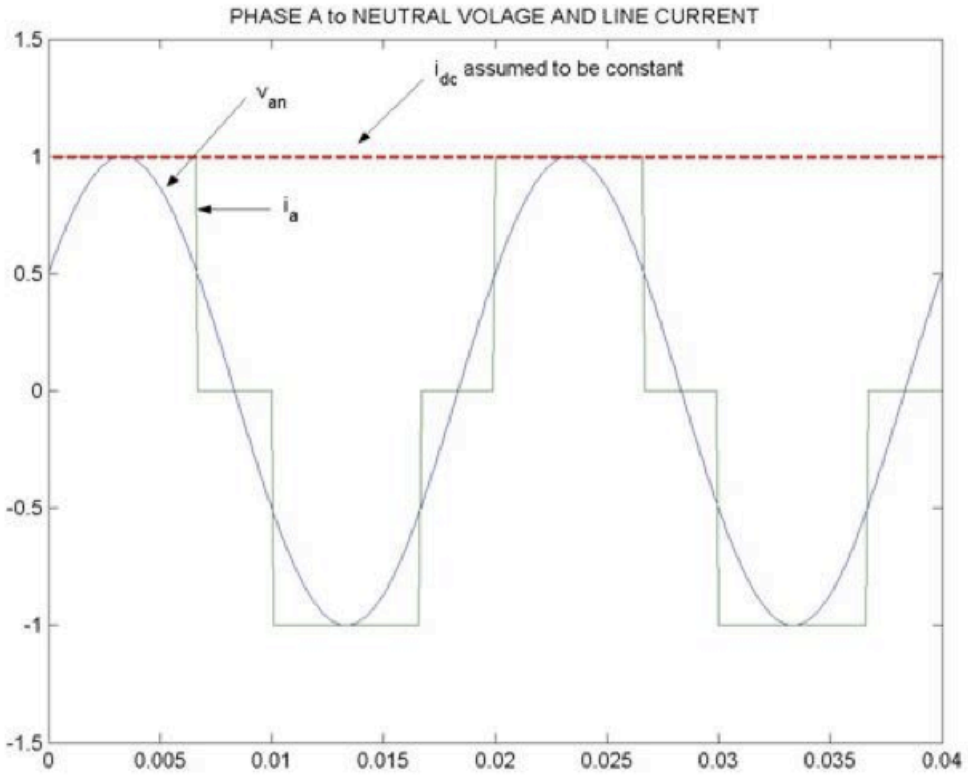
It is assumed here that as soon as a thyristor is switched on in the top or bottom half of the bridge (e.g. T1), then the previously conducting thyristor in that half (e.g. T6) switches off immediately. In actual practice, the current in the "outgoing" thyristor T6 dies down gradually due to source inductances (not shown here). This is known as commutation delay.



(click on image to enlarge)


AC & DC VOLTAGE WAVEFORMS
V_{ab} V_{ac} V_{bc} V_{ba} V_{ca} V_{cb} V_{ab} V_{ac} V_{bc} V_{ba} V_{ca} V_{cb}
T6,T1 ON
T1,T2 ON
T2,T3 ON
T3,T4 ON
T4,T5 ON
T5,T6 ON
T6,T1 ON
T1,T2 ON
T2,T3 ON
T3,T4 ON
T4,T5 ON
T5,T6 ON
DC Voltage Waveform
earliest instant when T1 can be fired
VOLTAGES
time
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04


VALVE VOLTAGE WAVEFORM
T1 ON
T1 ON
VOLTAGES
time
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04



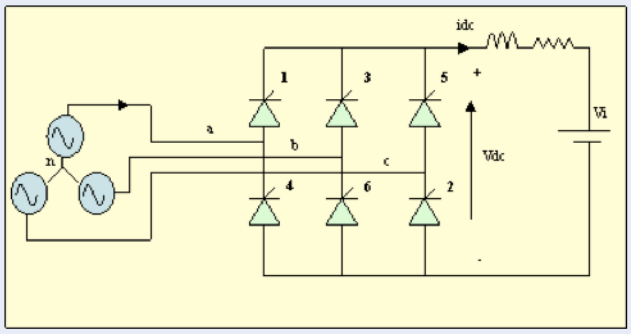
AC-DC converter control

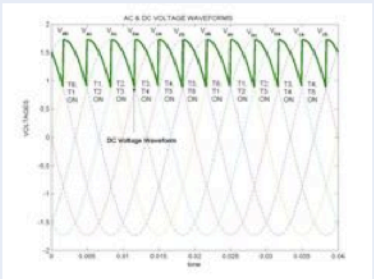
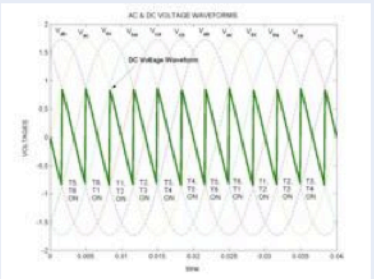
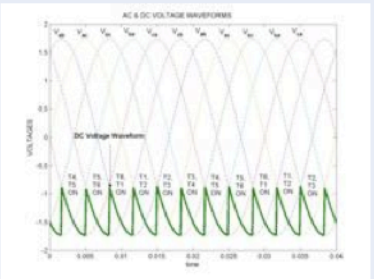
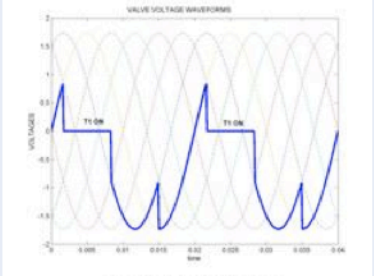
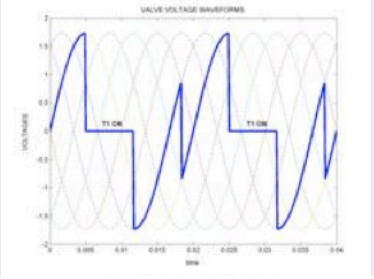
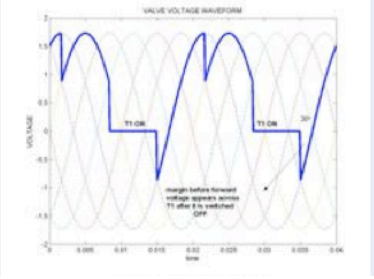
Now suppose the thyristors are not fired as soon as they are forward biased, but fired after some delay.

Let us define **delay angle** for thyristor T1 as the angular delay from the point at which Vac becomes greater than zero. Similarly the delay angle for the thyristors T2, T3..T6 is defined by the angular delay from the points at which Vbc, Vba, ... Vab become greater than zero respectively.

If all the thyristors are fired at a delay angle (a) of zero, then we get the waveforms as shown in the previous page. However, if we fire all thyristors at a delay angle of 30° we get the waveform shown below.

Similarly the waveforms for a = 90° and 150° are also shown.



a = 30°	a = 90°	a = 150°
		
		

(click on images to enlarge)

Recall that for a thyristor:

" Once i_{AK} falls to 0, positive V_{AK} should not be applied for a duration, toff (called turn off time), otherwise the thyristor may turn on again even without application of a gate signal ! "

Therefore if the margin is less than the turn off time, then a thyristor will turn on again **unintentionally**, i.e., without a firing gate signal (commutation failure). This is generally avoided as it disturbs the current and voltages in an unacceptable way. So an important condition for inverter operation is that a should not be too close to 180°.

Some other important issues:

1. Harmonics: Both AC side currents and DC side voltages have harmonics (what are the lowest order harmonics for these two quantities ?). Filters are usually necessary on AC and DC side to reduce harmonics.

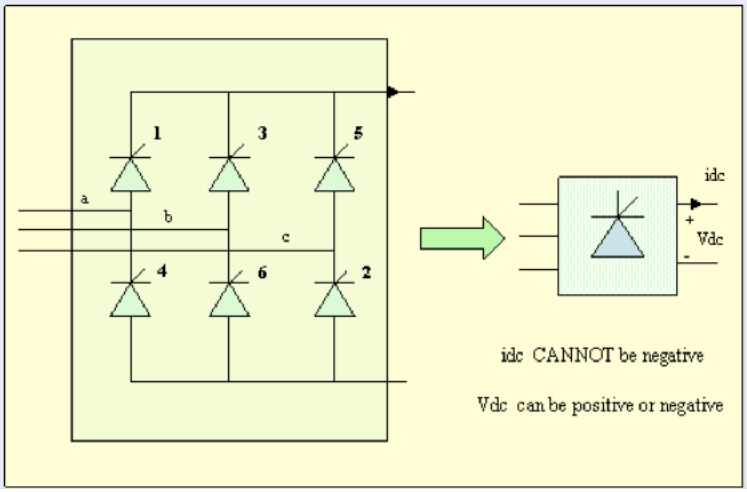
2. Reactive power: For reactive power drawal from AC side to be low, it is necessary to have a close to 0° for a rectifier and close (not too close) to 180° for an inverter.

HVDC link

In the previous page we have seen that a converter may be operated either as a rectifier or an inverter. For inverters, power is taken from the dc side and given to the AC side. Inversion occurs if Vdc becomes negative (note: current cannot become negative on the dc side since thyristors do not allow reverse conduction).

Vdc can be made negative if delay angle is > 90° and current is continuous -- it should not touch zero. The current can be kept continuous if dc source voltage Vi is more negative than the average negative value of Vdc (see previous page).

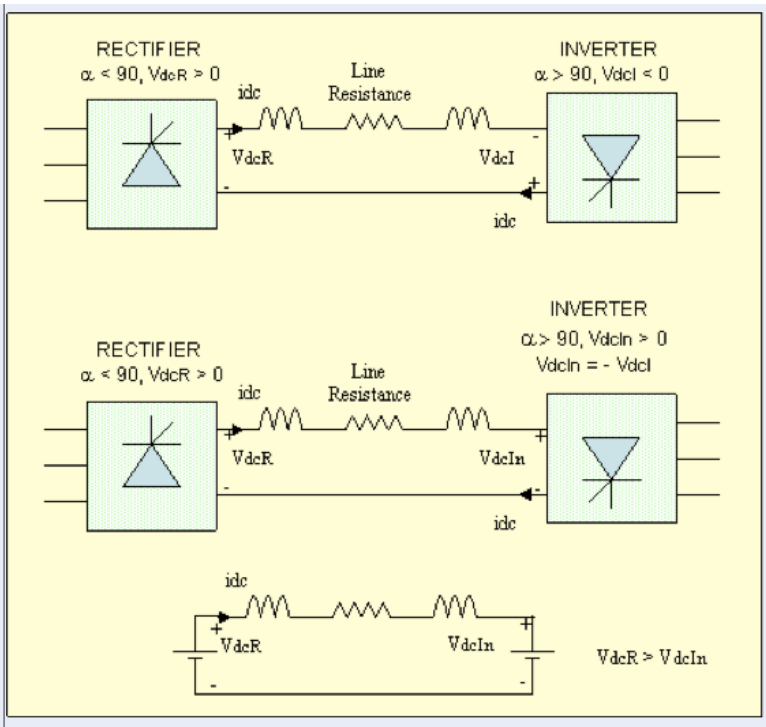
Let us represent a line commutated AC-DC converter as shown below:



An HVDC link consists of a rectifier and inverter connected together as shown in the figures on the right. Power flows from the rectifier to the inverter and is in effect transferred from the rectifier AC side to the inverter AC side. The rectifier voltage V_{dcR} plays the "role" of the dc source, Vi, required for inverter operation as discussed above.

The magnitude of the inverter and rectifier voltages is determined by the firing angle. Current (or power) can be changed by changing the magnitude of either or both voltages. However, since they are constraints on the maximum value of delay angles for inverter, so in practice current is controlled by controlling the rectifier dc voltage.

Since the magnitude of the dc side voltages are dependent only on AC side voltage magnitudes, delay angles, power flow is NOT dependent on AC side system frequency or phase angle at either end.

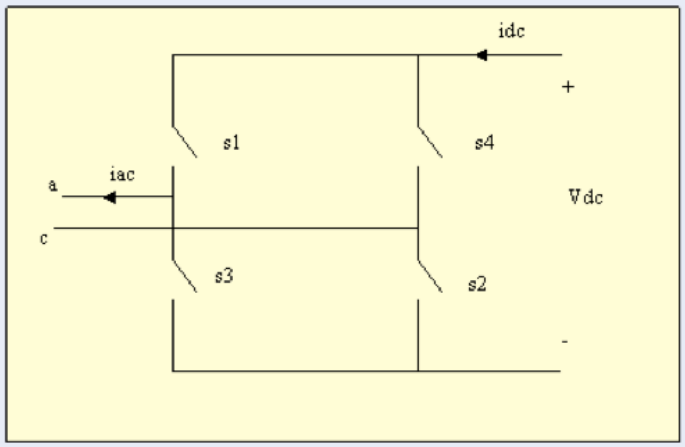


Note : AC side voltages are a must for the operation of a line commutated converter bridge. If AC side voltages are absent, the converter cannot be operated. This has significance in a power system: when the AC system voltages on either side are zero (due to a black out), power cannot be transferred via this link.

Can reverse the direction of power flow in an HVDC link ? how ?

AC-DC Voltage Source Converter: A brief introduction

The converter that we have studied is a "line commutated converter". The voltages required for commutation (turning off a device and switching on another) are obtained from the AC side. On the other hand, a **Voltage Source Converter** (VSC) uses a voltage source (or a capacitor whose voltage is maintained constant) on the DC side and switches with **turn off capability**. These switches can be turned on or off at will, if the voltage on the dc side is positive. The schematic of a single phase voltage source converter is shown below.



The voltage on the AC side is related to the DC side voltage depending on the switch positions:

s1 and s2 ON, s3 and s4 OFF : Vac = Vdc, idc = iac

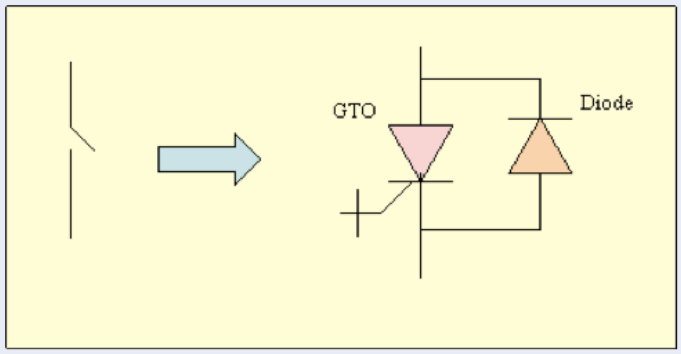
s3 and s4 ON, s1 and s2 OFF : Vac = - Vdc, idc = - iac

s1 and s4 ON, s3 and s2 OFF : Vac = 0, idc = 0

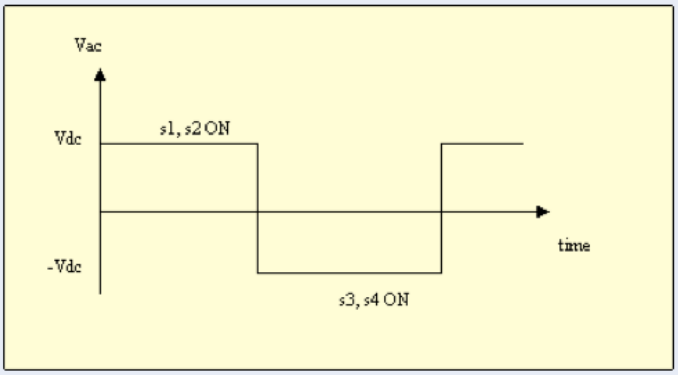
s2 and s3 ON, s1 and s4 OFF : Vac = 0, idc = 0

However, two switches in a leg (e.g. s1 and s3 in the figure above) should not be turned on simultaneously (why ?)

The direction of current on the dc side can be bidirectional, although dc voltage has to be unidirectional. The switches can be implemented using power electronic devices as shown below -- a device like a Gate Turn Off Thyristor (GTO) in parallel with a reverse connected diode. Note that devices like Gate Turn Off Thyristors can be switched off by a gate signal - a capability which an ordinary thyristor lacks. Devices like Insulated Gate Bipolar Transistor also may be used in a VSC instead GTOs.



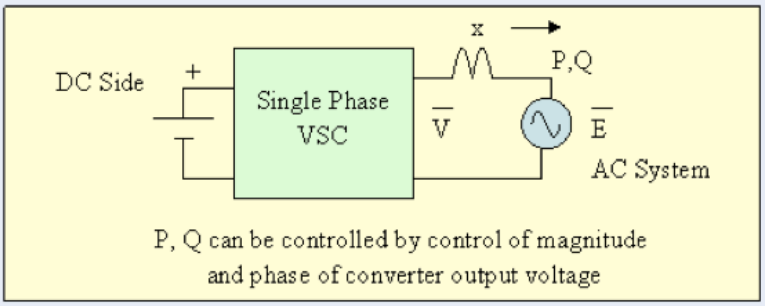
If s1 & s2 and s3 & s4 are switched alternately, then the voltage on the ac side is a square wave as shown on the right.



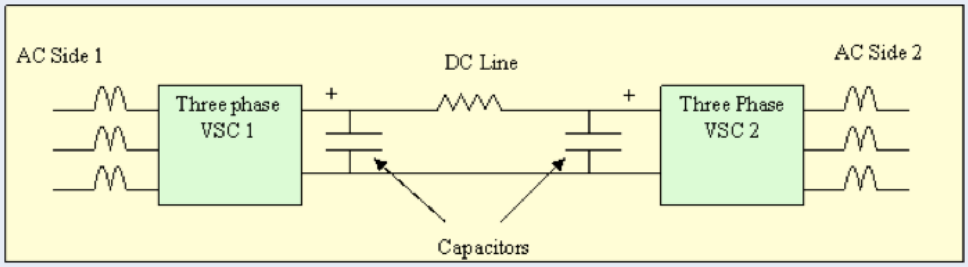
One could develop more elaborate switching schemes in order to yield ac side voltages with **lesser lower order harmonic content** as well as **control** the magnitude of the fundamental of the ac side waveform. These are called **PWM** (pulse width modulation) switching schemes. We do not cover them here, but an interested reader may consult any modern power electronics book for the same.

Note that phase angle of the fundamental of the converter ac side voltages may be easily controlled by delaying or advancing the instants of switching.

Thus we can control *both* magnitude and phase angle of the converter AC side voltage independently. This implies control of real and reactive power injected into the AC system:



A VSC has more flexibility from a power systems perspective due to independent control over magnitude and phase angle of the ac side waveforms. A schematic of an HVDC link based on VSCs is shown below.



Note that by appropriate control of voltage magnitude and phase angle at the AC terminals of the VSCs, one may draw/supply real and reactive power from the dc side. The two converter are controlled so that the real power drawn from AC side 1 is supplied to AC side 2 (minus the losses in the DC line resistance). If this balance is not maintained, then capacitor voltages cannot be maintained constant since they will get charged (or discharged) due to power surplus (or deficit) on the DC side.

Can power can be transferred via this VSC based link if the AC system at one end is blacked out ?

Recap

In this lecture you have learnt the following

- Special devices like Static VAR Compensators and tap changing transformers can be deployed in the network to improve voltage profile of a system
- Line -commutated rectifiers or inverters, employed in HVDC links, always absorb reactive power.
- Voltage control of a bus can be done by local reactive power injection.

Lecture 19 : Power Flow Control

Objectives

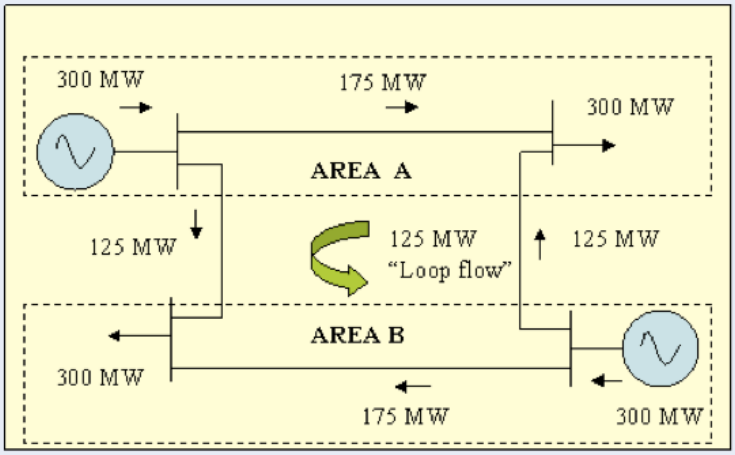
In this lecture you will learn the following

- Power transfer capability of a network may be limited due inability to control power flow in transmission paths.
- What are the ways by which one may control real power flows?

An important feature of interconnected AC systems is that power flow control through specific control paths is often quite difficult. This is because power flows in these paths as per Kirchhoff laws: the line parameters, topology of the network, generation and load location determine the value of flows. The power flows are not directly dependent on transmission line ownership, contracts, thermal limits or losses!

Inability to control power flows may result in situations wherein, some paths may be lightly loaded and some overloaded (to understand what one means by loadability see the discussion on line loadability in module 2).

The figure given on the right demonstrates a possible scenario wherein there is power flow in interconnecting tie lines of the two areas although each of them are individually self-sufficient. Thus the tie lines are unnecessarily loaded. This may be a problem if the capacity of the tie lines is limited and one actually wishes to transfer power from one area to another. This is because a part of the line capacities are utilized due to the undesired "loop flow".



Tie line power flow control by AGC was discussed in the previous module. While *cumulative* power exchange between the two areas through both tie lines may be controlled by adjusting the generated power in areas A and B, independent power flow control in each *individual* line connecting the 2 areas is not possible by generation control alone. In the above figure, cumulative power flow in the two tie lines is zero. It can be changed by reducing generated power in one area and increasing in another. However, the individual power in each line cannot be controlled in this fashion and depends on line parameters and Kirchoff's laws.

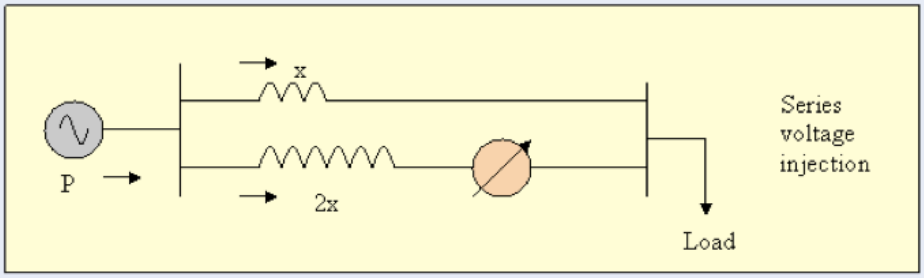
Therefore, special measures have to be taken in order to control power flow through individual transmission paths. Some of these are considered in the following lectures.

Power Flow control can be achieved by changing effective line parameters by connection of lumped series capacitors.

Series Compensation of lines

This involves changing the effective series reactance of lines by connecting capacitors in series with a line. Reducing line reactance also improves stability of a system (see module 2 for a discussion of large disturbance stability).

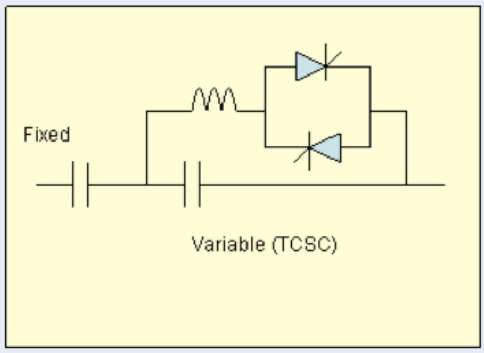
In the figure given below, the power flow in the branch with reactance 2x can be increased by compensating it with a series capacitor.



Since the amount of loading changes with time, the amount of series compensation may be varied by bypassing (shorting out) these capacitors when not necessary.

Normally a capacitor can allow for short duration over-rating for a few seconds. This allows insertion of larger capacitive reactance into a line for a short time in order to improve angular stability.

Alternatively, they may be controlled using power electronic controllers. For example, a TCR (discussed in the section on voltage control), may be connected in parallel to the series capacitor and its effective reactance may be controlled by controlling the firing angle delay. This device is called a **Thyristor Controlled Series Compensator (TCSC).**

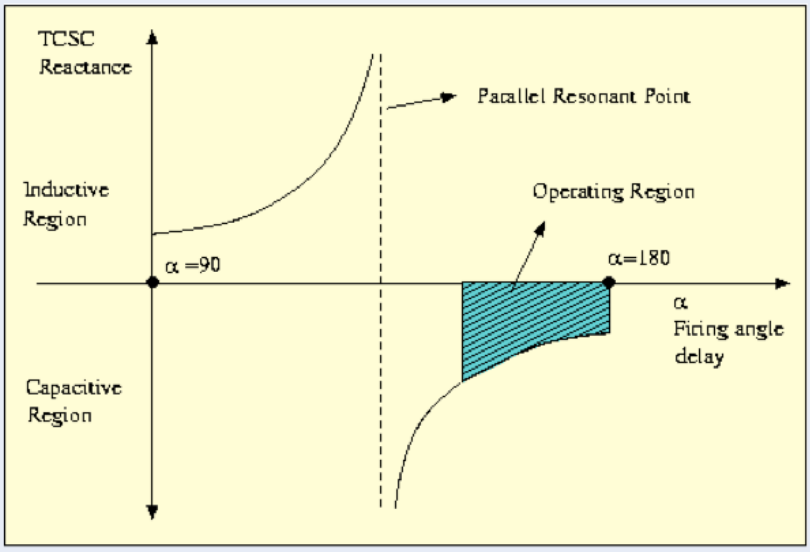


Thyristor Controlled Series Compensator

The effective reactance is the parallel combination of a fixed capacitor and variable reactance.

Parallel resonance may occur at certain values of firing angle. The characteristics of a TCSC are given on the right.

Why cannot we increase the effective reactance of the parallel combination to much larger values say upto the parallel resonance point?



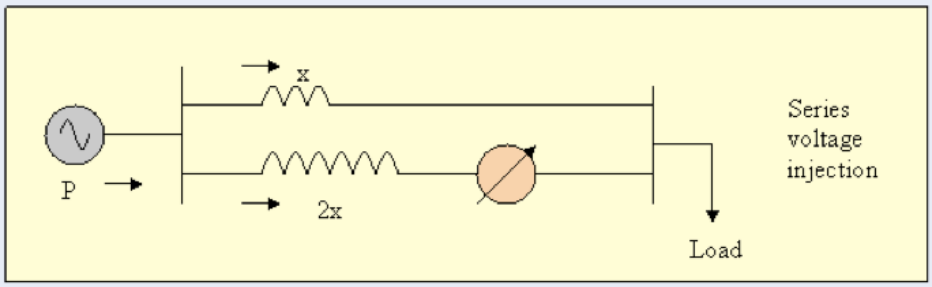
One of the major consideration of any series connected device is the need to bypass the device when *faults* occur nearby. Faults cause large overcurrent in a line, which in turn results in overvoltage across a capacitor. Therefore, Metal Oxide Varistors (which are nonlinear devices which have very low resistance when voltage magnitude is greater than a particular value) and circuit breakers are generally connected across series connected devices. Note that during faults, a series connected device is bypassed. However, immediately after the fault is cleared, it is necessary to insert maximum capacitive reactance in order to improve the stability of the rotor angular deviations which occur due to the fault.

Examples of controlled series compensation installations in India are the Raipur - Rourkela TCSC and the Kanpur Ballabhgarh TCSC. Details of the Kanpur-Ballabhgarh installation are given in the next module.

You may simulate the working of a TCSC using the MATLAB/SIMULINK file (TCSC3.mdl).

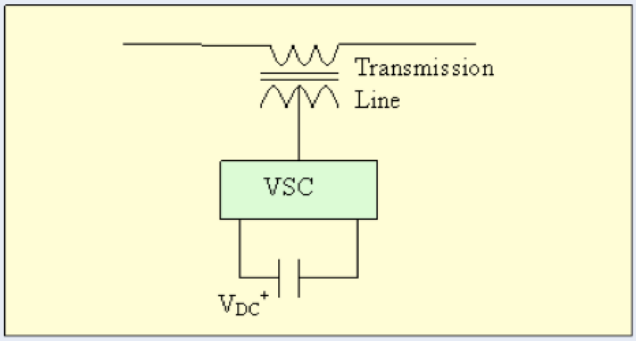
Series Compensation of lines

It is also possible to change power flows by injecting variable voltage in series with a line through a transformer as shown below.



This can be achieved by using power electronics based voltage source converters -VSC - (dc/ac converters). The schematic of this device is shown on the right.

When this device injects three phase reactive power in series with the line (no active power interchange), what is the nature of current on the dc side if the VSC is lossless? Does that explain why one can work with a capacitor (as shown) instead of a battery on the DC side?



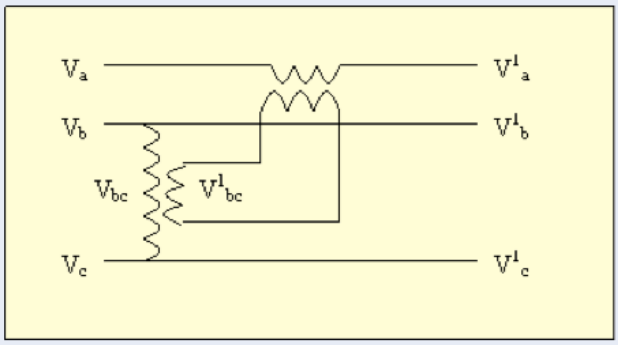
Devices like a TCSC or the device shown above are often called **"Flexible AC Transmission Systems"** controllers, because they increase the flexibility of AC system operation by allowing additional control.

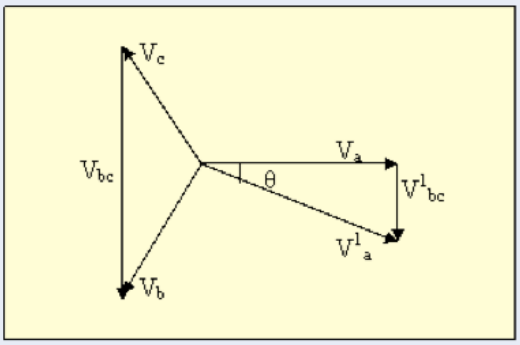
Series connected inductors can be connected to thermally limited lines to reduce power flow through them.

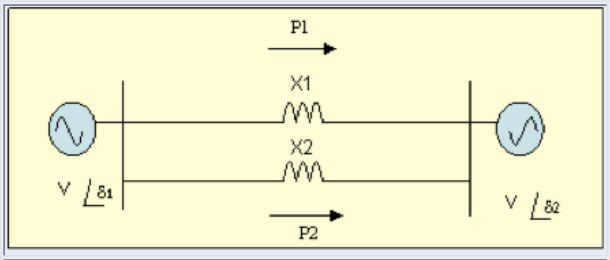
Phase Shifting Transformers

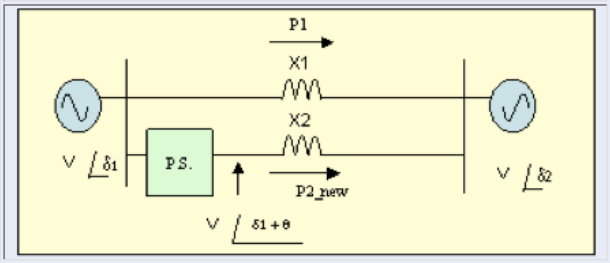
Another method of changing power flow through a transmission line is by use of a phase shifter.

A phase shifter introduces a small phase shift in the voltage at its two terminals by injection of a series voltage. The series injected voltage of one phase is in phase with the line voltage across the other 2 phases. This implies that the voltage injected in one phase is in **quadrature** to the bus voltage at one terminal of that phase. This small quadrature voltage injection mainly affects the bus voltage *angle* at the other terminal (which has superscript 1). The magnitude is not affected much if the series injected voltage is small. This is illustrated in the figures below. In the phasor diagram, the mechanism of obtaining a phase shift of q is shown. Similarly, a phase shift of -q can be obtained by reversing the injected voltage.









A phase shifter can alter the power flows in a transmission system.

In the figures shown on the left, for the given phasor angles at both ends, the power flow in line 2 can be altered. Thus the *sharing* of the total power flow :

$$\frac{P2_{new}}{P2_{new} + P1}$$

between the two lines can be altered. The expression for power flow in line 2 is given by,

$$P2_{new} = \frac{V^2 \sin(\delta_1 + \theta - \delta_2)}{x2}$$

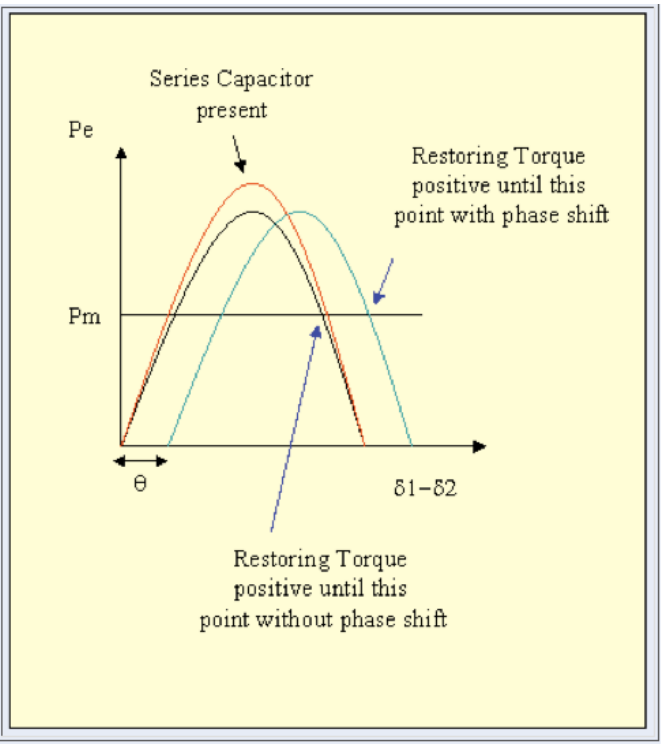
A phase shifter can alter the power versus (d1-d2) curve in a way that benefits stability.

In particular, by changing q when the rotor angular separation increases due to a disturbance, it can ensure that the "restoring torque" on the machines at the two ends is positive for larger values of (d1-d2).

In the figures shown above, if only line 2 is present, then the electrical power flow P versus (d1-d2) with and without a phase shift is as shown in the figure on the right. Note that a phase shifter does not change the maximum value of electrical power.

A device such as a capacitor connected in series with a line, on the other hand, increases the maximum value of power (and therefore restoring torque) but does not cause much shift in the point until which restoring torques are positive.

Both series connected capacitor and phase shifter, therefore enhance stability (the mechanisms are however different). Therefore they permit increase in actual line loadability.



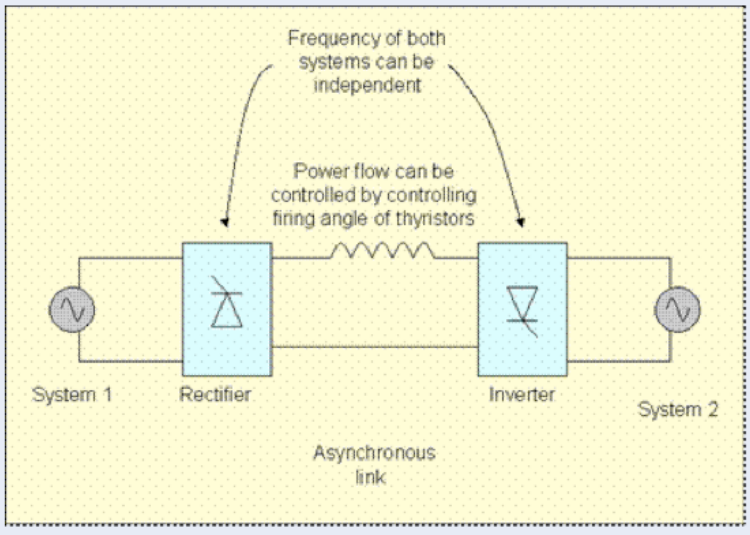
AC-DC-AC conversion

Power Flow can also be controlled by converting AC to DC by thyristor bridges or Voltage Source Converters and reconverting the DC to AC (at another bus) again by the same means. The converters are connected in shunt with the AC transmission system at 2 buses. If a substantial distance between the 2 buses is involved, then *high voltage* DC transmission is done. The power flow is a function of the converter firing angles and is practically independent of the voltage phase angle and frequency at either end.

Normally, DC links *embedded* in an AC system are mainly used to transfer power over long distances which is difficult with AC lines (see problems with long distance AC transmission in module 2).

DC links are also used to allow for controlled power inter-changes between two systems which are *not synchronously* connected is shown in the figure on the right.

You may simulate HVDC link using the MATLAB/SIMULINK file (hvdc.mdl).



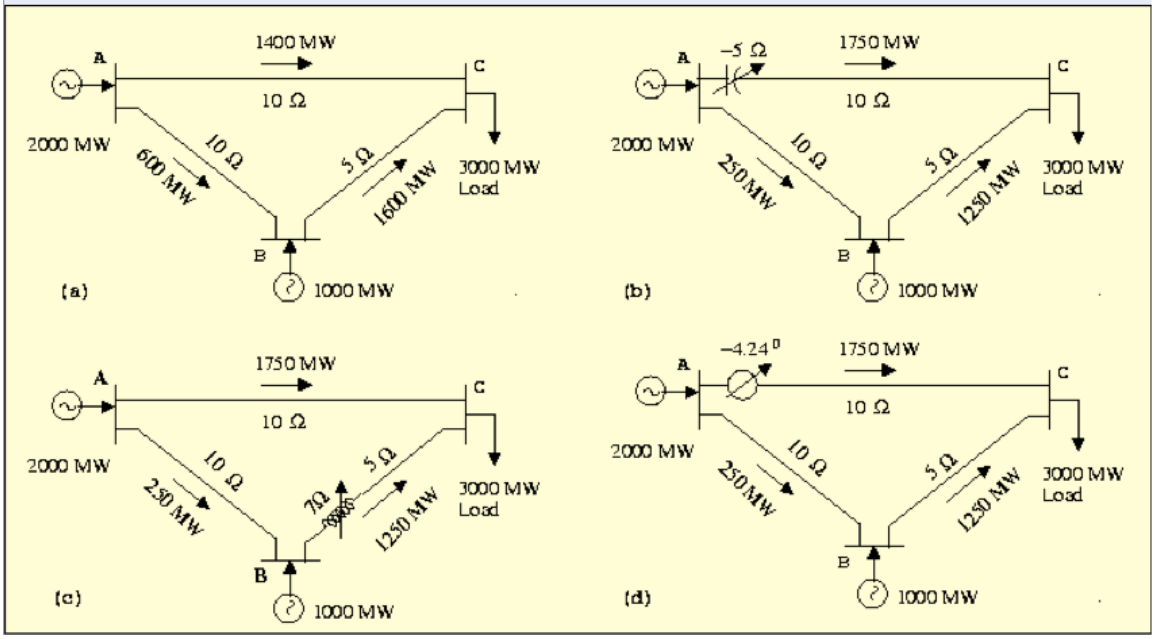
Note: If the two systems in the figure above were to be connected using **AC line(s)**, then the frequencies of both systems would have to be the same and the *total* power transfer from one system to another cannot be controlled even if variable reactance type devices like TCSC are connected. At best, such devices can control the sharing of power flow between parallel AC paths. The only way to control the *total* power flow in AC tie-lines is by controlling the generation/load in the two systems using AGC, which we learnt in the previous module.

All the options for power flow control involve substantial cost, but can allow for better utilization of network by not only allowing desired power flow through various paths but enhancing overall power transfer levels.

An example of power flow control in a meshed system using various devices is given next (**example**) .

This example is adapted from: *Hingorani.N.G*, " Flexible AC transmission", IEEE Spectrum, Vol. 30, Issue: 4, April 1993, pp.40- 45.

Consider an elementary case in which two generators are sending power to a load center through a network consisting of three lines in a meshed connection as shown in figure below [top left,(a)]. The lines AB, BC, and AC have continuous ratings of 1000 MW, 1250 MW, and 2000 MW, respectively, with emergency ratings of twice those figures. If one generator is rated at 2000 MW and the other at 1000 MW , a total of 3000 MW would be delivered to the load center. For the impedances shown, the three lines should carry 600, 1600, and 1400 MW, respectively. Such a situation would overload one line.



Power, in short, flows in accordance with transmission line impedances that bear no direct relationship to transmission ownership, contracts, or thermal limits. The difference between the free-flow path and the contract path is called "loop flow", and is usually characterized by a circulation of power that leaves the available capacity underutilized.

If, however, a capacitor whose reactance is -5 ohms, at the synchronous frequency is inserted in one line [top right,(b)], it reduces the line's impedances from 10 to 5 ohms, so that power flow through the lines will be 250, 1250, and 1750 MW, respectively. It is clear that if the series capacitor is adjustable (as in a TCSC), then other power-flow levels may be realized in accordance with the ownership, contracts, or thermal limitations.

Similar results may be obtained by the increasing the impedance of one of the lines in the same meshed configuration by inserting a 7 ohm reactor (inductor) in series with the line [bottom left, (c)]. Again, a series inductor that is partly mechanically and partly thyristor controlled could serve to adjust the steady-state power flows.

Another alternative is to introduce a phase shifter in the line A-C as shown in fig. (d)

Recap

In this course you have learnt the following

- Inability to control power flows on individual lines may restrict the utilization of a transmission network.
- Series compensation of line parameters by capacitors and phase-shifting transformers are conventional ways to control power flow.
- Power electronics based HVDC converters and controlled series compensation may also be used.

Lecture 19a : Use of Controllable Devices : An example

Objectives

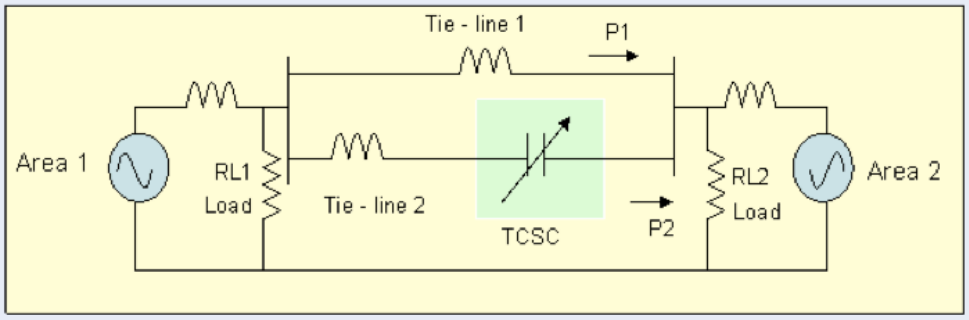
In this lecture you will learn the following

- The use of controllable devices with the help of an example

An example

In the previous lectures we have seen that power electronic based controllers like generator excitation systems, SVC, TCSC and DC links allow real and reactive power flow control. In this lecture, we try to understand how this can be achieved in practice, by considering a simple example. Most of the concepts in this lecture have been introduced previously. However, it is hoped that this lecture will not be a mere reiteration, but will bring clarity to most points.

Consider a two area system shown below, connected by two tie lines. A series controlled device (TCSC) is placed in one of the parallel lines. The lines have unequal impedance. Therefore steady state power flow through them is not equal.



We can utilize the power flow control ability of a TCSC to assist the system in the following tasks:

a) Regulate power or current in the parallel AC tie line in case it is thermally overloaded by diverting power to the line in which TCSC is connected.

Note: This function is "given up" in case the TCSC compensated line reaches its thermal limit or the TCSC itself reaches some limit (voltage, current or firing angle limit). In case *both paths* are thermally overloaded, a TCSC cannot help, and efforts will have to be made to reduce the cumulative power flow between the two areas by adjusting the generation (AGC) and/or loads.

b) Increase the "stiffness" of the tie-lines connecting the two areas by increasing series capacitive compensation for a short while after large disturbances. This will aid in improving angular stability -- keeping the two areas in synchronism.

c) Modulate the series capacitive reactance so as to make the relative angular motions (swings) settle down quickly.

The first function can be acheived "slowly", since thermal time constants are large. However, the last two functions require quick response capability which is possible since TCSC is power electronically controlled.

We now consider control systems which will help us achieve these strategies.

Control Strategy for Power Flow Control

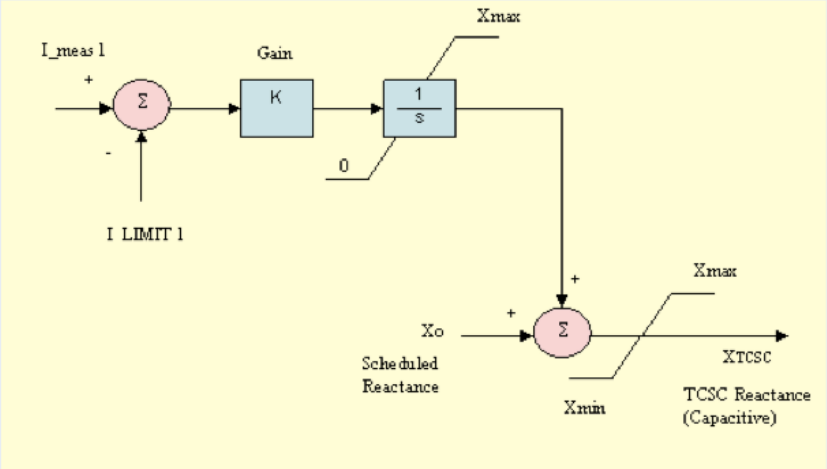
To divert power flow through the TCSC compensated line when a parallel path is overloaded, one would need to first sense that such a condition exists. Therefore the current in the parallel tie line is measured.

(One can also synthesize the current in the parallel path using only *local measurements* of current and voltage phasors at the TCSC location, provided all line parameters are known. How can one do it ?)

In case the current magnitude in the parallel tie - line is greater than the permissible value due to thermal limit, increase the power (or current) flow in the TCSC compensated line by increasing the TCSC capacitive reactance. Since the effective impedance of the TCSC compensated line is reduced, it will take on a larger share of the total power flow between the 2 areas.

A feedback control system which can achieve this is shown below. I_LIMIT_1 denotes the current limit of the tie line which is parallel to the TCSC compensated line. X_{max} and X_{min} denote the minimum and maximum controllable reactance of a TCSC in the capacitive region. The integrator has minimum and maximum limits of 0 and X_{max}. This means that as soon as the output of the integrator crosses these limits, then the integration is halted and resumes **only if** the sign of the input to the integrator is such that further integration will get the output back into the the range 0 to X_{max}.

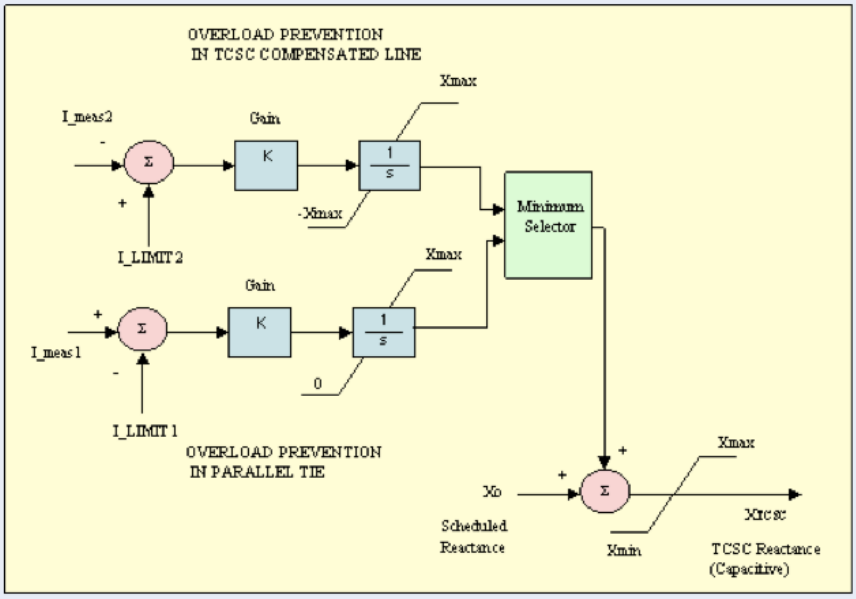
For example if the input to the integrator at a particular instant is -0.5 and the output is 0, then integration is stopped (the output of the integrator remains frozen at 0). It resumes only after the input becomes positive. Similarly if the input is +0.8 and output is X_{max}, then integration is stopped, and it resumes only when input becomes negative.



It can be inferred that if current exceeds the limit I_LIMIT1, then the output of the controller increases the value of TCSC reactance above its scheduled (or set point) value. On the other hand, if the limit is not exceeded, the integrator output will settle to the integrator lower limit, i.e., zero.

Control Strategy for Power Flow Control

In case the current in the TCSC compensated line (I_meas2) itself reaches its limiting value (I_LIMIT 2), the function of alleviating overload (if it exists) in the parallel path is given up, and TCSC reactance is *decreased* in order to reduce current in the TCSC compensated line. This feature can be incorporated by augmenting the TCSC controller as shown in the figure below.



Can you try to decode how this controller, which uses a minimum selector to decide the appropriate mode, works ? Can you think of any alternative scheme which implements these functions better?

This controller parameters can be designed to give a relatively slow response --- setlling times could be on the order of tens of seconds, since transient thermal limits are higher than the continuous thermal limits given by I_LIMIT1 and I_LIMIT2.

Auxiliary Control for improving angular stability

One could use the quick response capability of a TCSC to limit transient relative angular deviations between the two areas following disturbances, and make the transient angular oscillations "die" out.

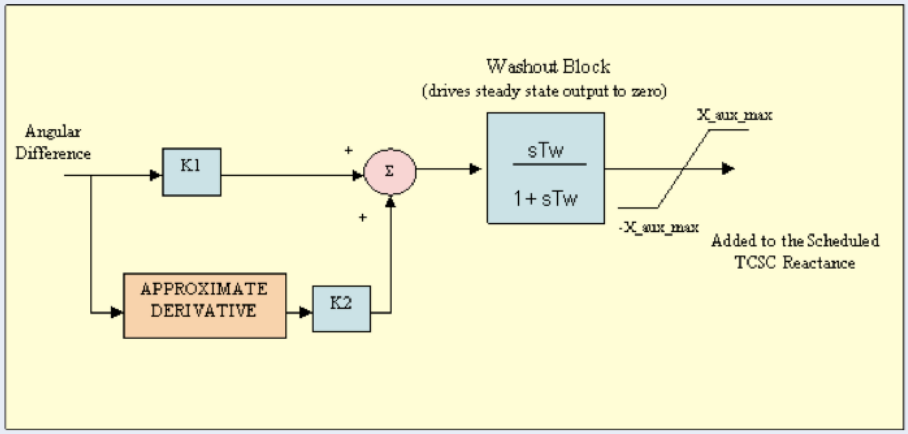
In other words, we wish to prevent potential loss of synchronism and also make the system settle down to a steady state as soon as possible.

A simple "intuitive" control strategy would be as follows:

1. Boost the electrical power flow from area 1 to area 2 if the angular difference between the voltage phasors at area 1 end of the transmission tie lines and the area 2 end increases. The increased electrical power flow between the two regions when angular deviation increases, retards the relative movement between generator rotors in one area with respect to the other. To boost the power flow, the capacitive reactance of the TCSC can be increased in proportion to the angular difference.

2. How does one make the transient angular swings settle down quickly ? Recall that rotor angular swings are analogous to the oscillations in a spring mass system. Oscillations die out in a spring mass system due to a retarding force which is approximately proportional to speed (i.e. viscuos friction). We can try to create similar retarding forces in a power system, by making the electrical power flow to be a function *of the rate of change of* angular difference (it is already a function of the angular difference).

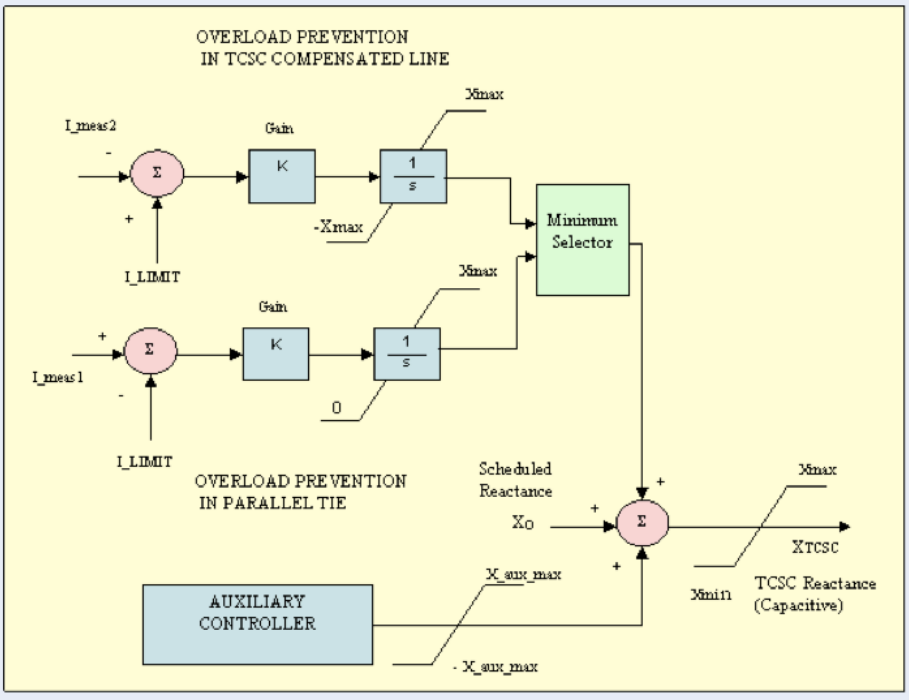
Thus, to reduce angular difference as well as to make the system "settle down" quickly, the TCSC reactance can be made a function of angular difference and rate of change of angular difference. This can be implemented by a controller schematic as shown below:



Important : This auxiliary controller is supposed to act only if there are transients corresponding to rotor relative motion (typical oscillation frequencies lie between 0.2-2 Hz). It has no steady state regulation function -- it should not "interfere" with the relatively slow control functions which were described earlier. The output of the controller has to be *gradually* driven to zero in steady state. Therefore a **washout transfer function** (see lecture 8a, Module 2), which has a steady state gain zero is used in cascade with this controller.

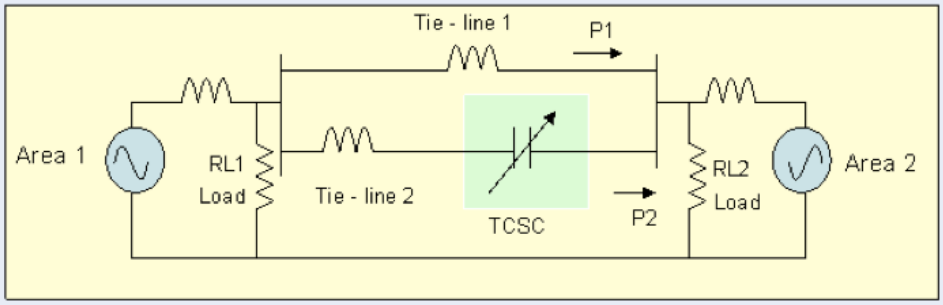
Caution: The control strategy given here is simplistic and is chosen to illustrate the concept; in practice, a more rigorous design approach has to be adopted.

The overall controller including the auxilairy controller look as follows:



Simulation Example

We can simulate using Runge Kutta method, a simple two machine system with local load as shown below:



Note that:

1. The two areas are connected by *two* tie lines one of which is compensated with a TCSC.

2. AGC and governor are not considered.

3. The impedance of the first line (j * X1 = j * 1.0) and the *compensated* impedance of the second (j * X2 - j * X0 = j*1.2 -j*0.2) pu are equal.

4. The continuous current (thermal) limits of the tie lines are 0.16 pu and 0.2 pu respectively.

5. Local loads RL1 and RL2 are purely resistive. RL1 = 1.579 and RL2 = 0.789.

6. Generator Powers (initial values) : 0.95 pu for both generators.

7. Data common to both generators: Inertia Constant H = 6 MJ/MVA, equivalent reactance (between generator internal voltage source and the load in the figure above) is Xs = 0.25pu

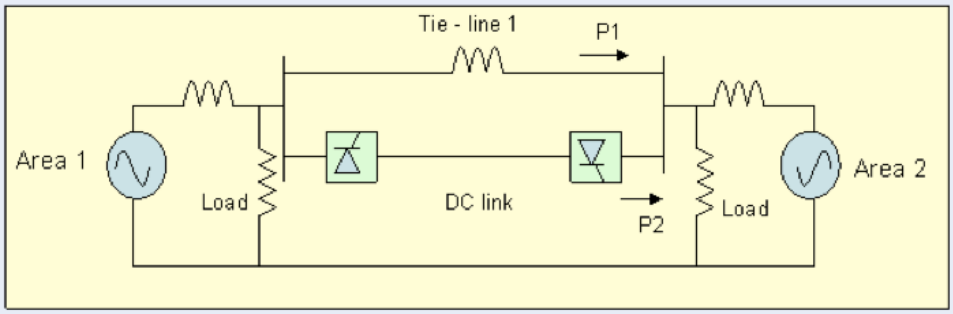
8. Modeling of generator swing equations and network for this simulation is similar to that given in the governor example of Lecture 14 (Module 3). However, governor is not modeled.

9. TCSC is modeled as a variable capacitive whose value is determined by the controller. The dynamics are associated with changing the firing angle and are not modeled explicitly.

Study the data (init.m) and the MATLAB/SIMULINK simulation (tcsc_controller.mdl) for a sudden 5% increase in generated power of one area and a 5% reduction in generated power in the other.

Note how the TCSC controller can divert power in the tie lines (provided its own limit is not reached !)

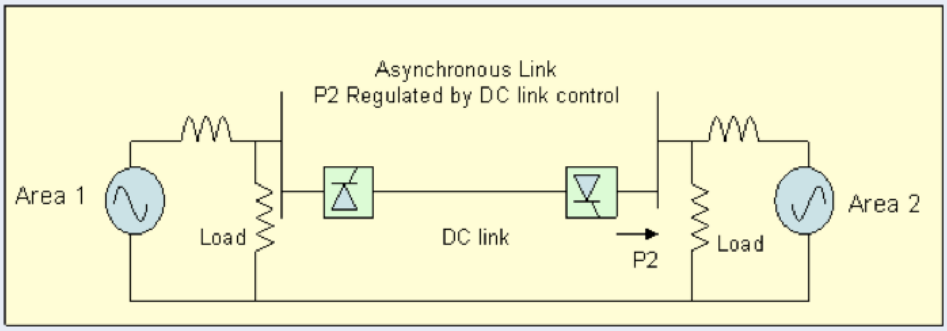
DC Link vs TCSC controlled AC link

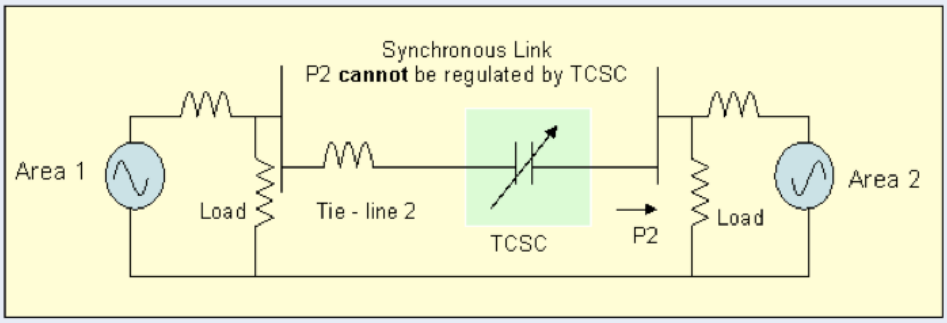


A TCSC controlled AC link and an HVDC link embedded in a synchronous grid can give equivalent functionallity.

> It must have occured to an intelligent reader that a DC link should also give us more degrees of freedom in control since we have *two* converters, thereby we have two firing angles which can be independently controlled. Alas, for thyristor based converters used in most DC links, a degree of freedom in control is lost due to the constraints placed on the reactive power drawal at both ends as well as the need to prevent firing angle angle at the inverter from straying too close to 180 degrees.

However, an HVDC link is able to provide an asynchronous tie in case other parallel AC paths are lost. Control of power between the two regions is regulated by the HVDC link itself. On the other hand, loss of synchronism is unacceptable for a system with a TCSC connected link. Moreover, a TCSC cannot regulate *cumulative* power flow between the two areas (AGC is required for this).





Recap

In this module you have learnt the following

- Using an illustrative example, we have seen how one can have more flexibility in system operation and control

Lecture 20 : Some Real Life Examples

Objectives

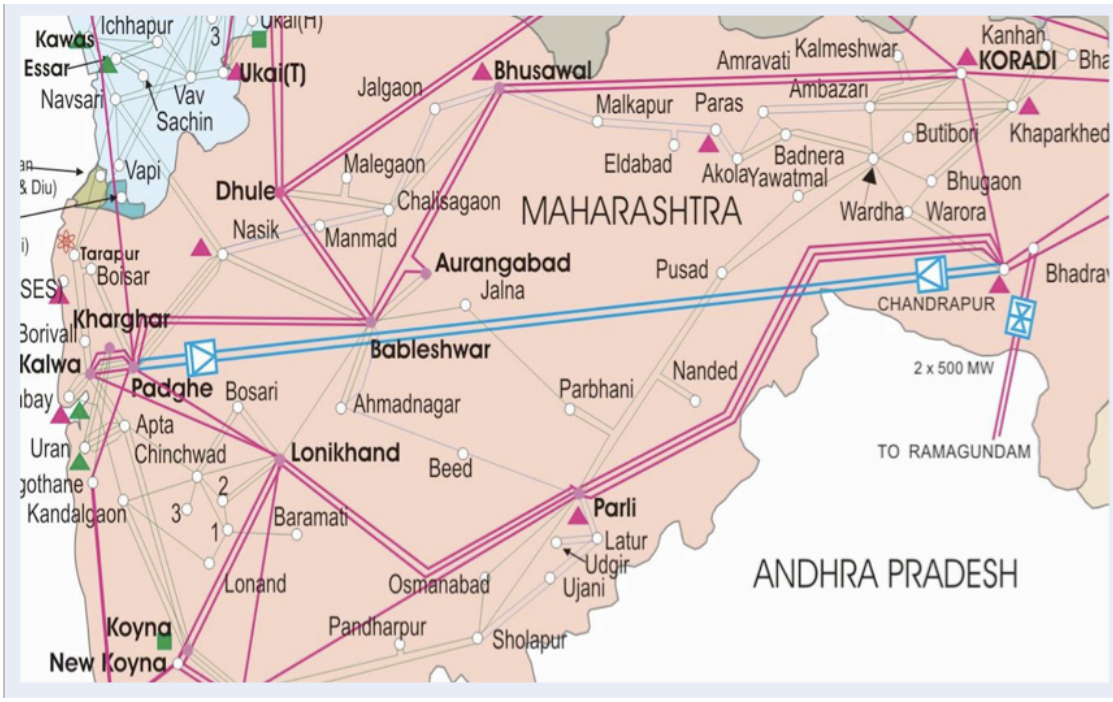
In this lecture you will learn the following

- Real life examples of a few power flow and voltage control devices.

Chandrapur Padghe HVDC link

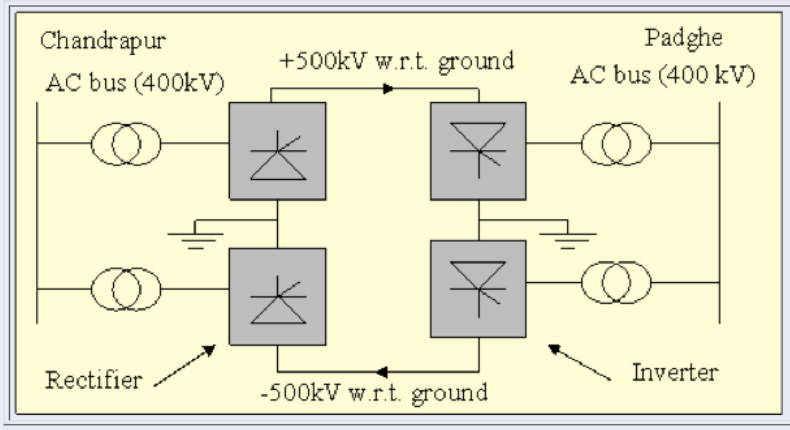
Maharashtra state, which is a part of the western regional grid of India, has most of its large generators in the eastern part of the state. The thermal power station at Chandrapur generates more than 2200 MW of power. However, most of the major loads lie in the western part of the state, i.e., in and around Mumbai, Nashik and Pune, which are at a distance of more than 600 km. Although 400 kV AC lines, are present they pose stability and voltage constraints due to the long distances involved. Therefore an HVDC link between Chandrapur and Padghe was planned and executed.

We have seen the power map of the western region of India. The figure below highlights the Maharashtra state grid. The 400 kV lines are in pink while the HVDC link is shown in blue. An asynchronous DC link between the Western and Southern region is also present (between Bhadrawati and Ramagundam).



The Chandrapur -Padghe DC link is primarily meant for point to point bulk power transfer. The overall capacity of the link is 1500 MW and consists of two sets of converters at Chandrapur (rectifier end) and Padghe (inverter end). The voltage between the two lines which form the link is maintained at about 1000 kV. The voltage of each line with respect to the ground is about +/-500 kV. The link also allows for power flow in the ground for a short while if one conductor which forms the return path is unavailable.

The power flows are set as per the schedule decided by a system operator, and implemented by closed loop control of the firing angles of both the rectifier(Chandrapur) and inverter (Padge).



Ratings of Link	Power Transmission Capability (MW)
Minimum Continuous Rating	1500
Nominal Rating	1500
2 hours overload rating	1650
5 second overload rating	2000
Rating under reduced DC voltage	1000

Fast control possible with power electronics also allows for the additional capabilities:

a) In case degraded insulation conditions exist, power can be transmitted at slightly lower voltages (losses will increase though).

b) The link can reverse power flow in case the system conditions so demand.

c) The HVDC link can act like an asynchronous link in case the eastern and western part of the state are not in synchronism and power flow can be controlled to keep frequency at one end constant.

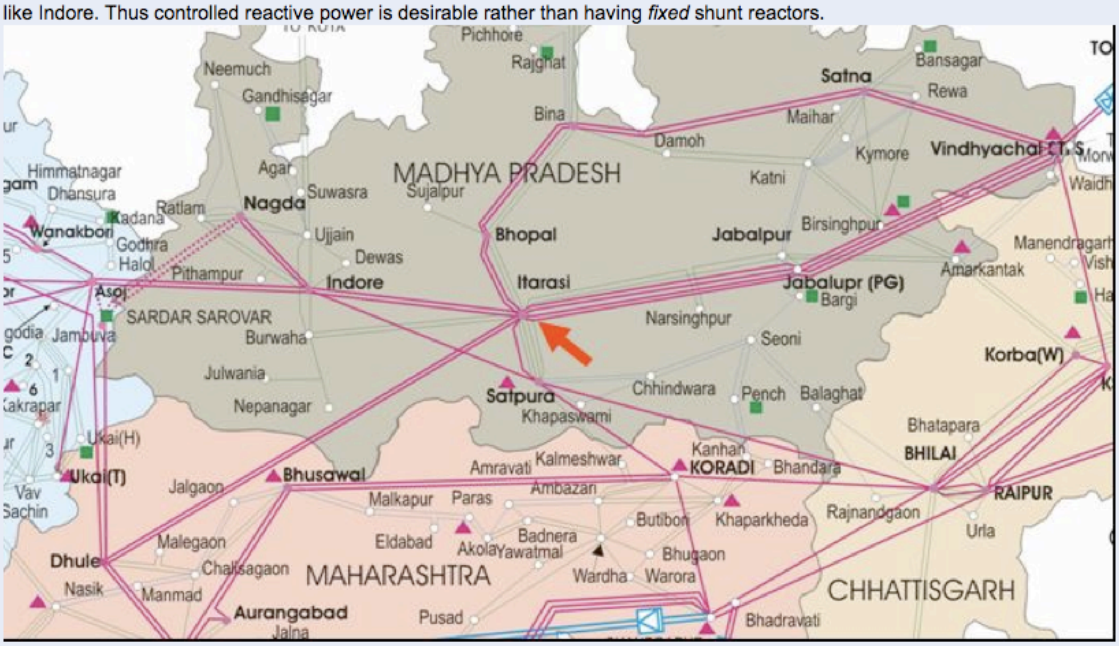
d) The reactive power absorbed at both rectifier and inverter can be increased if overvoltage conditions exist.

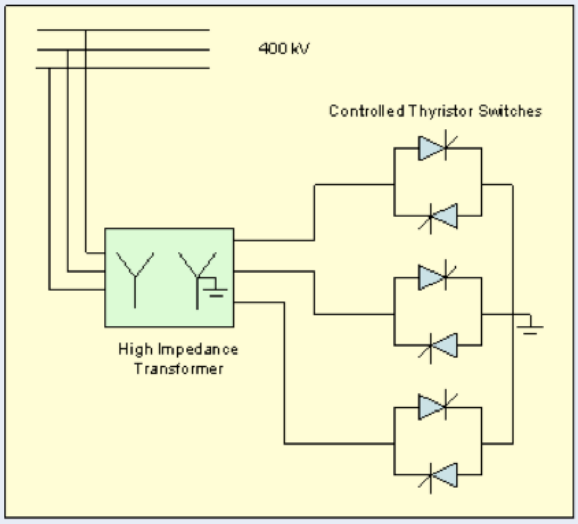
e) In case a parallel AC line (e.g., Chandrapur -Parli 400 kV line in the above figure) is tripped, it is possible to increase the power flow in the DC link to prevent instability or overload of other AC lines.

(Data obtained from "Operation, Maintainance and Protection Aspects of 1500 MW, +/- 500 kV Chandrapur-Padghe Bipole project", by R.S. Khandagale et al, Annual Workshop for Load Despatch Personnel and System Operators, Sept 2002, Mumbai)

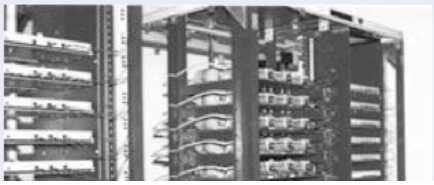
The Controlled Shunt Reactor at Itarsi

The Itarsi bus in the Western Regional grid of India has several 400 kV long lines (shown in pink in the Figure below) incident on it. These lines carry a large amount of power from the east to the west. However, to prevent over-voltages during *light* load conditions, reactive power has to absorbed at various buses. During heavy load conditions there is a sag of voltages at buses like Indore. Thus controlled reactive power is desirable rather than having *fixed* shunt reactors.

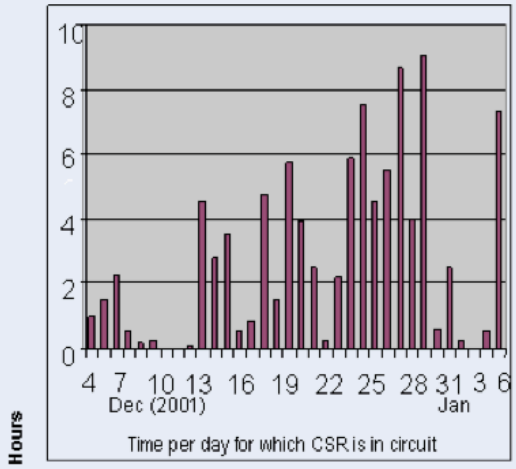




We have learnt about shunt reactive control using a thyristor controlled reactor (TCR). Usually, a TCR is connected to a high voltage bus using a transformer. However instead of having a separate reactor, the leakage reactance of a transformer, (which is designed to be high) itself may be utilized thereby achieving some economy. Such a device, denoted as a Controlled Shunt Reactor - CSR- has been designed and implemented by BHEL at Itarsi 400 kV bus in the Western Regional grid of India. The rating of the CSR is 50 MVAr.



Schematic of the CSR



Thyristor Valve for CSR circuit

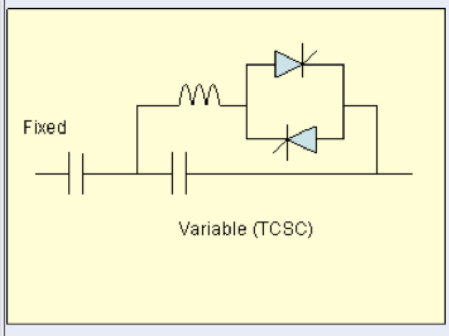
A CSR provides for controlled reactive compensation only when it is required (i.e. during light load conditions). Unlike fixed reactive compensation which results in voltage sags during heavy load conditions and reduction in power transfer capability, a CSR can be controlled or switched off during heavy load conditions. For example, if the voltage goes above 1.05 pu, then CSR is switched in. The typical response time is of the order of 10 ms.

(Details obtained from "Controlled Shunt Reactor : A member of FACTS family", S.V.N. Jithin Sundar et al, 11th NPSC, Bangalore, Dec. 2000, pp 63-68 and "Controlled Shunt Reactor", by Dipak Dutta, International Seminar and Tutorial on Power Transmission Research and Challenges, CPRI, Bangalore, 21-22nd Dec, 2005)

The Kanpur-Ballabhgarh 400 kV Thyristor Controlled Series Compensator

Based on system studies, a TCSC was planned on the 400 kV, 400 km line between Kanpur and Ballabhgarh, in the Northern Regional grid of India. This line is a part of one of the transmission corridors which evacuate power from the generation rich south-eastern part of the grid to the northern part. It was envisaged that the power carrying capability of the line could be securely increased from 400 MW to 600 MW. This would improve the grid capability in case of outages of parallel lines.

The power flow can be controlled by scheduling a certain value of the TCSC reactance. Also, the TCSC reactance may be controlled so as to improve angular stability. TCSC (a capacitor in parallel with a Thyristor Controlled Reactor -TCR) is used in conjunction with a fixed series capacitor bank as shown below. The parameters of the TCSC are given below:



System Parameters	Fixed capacitor	TCSC capacitor
Rated Voltage (line-line)	420 kV	420 kV
Nominal Reactive Power	151.6 MVAr	79.87 MVAr
Continuous Effective Impedance	31.5 W	10.4 W
Maximum effective dynamic impedance (because of the TCR)	31.5 W	26 W
Rated Current	1200 A	1600 A
Rated Continuous Voltage (across capacitor)	42.2 kV	16.6 kV
TCR inductance per phase	-	4.4 mH

Exercise:

1) Survey all the DC links existing in India

2) Study the following power flow controllers which have been installed in the world:

   a) TVA's Sullivan Static Synchronous Compensator (STATCOM)

   b) AEP's Inez Unified Power Flow Controller (UPFC)

   c) The Variable Frequency Transformer (VFT) at Langlois (Hydro-Quebec)

Note that (a) and (b) are based on voltage source power electronic controllers while (c) is a rotary device.

An interesting new concept in power flow control and series compensation is "**Distributed Static Series Compensator**". Can you find out more about this ?

Recap

In this course you have learnt the following

- Three real life examples of real/reactive power flow control devices: An HVDC link, a Controlled Series Reactor, and a Thyistor Controlled Series Compensator

We now summarize what we have learnt in this module :

a) Devices like generators can be made to absorb or supply reactive power by controlling their field voltages. Reactive power may be controlled so as to maintain terminal voltage of a generator constant.

b) Transmission Lines absorb or supply reactive power depending on loading. Cables invariably generate reactive power.

c) Tap Changing Transformers, Static VAR generators etc are devices which may be used to control voltages at key points in the system.

d) Power Flow Control and increased utilization of transmission lines is possible by series compensators. Power flow between two regions or across long distances can be achieved using DC links.

e) Power Electronics based variable shunt or series devices allow fast voltage and power flow control.

Congratulations, you have finished Module IV.

Module 5 : Real and Reactive Power Scheduling

Lecture 21 : Introduction

Objectives

In this lecture you will learn the following

- What is scheduling?
- Examples of control variables and constraints
- Objective functions

Control and Scheduling

In previous lectures, we have seen how various quantities like voltage, frequency and power flow are controlled. Speed governors, AGC, voltage and power flow regulators are essentially automatic feedback controllers which work without manual intervention and ensure that these quantities are maintained within acceptable ranges.

Most automatic controllers ensure that these quantities are *regulated, i.e.,* their values are maintained at a certain **set-point.** A set-point is a reference value which is given to these controllers by an operator. There exists some flexibility in deciding the exact value of the set point.

Consider the following examples:

1. A plant operator may increase the value of the voltage setting of a automatic voltage regulator from 1.0 pu to,say, 1.02 pu. This will increase the reactive power output of the generator. Note that while 1.0 pu is the nominal value of terminal voltage, it is not sacrosanct and an operator may change it to a value near-about it as long as it does **not** violate any limit (see capability of a generator).
2. Following a sudden load increase, governors respond and try to restore frequency by increasing generated power. Subsequently AGC ensures that tie line power flow and frequency is brought back to the nominal value. However, a system operator may wish to alter the sharing among various generators (say, to decrease overall cost of generation).
3. If due to line tripping, a system is in the alert or insecure state, then a system operator may reduce power flow through certain lines by changing the power output of different generators.

Manual control actions by a system or plant operator which essentially change the set point of various automatic controllers is termed as "scheduling". Scheduling may be done in order to achieve a certain economic objective or may supplement the existing automatic controller action in ensuring secure system operation.

Control Variables

A system or a plant operator has the flexibility to set several variables (not only the generator power) in order to meet certain objective(s).

A tentative listing of such variables are:

a) Real Power Output of generators which are on-line. This is done by changing the load references of the prime-mover control systems -- also known as "scheduling" or "tertiary control". Primary control (governors) and Secondary control (AGC) settings are not usually changed by a system operator.

b) The decision of whether to engage a generator, at what time and for how long. This is called the unit commitment problem.

c) Reactive Power output of generators (obtained by changing the voltage reference of Automatic Voltage Regulators).

d) Tap settings of transformers.

e) Phase shift setting of phase shifting transformers.

f) Circuit breaker positions for switching in or out Shunt Reactors/Capacitors.

g) Circuit breaker positions for bypass or insertion of Series Capacitors in transmission lines.

h) HVDC power flow.

i) References of Static VAR compensators, and other Power Electronic Controlled devices.

These variables may be set by an operator based on certain objectives. All these variables -- we shall call them *control* variables -- have limits, some of them take on discrete values while others can be varied continuously.

The freedom to set all control variables may not be there in the hands of one system operator, but may be in the hands of many operators who belong to different entities.

Constraints on System Operation

While an operator has the flexibility to set some or all control variables listed in the previous slide, it has to be done with utmost care so as to not violate any constraint. Some constraints are given below:

a) Kirchhoff Voltage and Current Law should be satisfied.

b) Load/Losses and generation should be matched at near-about nominal frequency.

c) Voltages at buses or along a transmission line should be within limits.

d) The transmission lines should be loaded below their thermal limits.

e) Phase angles differences between 2 nodes should not be too large -- the angular stability constraint.

f) The settings of all controllable devices should be within their available range, eg, Generator output should be within the capability curve.

g) Constraints of energy supply systems (water usage in hydel-plants, availability of fuel etc.)

h) Constraints on turbine and boiler operation may restrict the power output range (minimum and maximum) in which they can operate.

In general constraints can be classified as equality constraints which usually can be expressed as algebraic equations as follows :

$$g(x,u) = 0$$

where, u : control variables (which can be set by an operator) x : auxiliary variables like voltage and phase angle at buses

and inequality constraints :

$$h_{\min} \leq h(x,u) \leq h_{\max}$$

An example of an equality constraint equations are the real and reactive power balance equations at every node (or equivalently Kirchhoff's law).

An example of inequality constraint is that voltage magnitude at a load bus should be within minimum and maximum limits.

Objective Functions

There is no unique way that an operator may run a power system. For example, if five generators are to meet a certain load demand, there are many ways that they can be scheduled to share the load.

The main aim of a system operator is to achieve certain objective(s) by appropriately setting the control variables, and satisfying all constraints. Rather than making random adjustments, a system operator will attempt to make optimal settings according to some objective (or "cost") function. A cost function is a mathematical function of the control and auxiliary variables. The control and auxiliary variables are also constrained as discussed in the previous lecture. The basic optimization problem can be therefore formulated by the following equations :

Minimize

$J(x,u)$

subject to

$g(x,u) = 0$

and

$h_{\min} \leq h(x,u) \leq h_{\max}$

Common objective functions are:

a) Schedule generation to minimize generation Cost.

b) Schedule real and reactive power controls to minimize Active Power Losses.

c) Unit commitment (decide whether to engage a generation unit and for what duration based on cost and prime mover/energy supply constraints).

The time frame of optimal flow can vary; the unit commitment problem involves several days or months, while generation scheduling based on cost and active loss minimization can be "online", i.e. of the order of an hour. Strictly speaking, all the objectives listed above are coupled, i.e., active power losses affect generation requirement which in turn affect their schedule and also affects unit commitment. However, these couplings are not very strong and in order to make the problems tractable, these
are decoupled. For example, in the generation cost minimization problem ("economic dispatch"), only generators which are online (decided a priori by the solution of the unit commitment problem) are considered and losses are neglected.

Since generation, transmission and distribution, and loads are not necessarily owned by the same entity, an objective function depends on the entity which is carrying out the optimization. In systems where generation, transmission and distribution is a part of a single entity -a "vertically integrated" utility, a system operator would try to minimize cost of electricity to a consumer while ensuring that the transmission system and generators can recover their costs (including a rate of return on investment). In systems which are not vertically integrated, where there are several "players", the problem becomes more complex.

We shall discuss the mathematical procedure for optimizing objective functions under equality and inequality constraints as well as discuss the nature of objective functions in the following lectures.

Recap

In this lecture you have learnt the following

- What is scheduling as opposed to control ?
- Examples of control variables, constraints and cost functions.

Lecture 22 : Real Power Scheduling

Objectives

In this lecture you will learn the following

- Consider a simple real power scheduling problem
- Objective functions related to the real power scheduling problem

Merit Order Dispatch

The simplest real power scheduling problem can be formulated by assuming the following:

a) The generators which are "committed" are known (the issue of whether to engage a generation unit and for what duration, based on cost and prime mover/energy supply constraints is **not** considered here).

b) The cost of generation (per kW-hr) for various generators is a constant.

c) There are no inequality constraints considered except for the maximum power output of various generators.

d) Power losses are neglected and the only equality constraint to be satisfied is that the *total load is equal to the generation*.

Now, schedule the generation so that the **cost per kW-hr is minimized.**

Let us illustrate the problem and its solution by a simple example:

Suppose there are three generators, say, G1, G2 and G3 with maximum power output of 200 MW, 250 MW and 300 MW respectively.

The cost of energy for the three generators are 2000, 3000 and 2500 Rs/ MW-hr respectively.

The total load of 550 MW is to be shared by the three generators. Find the least cost schedule.

The solution for this problem is easy. We just order the generators as per their cost and utilize the cheapest generator fully. Since the cheapest generator when load fully cannot accomodate all the load demand, the next cheapest generator is utilized to the extent that the load demand is met or upto its maximum output.

Therefore the obvious schedule is:

G1 : 200 MW, G3 : 300 MW and G2 : 50 MW.

The average cost of electricity is C = (200*2000+300*2500+50*3000)/550 = 2364 Rs/MW-hr

Economic Dispatch

In the previous problem we assumed a constant cost of energy in Rs/MW-hr for the generators irrespective of the power output of generators. In actual practice, there is a fixed and variable cost component. As a result, cost is a function of power generated.

A cost function could have the form:

$C = 40*P1 + 5 * P1^2$ Rs/hr,

where P1 is the actual power generated.

Consider two generators with the following cost functions:

$C1 = 50 *P1 + 3 * P1^2$

$C2 = 50 *P2 + 2.5 * P2^2$

Let us assume that the generators (G1 and G2) do not have any maximum limits. Find the least cost schedule.

Total cost is given by : C1+C2 which is to be minimized.

subject to P1+P2 = 550

We shall soon learn a formal (generalized) mathematical procedure to solve this problem. However we shall try to solve the problem in an informal way now.

Since P1+P2=550, P1=550-P2. Substituting this in the cost equation we obtain:

$C=C1+C2 = 50 *P1 + 3 * P1^2 + 50*(550-P1) + 2.5*(550-P1)^2$

This yields, $C=C1+C2= (50 * 550 + 2.5 * (550)^2) -2.5 * 2 * 550 * P1+5.5 * P1^2 = 783750-2750 * P1+5.5 * (P1)^2$

The cost will be minimum when dC/dP1 =0. Therefore, we obtain 2750=11*P1 which results in P1=250 MW. P2 consequently is 300 MW.

Total cost is C = Rs 4,40,000 /hr.

The *price* of electricity for a consumer in Rs per MW-hr is Total Cost / Total Demand = C/550 = Rs 800 /MW-hr.

It is interesting to note here that dC1/dP1 evaluated at P1=250 MW equals Rs 1550 /MW-hr

and dC2/dP2 evaluated at P2 = 300 MW equals Rs 1550 /MW-hr. Both are equal! Is this a coincidence?

Economic Dispatch : Some Issues

In the previous slide we have not considered several major complicating factors in the economic dispatch, viz, limits on the generator power output, power losses etc. However, even if we are blind to these complications for the time being, we can consider some issues about pricing of power.

In a vertically integrated utility wherein a single entity has full control of all aspects of generation, transmission and distribution, a system operator who performs economic dispatch obtains the actual cost functions of various generators (including fixed and variable costs - and allowing for a certain *regulated* rate of return on investment or profit).

Utilising economic dispatch, the operator tries to minimize costs and thereby minimize the price of electricity for a consumer (consumer may be charged a fixed rate based on estimates of this price over a long period).

Another situation is as follows: generation, transmission and distribution are not a part of the same utility. Generation belongs to several entities and is open to competition. In this situation, actual generation cost need not be available to a system operator. The entities who own the generators, submit their "costs" as a function of power generated. These "costs" are not necessarily the true costs (including the *regulated* rate of return on investment). In other words, the generation owners can decide their own profits! However, if the price of generation is not competitive (i.e. it is too costly), then less power will be scheduled from that generator. Therefore, market forces determine the price of electricity. This, of course will work only if total generation is more than the load demand and other constraints on scheduling (like power flow limits on transmission lines) do not apply.

The overall objective is still the same: to deliver power to a consumer at the minimum price while satisfying various constraints (including serving a given load demand). It is assumed here that all loads buy power at the same price. It may be also possible for independent generators and loads to have independent bilateral contracts at a pre-determined price.

Another assumption is that load demand is given and is independent of the final price which is obtained after the optimization. Actually, load demand can be a function of this price. If the price of electricity is too high some consumers are likely to reduce their demand for power (this is also known as *demand elasticity*).

Recap

In this lecture you have learnt the following

- How does one schedule generation so as to obtain minimum cost ?
- Two simple examples relating to economic dispatch.
- Some issues in power pricing

Lecture 23 : Optimization

Objectives

In this lecture you will learn the following

- How does one solve a constrained optimization problem ?
- An illustrative example

Unconstrained and Constrained Optimization

In an *unconstrained* system, the usual approach to minimize the cost function is to set the function derivatives to zero and then solve for the control and auxiliary variables from the set of resulting equations, i.e., if we wish to

Minimize J(u)

then the solution is obtained by setting the partial derivative of J with respect to every control variable u to zero. The number of equations will be equal to the number of control variables.

Suppose we wish to minimize a cost function subject to some equality and inequality constraints. If constraints are specified, then the the above-mentioned procedure will not work because the solution obtained from the procedure may not satisfy the constraints. Therefore an alternative method is required in order to take constraints into account.

Let us first consider the case wherein **only equality constraints** are present:

Minimize

$J(x,u)$

subject to

$g(x,u) = 0$

In such a case the procedure is to form a composite cost function

$C = J(x,u) - l' * g(x,u)$

where l' is a row vector of the variables called the Lagrange Multipliers. The number of multipliers is equal to the number of equality constraints $g(x,u)=0$.

The composite cost function is minimized by treating the Lagrange mulitpliers as additional variables. *This solution is a minimum solution which satisfies the equality constraints.*

We shall now illustrate this using a simple example.

Equality Constrained Optimization

We will use the same example as in the previous lecture, but solve it by using the formal Lagrange Multipliers formulation.

Consider two generators with the following cost functions:

$$C1 = 50 * P1 + 3 * P1^2$$

$$C2 = 50 * P2 + 2.5 * P2^2$$

Let us assume that the generators (G1 and G2) do not have any maximum limits. Find the least cost schedule if a load demand of 550 MW is to be met.

Total cost is given by : C1+C2 which is to be minimized.

subject to P1+P2 - 550 =0

The composite cost function is given by : C = C1+C2 - λ * (P1+P2-550)

Since there is only one constraint equation, λ is a scalar. In this example, P1 and P2 are the control variables and there are no auxiliary variables (to keep things simple!).

Now by differentiating with the reference to the control, auxiliary (there are none in this example) and λ, and equating all derivatives to zero, we obtain :

dC/dP1 = 50 + 6* P1 - λ = 0 , dC/dP2 = 50 + 5* P2 - λ = 0, and dC/dλ = - (P1+P2-550) = 0, or equivalently

$$\begin{bmatrix} 6 & 0 & -1 \\ 0 & 5 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} P1 \\ P2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -50 \\ -50 \\ 550 \end{bmatrix}$$

These are three independent equations in three variables and can be solved by pre-multiplying the column vector on the RHS by the inverse of the matrix on the LHS. Verify that that P1=250 MW, P2=300 MW, and λ = Rs 1550 / MW-hr is the solution.

Also note that since at the optimum solution, dC/dP1 = dC1/dP1 - λ =0 = dC/dP2 = dC2/dP2 - λ. In other words, at the optimum solution, dC1/dP1 = dC2/dP2 = λ

The equality of dC1/dP1 = dC2/dP2 was an observation that we made in the previous lecture too.

It is not difficult to see why this is true at the optimum (least cost) solution. If dC1/dP1 > dC2/dP2 at a certain value of P1 and P2, then it would be possible to reduce total cost by slightly increasing P2, and reducing P1 by the same amount. Therefore these values of P1 and P2 CANNOT be the least cost solution. The same can be argued for values of P1 and P2 such that dC1/dP1 < dC2/dP2.

Only if dC1/dP1 = dC2/dP2 can we state that no improvement in cost will result for (small) changes in P1 and P2.

General Constrained Optimization with Inequality Constraints

What happens variables are constrained by an *inequality* constraint ?

An inequality constraint usually arises when one wants to place some limits on the value of a variable. For example, power output of a generator is limited by its capacity. A simple and *approximate* way of ensuring that limits are not hit (i.e. inequality constraints are not violated) is to augment the composite cost function discussed earlier, so that costs become extremely high if any limit is violated.

Consider the following problem:

Minimize

$$J(x,u)$$

subject to

$$g(x,u) = 0$$

and

$$h_{\min} \leq h(x,u) \leq h_{\max}$$

The last equation can be rewritten (for convenience) as follows:

$$\begin{aligned} h(x,u) &\leq h_{\max} \\ -h(x,u) &\leq -h_{\min} \end{aligned}$$

Therefore all the inequality equations can be re-written as: $q(x,u) \leq q_{\max}$

Now, the procedure for miminizing J subject to the constraints is to form a composite (augmented) cost function as follows:

$$C = J(x,u) - \lambda' * g(x,u) + p(q(x,u) - q_{\max})$$

Let us define $t = q(x,u) - q_{max}$.

The function *p(t)* is chosen such that its value is very small if *t* is less than zero, otherwise it is very large.

A "penalty" function like

$p(t) = e^{\beta t}$ where b >0 is suitable for this purpose. (How does one decide the value of b ?)

General Constrained Optimization with Inequality Constraints

Let us again consider the previous example, but now we will also incorporate inequality constraints.

Consider two generators with the following cost functions:

C1 = 50 *P1 + 3 * P1 *P1

C2 = 50 *P2 + 2.5 * P2 *P2

Let us assume that the generators (G1 and G2) have maximum limits: P1 < 290 MW and P2 <290 MW

Find the least cost schedule if a load demand of 550 MW is to be met.

Clearly, the solution without the inequality constraints is not feasible since P2=300 MW violates the constraint P2 <290 MW. The correct solution can be obtained by following the procedure outlined below (can you guess the answer right away ?)

Total cost is given by : C1+C2 which is to be minimized, subject to P1+P2 - 550 =0 and P1 < 290 MW and P2 <290 MW

The augmented cost function considering equality and inequality constraints is

C = C1+C2 - l * (P1+P2-550) + p(P1 -290) + p(P2- 290)

$p(t) = e^{\beta t}$, where b is chosen to be 3.

The minimum value of C is obtained by calculating the derivative with respect to the variables, P1, P2 and l.

.As a result we obtain the following (nonlinear) equations:

$$\begin{bmatrix} 6 & 0 & -1 \\ 0 & 5 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} P1 \\ P2 \\ \lambda \end{bmatrix} + \begin{bmatrix} 3 * e^{3*(P1-290)} \\ 3 * e^{3*(P2-290)} \\ 0 \end{bmatrix} - \begin{bmatrix} -50 \\ -50 \\ 550 \end{bmatrix} = 0$$

These equations are non-linear and cannot be solved by direct matrix inversion. In such circumstances, one can try to employ numerical techniques like Gauss-Siedel or Newton-Raphson method. Alternatively one can use a "gradient descent method', in which the LHS of the above equation is evaluated at a guess value, say P1 = 250 MW , P2 = 250 MW and l = 1550. Note that the LHS is the "gradient" of the cost function, i.e.,

$$\nabla C_k = \begin{bmatrix} 6 & 0 & -1 \\ 0 & 5 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} P1_k \\ P2_k \\ \lambda_k \end{bmatrix} + \begin{bmatrix} 3 * e^{3*(P1_k-290)} \\ 3 * e^{3*(P2_k-290)} \\ 0 \end{bmatrix} - \begin{bmatrix} -50 \\ -50 \\ 550 \end{bmatrix}$$

Recall from vector calculus that the gradient of a function points towards the direction of maximum change of the function for small changes in the value of the variables. The trick is to change P1, P2 and l in a direction which is negative to the gradient direction, i.e.,

$$\begin{bmatrix} P1_{k+1} \\ P2_{k+1} \\ \lambda_{k+1} \end{bmatrix} = \begin{bmatrix} P1_k \\ P2_k \\ \lambda_k \end{bmatrix} - \alpha \nabla C_k$$

a is chosen to be a small value. One keeps updating the values of P1, P2 and l in this fashion ("descending") till there is very little difference between the values of variables for 2 consecutive iterations. Consider a MATLAB program which implements this (you may copy this program and run it):

```
A=[6 0 -1;0 5 -1;1 1 0];
B=[-50;-50;550];
x=[250;250;1550];
alpha=0.01; beta=3;
for k=1:2000
C=[beta*exp(beta*(x(1)-290)); beta*exp(beta*(x(2)-290));0];
grad=A*x+C-B;
xold=x;
x=x-grad*alpha;
if norm(x-xold,2)<0.01
break
end
end
```

What values of x does it converge to? Do the generator powers remain within the limits specified ? If not why ?

What is the effect of varying a and the value of b ? What if beta is chosen to be a) too low or b) too high ?

Recap

In this lecture you have learnt the following

- How to solve a constrained optimzation problem using Lagrange Multipliers ?
- Handling inequality constraints using penalty function approach.

Lecture 24 : Real and Reactive Power Scheduling

Objectives

In this lecture you will learn the following

- Cost minimization considering network constraints and losses.
- An illustrative example

Optimal Power Flow

Until now, we have considered fairly simple examples pertaining to the economic dispatch problem. In an actual system, there are many variables (other than generated power) that need to be scheduled. These may be set under the direction of a system operator. If the quantity which is to be scheduled belongs to an independent entity, a system operator may "buy" the quantity based on the price at which it is sold.

The formulation of the minimum cost problem, by including the network, reactive power demand, voltage and current constraints is known as an "optimal power flow". Note that optimal power flow takes into account of losses in the network.

Let us consider an optimal power flow problem in which :

a) Control variables (which can be scheduled independently) are the generated real and reactive power

b) Load demand is fixed

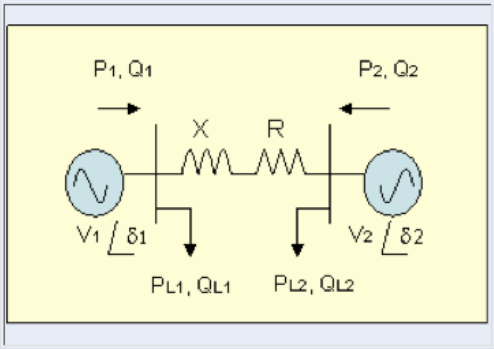
c) Cost is a function of only the real power (this assumes that the service of providing reactive power for maintaining voltages is free!)

d) Real powers (control variables) are constrained by maximum limits.

e) Voltage magnitudes (auxiliary variable) is constrained by maximum and minimum limits.

In actual practice, several additional constraints and control variables are present, but for the sake of simplicity let us consider only the ones mentioned above.

Optimal Power Flow - An example



Consider a 2 generator system which supplies 2 loads at buses 1 and 2. Compute the minimum cost real and reactive power schedule of the generators if the cost of real power (power is in per-unit) is:

$C1 = 50 * P_1 + 300 * P_1^2$

$C2 = 50 * P_2 + 250 * P_2^2$

Reactive power is assumed to be free. The following limits are applicable: Voltages at both buses should be between 0.975 pu and 1.025 pu. Generated real power for each generator should be less than 3 pu. Reactive power capability is assumed to be unlimited.

$R+jX = 0.02 + j0.1$, $P_{L1} + jQ_{L1} = 4.0 + j*2.0$, $P_{L2} + jQ_{L2} = 1.5 + j*0.75$ pu

The problem can be stated formally as follows:

Total cost is given by : C = C1+C2 which is to be minimized.

subject to

$P_1 - P_{L1} - V_1^{2*}\cos f / Z - V_1 * V_2 * \cos(f + d1 - d2) / Z = 0$

$P_2 - P_{L2} - V_2^{2*}\cos f / Z - V_1 * V_2 * \cos(f + d2 - d1) / Z = 0$

and

$V_1 - 1.025 < 0$, $V_2 - 1.025 < 0$,

$0.975 - V_1 < 0$, $0.975 - V_2 < 0$,

$P_1 < 3.0$, $P_2 < 3.0$

where $Z = |R+jX|$, and $f = \arctan(X/R)$

Note that there are implicit equality constraints which are present due to real and reactive power balance equations at the 2 nodes. The auxiliary variables are V_1, V_2 and $d1-d2$. Note that $d1$ and $d2$ cannot be obtained as independently and uniquely since they appear as $(d1-d2)$ in all equations (i.e., if $d1$ and $d2$ satisfy the equations, then $d1+g$ and $d2+g$ also satisfy them, g being an arbitrary choice).

Optimal Power Flow : An example (contd.)

To obtain the solution to the problem, we first form an augmented cost function which factors the equality and inequality constraints as follows:

$C = 50 * P_1 + 300 * P_1^2 + 50 * P_2 + 250 * P_2^2 -$

$l1 * (P_1 - P_{L1} - V_1^{2*}\cos f / Z + V_1 * V_2 * \cos(f + d1 - d2) / Z) -$

$l2 * (P_2 - P_{L2} - V_2^{2*}\cos f / Z + V_1 * V_2 * \cos(f + d2 - d1) / Z) +$

$p(P_1-3.0)+p(P_2-3.0)+p(V_1-1.025)+p(V_2-1.025)+p(0.975-V_1)+p(0.975-V_2)$

where the penalty function has the form : $p(t) = e^{\beta t}$, b being set to 100.

We then set all the partial derivatives to zero and solve the resulting equations to obtain the values of the variables which result in the minimum value of C. The equations to be solved are *non-linear algebraic equations* and can solved by a variant of Gauss Siedel method. (You may try the same using gradient descent method which was discussed in the previous lecture). A matlab program to do can be downloaded here: opf.m . This may be run by typing opf at the MATLAB prompt.

The solution and comments

The final schedule is calculated to be:

P_1 =2.5982 , P_2=2.9413 , Q_1=1.1795 , Q_2 =0.7677

The voltages are V_1= 1.0062, V_2 = 1.0263, and the phase angular difference δ_1-δ_2 = -8 degrees.

The power flow from bus 2 to bus 1 equals 1.4413 pu, and the loss is 0.0395 pu

Comments:

1) It is cheaper to import some power from generator 2 for the local load (4 pu) at the bus 1.

2) However, this implies that there will be some loss due to transmission which will increase the cost.

3) Therefore we have to tradeoff : obtain power from a cheaper remote generator and at the same time ensure losses due to transmission are low.

4) Voltages at the 2 ends can be adjusted so as to reduce losses. Obtaining power at voltages above the nominal results in lower losses (due to lesser current) but then voltages have to be restriced within the limits.

5) Voltage at bus 2 is slightly higher than the specified limits. This can be corrected if the penalty function parameter β is increased; however, this may cause divergence in the numerical method used for solving the optimization (try doing it with the program given to you).

Optimal Power Flow : Security constrained Optimal Power Flow

In a realistic power system, the following **additional** points have to be considered.

a) Reactive power capability is not free and unlimited.

Instead of specifying the reactive power schedule, an operator may prefer to schedule the voltage at the buses at which reactive power is controllable.

b) There may be additional controllable variables (possibly having service costs), like tap settings of tap changing transformers, SVC output, TCSC reactance , HVDC power flow etc. Each controllable variable is likely to have a limit which is dictated by the associated equipment.

c) Current in the transmission line has to be within values as dictated by the thermal limit.

d) **Security Constraints:** Even if we have ensured that we have a least cost real and reactive power schedule for a given newtork and load condition, we would also like to ensure that this schedule is "secure". This means that *if* there is a disturbance when the system is operating as per this schedule, it should be :

1) stable, i.e., it should "settle down" after a credible(realistic) disturbance to a post-fault equilibrium (see Module 2 for a discussion of stability).

2) transmission line voltage and current constraints are not violated in the post-disturbance steady state (which may be different from the pre-disturbance steady state due to loss of a component like transmission line).

Incorporating security constraints rigourously is not an easy task since there can be a large number of realistic disturbances which can take place. Restricting pre-disturbance power flows or phase angular differences across lines is a convenient way to specify Angular Stability constraint.

.An optimal power flow which includes security constraints is called a "*security constrained optimal power flow*".

If an optimal power flow study is carried out *without* considering security constraints, i.e., the constraints imposed by the steady state / dynamic behaviour following a potential contingency, then this may call for changes (or usually some minor adjustments) in the final "schedule" in order to satisfy security constraints. This is known as preventive re-scheduling (we shall see an example of this in the next module).

Recap

In this lecture you have learnt the following

- Optimal Power Flow: Cost minimization, by considering network constraints and losses.
- Illustration of optimal power flow by an example.

We now summarize what we have learnt in this Module

1) A system operator can schedule several quantities (control variables) which ensure that the system operates with minimum cost of energy, while satisfying constraints of equipment and stability.
2) Apart from generated real power there are several other control variables, which may be available for scheduling - at a service cost.
3) In systems where generation is not a regulated monopoly, the price of electricity need not necessarily reflect actual costs, but may be sold at a price determined by load demand and competition.
4) Real and reactive power scheduling problems can be formulated as constrained optimization problems.

Congratulations, you have finished Module V.

Module 6 : Preventive, Emergency and Restorative Control

Lecture 25 : Introduction

Objectives

In this lecture you will learn the following

- Nature of Control Actions in a Power System
- Brief recap on operating states and control actions
- The role of the load dispatch center

Nature of Control Actions in a Power System

What we have learnt so far can be broadly classified as follows:

1. Frequency, voltage and power flow control
2. Real and Reactive power scheduling

Essentially, these are "routine" control actions of a manual controller (i.e., a system/plant operator) or an automatic controller (like a generator terminal voltage regulator).

These actions ensure a good quality supply (i.e., near-constant voltage and frequency) at a low cost.
However, an important class of control actions are appropriate when a system is not in a "normal" state. To understand this let us revisit a previous lecture on various operating states of a power system.

Operating states and nature of control actions (revisited)

The state (or condition) of a power system can be judged from the answers to the following questions:

- Is the demanded load being met ?(i.e., is there a real and reactive power balance?)
- Are all equipment within their current and voltage limits?
- Can the system withstand stresses due a possible contingency (leading to a loss of equipment)?

Definition of states and control actions*

System operation in steady state is governed by equations which express:

(1) Real and Reactive power balance at each node (**Equality Constraints**)

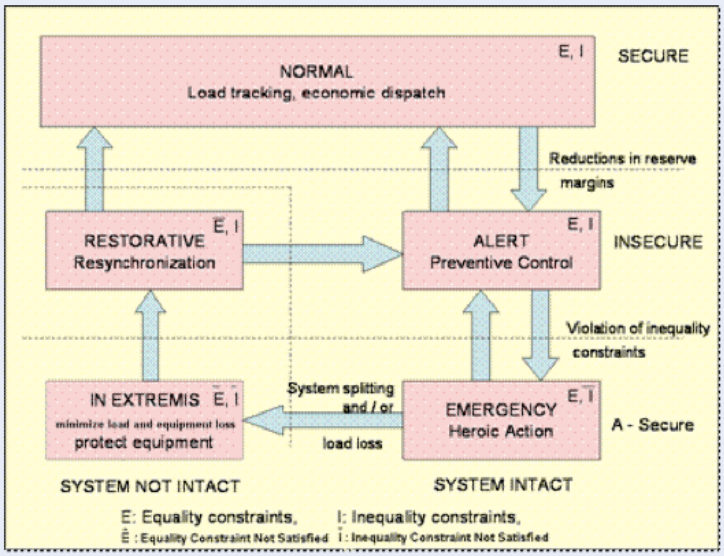
(2) Limitations of physical equipment, such as currents and voltages must not exceed maximum limits (**Inequality Constraints**)

Normal (Secure) State	Here all equality (E) and inequality constraints (I) are satisfied. In this state, generation is adequate to supply the existing load demand and no equipment is overloaded. Also in this state, reserve margins (for transmission as well as generation) are sufficient to provide an adequate level of security with respect to the stresses to which the system may be subjected. The latter maybe treated as the satisfaction of security constraints.
Alert (Insecure) State	The difference between this and the previous state is that in this state, the security level is below some threshold of adequacy. This implies that there is a danger of violating some of the inequality (I) constraints when subjected to disturbances (stresses). It can also be said that security constraints are not met. Preventive control enables the transition from an alert state to a secure state.
Emergency State	Due to a severe disturbance, the system can enter emergency state. Here (I) constraints are violated. The system, would still be intact., and emergency control action (heroic measures) could be initiated to restore the system to an alert state. If these measures are not taken in time or are ineffective, and if the initiating disturbance or a subsequent one is severe enough to overstress the system, the system will breakdown and reach "In Extremis" state.

* adapted from Power System Dynamics - Stability and Control, K.R. Padiyar, Interline Publishing Pvt. Ltd., 1996

Definition of states and control actions (contd...)

In Extremis State	Here, both (E) and (I) constraints are violated. the violation of equality constraints implies that parts of the system load are lost. Emergency control action should be directed at avoiding total collapse.
Restorative State	This is a transitional state in which (I) constraints are met from emergency control actions taken but the (E) constraints are yet to be satisfied. From this state, the system can transmit to either the alert or the normal state depending on the circumstances.



In further developments in defining the system states, the power system emergency is defined as due to either a

a) *viability crisis*, resulting from an imbalance between generations, load and transmission whether local or system-wise, or

b) *stability crisis*, resulting from energy accumulated at sufficient level in swings of the system to distrupt its integrity

'In Extremis' state corresponds to a system failure characterized by the loss of system integrity involving uncontrolled islandings (fragmentation) of the system and/or uncontrolled loss of large blocks of load.
It is obvious that the objective of the emergency control action should be to avoid transition from emergency state to a failure state (In Extremis).

Load Dispatch Centre

A load dispatch center (or more appropriately, an *Energy Management Centre*) enables a system operator and supporting engineers to monitor a power system in real time and includes facilities to capture the current operating state of the system and instruct a generating plant or any other controllable system components so that a system operates with good quality and security.

The major components of a load dispatch centre are:

a) Engineers for carrying out scheduling and monitoring

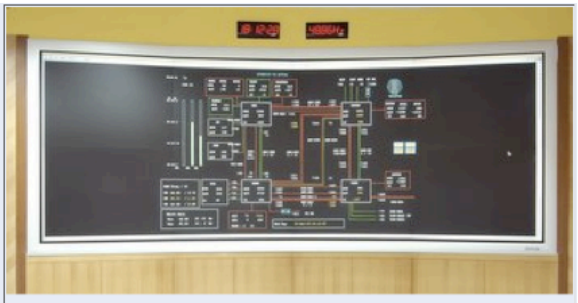
b) Software programs to carry out monitoring and scheduling functions

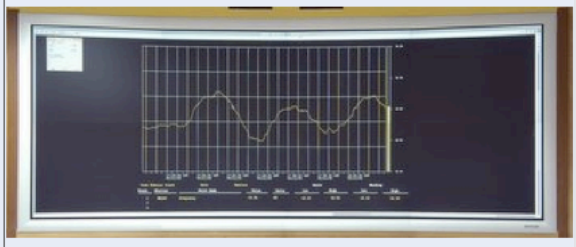
c) Displays for adequate visualization

In a previous lecture in Module 1, we saw that there is a hierarchy of controls in a power system.

A Central Load Dispatch centre is at the top of the hierarchy and oversees the operation of the entire grid using a "Supervisory Control and Data Acquisition" (SCADA) system.

A SCADA system obtains data from various levels in the control hierarchy and displays it in a meaningful way (like a one line mimic diagram on a large LCD screen).





(pictures courtesy from WRLDC, Mumbai)

The data is used to feed an Alarm Management systems which indicates to the monitoring engineers, an abnormality in the system.

The software tools in an Load Dispatch Centre process data obtained from field measurements to identify the topology of the network in real time (using the "on" or "off" data of circuit breakers and isolators). Along with voltage and current measurements, this data is used to estimate the "state" of the system.
Since the number of components need to be monitored are very large, a sophisticated digital processing of the data is required. If data is simply displayed, then an operator has to use his past experience to co-relate the displayed data and the system state and even take remedial actions.

However, if data is used for automatic estimation of the state and to compute the response of a system to a potential disturbance (contingency analysis), then reliability of the measured data needs to be ensured.

This process is known as "Power System State Estimation" which we shall consider in the next lecture.

Recap

In this lecture you have learnt the following

- Preventive Control, Restorative Control and Emergency Control.
- The role of the Load Dispatch Center and what it comprises of.

Lecture 26 : Power System State Estimation

Objectives

In this lecture you will learn the following

- The fundamentals of Power System State Estimation
- Handling errors and bad data in static state estimation
- Dynamic Measurements

Power System State Estimation

Power System State Estimation is a process whereby telemetered data from network measuring points to a central computer, can be formed into a set of reliable data for control and recording purposes.

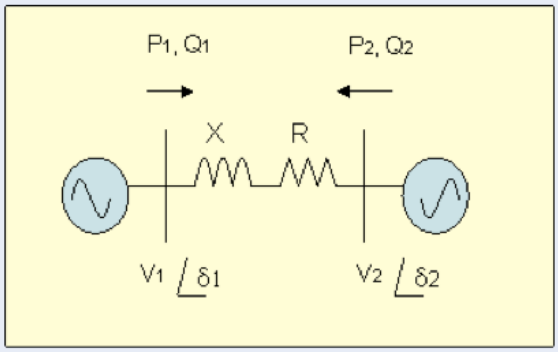
A *static* state estimate is obtained from measurements taken within a time interval of about 0-5 s. This is the commonly used state estimator. Obviously, a state estimator of this type essentially gives a steady state snapshot of the system.

A *dynamic* state estimate is obtained from measurements in a relatively shorter time (say 0.01 s). Moreover, all such measurements are synchronised or "time stamped" using a common clock and communicated from geographically distant locations to a load dispatch centre. These measurements could be used for advanced control schemes which we shall see later.

The main concern in state estimation is the reliability of the measured data. Usually to minimize the errors, the data is crosschecked using more measurements than necessary (redundant measurements). This is done in a systematic fashion as discussed in the following slides.

Static State Estimation Procedure

We will try to understand the procedure in an informal fashion. Consider the simple system shown below:



In the above system we have obtained the following measurements: $P_1, Q_1, P_2, Q_2, V_1, V_2$: i.e, the real and reactive flows on both ends of the line and the voltage magnitudes at the two ends. If d1-d2 is known then, the system is fully "estimated" (i.e., nothing more is left to be known). We assume here that the parameters of the line are known fairly accurately.

If all measurements are accurate, then it should be possible to obtain d1-d2 quite easily by the following formula:

$P_1 = V_1 * V_1 * \cos f / Z - V_1 * V_2 * \cos(f + d1\text{-}d2) / Z$

where, $Z = |R+jX|$, and $f = \arctan(X/R)$

However if there is a measurement error in P_1, V_1 or V_2, then d1-d2 will be inaccurate. So it is best to cross-check the results with other formulae:

$Q_1 = V_1 * V_1 * \sin f / Z - V_1 * V_2 * \sin(f + d1\text{-}d2) / Z$

However if there is a measurement error in P_1, V_1 or V_2, then d1-d2 will be inaccurate. So it is best to cross-check the results with other formulae:

$Q_1 = V_1 * V_1 * \sin f / Z - V_1 * V_2 * \sin(f + d1 - d2) / Z$

$P_2 = V_2 * V_2 * \cos f / Z - V_1 * V_2 * \cos(f + d2 - d1) / Z$

$Q_2 = V_2 * V_2 * \sin f / Z - V_1 * V_2 * \sin(f + d2 - d1) / Z$

If R+jX= 0.01 + j0.1, then check that the following set of measurements will give consistent values of d1-d2.

$P_1 = 5.1$; $P_2 = -4.82$; $Q_1 = 0.83$; $Q_2 = 1.82$; $V_1 = 1.0$; $V_2 = 1.0$;

Verify that d1-d2 = 30

If d1-d2 obtained using different expressions are inconsistent, it means that error exists in one or more of the measurements. Thus, using redundant measurements Q_1, P_2 and Q_2 we can check for presence of errors (which will invariably exist).

For example, one or more of the following set of measurements have some error.

$P_1 = 4.8$; $Q_1 = 0.789$; $V_1 = 1.0$; $V_2 = 1.0$;

d1-d2 = 28.19 when substituted in P1 but -17.96 when used in Q1.

Now the question arises: how do we get a correct estimate of d1-d2 ?

Handling Errors and Bad Data in Static State Estimation

One possibility is to take the average value of d1-d2 obtained from formulae for P1 and P2. This way, one does not place too much faith on one measurement.

Alternatively one could try to obtain a value of d1-d2 such that plugging that value will minimize the following error:

$$J = k1 * (P_1 - V_1 * V_1 * \cos f / Z + V_1 * V_2 * \cos(f + d1 - d2) / Z)^2 + k2 * (Q_1 - V_1 * V_1 * \sin f / Z + V_1 * V_2 * \sin(f + d1 - d2) / Z)^2$$

$$+ k3 * (P_2 - V_2 * V_2 * \cos f / Z + V_1 * V_2 * \cos(f + d2 - d1) / Z)^2 + k4 * (Q_2 - V_2 * V_2 * \sin f / Z + V_1 * V_2 * \sin(f + d2 - d1) / Z)^2$$

where, k1,k2,k3 and k4 are positive weights. Note that J is always greater than or equal to zero. Minimization of J, results in the "least square" deviation from the measured values.The main problem, however, is to chose the weights k1, k2, k3 and k4 appropriately. For similar quantities (power, reactive power) one can choose weights to be equal. Minimizing J requires us to use an optimisation procedure as outlined in the previous module.

Both the approaches (1. averaging and 2. minimizing J) are likely to work well if errors are small in magnitude. If there exist gross errors in the data (which occur, say, due to failure of communication), then the above approaches may fail.

For example, if the measured values are P1 = 5.1; P2 =0 ; Q1 = 0.83; Q2 = 1.82; V1 = 1.0 ; V2 = 1.0; then it is obvious that some major flaw exists in the data.

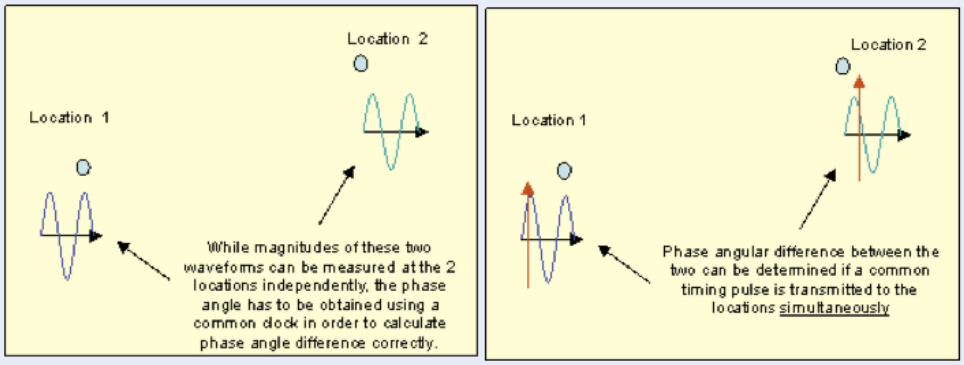
Can you find a reasonable way to identify grossly erroneous measurements ("bad data detection") ?

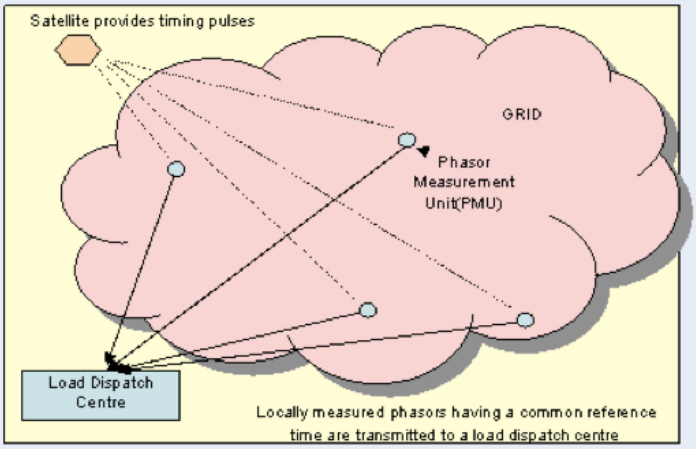
Question: Can d1 and d2 be estimated individually ? Why?

Dynamic Measurements : Wide Area Measurements

Dynamic state estimation allows data acquisition at a faster rate (say several samples within a cycle of 20 ms). This data can be obtained from various locations which may be geographically far apart. Since data is measured and transmitted at a fast rate, it is possible to capture changes as they evolve in time e.g., transients like angular oscillations (see Module 2).

These measurements are also *synchronised*. Synchronisation is made possible by common timing signals provided to the various measuring units at different locations by a satellite. Therefore the voltage/current phasors at different locations can be measured with respect to a common reference and communicated to a load dispatch centre, facilitating the direct calculation of angular differences. In *Static State Estimation,* measurements are usually not synchronised and phase angular differences are obtained not by direct measurement, but estimated from real and reactive power measurements.





Use of these phasors for line and equipment protection is conceivable. Moreover, these measurements may be used to detect or predict angular instability when a system is actually undergoing a transient. This can allow for quick control actions to avoid complete failure of the grid (blackout). At present, these systems are not widely deployed, but in future they are likely to find greater use in real time control and protection of power systems.

Recap

In this lecture you have learnt the following

- Power System State Estimation is a process whereby data from network measuring points to a central computer, can be formed into a set of reliable data for control and recording purposes.
- Dynamic state estimation allows time synchronised data acquisition at a faster rate.

Lecture 27 : Normal and Alert State in a Power System

Objectives

In this lecture you will learn the following

- Different states in a power system
- Schematic of Security Assesment Procedure
- Preventive Re-scheduling of generation

A Power System in the Normal State

Once a system operator has the *static* estimate of all the system variables (voltage, current phase angular differences), he may wish to check whether the state can be characterised as a normal, alert, or emergency state.

While dynamic state information may also available, a system operator may not be able to directly utilize it since the time frame to do so may be limited (for example loss of synchronism may take place within seconds and even if an operator sees it happening, he may not be able to take corrective action). Therefore dynamic measurements can be made use of mainly by *automatic* control or protection strategies.

For the time being we restrict our discussion to static state estimation.

If all the equipment in the system are within their respective limits, then a system could be in the normal or alert state. If a system can withstand *potential* contingencies (like a fault followed by line tripping or a generator trip) without equipment limits being violated or without losing stability, then we say that the system is in a normal or "secure state". A network configuration or loading state which can withstand an element outage without loss off supply to any load is called "n-1" secure. Otherwise we classify the system as being "insecure", i.e., in the alert state.

By a potential contingency we do not mean that the contingency has occured, but has a finite chance of occuring. The classification of secure and insecure is done by simulating (mimicking) contingencies on a computer.

Normal and Alert state

To distinguish between a normal state and an alert state, a system operator carries out the following studies using the network configuration, load and generation values obtained from a static state estimation procedure:

a) Static Security analysis :This involves checking for equipment limit violations, *if* one of the elements of the network/load /generation configuration existing at that point of time were to be tripped due to some contingency. Note that this element is not actually tripped by an operator, but only simulated using a computer program (essentially a load-flow study which computes the steady state power flows in transmission lines, generator real and reactive power output, and voltages at various nodes for such a tripping).

b) Dynamic Security analysis : This involves checking the stability of the system, *if* one of the elements of the network/load /generation configuration existing at that point of time were to be tripped due to some contingency. The exact nature of the contigency can impact the transient behaviour. For example, the contingency could be due to a single phase to ground fault which results in protective action (circuit breakers disconnecting the faulted element) within, say, 0.1s. Note again, that this element is not actually tripped by an operator, but only simulated using a computer transient analysis program (which essentially does a numerical integration of the differential equations which describe the system). A computer program which checks for angular stability requires a significantly large amount of computation time. Therefore, it is not implemented in most load dispatch centres at present.

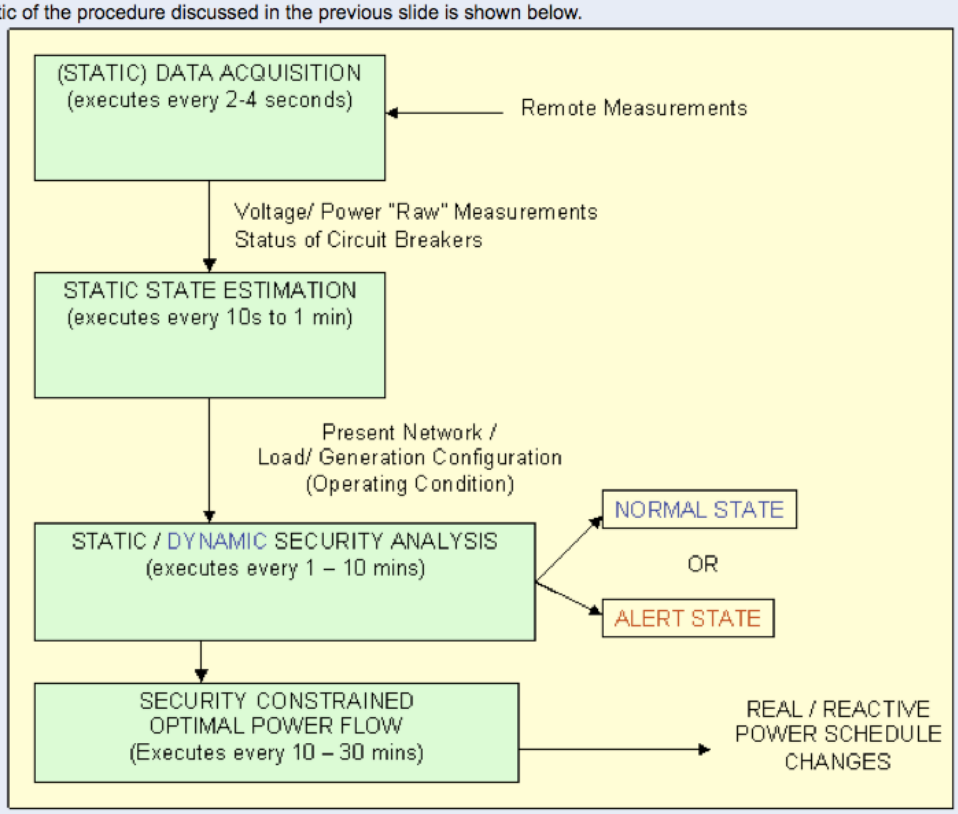
It is important to carefully choose the element whose outage is to be simulated since the number of elements in a power system are too numerous for all of them to be considered one by one. Usually a set of critical elements are chosen by some rough screening based on an operator's experience and the security analyses are carried out for the outage of these elements.

If the security analysis shows that the system is secure, it is classified as a normal state. If the state is normal, then a system operator may wish to do some minor changes in real and reactive scheduling (from an economic perspective), if such flexibility exists. However any such change should not bring the system out of the secure state.

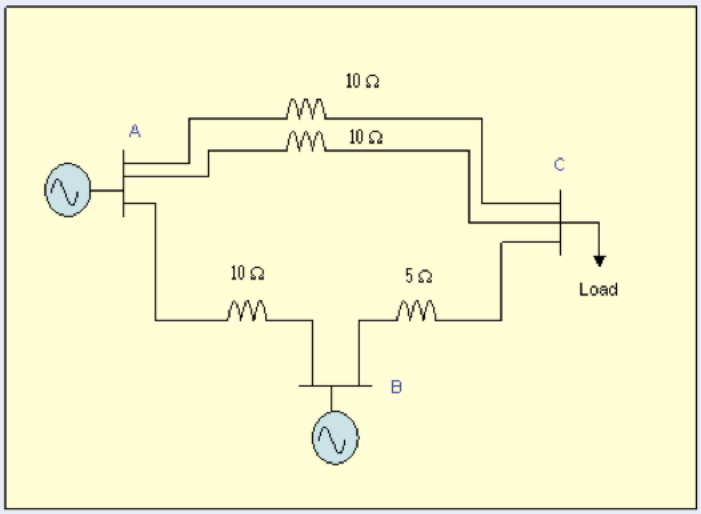
If the system is not secure (alert), then the operator *has to* try to steer it into the secure state by real or reactive power re-scheduling (**Preventive Control**). However, note that this re-scheduling is done to improve security and may result in higher cost if cheaper generators are asked to "back down" their generated power while costlier ones are ramped up. Therefore, even if preventive control is to be done, it should be done in a way which will minimize any cost increase while simulateneously ensuring security.

This is done using a *security constrained* optimal power flow program (discussed in the previous module).

A schematic of the procedure discussed in the previous slide is shown below.



An Example



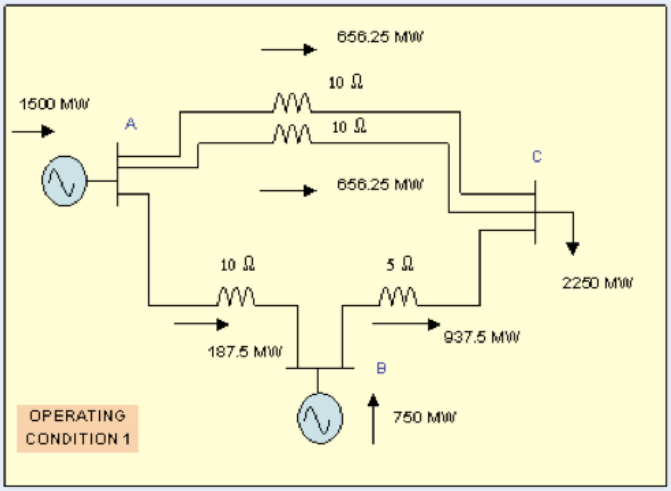
Two Generators supply a load at bus 'C' via transmission lines. It is assumed for simplicity that voltages at all buses are equal to the nominal value (1.0 pu). Also, we assume that $\sin(\delta_{diff}) = \delta_{diff}$ and $\cos(\delta_{diff}) = 1$, where δ_{diff} is the phase angle difference between the voltages at any 2 buses. This simplifies the circuit solution (load flow!) considerably.

Moreover, under these assumptions, power flow is directly proportional to the line current magnitude.

We now attempt to assess the static security of the given system.

We assume that the thermal limits of all the lines are equal and they dictate that the power flow should not exceed 1500 MW. There are other limits due to voltage & stability. However, we restrict our discussion to thermal limits only.

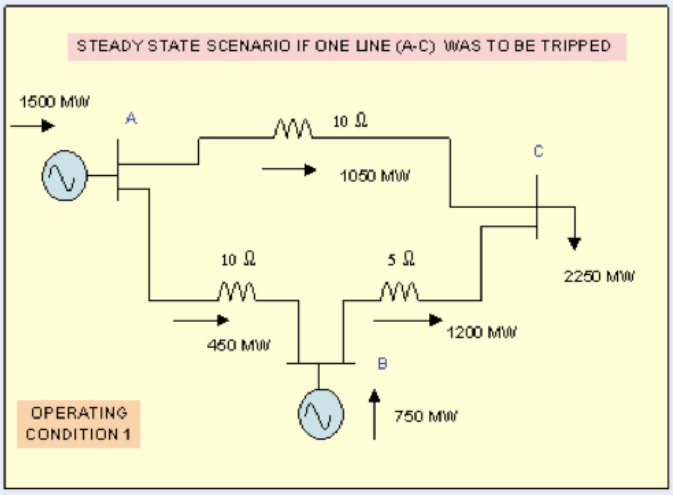
An Example : Case I



Here the generator at A generates 1500 MW and the one at B generates 750 MW.

Under this operating condition, the flows for all lines are below the thermal limits (1500 MW for every line).

We are interested to know what happens if one of the lines trips.

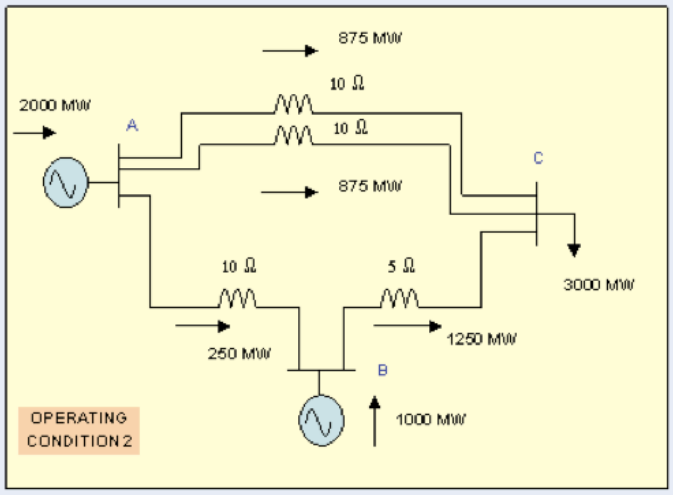


The steady state flows subsequent to the loss of one line between A & C are shown in the figure on the left.

One can verify the loss of any one line will not cause any of the remaining lines to overload.

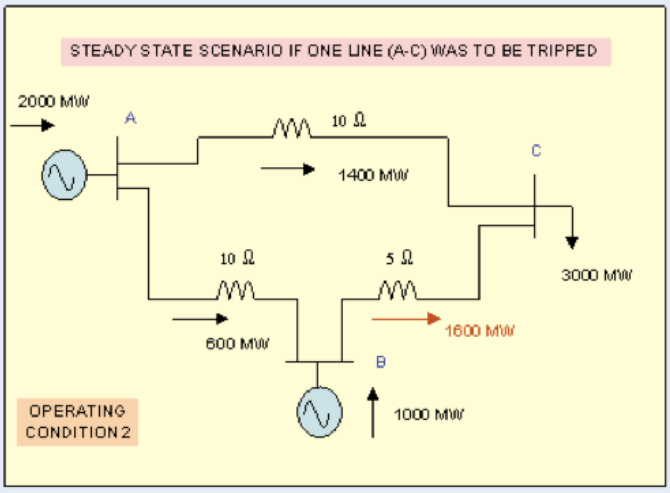
Thus the operating condition is said to be **steady state secure** (or normal) for a line outage contingency.

An Example : Case II



In this operating condition, the generator at A generates 2000 MW and the one at B generates 1000 MW.

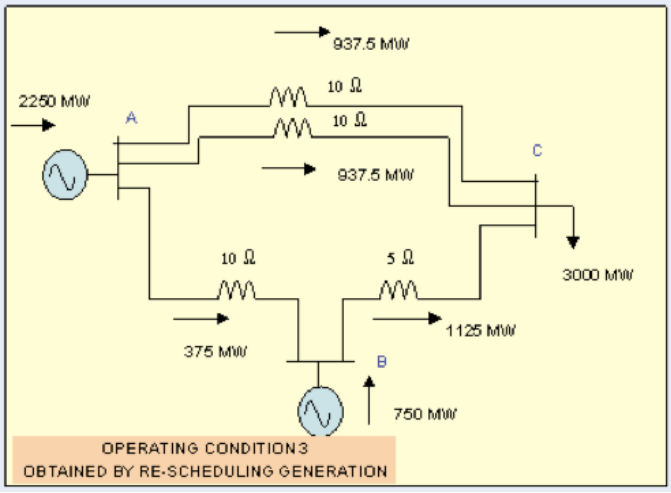
Under this operating condition, the flows for all lines are below the thermal limits (1500 MW for every line).



We are interested to know what happens if one of the lines trips.

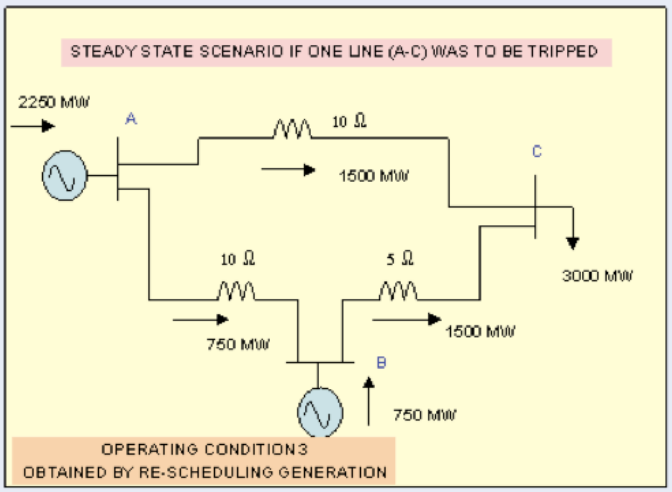
One can verify the loss of any one line will cause line B-C to overload.

Preventive Re-scheduling of generation



It is clear that operating condition 2 is not a secure operating condition (i.e. the system is in 'alert state')

Therefore in order to bring the system back to normal state, a system operator has to re-schedule the generation.



An adjustment of 250 MW between the generators can ensure that the system is secure, as is evident from the power flows for the contingency of line A-C outage.

Note, however, re-scheduling may increase the cost, if generation at A is costly.

This is the price one has to pay for improved security.

A concluding note: System security cannot be assessed by only considering post-contingency *steady state* power flows (as is done in the example presented). A system could be unstable (**angular and voltage instability were discussed in module 2**) for a disturbance even if a post - disturbance *steady state* exists and power flows and voltages for that steady state are within equipment limits. If a system is unstable, it will not settle down to that steady state.

The assessment of dynamic security (stability) is a more complex task as it requires numerical integration of the system dynamic equations (e.g. swing equations of all generators). This is a computationally intensive and many probable contingencies have to be considered. Direct numerical integration of differential equations can be avoided if one uses criteria like "equal area criterion" to adjudge system angular stability. However, equal area criterion cannot be extended in a straightforward manner for multi-machine systems with detailed models of all system components. Therefore quick assessment of dynamic stability is still a challenge to system engineers.

Recap

In this lecture you have learnt the following

- Classification of system into normal or alert state
- An example to illustrate preventive control

Lecture 28 : Emergency Control

Objectives

In this lecture you will learn the following

- Transition from an alert state to an emergency state
- Emergency Control
- Angular Instability

Transition from an alert state to an emergency state

If a system operator infers from the operating data that a system is in an alert state, then he takes preventive control actions to bring the system back to a normal state. However, it is possible that the system operator is unable to act in time before a contingency actually occurs. A grid may even operate insecurely (in an alert state) due to a high cost of preventive control or due to inadequate reserve margins. However this situation is undesirable since it may lead to blackouts (if emergency control actions fail) which can cause great economic loss. The classification of a system state as a normal or alert state is based on simulating some disturbances. Often, even though the system has been classified as being in a normal state, *several improbable* disturbances, which would not have been analyzed for doing this classification, take place.

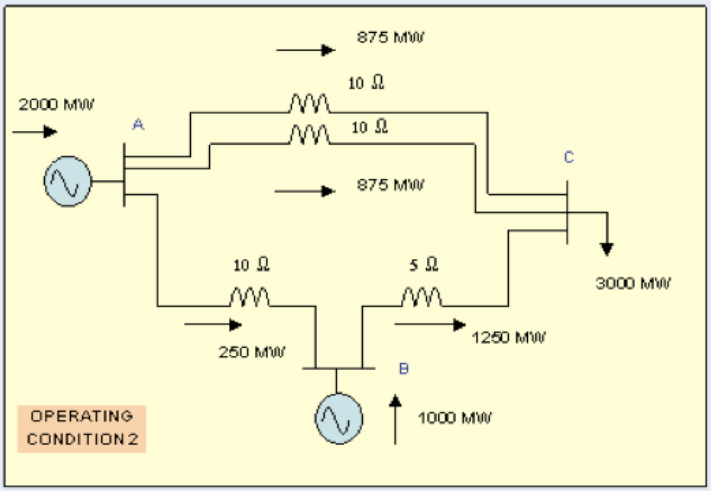
Therefore the system can transit from a perceived alert state to an emergency state if no preventive controls are exercised and a contingency occurs, or may directly transit to an emergency state from a perceived normal state if an unanticipated sequence of several contingencies occur. If the system does go into an emergency state some equipment limits are exceeded which may cause further tripping of equipment, thereby worsening the situation and may cause a complete blackout.Emergency control actions (manual or automatic) are required to retrieve the situation. If there is a thermal overload of an equipment then there is some time to act and quick "heroic action" from a system operator would be needed. However in most cases one has to rely on automatic controls to quickly respond to such a situation.

Some emergency control actions are :

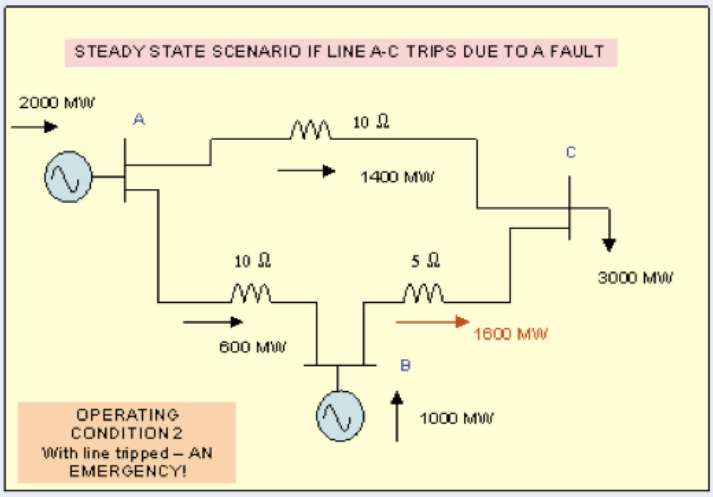
- Generator / Load tripping or fast reduction of generated or load power.
- Control of voltage and power flow control devices (discussed in Module 4)

Transition from Alert to Emergency State

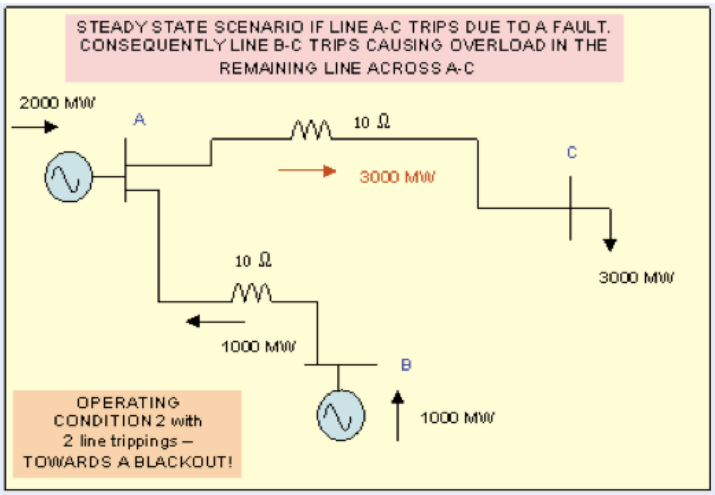
Let us consider a power system in an alert state as discussed in the previous lecture and which is shown below.



We have seen in the previous lecture that if the line A-C is tripped then the line B-C becomes thermally overloaded - an emergency condition.



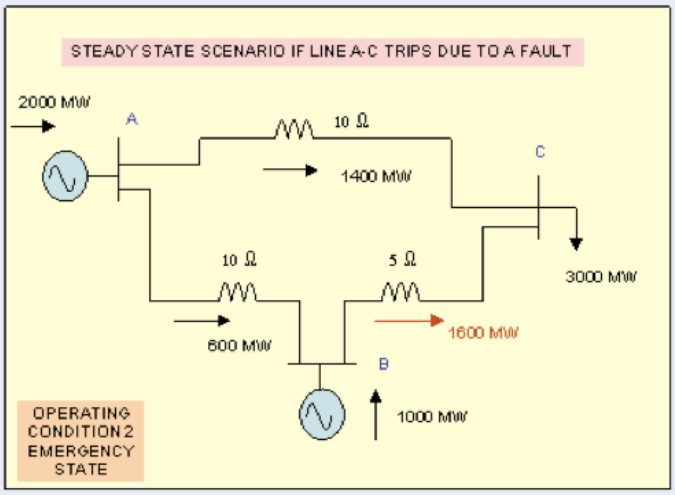
The thermal overload on line B-C will cause it to trip too! This will cause the remaining line between A-C to trip too causing a complete loss of load serving cabability (a **black out**).



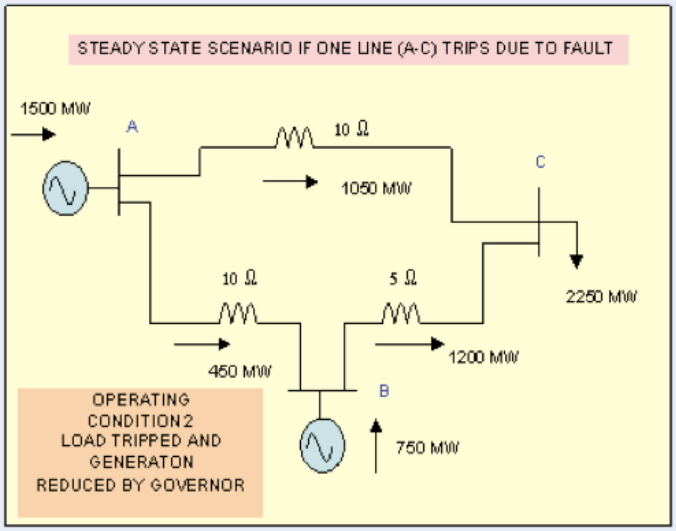
Emergency Control

We saw in the previous slide that a system in an alert state may cascade into an emergency and subsequently into a total blackout if no control actions are taken.. Emergency control measures can try to arrest this. Since most equipment can withstand a short-time thermal overload, there is a small window of time in which some manual emergency measures can be executed. For other emergency situations (like instability), time may be too short and predesigned automatic emergency measures are necessary. One may consider the following alternatives:

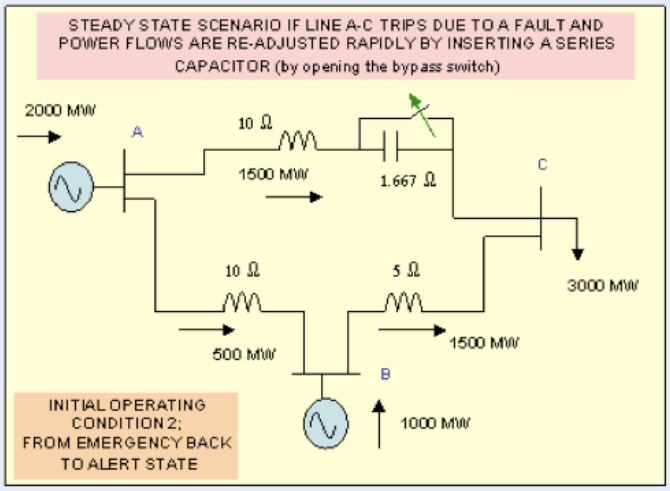
a) Control of generation
b) Tripping of generation or load
c) Re-routing of power flows



We consider the option of generation/load tripping. Consider the system in the alert state as shown on the left.



If now, the load is reduced from 3000 MW to 2250 MW, then the line flows will be as follows. Note that as soon as the load is tripped, the frequency will rise due to load-generation imbalance. The governors will therefore come into play and reduce the power output of generators by an amount depending on the droop settings. Subsequently, generation may be adjusted by AGC or manually. In the figure given here, the final generation is reduced proportionally.



Another alternative is to bring to the circuit devices like a TCSC (Thyristor Controlled Series Compensator) or a switched series capacitor which can rapidly adjust the line impedance (as shown in the figure). This will re-route power from the "congested" path B-C to the path A-C.

Other Emergencies

For simplicity, we have restricted our discussion of alert and emergency states arising from *line thermal overload.* However, it should be recognised that many disturbances may lead to other equipment limits being violated.

For example, a sudden load tripping may cause overvoltages in long EHV lines especially if the transmission line loading goes significantly below the Surge Impedance Loading. This may cause line tripping on overvoltages. Emergency control can be in the form of insertion of shunt reactive power absorbing devices (see Module 4 for the various voltage control devices in a system).

Another class of alert and emergency situations which we have not considered so far are due to stability problems. These are as follows:

1) A sudden loss of generation or load due to some fault. This may cause a significant **variation in the system frequency** if the load-generation imbalance is significant. Large frequency deviation may cause tripping of generators (to avoid turbine damage) and may eventually lead to a complete blackout. A small imbalance can be corrected by governor action if there is some reserve margin of generation left. Emergency Control may entail load tripping (also called Under-frequency Load Tripping) if generation is less than load. If load is less than the generation, then generator(s) may be tripped.

2) Large disturbances may cause **Angular Instability** (see Module 2). A synchronous grid cannot be operated in an integrated fashion if the generators fall "out of step". As discussed in module 1 and 2, if all generators are not in synchronism and are still connected to each other, then large variations in voltage and power result which can damage equipment and render inter-connected operation unviable.

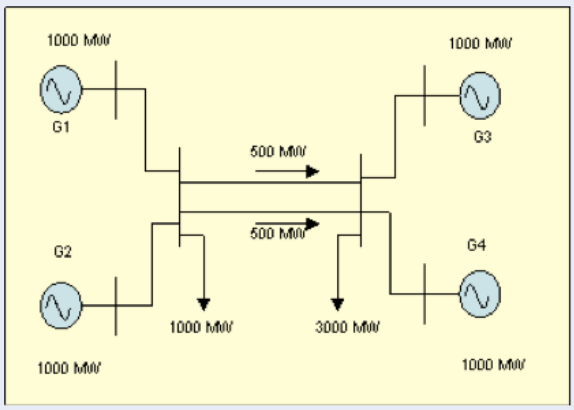
Fast emergency control (i.e, insertion of series capacitors, boosting the excitation of generators) can try to prevent angular instability, but if angular separation becomes too large and keeps increasing, the system has to be split into islands such that generators within an island are in synchronism. However, the emergency is not over - within an island load-generation has to be balanced to avoid large frequency variation. The splitting into islands may be uncontrolled (due to action of protective systems like distance relays) and sometimes very few, if any, islands survive, causing a blackout.

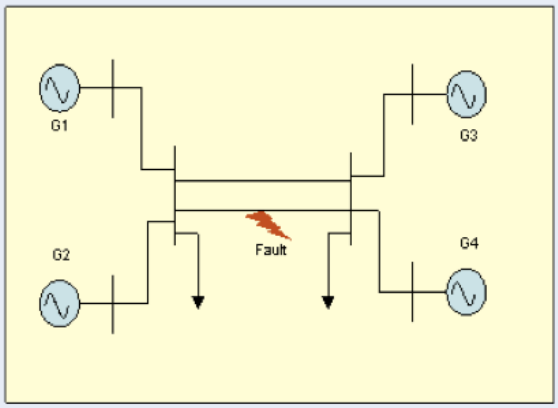
Emergency control measures under loss of synchronism conditions includes activation of a *pre-designed* islanding scheme (controlled system separation), and measures like under-frequency load-shedding within islands with low generation.

3) Weakening of transmission system along with heavy reactive power demand and low reactive power generation margin may cause **voltage instability** (Module 2). Emergency Control in the form of Under-voltage tripping of loads, disabling of tap changers in transformers (see **voltage instability example** in Module 2) etc. may often be able to retrieve the situation.

Angular Instability and Emergency Control

Let us now consider a situation wherein an emergency is caused by loss of synchronism between generators. Consider a four generator system shown (below/right/left). We shall assume that the loads at the 2 buses shown in the figure are not voltage dependent and that losses in the system are neglible. Suppose that a fault occurs (e.g., a short circuit between phase to ground) on one line which is carrying 500 MW) This is detected by relays at both ends of the line and they send a trip signal to the circuit breakers which disconnect this line. This is usually done in a very short time (about 100 ms). The loss of this line causes power to get diverted to the parallel line.





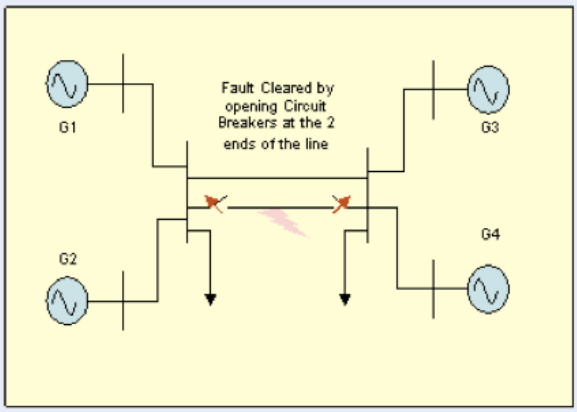


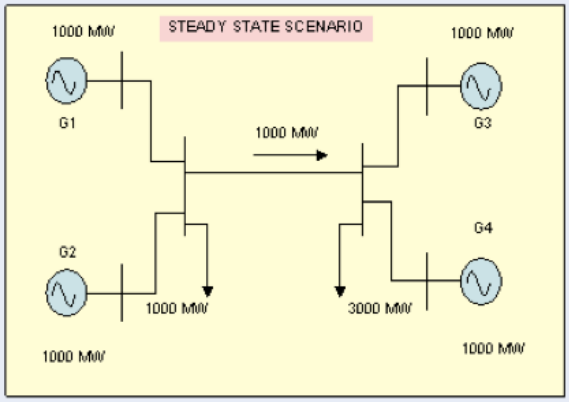
Photo taken from "OPERATIONAL EXPERIENCE OF TCSC ON 400KV RAIPUR–ROURKELA D/C LINE " by A. K. Datta et al, International Seminar and Tutorial on Power Transmission Research and Challenges, CPRI, Bangalore, 21-22nd Dec, 2005.

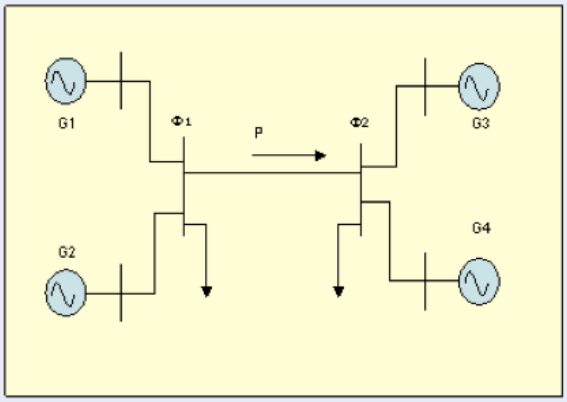
Angular Instability and Emergency Control : Transient Phenomena

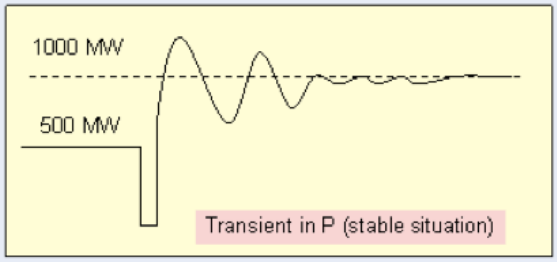
While the steady state scenario following loss of line does not cause violation of equipment limits, the question arises: Will the system reach a steady state ?

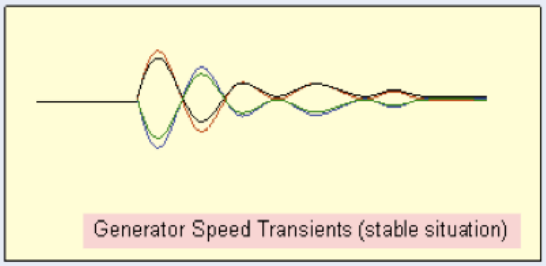
The worrisome aspect of this disturbance is that the transient behaviour of the relative generator speeds after the disturbance may not be stable. During the fault, generator speeds deviate due to a sudden change in the electrical power due to the fault. After the line is tripped, the generator speeds do not directly go into steady state because of the deviation caused due to a fault. However, if the relative angular differences between the machines are not too large, then the electrical torques try to pull all the generator speeds together in synchronism, i.e., the system is angular stable (for a detailed discussion on angular stability, refer to Module 2).

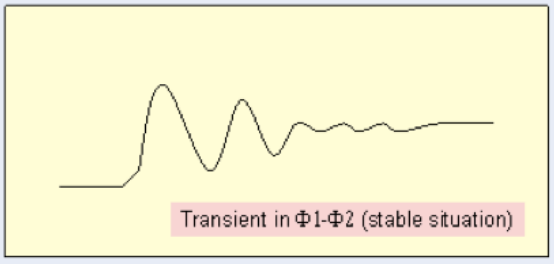
The typical waveforms of generator speeds, power flow (P) and the phase angular difference between the bus voltages at both ends of the line shown below. The power oscillates and eventually settles at a constant value after some time (damper windings in generators can contribute to damping these oscillations). Also, all generator speeds reach the same value.







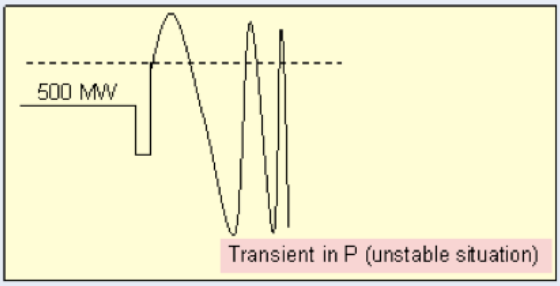


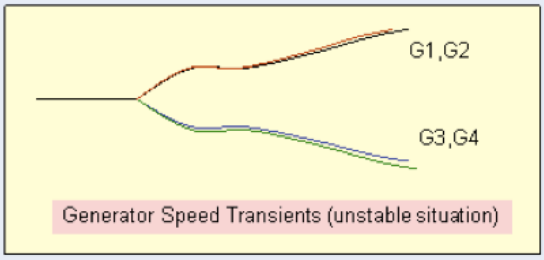


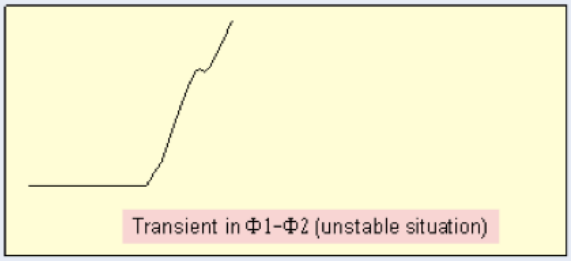
where f1-f2 = bus voltage angular difference

Angular Instability and Emergency Control : System Separation

We saw the typical waveforms when a system is angular stable after a large disturbance like a fault. If relative speed and angular deviations are excessive then the restorative torque may become negative. Under this situation, the generators are unable to pull back into synchronism (angular instability). While the generators are all connected to one another, the speeds do not settle down to a common value, as shown below. This causes large changes in power and voltages in the interconnecting lines which cannot be tolerated for long. It becomes necessary to split (island) the system such that generators with large differences in speeds are not connected to each other.

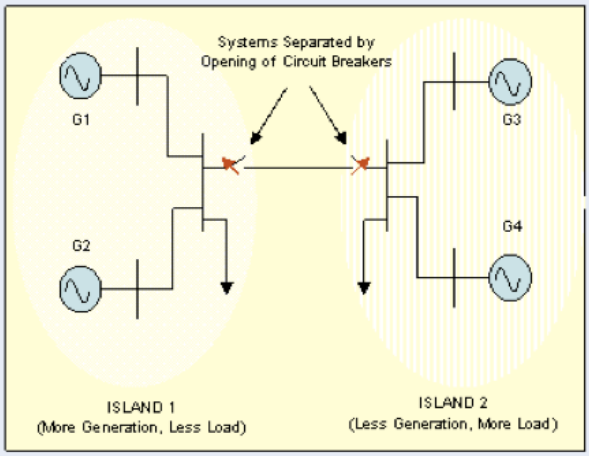






Usually, distance relays mistake these large power and voltage swings for a fault and cause interconnecting lines to trip. Thus a "natural" uncontrolled spiltting occurs. It would be desirable if the evolving transients were swiftly and correctly diagonised as being unstable and system splitting is done "gracefully", i.e., in a controlled fashion by tripping a preselected set of lines. If wide-area synchronised measurements (discussed earlier in this module) are available, then they can aid the diagnosis of the evolving transients.

For the example which we are considering, the tripping of the line shown splits the system into 2 islands. One of the islands has less generation. In order to prevent a large and sudden drop of frequency, some load is shed by sensing the frequency (under-frequency load shedding). Stabilising the voltage and frequency in an island is not easy due to the reduced cumulative generator inertia (due to lesser number of generators) and possibly lower real and reactive power reserve margins. In an over-generated island, generation has to be reduced quickly to stabilze frequency.



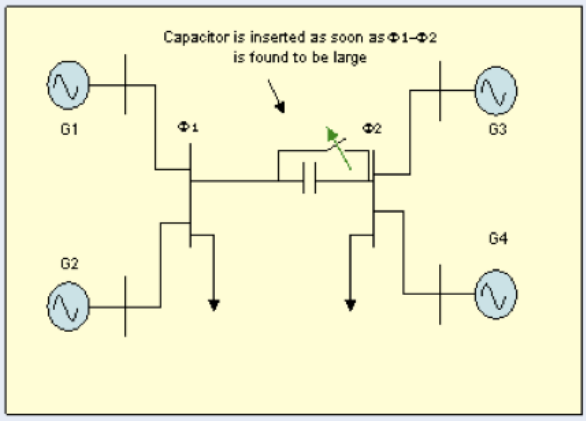
Angular Instability and Emergency Control : Concluding Remarks

One may wonder, why *preventive* control could not be applied to this situation. Actually, it can be applied. A situation like this can be anticipated before it happens and preventive rescheduling can be done so that power flows can be reduced in certain lines. (We saw in Module 2 that angular instability can be avoided by reducing line power flow). However simulating (mimicing) several such situations on a computer (so that appropriate preventive action can be contemplated) --- also known as *dynamic security assessment* --- is time consuming since it involves computationally intensive numerical integration of swing equations.

We have discussed emergency measures (splitting into islands) that are taken to prevent a complete blackout if angular instability occurs. Are there emergency control measures which can try to avoid angular instability so that the system need not have to be split at all?

Yes. Usually there are a few equpiment with unutilised operating ranges (margins) which can be quickly brought into the picture. For example, an excitation system in a generator can (for a short time) boost the voltage applied to the field winding. Also, power flow control devices can quickly change power flow such that restorative torques are maximized. Maximum use must be made of short-time transient limits of these devices, which are higher than the limits in steady state. Thus, most devices virtually have a higher operating range during transients, which can be gainfully employed under emergency situations.

A possible emergency action is illustrated below. A sudden change in phase angular separation between the two ends of a line can be made to trigger opening of a bypass breaker shown in the figure. The insertion of a series capacitor boosts the transient electrical power flow through the line due to reduced effective reactance, which in turn decelerates the relative speed between the generator groups G1,G2 and G3,G4.



Recap

In this lecture you have learnt the following

- Application of Emergency Control Measures for thermal overload conditions as well as angular instability
- Illustrative examples

Lecture 29 : Emergency Control : An example

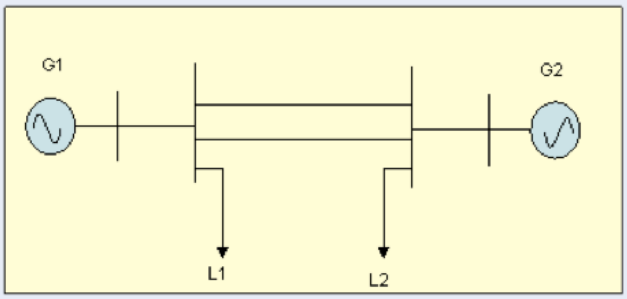
Objectives

In this lecture you will learn the following

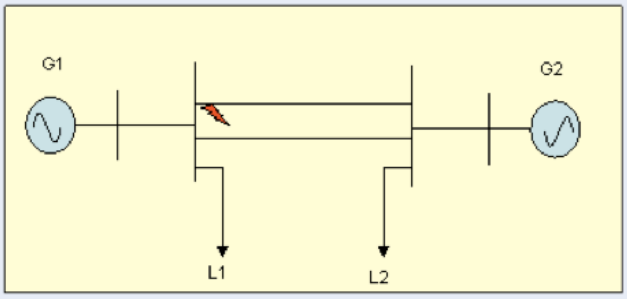
- An example to illustrate the system angular instability and islanding

A simple 2 machine example

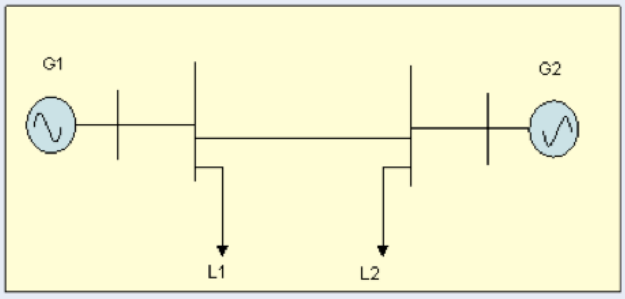
Consider the two machine system shown below:



Suppose it is subject to a fault on one of the interconnecting lines.



This fault is cleared by tripping the lines using Circuit Breakers which are triggered by protective relays



What are the possible consequences of such a disturbance ?

Possible Consequences

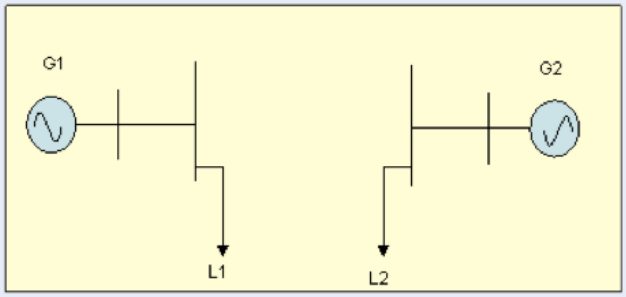
The possible consequences of a large disturbance like a fault (followed by line clearing) can be :

a) The system settles to a new acceptable equlibrium after some initial transients die down.

b) The system settles to a new equilibrium, but the equilibrium is violative of some steady state equipment limit (leading to tripping out of that equipment).

c) The system does not attain a new equilibrium due to angular or voltage instability.

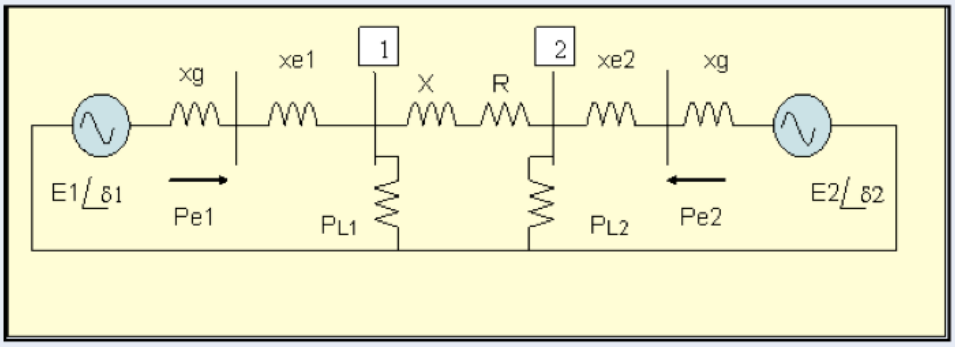
Voltage instability leads to unacceptably low voltages (which can be corrected by under-voltage load shedding), while angular instability (loss of synchronism) leads to violent excursions in current, voltage and power leading to equipment damage. Therefore, the generators which have lost synchronism have to be disconnected from each other. This may occur "naturally" due to distance relays (which mistake the large variations in voltage and current for a fault and trip the interconnecting lines), or intentionally -- controlled system separation --- by action of specially designed out of step relaying schemes. This situation is shown below.



The resulting two islands may have substantial real and reactive power deficit or surplus. This may result in decline or rise of frequency and//or voltage. Therefore excitation control, governors, generator overspeed control, and under-frequency/under-voltage load shedding are essential to make the islands stable.

We now anaylse this disturbance ...

In order to understand the various possibilities, we consider the system shown below which is modelled as follows:



A **generator** in this example is modelled as a voltage source (constant magnitude) behind a reactance. The angle dynamics is described by the swing equation:

$$\frac{d(\omega_i - \omega_o)}{dt} = \frac{d\omega_i}{dt} = \frac{\omega_B}{2H_i}(P_{mi} - P_{ei})$$

$$\frac{d\delta_i}{dt} = (\omega_i - \omega_o)$$

where, i = 1 and 2 for generator 1 and 2 respectively. $\omega_i - \omega_o$ is the speed deviation from nominal, $(P_{mi} - P_{ei})$ is the difference in mechanical input and electrical output power. The electrical power is obtained from the circuit solution for the figure shown above. ω_B is the base (nominal frequency) and H_i is the inertia constant.

The value of the rotor angle and speed is obtained by **numerical integration** (e.g. Runge Kutta Method) of the equations.

Caution: In actual practice, a generator model is much more complicated due to the dynamics of stator and field fluxes and the excitation system.

The **loads** are assumed to be resistance type (no frequency dependence and unity power factor).

The values of various parameters shown in the figure and the swing equations are:

PL1 = 0.63 pu, PL2 = 1.27 pu, xe1=xe2=0, xg = 0.25 pu, H1=H2=6 MJ/MVA, wo=wB=2*p*50 rad/s

Initial operating conditions: V1 = V2 = 1.0, and Pm1 = Pm2 = 0.95 pu

Note that initially power flows from bus 2 to bus 1 via **two parallel lines** (shown in the figure above as one *equivalent* line with impedance R+jX)

Prefault : Two identical lines in parallel : R = 0, X = 0.5,

Three Phase Fault : On one of the parallel lines at bus 1, and lasts for Tclear seconds

Post Fault : The faulted line is tripped. Therefore after the fault, R = 0 and X = 1.0

Results

We perform numerical integration using the MATLAB/SIMULINK files init.m and emergency.mdl, which can be downloaded from here. The various possibilities are shown below:

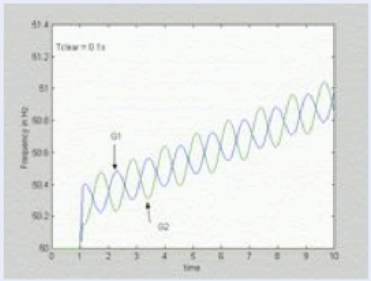
Case 1: Tclear = 0.1 s

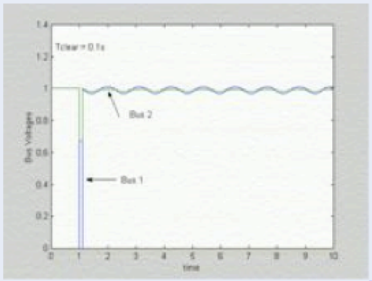
In this case, the disturbance does not lead to a loss of synchronism. However power, frequency and voltage undergo "swings" which are caused by relative motions of the generator rotors. These swings are acceptable and will usually die down due to effect of damper and field windings in a generator (not modelled here).*

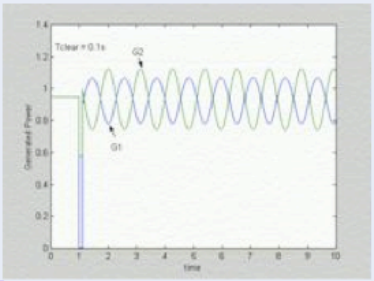
*Sometimes (not in the example shown here), these swings do not die down but grow with time. This is due to the effect of large gain feedback controllers - like voltage regulators in a generator excitation system.

Note: Due to slightly lower voltages at the buses 1 and 2 after tripping of the line, the total power drawn by the resistive loads decreases. Since the mechanical power is not changed, the frequency of both generators keeps increasing.

The frequencies will settle down -- to a value greater than 50 Hz -- if the loads are frequency dependent or if governors adjust the mechanical input to the generators (not considered here).

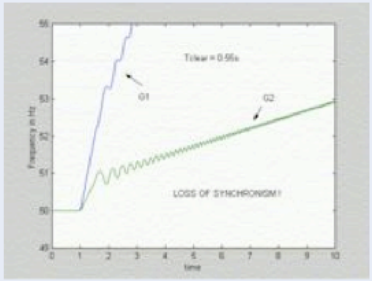


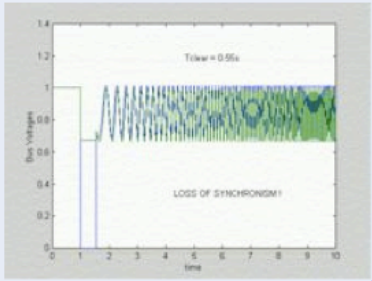


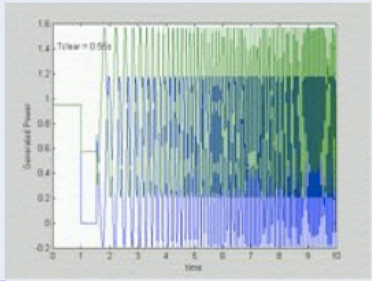


Case 2 : Tclear = 0.55 s

In this case, the disturbance causes the two generators to lose synchronism. The frequencies of the two generators "separate out" and electrical power and voltage undergo violent pulsations. (Normally this will not be allowed to continue).





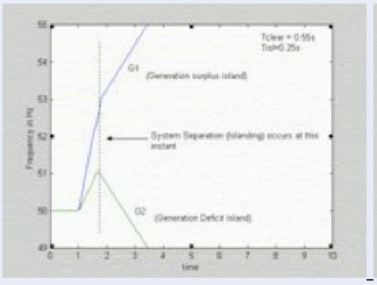


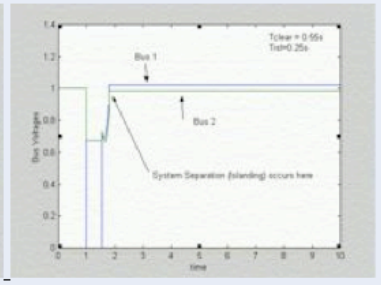
Results (contd.)

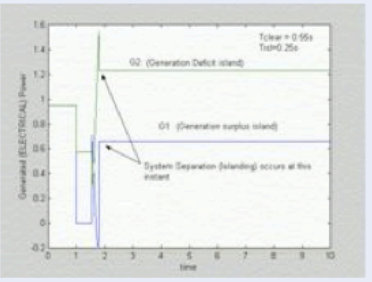
Case3 : Tclear = 0.55s, Tisl=0.25s

In this case, the disturbance leads to a loss of synchronism, but the two generators are separated by disconnecting the remaining line connecting them 0.25s after the faulted line is cleared.

Frequency in the "islands", one of which has excess load and the other which has surplus generation, changes very rapidly due to the large imbalance, calling for quick measures.



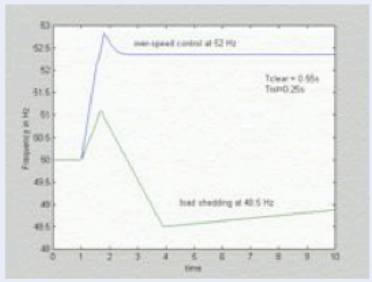


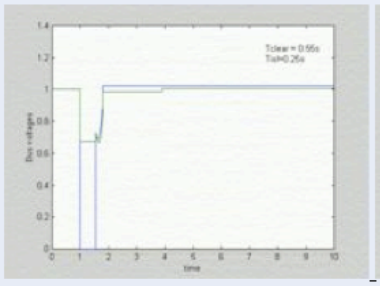


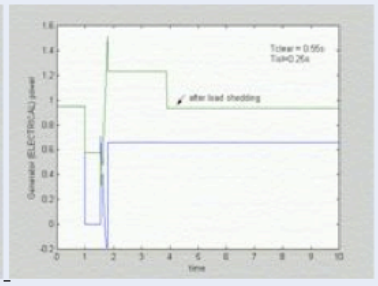
(click on figures to enlarge)

Case 4 : Tclear = 0.55s, Tisl=0.25s, Frequency Controls enabled

The effect of two **(idealized)** emergency frequency control schemes (see SIMULINK block diagram emergency.mdl for details) is shown here: Shedding of some load when frequency < 48.5 Hz, in the under-generated island, and *fast acting* overspeeding control to reduce mechanical power at frequency > 52 Hz in the over-generated island, prevents large frequency deviations.







Recap

In this lecture you have learnt the following

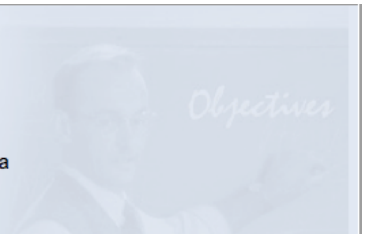
- An example to illustrate the system angular instability and islanding.
- If generators within a grid lose synchronism, then they have to be disconnected from each other.
- The separate subsystems (islands) may not survive if adequate frequency control measures are not in place.

Lecture 30 : A Blackout !

Objectives

In this lecture you will learn the following

- Definition of blackouts and their occurences.
- A detailed description of the January 2001 Collapse of Northern Regional Grid of India



When can a blackout occur ?

In the previous lecture we saw how a system can go from an alert state to an emergency state because of equipment limit violations or instability. The first line of defence against these undesirable transients is to carry out static and dynamic security analysis (using computer simulations) in order to check the system behaviour under different contingency situations (if they were to occur). Preventive Control actions ensure that any such contingency (if it occurs) does not lead to equipment limit violation or instability.

Emergency control actions come into play if an actual disturbance takes place and is evolving into an equipment limit violation or instability, or both. These actions try to prevent an emergency situation from deteriorating into a near-complete loss of generation and load (a **blackout** !).

Inspite of security analysis and preventive actions (done during actual operation), and emergency control actions (usually pre-designed offline), blackouts do occur. It should be understood that because of the large number and diversity of equipment in a power system, *every disturbance and contingency scenario cannot be anticipated* . Moreover there may be mal-operation of protective equipment and relays.

Therefore, blackouts have occured from time to time in the world. In this lecture we describe a real life event - the January 2001 blackout of the northern regional grid of India.

The January 2001 collapse of the Northern Regional Grid of India

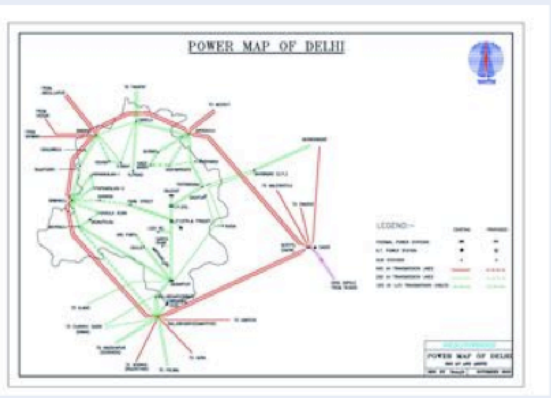
North India enters new millennium shivering in the dark (Indian Express, Wednesday, January 3, 2001)
NEW DELHI, JANUARY 2: Virtually the entire northern India, from Jammu and Kashmir to Uttar Pradesh, faced its worst ever blackout in the last five years as seven states went without power for nearly sixteen hours since around four in the morning on Tuesday.....

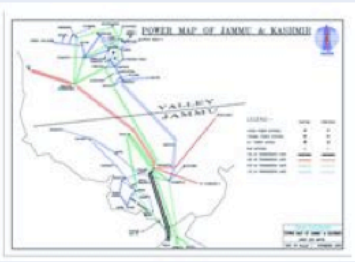
Overview of the Northern Regional Grid (adapted from the REPORT OF THE COMMITTEE ON GRID DISTURBANCE IN NORTHERN REGION ON 2ND JANUARY, 2001 authored by V.V.R.K.Rao & R.N.Srivastava -http://powermin.nic.in/reports/report_disturbance_northern_rg.htm, and the text of the submission by Power Grid Corporation of India Ltd (PGCIL) before the Central Electricity Regulatory Commission on 15-1-2001)

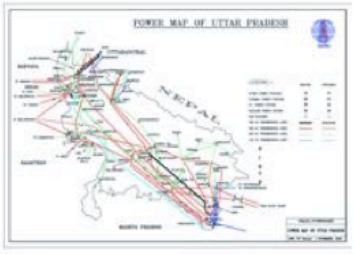
Northern Regional Grid is the 2nd largest inter-connected network (synchronous grid) in the country. As on 1st January, 2001 Northern Regional Power System had an Installed capacity of 27042 MW comprising of 8311 MW Hydro (31%), 17597 MW thermal (including GT and Diesel plants) (65%), and 1130 MW Nuclear (4.0%) and balance 4 MW wind. The share of state sector (16190 MW) and central sector (10852 MW) in the total capacity was 60% and 40%. The Region has a hydro-thermal mix of 31:69.During the period up to December, 2000, the Northern Region had been facing peak shortage of 10% and energy shortage of about 6%. Against a requirement of 98 billion units of energy, the actual met had been about 92 billion units. The peak demand met was of the order of 19860 MW against a restricted peak demand of 21740 MW. The shortages vary from month to month.

The Northern Regional Power System has major thermal power stations located at the coal based pit heads at Singrauli, Rihand, Obra and Anpara. The Region has major hydro electric power stations located in the Himalayan belt, such as Bhakra, Dehar, Pong, Chamera, Baira-Siul and Salal, Uri and the hydro power stations in the Yamuna region of UP. A number of thermal power Stations are also located near the load centres, such as Tanda, Unchahar, Panki, Dadri, Parichha, Harduaganj, Badarpur, Indraprastha, Rajghat, Panipat, Ropar, Bhatinda, Suratgarh and Kota and also the Rajasthan and Narora Atomic Power Stations. Also the Region includes Gas Turbine Power Stations at Auraiya, Anta, Dadri, Indraprastha and Srinagar.

The large coal pit head thermal power stations are located in the extreme South-Eastern part of the Regional grid. Therefore, there is a large flow of power from the South-Eastern part to the Central and Western parts of the grid round the year. During winter months when the flows dwindle to their annual minimum value, many of hydro stations are shut down during night off peak hours. To handle the bulk transmission of power, a point-to-point high voltage direct current link, viz. ± 500 kV HVDC Rihand-Dadri bipole with a capacity of 1500 MW has been established which operates in parallel with an extensive 400 kV A.C. transmission system and the underlying 220 kV network. There also exists a high voltage 500 MW capacity back-to-back DC link between Singrauli in Northern Region and Vindhyachal in the Western Region over which power exchanges are carried out between the two regions. A 2x140 MVAR SVC (Static Var Compensator) is also provided at 400 kV substation at Kanpur which helps in improving the stability of the system. For a grid diagram of the system, view the same below (courtesy NRLDC website : www.nrldc.org).







(the above maps can also be viewed from this site : http://nrldc.org/nrldc/powermaps.asp)

Situation prior to the collapse - An alert scenario

The following major links connect the large generation stations (total capacity > 6 GW) in the south-eastern part of the grid to the load centres in Western UP and Delhi.

a) Rihand Dadri (±500 kV, 1500 MW) HVDC link. Power is transmitted on 2 wires which are ± 500 kV with respect to the ground (this is commonly known as a bipole configuration).
b) 400 kV (AC) Singrauli- Kanpur-Ballabhgarh-Dadri
c) 400 kV (AC) Singrauli-Lucknow-Moradabad-Muradnagar-Dadri
d) 400 kV (AC) Singrauli-Vindhyachal-Kanpur-Agra-Ballabhgarh
e) 400 kV (AC) Obra-Panki-Muradnagar
f) 400 kV (AC) Anpara-Unnao-Agra-Muradnagar

What we show here is only a "reconstruction" of what probably happened on that fateful night, based on manual observations and recorded measurements. Since all measurements and observations are not necessarily synchronised with a common clock, some amount of correlating or "pieceing together" of evidence is required. However some uncertainty is expected in the final sequence. A rigourous approach would involve simulation (numerically evaluating the response) of the entire sequence on a computer and correlating the same with the observed data. However, this is not shown here.

Time (Hrs.)	Event
1st Jan, 2001	**1st Jan, 2001**
23:21, 23:33 and 23:41	Transient faults on the operating pole-1 of HVDC Rihand-Dadri (POWERGRID) line. The line continued in operation with successful auto-re-starts.
23:49	Singrauli and Rihand Stations were asked to back down from 1840 MW to 1620 MW and from 910 MW to 810 MW respectively.
2nd Jan,2001	**2nd Jan,2001**
00:02	Transient faults on the operating pole-1 of HVDC Rihand-Dadri line. The line continued in operation with successful auto-re-starts.
00:17	Pole-1 of HVDC bi-pole was put on reduced voltage mode of operation thereby reducing power flow on this line from 750 MW to 500 MW.
01:05	400 kV Obra-Panki S/C (UPPCL) line tripped on fault and remained under breakdown.
01:07	Frequency goes above 51.0 Hz, NRLDC asked Singrauli and Rihand Stations to back down from 1620 MW to 1540 MW and from 810 MW to 730 MW respectively.
02:45 02:47 03:26 03:41 04:31	400 kV Agra – Ballabhgarh line tripped on Y-phase earth fault and auto reclosed.
03:11	400 kV Panki – Muradnagar developed fault. Circuit Breaker at Panki end did not operate resulting in back-up protection operation of Main bus A. 400 kV Panki – Kanpur II and ICT -II (inter-connecting transformer) at Panki connected to this bus also tripped.
03:12	400 kV Unnao – Agra line also tripped on fault.
03:18	System frequency rose to 51.2 Hz.

Situation prior to the collapse - Alert to Emergency

03:31	220 kV Narora Atomic Power Plant – Muradabad tripped on fault.
03:35	220 kV Panki – Fatehpur tripped on fault.
03:33 03:54	400 kV Agra – Kanpur line also tripped on B-phase earth fault and auto reclosed.
03:50	Load dispatcher asked Singrauli Generators to further backdown from 1480 to 1320 MW.

At 04:38 hrs, 220 kV Panki-Fatehpur line tripped on fault.

With all these outages, the system could not be kept in synchronism and the following lines tripped (as a consequence of relay response to an out of step situation) and the system separated naturally into 2 subsystems.

04:38 400 kV Lucknow – Moradabad, 400 kV Kanpur – Agra, 400 kV Kanpur – Ballabhgarh, East – West separation of Northern Grid took place.

The separation of the system into 2 islands need not have led to a complete blackout.

Since one island had excess generation, quick tripping of a few generators or govenor action could have stabilised frequency. In the under-generated island, under-frequency load shedding would have stabilised the frequency. However neither of these emergency measures were in place. Therefore,

04:40	Frequency of western part dipped and sub-system collapsed.
04:40 – 04:44	Frequency of eastern part shot up to more than 52.5 Hz. All running machines in eastern part tripped on over frequency and sub-system collapsed

Practically, whole of North India was plunged into darkness. A few generators survived by separating themselves from the rest of the grid and kept running supplying only their "house load".

Some of the major causes of the blackout can be summarised as follows:

a) Long duration of equipment (converter transformer) outage in the HVDC link
b) Heavy pollution and fog causing insulation flashovers.
c) Inadequate preventive control
d) Inadequate emergency control

After the blackout, it took almost 16 hours to fully recover and reconnect the grid. What are the issues in the restoration of the grid ?

We shall study restoration issues in the next lecture. In the meanwhile, can you carry out a survey of the recent blackouts which have taken place in the world ?

Recap

In this lecture you have learnt the following

- The various factors that contribute to a blackout.
- The sequence of events that led to the 2001 blackout in North India.

Lecture 31 : Power System Restoration

Objectives

In this lecture you will learn the following

- Restoration of a power system after a blackout has taken place.
- Problems associated with restoration.
- An Illustrative Example

After a blackout

If a blackout (a near total loss of generation and load) takes place, efforts have to be taken to bring back the system to a normal state at the earliest. It may surprise you to know that this (black starting!) is not an easy task. We shall see why in this lecture.

Once a generator is tripped, restarting it requires a significant amount of power. Power is required for 2 types of activities:

a) Survival Power: For emergency lighting, battery chargers etc. Usually the requirement is 0.3% of the generator capacity.

b) Startup Power: For starting power plant auxiliaries (pumps etc.) Interestingly, nuclear and thermal units require approximately 8 % of the unit capacity for auxiliaries alone! Therefore, a 500 MW generator requires approximately 40 MW for running its auxiliaries.

Hydro and Gas units, on the other hand, require only about 0.5-2% of unit capacity for auxiliaries and can be started usually from in-house DG sets.

The major steps required for restoration are:

a) Islands which have survived need to be stabilised for frequency and need to be used for starting other units

b) Hydro/Gas units which require less startup power need to be started using in-house DG sets.

c) Larger thermal units need to be fed "startup power" from: 1) Islands which have survived 2) Blackstarted generators 3) Other synchronous grids (temporarily)

d) Started units are synchronised with one another.

e) Loads and Generation have to be matched as much as possible to avoid large frequency variations. Governors have a major role in stabilizing frequency in an island.

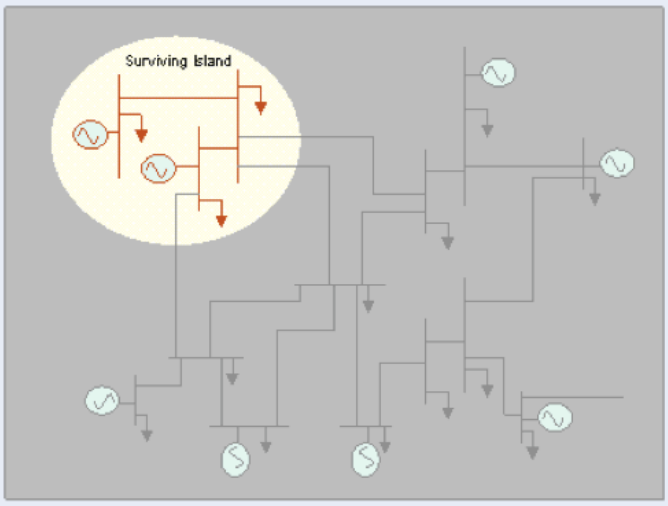
Problems in Restoration

a) Securing Islands

After a blackout a few islands may survive due to separation of the system in time. A few hydro or gas generators could be blackstarted using in-house D-G sets. Therefore some small pockets will be there in the otherwise blacked out grid wherein generators are supplying some loads. However, the situation in these islands is usually precarious due to the small number of generators within the island (having very little cumulative inertia).

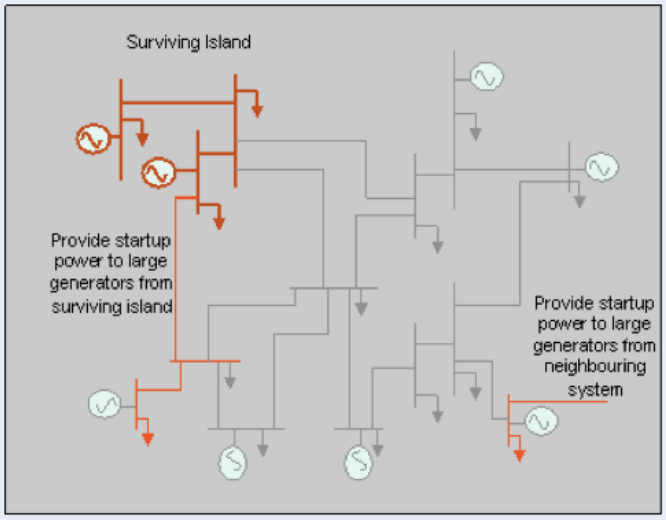
Recall that the initial rate of change of frequency is determined by *cumulative machine inertia* and the initial load-generation imbalance, while the final settling frequency is determined by the governor and load frequency characteristics (see Module 3).

Therefore if the load in the island is fluctuating (for instance, traction loads), the rate of change of frequency within the island due to fluctuating loads may be quite large -- large enough for the island to collapse due to excessive frequency variations - causing generators to trip. Therefore control of generated power (by governors) and frequency based tripping or energisation of load is important.



Black-starting of large generators is done by availing startup power from other started generators or islands. Startup power may also be availed from neighbouring synchronous grids if an AC transmission link exists (normally disabled). Unfortunately, startup power cannot be availed via DC links (which use AC line voltages for commutating thyristors), because AC voltages are not available in the system which is blacked out.

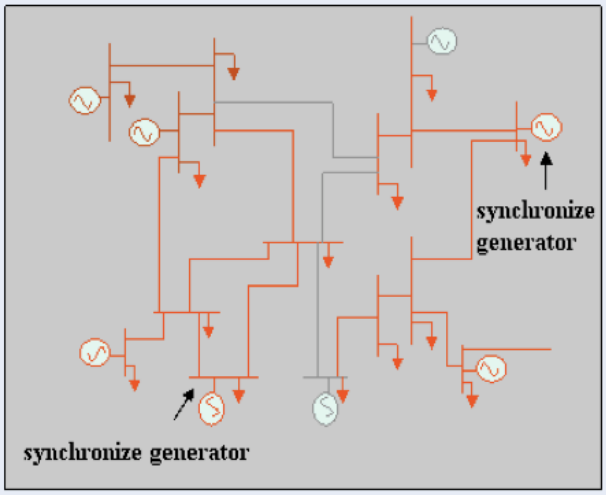
Therefore a generator at Vindyachal (near the border of the western region and northern region grid of India, which are not synchronised but exchange power through DC asynchronous links during normal conditions) can avail startup power through *an AC line* from the northern grid.



Problems in Restoration

b) Extending Power to Loads from Generators which are black-started

The next step in power system restoration is to supply loads from black-started generators. Some of these loads may be in the form of the startup (auxiliary) loads of other larger generating plants which need to be black-started.



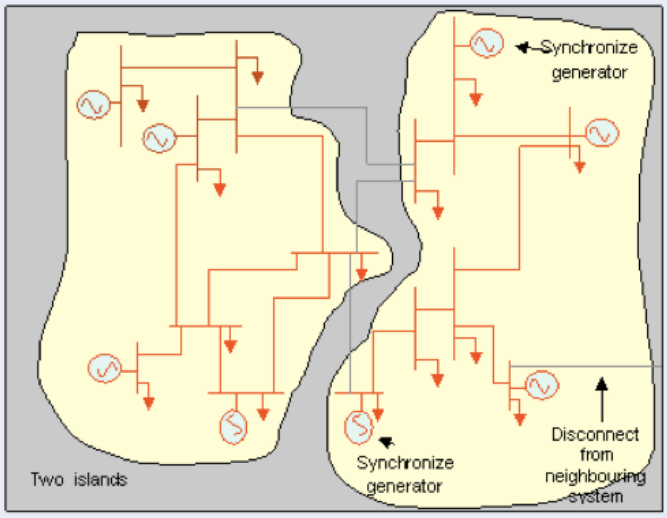
These loads are supplied via transmission lines. Enregising a transmission line initially without any load can cause **over-voltages** (why ?). This is avoided by:

1. Energising fewer high voltage lines
2. Operating generators at minimum voltage levels (by keeping filed excitation low)
3. Deactivating switchable capacitors
4. Connecting shunt reactors and tertiary reactors
5. Adjusting of transformer taps
6. Pick up loads with lagging power factor
7. Charging more transformers
8. Charging shorter lines
9. Operating synchronous condensers / SVCs where available
10. Avoiding charging lines with series capacitors

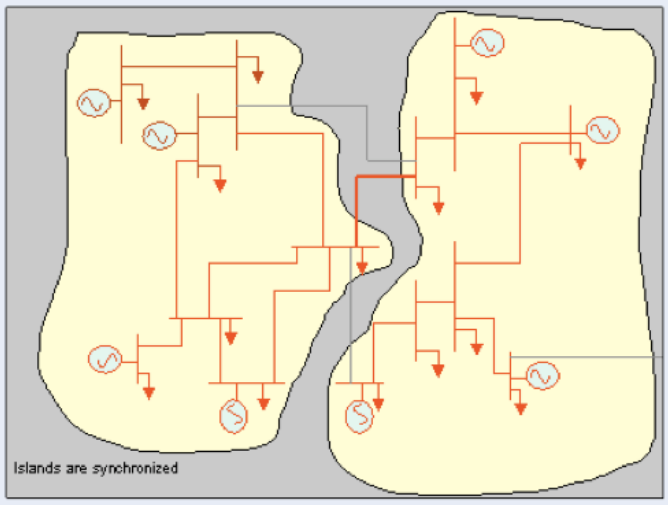
Problems in Restoration

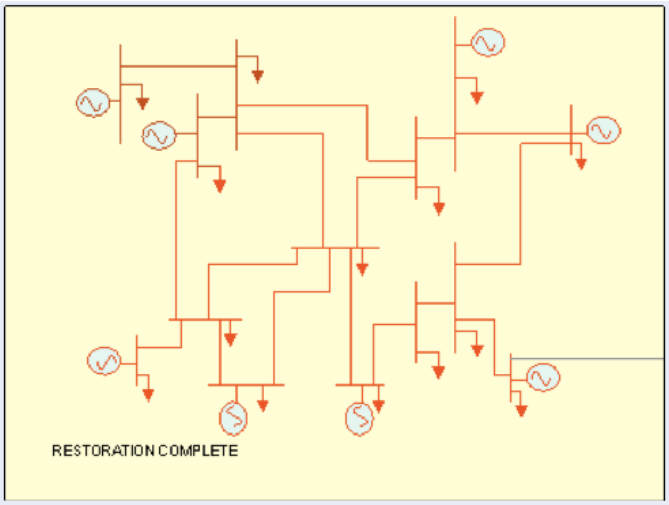
c) Re-integrating the grid

As mentioned before, some islands which have been secured should be connected with each other so that the system cumulative inertia increases, a better generation-load balance can be achieved by encompassing a larger set of loads and generators, and better redundancy in transmission and generation is achieved.



Note that an important step in reconnecting islands to one another is "synchronisation". While each generator has synchronising facilities, the interconnection of two islands may have to be done at some bus in the network wherein such facilities are available. The basic requirements for successful synchronisation of two systems are the same as those for an individual generator connected to a large grid (see Module 2). In particular, the frequencies should be practically the same and phase angular difference at the instant of connection should be small. If two systems are connected at an inappropriate instant, then the generators in both islands will not synchronize, and the situation will be akin to an out-of-step scenario; the link will have to be disconnected.





Restoration : A practical example

The **sequence** of actual restorative actions which were taken after the Northern Grid Collapsed in January 2001 are given below.

Details of the blackout are given in the previous lecture. Again, what we show here is only a "reconstruction" of what probably happened, based on manual observations and recorded measurements. Since all measurements and observations are not necessarily synchronised with a common clock, some amount of correlating or "pieceing together" of evidence is required. However some uncertainty is expected in the final sequence. A rigourous approach would involve simulation (numerically evaluating the response) of the entire sequence on a computer and correlating the same with the observed data. However, this is not shown here.

The system was restored in two parts. The south-eastern part was restored by taking power from Western Region (a neighbouring grid) through an AC line which runs parallel to the HVDC back-to-back station at Vindhyachal.

The western part was restored after starting Bhakra (hydro) machines. Chronology of restoration process of Eastern and Western parts is given below :

Restoration of Eastern Part (2nd January 2001)

At 04:59 one machine of Rihand (Hydro generation) was started.

At 05:10 hrs startup power from Western Region (WR) was extended to Singrauli generators through AC bypass of HVDC back-to-back station at Vindhyachal.

05:38 hrs WR power was further extended to Rihand Generators by charging 400 kV Singrauli – Rihand I.

06:32 hrs WR power was also extended to Kanpur by charging 400 kV Vindhyachal – Kanpur line.

At 07:01 hrs 400 kV Kanpur – Panki I was charged.

At 07.42 hrs. Rihand (H) - Anpara 132 kV line charged and Anpara unit cleared for light up at 08.15 hrs.

The power was extended to Mainpuri at 12.38 hrs.

Between 09:11 and 11:20 hrs following units were synchronized:

Singrauli unit 6 (500 MW) at 09:11hrs

Rihand unit 1 (500 MW) at 09:54 hrs.

Singrauli unit 5 (210 MW) at 10:48hrs

Singrauli unit 3 (210 MW) at 11:20hrs

Since there was considerable delay in getting Bhakra (hydro) power at Dadri, WR Power was extended from Kanpur to Agra at 11:44 hrs by charging 400 kV Kanpur – Agra line and further to Ballabhgarh and Dadri at 11:57 hrs and 12:01 hrs respectively. However, 400 kV Ballabhgarh – Dadri I tripped on over voltage at 12:17 hrs.

Ballabhgarh Interconnecting Transformer (ICT) was charged to extend power to Ballabhgarh at 12:18 hrs. 400 kV Agra – Ballabgarh line tripped at 12:39 hrs

Anpara unit-4 synchronised at 12.50 hrs.

Restoration of Western Part

As specified in Black Start Procedures, immediately after the grid failure Bhakra hydro station attempted two times to revive the system at 05:05 hrs. and 05:40 hrs. but the machines could not be stabilized.

Unit 4 at Bhakra (L) started at 06:00 hrs.

At 07:07 hrs. power was extended to Panipat by charging 220 kV Bhakra – Ganguwal – Dhulkote – Panipat sections.

At 07:28 hrs Panipat – Dadri line was charged. This line however, tripped on over voltage. The line was again charged at 07:33 hrs but tripped immediately on over voltage.

The line was again charged from Panipat end at 08:47 hrs utilizing an open circuited 220/400 kV transformer at Dadri as a reactor to limit over voltage. Dadri – Ballabhgarh I was charged at 09:08 hrs but tripped immediately along with Panipat line on over voltage. At 09:19 hrs Panipat – Dadri line was again charged. Dadri Gas Turbine generators synchronized and started generating 200 MW.

The following 400 kV lines were charged between 09:29 to 09:54 hrs from Dadri end

Dadri – Ballabhgarh 09:29 hrs, Dadri – Murad Nagar 09:39 hrs (tripped at 10:10 hrs), Ballabgarh - Agra 09:48 hrs.

Dadri – Mandola I 09:54 hrs, Agra – Auraiya 10:00 hrs

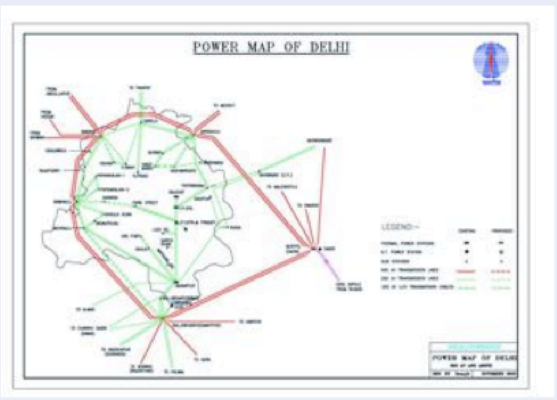
At 10:16 hrs Bhakra island collapsed due to under frequency.

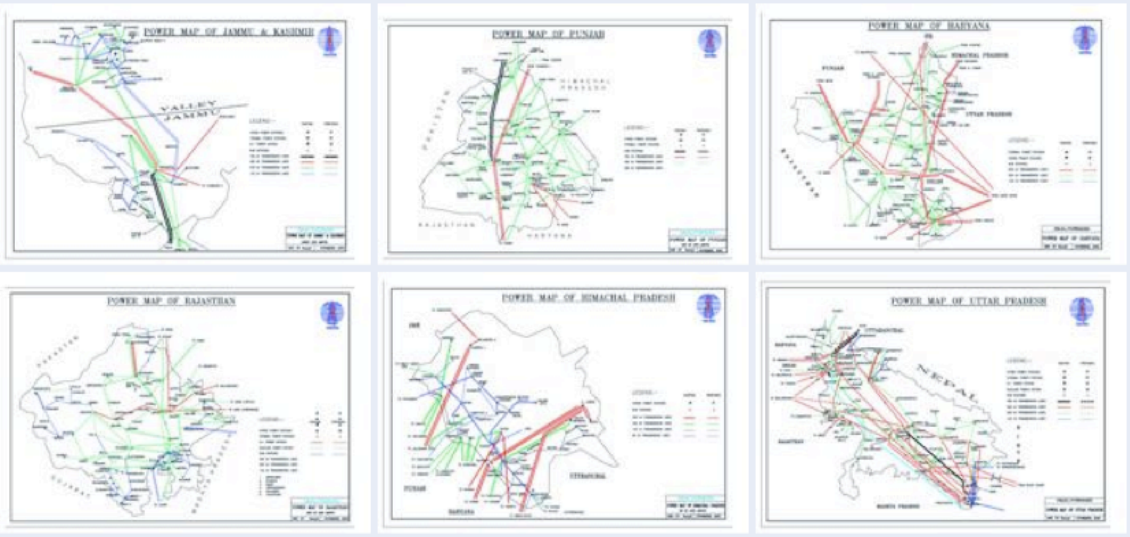
At 10:33 hrs 1st unit at Bhakra(R) station started but tripped at 10:42 due to large fluctuations in loads. It was again started at 10:46 hrs but again tripped at 10:52 due to same reasons as above. The unit was again started at 11:00 with manual regulation of load by locking the load limiter in steps. Power was extended to Panipat at 11:14 hrs. At 12:04 hrs 220/400 kV Inter-Connecting Transformer 2 was charged.

Power to Badarpur was extended from Bhakra system through Panipat - Charkhi Dadri - Ballabhgarh at 13:08 hrs.

Bhakra power was not extended to Dadri as it received WR power from Ballabhgarh.

The eastern and western parts of the grid were synchronized at 13:32 hrs.





(the above maps can also be viewed from this site : http://nrldc.org/nrldc/powermaps.asp)

Recap

In this lecture you have learnt the following

- Nature of problems faced when restoring the system after a blackout.
- An example : Restoration of the Northern Grid after the 2001 blackout.

We now summarize what we have learnt in this module :

a) Other than routine scheduling of real and reactive power flows based on economic and power quality criteria (i.e. maintaining a good voltage and frequency), a system operator monitors the health of the system by analysing measured data during real operation

b) The system operator "checks out" the health of the system by mimicking the system behaviour (for different disturbances) on a computer for a given operating scenario (obtained from measurements). This is called security assessment.

c) Based on computer simulations, if the system is percieved to be in an alert state (i.e., a system will not be able to withstand credible contingencies), then operator carries out preventive control actions like generation re-scheduling and load curtailment.

d) Due to inadequate preventive control or lack of anticipation, a system may go into an emergency if contingencies occur.

e) Emergency Control actions like generator or load tripping may prevent total collapse. Failure of emergency control actions may result in a complete blackout.

f) Restoration of a power system after a blackout is not a trivial process and requires prior planning to prevent frequency or voltage problems. During the initial stages of restoration, a system is particularly vulnerable to these problems.

Congratulations, you have finished Module VI.

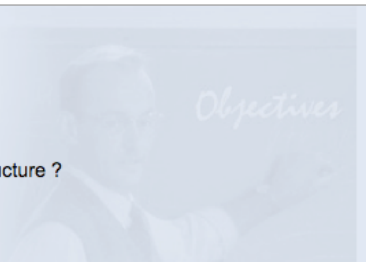
Module 7 : Power System Structures

Lecture 32 : A vertically integrated utility

Objectives

In this lecture you will learn the following

- What is a vertically integrated utility?
- What are the conditions that have led to changes in the traditional power system structure ?



Power System Structure

Power systems traditionally have been what are known as "vertically integrated utilities". In this type of structure, one utility handles the all functions of generation, transmission and distribution within a certain geographical area. The operation and coordination of such a system is somewhat simple, since all functions are controllable by a system operator. The operational objectives were to provide quality power (voltage and frequency nearly constant) to a consumer, while ensuring reliability and overall economy (low cost).The price of power was "regulated" and based on actual costs.

An alternative is to treat power as a tradeable commodity. The functions of generation and in many cases, distribution, are open to private participation. While the "technical objectives" are similar to those in a vertically integrated utility, the price is not regulated, but depends on market forces and competition between the participants. In a generation deficit scenario, price may still need to be regulated. Alternatively, the amount of loads should be price sensitive or else prices will spiral upwards. The cost of use of transmission lines (to which all players will have "open access" subject to the transmission constraints) would also be regulated. Therefore a "regulator" would still be required. However, a regulator would be an independent body. An independent system operator would perform the co-ordination functions required to operate the system reliably and ensure that voltage and frequency are within limits. The real and reactive power resources required to maintain voltage, frequency and reliability may be "purchased" and charged to all the players in a fair manner.

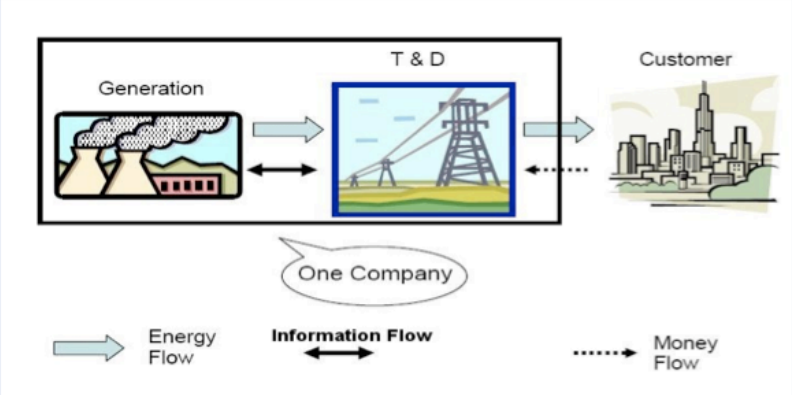
We shall discuss these structures and their impact on power system operation in the following lectures. We begin by considering in detail, a vertically integrated utility.

Structure of a traditional Vertically Integrated electric industry

The electric power industry has over the years been dominated by large utilities that had an overall authority over all activities in generation, transmission and distribution of power within its domain of operation. Such utilities have often been referred to as vertically integrated utilities. Such utilities served as the only electricity provider in a region and were obliged to provide electricity to everyone in the region.

The typical structure of a vertically integrated electric utility is shown in figure below. In the figure, the money flow is unidirectional, i.e. from the consumer to the electric company. Similarly, the information flow exists only between the generators and the transmission systems.

In vertically integrated utilities , it was often difficult to segregate the costs involved in generation, transmission or distribution. So, the utilities often charged their customers an average tariff rate depending on their aggregated cost during a period.



The state electricity boards (SEB) in India were examples of a vertically integrated utility; they are now being re-structured.

Characteristics of a traditional Vertically Integrated electric industry

■ Monopoly Franchise	Only the local electric utility can produce, move, or sell commercial electric power within its service territory.
■ Obligation to serve	The utility must provide service to all electric consumers in its service territory, not just those that would be profitable.
■ Regulatory Oversight	The utility's business and operating practices must confirm to guidelines and rules set down by government regulators. The utility's rates are set in accordance with government regulatory rules and guidelines. The utility is assured a fair return on its investment, if it confirms to the regulatory guidelines and practices.

Why was a traditional electric utility industry structured the way it was?

During early days of the electric power industry, governments favoured a regulated monopoly - vertically integrated utility structure. The reasons are manifold:

- This offered a risk free way to finance the creation of electric industry. Establishment of electric industry required large capital for infrastructure building. Thus for the purpose of risk minimization, a local monopoly and stable market was assured. The utility leaders could focus on building up their systems without having to worry about the competitors undercutting the prices to gain market share etc.

  To prevent exploitation of consumers due to monopoly, the government brought in regulation.

- This legitimized the electric utility business. Government franchises and regulation clearly implied to a possibly skeptical public that civic leaders thought electricity was a good thing.

- It gave electric utilities recognition and support from the government, which was necessary to solve problems like 'Right of Way' (i.e. the "right" to an exclusive corridor to bulid a transmission line).

During the nineties, many electric utilities and power network companies world-wide have been forced to change their way of operation and business, from *vertically integrated* mechanisms to open market systems. This can be specifically observed in countries like UK, Sweden, Finland, Norway, US and some countries of South America. The reasons for change have been many and have differed over regions and countries. We shall study of these developments in the next lecture.

Conditions which are leading to changes in traditional power system structures

Basic motivation for changes in power industry scenario

There are many reasons that are leading to restructuring of power systems. One force was the change in generation economies of scale that occurred throughout the 1980's. Traditionally, electric utility systems evolved with the central station concept because of significant economy of scale in power generation. Very large generators produced power at less than half the cost per kilowatt of small generator units, and the bigger the generator, the more economical the power it produced. For the reasons stated below, the shift in economy of scale was observed:

- Technological innovation improved the efficiency of small units for gas turbines, combined cycle, hydro and fuel cells over that of large ones.
- Improvements in materials, including new high temperature metals, special lubricants, ceramics, and carbon fiber, permit vastly stronger and less expensive small machinery to be built.
- Computerized control systems have been developed that often reduce the number of on-sight personnel to zero.
- Data communications and off-site monitoring systems can control the units from remote operations centers, where one central operator can monitor a dozen units at various sites, as if present at each.

Thus in *some* instances, it is possible to build new power plants that could provide energy at a lower price than what customers were paying. It became possible for the industrial and commercial users of electricity to build and operate their own plants also sell the excess power to small customers especially in generation deficit areas.

The reasons for restructuring

The reasons for initiating the idea of deregulation (**we will henceforth use the word *deregulation* to describe changes in power system structures; however, it will be clear that these changes involve changes in regulations rather than deregulation!** In some countries, these changes are also described as "liberalisation") in power industry are many.

Following are the main reasons:

1. The need for regulation changed.

More fundamental than any other reasons for change was the fact that the basic needs for regulation of electric industry had died away before the end of 20 th century. First, the original need for regulation, which was to provide risk free finance to build the infrastructure, did not exist anymore. Second, most of the the major electrical infrastructure was paid for, decades ago. The revenues gained by the electric utilities was invested to renew their system, and the level of risk in doing so was less as compared to that existed in the initial era. Being a proved technology, the risk involved in investing money in such a technology was nullified. The electricity could now be thought of as an essential commodity, which can be bought and sold in the marketplace in a competitive manner, just like other commodities.

2. Ideological Reason : Privatization

Usually the motive was the government's firm conviction that private industry could do a better job of running the power industry. This belief, of course came from better privatization experiences of the other industries. Deregulation does not necessarily have to be a part of privatization efforts.

3. Cost is expected to drop

Competition is expected to bring innovation, efficiency, and lower costs.

contd ...

4. Customer focus will improve

Although monopoly franchise utilities have an obligation to serve all customers, that does not promote the pro-active attention to customer needs. A monopoly franchise utility listens to its customers when they explain their needs, and then responds. A competitive electric service company *anticipates* customer's needs and responds in advance. The technological advances that will be applied under deregulation, address customer service. More important gain of competition in the electricity market is the customer value rather than lowering the cost.

5. Encouragement for innovation

The regulatory process and the lack of competition gave electric utilities no incentive to improve on yesterday's performance or to take risks on new ideas that might increase customer value. If a new idea succeeded in cutting costs, the utility still made only its regulated rate of return on investment; if it didn't work, the utility would usually have to 'eat' a good deal of the failed attempt, as imprudent expenses. Furthermore, why would a regulated utility want to use new ideas to lower its costs under a regulated rate of return framework?

Under deregulated environment, it was felt that the electric utility will try to innovate something for the betterment of service and in turn save its costs and maximize the profit. By means of this, the utility will try to ensure that it will maintain its customer base in spite of competition.

Recap

In this lecture you have learnt the following

- What is a vertically integrated utility
- The structure of a vertically integrated utility
- Both technological and economic reasons have lead to changes in the traditional power system structure

Lecture 33 : Structure of a Deregulated Industry

Objectives

In this lecture you will learn the following

- Structure of a deregulated industry.
- Different entities in a deregulated industry
- Ancilliary Service Management

Overview of A Deregulated Industry

Unbundling of traditionally vertically integrated utility.

One of the principal characteristics of a competitive structure is the identification and separation of the various tasks which are normally carried out within the traditional organization so that these tasks can be open to competition whenever practical and profitable.

This process is called *unbundling* .

An unbundled structure contrasts with the so-called vertically integrated utility of today where all tasks are coordinated jointly under one umbrella with one common goal, that is, to minimize the total costs of operating the utility.

One of the first steps in the restructuring process of the power industry has been the separation of the transmission activities from the electricity generation activities.

The subsequent step was to introduce competition in generation activities, either through the creation of power pools, provision for direct bilateral transactions or bidding in the spot markets.

On the other hand, the transmission system has significant economics of scale (i.e., it is economical to have a common bulk transmission system of a large capacity rather than individual small capacity transmission links). Consequently it was suited to be a natural monopoly and a separate entity. It was felt necessary to introduce regulation in transmission so as to prevent it from overcharging for its services. The transmission system thus became a neutral, natural monopoly subject to regulation by public authorities. And to overcome the monopolistic characteristic, the trend has been to establish new legal and regulatory frameworks offering third parties "open access" to the transmission network subject to technical constraints.

An important point to note is that the restructuring process was however not uniform in all countries. While in many instances, it started with the breaking up of a large vertically integrated utility, in certain other instances restructuring was characterized by the opening up of small municipal monopolies to competition.

In brief, Electric utilities are expected to split apart into unbundled companies, with each utility re-aligning itself into several other companies that respectively focus on each part of the new industry, i.e., power delivery and retailing. This is known as Disaggregation.

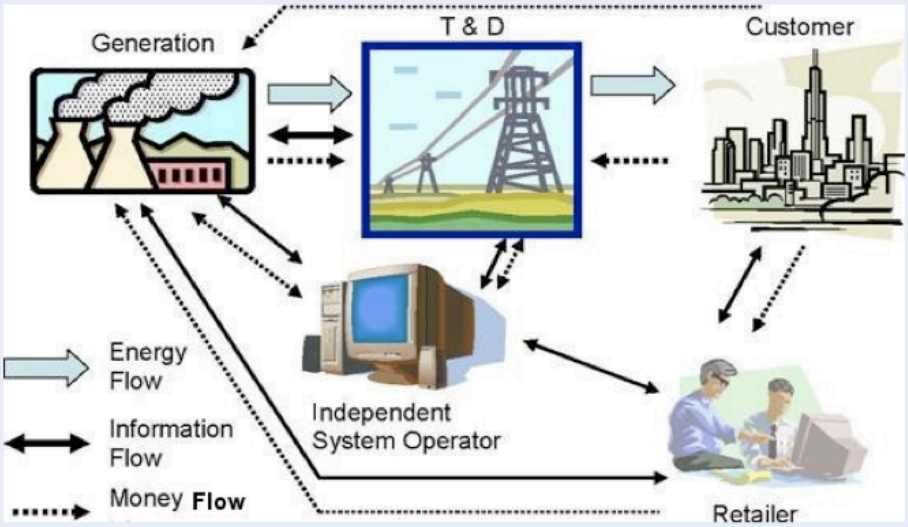
Under deregulation, the vertically integrated utility, one giant company that generates, transmits, distributes and sells electricity in coordinated manner will become thing of the past. To function in an open access system, such utilities will have to rearrange their operational organization to match the unbundled functions they must perform. Each part of the company will need to work in its new form. Generation will have to compete in the competitive power generation market place. T & D will have to operate as an open provider of delivery services. Competition will be present in retailing.

Generally, the governments advocating deregulation want competition in energy production, and they want to see significant levels of customer choice in the retail market for electricity. At the same time, it recognizes that it is best to have only one transmission and one distribution system in any one area. Therefore, the purpose of deregulation is to restructure the electric industry so that power production and retail sales are competitive, while delivery is still a regulated, monopoly franchise business.

Structure of deregulated industry

The figure below shows the typical structure of a deregulated electricity system with links of information and money flow between various players.

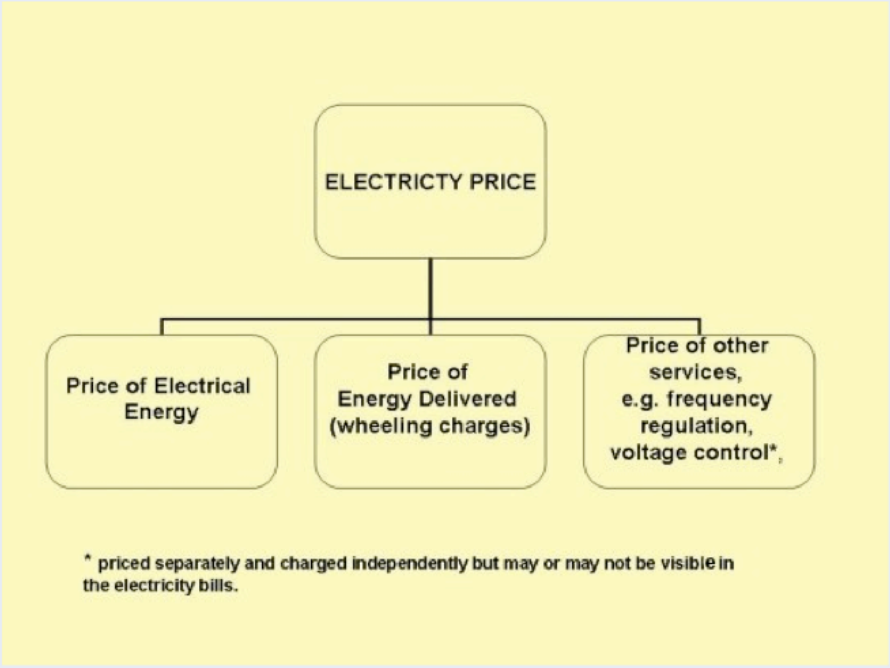
The configuration shown in the figure is not a universal one. There exist variations across countries and systems.



A system operator is appointed for the whole system and it is entrusted with the responsibility of keeping the system in balance, i.e. to ensure that the production and imports continuously match consumption and exports. Naturally, it was required to be an independent authority without involvement in the market competition nor could it own generation facilities for business. This system operator is known as **Independent System Operator (ISO).**

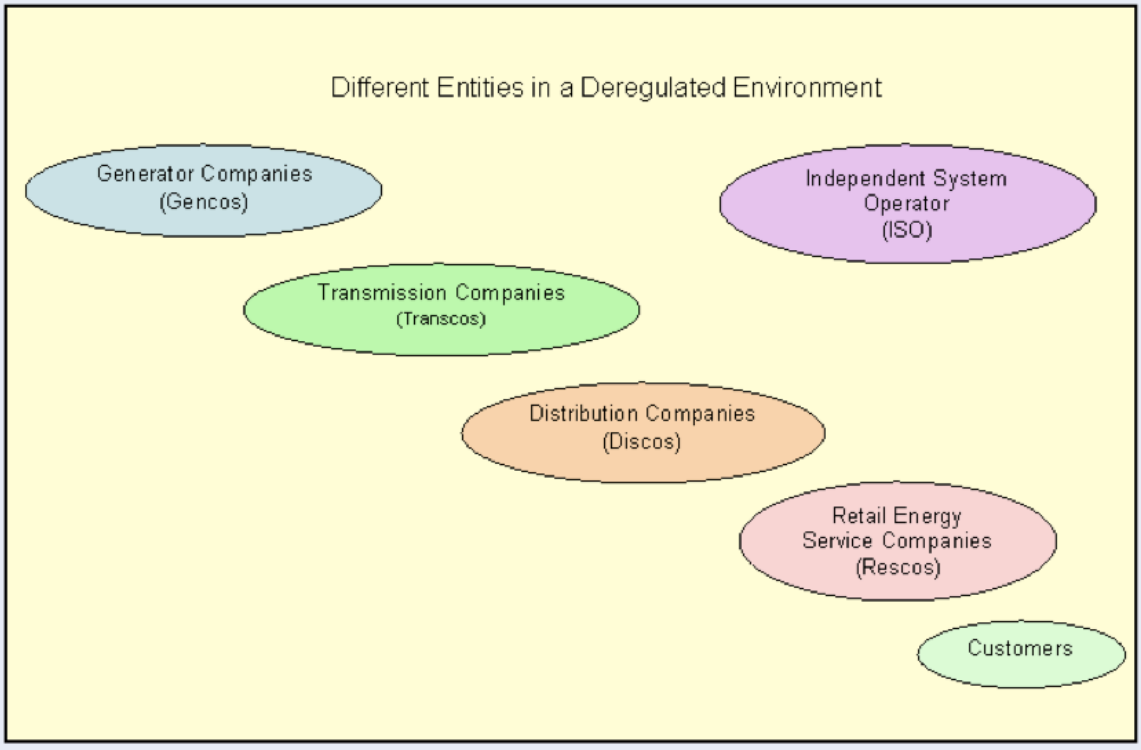
Different power sellers will deliver their product to their customers (via retailers), over a common set of T & D wires. These operations are supervised by an independent system operator (ISO). The generators, T & D utility and retailers communicate with the ISO. Mostly, customers communicate with a retailer, demanding energy. The retailer contacts the generating company and purchases the power from it and makes it transferred to its customer's place via regulated T & D lines. The ISO is the one responsible for keeping track of various transactions taking place between various entities. A customer can also enter into a bi-lateral contract with a generator directly for supply of the required energy.

In the vertically integrated environment, the electricity bill consisted of a single amount to be paid towards the generation, transmission and all other costs. But, in the restructured environment, the electricity price gets segregated into the following:



Different entities in deregulated environment :

The introduction of deregulation has brought several new entities in the electricity market place, while on the other hand redefining the scope of activities of many of the existing players. Variations exist across market structures over how each entity is particularly defined and over what role it plays in the system. However, on a broad level, the following entities can be identified as shown in the figure below.



■ Genco (Generating Company)	Genco is an owner-operator of one or more generators that runs them and bids the power into the competitive marketplace. Genco sells energy at its sites in the same manner that a coal mining company might sell coal in bulk at its mine.
■ Transco (Transmission Company)	Transco moves power in bulk quantities from where it is produced to where it is delivered. The Transco owns and maintains the transmission facilities, and may perform many of the management and engineering functions required to ensure the system can continue to do its job. In most deregulated industry structures, the Transco owns and maintains the transmission lines under monopoly franchise, but does not necessarily operate them. That is done by Independent System Operator (ISO). The Transco is paid for the use of its lines. In some countries, Transo itself acts as a system operator.
■ Disco (Distribution Company)	It is the monopoly franchise owner-operator of the local power delivery system, which delivers power to individual businesses and homeowners. In some places, the local distribution function is combined with retail function, i.e. to buy wholesale electricity either through the spot market or through direct contracts with gencos and supply electricity to the end use customers. In many other cases, however, the disco does not sell power. It only owns and operates the local distribution system, and obtains its revenues by 'renting' space on it, or by billing for delivery of electric power.
■ Resco (Retail Energy Service Company)	It is the retailer of electric power. Many of these will be the retail departments of the former vertically integrated utilities. Others will be companies new to the electric industry that believe they are good at selling services. Either way, a resco buys power from gencos and sells it directly to the consumers.

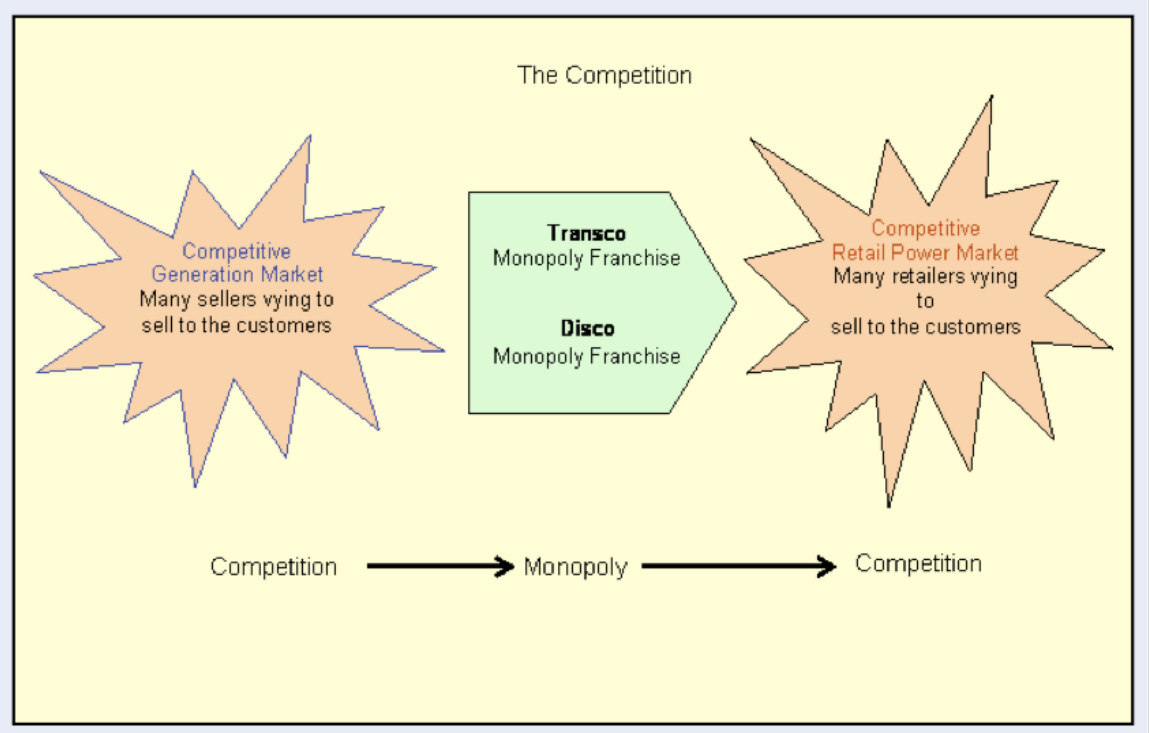
■ **ISO** **(Independent System Operator)**	The ISO is an entity entrusted with the responsibility of ensuring the reliability and security of the entire system. It is an independent authority and does not participate in the electricity market trades. It usually does not own generating resources, except for some reserve capacity in certain cases. In order to maintain the system security and reliability, the ISO procures various services such as supply of emergency reserves, or reactive power from other entities in the system.
■ Customers	A customer is an entity, consuming electricity. In deregulated markets, a customer has several options for buying electricity. It may choose to buy electricity from the spot market by bidding for purchase, or may buy directly from a genco or even from the local distribution company.

The competition :

In a deregulated environment, two levels of competition exist or are encouraged. At what can be termed as wholesale level, gencos produce and sell bulk quantities of electric power. Power is typically sold in bulk quantities to other companies or very large industrial customers, through some deregulated power market mechanism. The gencos bid their power at the marketplace so as to maximize their profits.

Locally, retail delivery is accomplished by retailers, who compete for the business of the consumers in the area by offering low price, good service and additional service features.

Thus, a restructured, completely competitive electric industry is a sandwich of competition above and below a power delivery system. This structure can be conveniently divided into wholesale and retail levels. The important thing to note is that the power delivery i.e. transmission and distribution remains a monopoly. This is shown in the figure below.



Ancillary Service Management

Ancillary services are defined as all those activities on the interconnected grid that are necessary to support the transmission of power while maintaining reliable operation and ensuring the required degree of quality and safety. This requires that a independent system operator should have adequate leverage to implement "normal" functions like frequency and voltage regulation, but also carry out actions of preventive , emergency and restorative control actions as discussed in module 6.
In deregulated power systems, transmission networks are available for third party access to allow power wheeling, and spot markets for electricity have been developed in many countries. In such an environment, **ancillary services are no longer treated as an integral part of the electric supply**. They are unbundled and priced separately and system operators have to purchase ancillary services from ancillary service providers. The following are some examples of anciliary services which are procured and mamaged by an ISO.

Recap

In this lecture you have learnt the following

- Structure of a deregulated industry.
- Different entities in a deregulated industry
- Ancilliary Service Management

Lecture 34 : Indian Scenario

Objectives

In this lecture you will learn the following

- Indian Scenario
- Electricity Act 2003

Indian Scenario: Ownership Issues

In India, the power sector was mainly under the government ownership (>95% distribution & ~98% generation) under various states and central government utilities, till 1991. The remarkable growth of physical infrastructure was facilitated by four main policies:

1) centralized supply and grid expansion
2) large support from government budgets
3) development of sector based on indigenous resources
4) cross subsidy

Cross-Subsidization means that a certain class of customers are charged higher prices for energy usage, while another class is charged less. This is done with a social objective in mind

In mid 1990s, Orissa began a process of fundamental restructuring of the state power sector. This consisted of a three pronged strategy of:

1) Unbundling the integrated utility in three separate sectors of generation, transmission and distribution,
2) Privatization of generation and distribution companies and,
3) Establishment of independent regulatory commissions to regulate these utilities. Meanwhile, some moderate steps were taken towards reforms until the Electricity Bill 2003 was approved by Parliament in May 2003.

The Electricity Act 2003:

The conceptual framework underlying this new legislation is that the electricity sector must be opened for competition. The Act moves towards creating a market based regime in the power sector. The Act also seeks to consolidate, update and rationalize laws related to generation, transmission, distribution, trading and use of power. It focuses on:

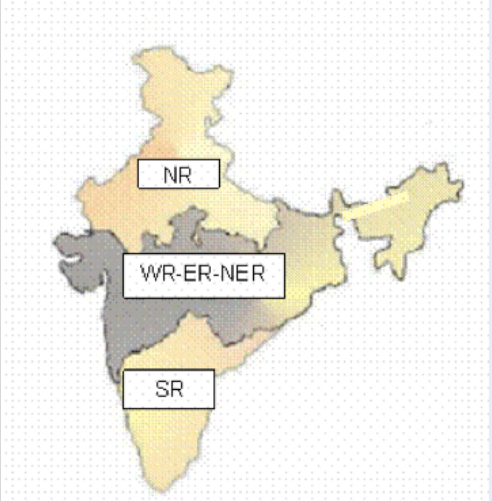
- Creating competition in the industry
- Protecting consumer interest
- Ensuring supply of electricity to all areas
- Rationalizing tariff
- Lowering the cross-subsidization levels.

Some of the major provisions of the Electricity Act are:

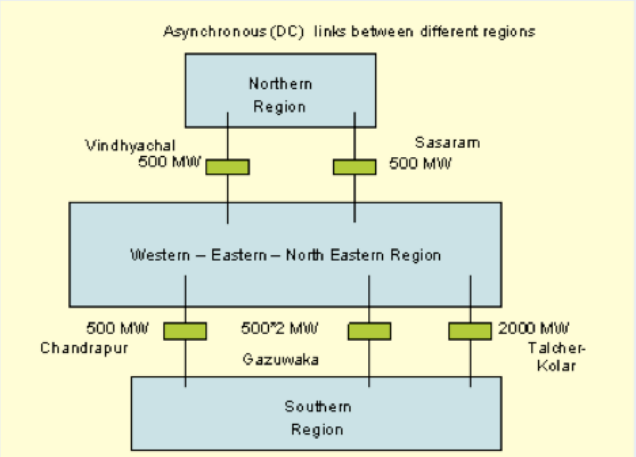
- Elimination of licensing for setting up a generating station, subject to compliance with technical standards. This excludes Hydro-Electric power station
- Removal of captive power plants from the ambit of licensing and other permissions
- Provision for issuing more than one license for transmission and distribution in the same geographical area.
- Provision of 'Open Access' with respect to transmission for all generators (subject to technical constraints)
- Introduction of a spot market for bulk electricity
- Unbundling of the SEBs on the basis of functions (Generation, Transmission and Distribution)
- Compulsory metering of all consumers in order to improve accountability
- State Governments will have the freedom to decide the sequence and phases of restructuring, and also retain the integrated structure of the SEB for a limited period.

Indian Scenario: Power Transmission System

As mentioned in the first Module, the Indian Power System is made of 3 synchronous grids, the Northern Region, Western-Eastern-North Eastern Region (W-E-NE R), and the Southern Grid. The installed capacity is greater than 110 GW and a demand of 70 GW is met (2005). The synchronous regions are interconnected by asynchronous (HVDC) ties. Therefore, it is possible to operate the three grids independently since the power through the HVDC links can be independently controlled.



Synchronous Grids in India (2005)



All DC links as shown in the figure are back to back, except for the one at Talcher-Kolar, which is a long HVDC line.

Most of the power generation is likely to come up in the coal rich central-eastern region and the hydro-potential rich north eastern region. Therefore adequate capacity transmission is necessary to transmit this power to the load centres.

Plan for a National Grid

The plan for formation of national grid includes synchronous interconnection of the Northern Region with the W-E-NE-Region around 2006. Synchronous operation will allow for subsequent interconnection at several other locations using AC lines, thereby increasing the power exchanges between regions. Of course, one may increase the number and capacity of asynhronous links and achieve the same thing. Eventually, it is a tradeoff between the technical considerations and costs.

What are the challenges in the integration of 2 large systems synchronously ? Among the problems which one will encounter are control of power flows between the interconnected regions and ensuring the maintainance of synchronism between interconnected generators subsequent to disturbances.

Challenges of Synchronous Operation : A practical example of emergency control

The Talcher - Kolar HVDC link is a high capacity link which transfers power from Talcher in the Eastern Region to Kolar in the Southern Region. In case the link trips, the W-E-NE Region is left with surplus power. Most of the surplus power rushes towards the western region (which has a large load) through relatively weak AC tie lines. The ensuing power swings (relative motion between generators) are not stable and the system separates into two regions due to loss of synchronism.

Therefore, in order to prevent ER-WR system separation, the following emergency scheme was concieved:

In case due to an unexpected contingency, the heavily loaded HVDC link trips, *then a few generators at Talcher are also tripped* immediately. This ensures that the W-E-NER system is not left with too much surplus power.

Note that the other asynchronous links to the Southern Region (e.g. the one at Gazuwaka) are of limited capacity and *cannot* suddenly push the excess power to the Southern Region unless they are lightly loaded. Moreover power flow through these links is *regulated*. Therefore, power can be ramped *slowly* by manual intervention. However, a strategy specifically suited to deal with this situation can be concieved.

The proposed plan for synchronous operation of NR and WR-ER-NER brings out interesting questions:

What will be the role of the existing back-to-back HVDC links (which asynchronously interconnect the two systems at present), when NR and WR-ER-NER systems are synchronized ?

Can the back-to-back HVDC links regulate the *total* power exchange between the two regions once they are synchronized ?

Recap

In this lecture you have learnt the following

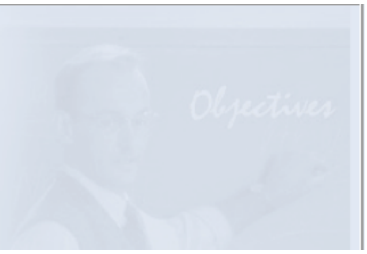
- Indian Scenario
- Electricity Act 2003

Lecture 35 : Concluding Notes

Objectives

In this lecture you will learn the following

- A summary of what we learnt in the entire course
- What are future challenges and trends in power system operation and control ?



We have come to the concluding part of this course. We started out in this course by reviewing the basic power system structure, operational objectives and the ways to achieve them.

We re-iterate the basic objectives here in a concise sentence

A power system should supply *quality* power, on demand, to a consumer *economically* and with minimal ecological impact.

Quality power implies that a consumer is given a supply with a practically constant voltage magnitude and frequency. The supply is also reliable, which means that barring a very large (and rare) disturbance or a sequence of disturbances, "the lights are always on".

The power system operation is controlled by a mix of automatic feedback and manual control actions. Real time operation of a power system is monitored by a "system operator" who has several "schedulable" quantities at his disposal. These quantities are scheduled to maximize economic benefit for a consumer and also to ensure security.

So is there any thing more to be "done" ? Yes. Although we have learnt all about operational and control objectives, there is scope to improve the way a system is operated. With concerns about ecological impact, it has become difficult (and expensive) to build new transmission lines and add generation. As a result, the focus is to utilize our resources even more optimally, while still satisfying all constraints. With changes in power system structure ("de-regulation") from a vertically integrated utility structure to a market based one, there is a greater need for flexibility to accomodate "power transactions" and also ensure fair access of common resources (such as the transmission network) for all players.

These challenges can be tackled with newer technologies -- we will discuss some of these soon -- and by better analytical techniques.

Some New Technologies

Often new ideas which have been proposed either recently or long before become viable economically due to advances in materials and manufacturing techniques. Amongst the recent technologies which are likely to impact power system *operation and control* are:

a) **Wide Area Measurement Systems (WAMS):** We introduced Wide Area Measurement Systems in Module 6. This communication technology can allow for fast control. Most automatic controllers in a power system use locally measured feedback systems for control, e.g., Automatic Voltage Regulators and Governors. In fact, AGC, which uses feedback signal from the tie -lines to control generation, is the **only** control which uses non-local measurements. AGC itself is a slow control (it acts over several minutes) and therefore the communication requirements are quite modest. It is expected that fast wide area measurements may allow for superior emergency control schemes which need to be executed in a very short time.

b) **Power Electronics :** Although HVDC has been around for a fairly long time, new converter topologies using voltage source converters offer a promise of increased flexibility in control. These converters use devices (like Gate Turn-off Thyristors) which can be turned off by a gate signal (unlike thyristors). This implies that there is an extra degree of freedom in their control. Voltage source converters as well as variable impedance controllers based on thyristors, like TCSC, which we learnt in Module 4, are now commercially available.

Can you do a survey of recent literature on both these technologies ?

Needless to add, there are and will be several other technologies which can revolutionalize the way power systems are operated. However, along with new technologies, there is a need to develop appropriate analytical tools to *apply* these technologies optimally.

Analysis Tools

A major challenge in future power system operation and control is the development of appropriate analytical tools to apply the new technologies and also take into account power system restructuring. The challenge is to also utilize the existing resources optimally.

Some of the problems that require good analytical tools (to aid a system operator) are as follows:

a) To determine whether or not the system is operating in a secure condition.

b) What are the optimal preventive control actions if one is insecure?

c) With the advent of WAMS, how does one use it to do emergency control ? How to design emergency control logic/schemes like islanding, generation tripping etc.?

d) How does one use the extra transient margins of existing controllers in the system including HVDC and other power electronics based controllers for improving transient behaviour?

e) What is the optimal sequencing of restorative actions to minimize restoration time ?

f) What is the fair method of allocation and pricing of common resources such as a transmission system capacity in the deregulated power system structure ?

g) What are the optimal operating strategies for individual entities in a deregulated environment ?

Although computational power has increased over the years, most of the above problems are difficult to solve due to the large size of power systems.

Power Systems Operation and Control- The future

It is unlikely that for the coming few decades, the interconnected grid structure of a power system will change significantly. It is possible that a large number of smaller generators - using gas, solar energy, wind energy - will get added on at lower voltage levels (distributed generation) depending on the availability of technologies which will make building and operating smaller generators more economical.

Power system operation and control will become more complex, although it will be aided by technologies like power electronics, computers and communications. Utilization of existing resources more optimally would imply that operating conservatively with large security margins will be a thing of the past. This also means that system operation will have to rely more on automation than on operator experience and intuition.

There are exciting times ahead !

Recap

In this lecture you have learnt the following

- A summary of what we learnt in the entire course
- What are future challenges and trends in power system operation and control ?

Congratulations, you have finished Lecture 36, Module7 and the course !

I hope you have enjoyed it!